

Fixed-Space Prime Compatibility in Burnol's Sonine Quotients

Complete working manuscript, version 1.26
Finite sampling, endpoint recovery, and the remaining graph defect

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September 21, 2026

Abstract

We study fixed-space prime actions on Burnol's Sonine quotients and finite Weil forms. Source covariance, compact Mellin division and Fredholm reduction identify arithmetic boundary and range defects. For the repaired fixed-order prolate source, tail, endpoint, correlation and domain estimates establish weak G1. Its polynomial Fourier projection recovers the physical endpoint and has an exponentially small ordinary operator residual. These estimates do not identify the lowest eigenspace or control a growing source block.

The centered Hardy sampler with an endpoint shell preserves ordinary Gram and Weil-form limits while retaining an explicit graph defect. Reflected tests and sharp cuts expose endpoint and complex-strip obstructions. A finite-core metric criterion bypasses the global arithmetic operator realization. Full-space projection preserves strip-product bounds and Weil transfer, but its signed commutator remains uncontrolled.

For the canonical semilocal Weil operator, signed Schur certificates prove complete positivity and ground ordering at $\lambda = 3$. At $\lambda = 4$ the entire tail $|n| > 16$ is bounded below by its positive archimedean energy. A structured inverse estimate retains this certified tail metric in the low-mode correction, with complete remote-row and dimension-aware error bounds. Finite-support trials certify a zero head-error term at this window. Parity signs sharpen the complete inverse iteration, with finite-prefix bounds retaining all remote rows. Complete mixed-term estimates rule out the first inverse degree with the saved head certificate, including its positive update. Retaining that update gives a monotone family of complete inverse upper bounds for subsequent degrees. Energy-scaled frozen trials now certify positivity of the entire even form at this window, with all residual cross-Grams and every omitted row retained. A complete odd-complement certificate and a repaired single direction now prove positivity of the entire form at $\lambda = 4$, with exact fixed-window error $\varepsilon_4 = 0$. The present scalar far majorant requires exponentially growing cutoffs as the window grows. Strong cancellation persists in the exact source complement. Every complete eigenfunction has zero physical endpoint. A cofinal complete-Weil lower error tending to zero would imply RH directly, but remains unproved.

Status of this version. Version 1.26 proves positivity of the complete Weil form at $\lambda = 4$. A complete 15-dimensional complement certificate, a repaired odd trial supported through 8192, and a sharp matrix inequality close the remaining odd-sector sign. The frozen directional witness is verified at 1024 and 1280 bits with every omitted Fourier row bounded. Together with the previous even certificate this gives the exact fixed-window error $\varepsilon_4 = 0$, not a cofinal estimate. Weak G1, all 72 historical dispositions, the physical endpoint and the explicit sampler graph defect are retained. The independent arithmetic estimate as $\lambda \rightarrow \infty$ remains open; G2 and RH are not proved.

Reading conventions. The Hermitian product is linear in the first variable: $\langle x, cy \rangle = \bar{c} \langle x, y \rangle$. In the Sonine development the multiplicative cutoff is $R > 1$. In the finite Weil development the multiplicative cutoff is $\lambda > 1$ and its logarithmic length is $L = 2 \log \lambda$. The source graph functional η_B , the equal-coefficient boundary functional η_∂ , and physical evaluation $\delta = L^{-1/2} \eta_\partial$ are different objects. The smoothed Cauchy value is $\mathcal{E}(F) = \sigma F(\sigma)$. An off-line root space is used only under a contradiction hypothesis; no off-line zero is asserted to exist. Independence from a zero list is a property of a construction, not a substitute for proving its estimates.

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1 Objective and correction

Fix a prime number p and write

$$\lambda_\rho = p^{1/2-\rho}.$$

The target is a fixed positive Hilbert-space action that retains this multiplier on the arithmetic modes attached to the nontrivial zeros, with positivity or contractivity proved independently of their locations.

A convention correction is essential. Burnol's Z_w is defined by a complex bilinear pairing, whereas the project uses a positive Hermitian product linear in the first variable. If k_w denotes the Hermitian Riesz vector, then

$$k_w = Z_{\bar{w}}.$$

The project mode is therefore

$$v_\rho^R = k_{\bar{\rho}}^R = Z_\rho^R,$$

not Z_ρ^R . With this convention the original compression has the correct critical-line phase. The historical reciprocal-phase rejection resulted from a second, erroneous conjugation.

2 Burnol framework

Work in

$$K = L^2((0, \infty), dx),$$

with Hermitian inner product linear in the first slot. Let

$$If(x) = x^{-1}f(1/x), \quad G = I\mathcal{F}_+I,$$

where

$$(\mathcal{F}_+f)(x) = 2 \int_0^\infty \cos(2\pi xt) f(t) dt$$

is the unitary cosine transform, initially on integrable square-integrable functions and then by its L^2 extension. The Mellin conventions are

$$M_0f(s) = \int_0^\infty f(x)x^{s-1} dx, \quad c_0(s) = \pi^{-s/2}\Gamma(s/2), \quad Mf(s) = c_0(s)M_0f(s).$$

These formulas are first used where the integrals converge and elsewhere by the Sonine continuation. In particular $M(E\psi) = c_0\zeta M_0\psi$. The evaluation and division properties of the completed Sonine space are the framework inputs from [1, 2, 3]. They do not include the additional quantitative assumptions below. For $R > 1$ define

$$H_R = \{f : \text{supp } f, \text{supp } Gf \subset (0, R]\},$$

$$V_R = C_c^\infty((1/R, R)), \quad \mathcal{D} = x \frac{d^2}{dx^2} x,$$

and

$$E(\psi)(x) = \sum_{n \geq 1} \psi(nx) - \frac{1}{x} \int_0^\infty \psi(t) dt.$$

The arithmetic co-Poisson space and quotient are

$$W_R = \overline{\{E(\mathcal{D}\phi) : \phi \in V_R\}}, \quad Q_R = H_R \ominus W_R.$$

Let S_R and P_R be the orthogonal projections onto H_R and Q_R respectively.

For a nontrivial zero ρ , Burnol's construction gives a nonzero vector

$$v_\rho^R = Z_\rho^R = k_{\bar{\rho}}^R \in Q_R.$$

Put

$$v = v_\rho^R, \quad v_+ = v_\rho^{pR}, \quad b = v_+ - v.$$

Because the ambient Sonine spaces are nested and the two vectors represent the same evaluation on H_R ,

$$S_R v_+ = v, \quad b \perp H_R, \quad \|b\|^2 = \|v_+\|^2 - \|v\|^2. \quad (1)$$

The unitary dilation is

$$(D_p f)(x) = p^{-1/2} f(x/p),$$

and the completed Mellin transform obeys

$$M(D_p f)(s) = p^{s-1/2} M(f)(s).$$

Quantitative evaluator inputs and compact division

Assumption 2.1 (Evaluator estimates used in the asymptotic sections). For each fixed critical-line parameter w , the chosen completed-Mellin Riesz vectors obey $\|v_w^R\|^2 = C_w \log R + O_w(1)$, $C_w > 0$. For fixed distinct critical parameters their cross inner product is $O_{w,z}(1)$. For a fixed off-line parameter with $d = |\Re w - 1/2|$, $\|v_w^R\| \asymp_w R^d$. For each fixed jet order m , the corresponding ambient jet Gram matrix $V_{w,R}$ satisfies $\det V_{w,R} \ll_{w,m} R^{2md}$. Where raw representatives are used below, their local expansion and Sonine projection error are assumed to have the stated uniform bounds, including the $O_w(R^{-1})$ projection correction.

These quantitative statements were attributed to earlier calculations in the previous draft without a complete derivation in the supplied files. They are not consequences of completeness or minimality alone. Sections using their rates are conditional on this assumption. Exact compression, covariance, and projection identities do not require it. Proposition 8.2 below proves the off-line *upper* norm bound directly and uses only nested nonzero evaluation for a lower bound. It does not prove the matching asymptotic, the critical-line estimates, or the projection-error assertions above.

Lemma 2.2 (Compact Mellin division). *Let $\theta \in C_c^\infty((1/R, R))$, let $\Theta = M_0 \theta$, and let $w \notin \{0, 1\}$. There exists a source η in the same compact support interval with $M_0 \eta(s) = \Theta(s)/(s-w)^r$ if and only if Θ vanishes to order at least r at w . If $\Theta(0) = \Theta(1) = 0$, the divided source has the same two moments. Moreover $\mathcal{D}V_R$ is exactly the compact smooth source class with these two moments zero.*

Proof. For one division choose b below the support and set

$$\eta(x) = -x^{-w} \int_b^x t^{w-1} \theta(t) dt.$$

The function vanishes below the support. It vanishes above it precisely when $\Theta(w) = 0$, and then is smooth and compactly supported in the same interval. It solves $-x\eta' - w\eta = \theta$. Integration by parts gives $(s-w)M_0\eta = \Theta$. Repeat the argument for higher order; the successive removability conditions are exactly the r vanishing jets. Necessity follows because a compact source has an entire Mellin transform. Evaluation at 0 and 1 proves moment preservation when $w \notin \{0, 1\}$. Finally $M_0(\mathcal{D}\phi) = s(s-1)M_0\phi$. Conversely divide first by s and then by $s-1$ using the two zero moments and the same compact construction. The resulting ϕ satisfies $\mathcal{D}\phi = \theta$. $\square$

Lemma 2.3 (Finite-jet realization and passage to the closed kernel). *Fix $w \notin \{0, 1\}$ and a finite order m . Admissible compact sources realize arbitrary jets $(\Psi(w), \dots, \Psi^{(m-1)}(w))$. If their associated functionals extend continuously to W_R , the intersection of their kernels in the source core is dense in the intersection of their kernels in W_R .*

Proof. For $\psi = \mathcal{D}\phi$, multiplication by $s(s-1)$ is an invertible triangular map on the jet at w . The distributions $x^{w-1}(\log x)^j$, $0 \leq j < m$, are linearly independent on every open source interval. Hence the jet map from compact smooth ϕ is onto: otherwise a nonzero linear combination of those distributions would annihilate all test functions. Choose core vectors $u_1, \dots, u_m$ whose joint jets are the coordinate vectors. If f_n are core approximants to a vector f with zero jet, replace them by $f_n - \sum_j \ell_j(f_n)u_j$. Continuity makes the correction tend to zero and each corrected approximant has zero jet. $\square$

3 The corrected fixed compression

Define

$$T_{R,p} = P_R D_p|_{Q_R}.$$

It is a contraction independently of RH. The corrected exact boundary identity is

$$(T_{R,p} - \lambda_\rho I)v_\rho^R = -P_R D_p(v_\rho^{pR} - v_\rho^R). \quad (2)$$

Indeed the first-slot Riesz convention gives

$$S_R D_p v_\rho^{pR} = \lambda_\rho v_\rho^R,$$

and (2) follows from (1).

Under Assumption 2.1, the critical-line evaluator norm law is

$$\|v_\rho^R\|^2 = C_\rho \log R + O_\rho(1), \quad C_\rho > 0.$$

Hence (1) gives $\|b\| = O_{\rho,p}(1)$ and

$$\frac{\|(T_{R,p} - \lambda_\rho I)v_\rho^R\|}{\|v_\rho^R\|} = O_{\rho,p}((\log R)^{-1/2}). \quad (3)$$

Thus every fixed critical-line mode is asymptotically retained with the correct multiplier. The all-zero version of this estimate remains unproved and is RH-strength.

4 New source covariance at adjacent scales

The next lemma is a project deduction from the exact co-Poisson identities. It does not use a zero location.

For a source function define

$$(\tau_p \phi)(x) = \phi(x/p).$$

A direct calculation gives

$$\mathcal{D}\tau_p = \tau_p \mathcal{D}, \quad D_p E(\psi) = p^{-1/2} E(\tau_p \psi). \quad (4)$$

Consequently

$$D_p W_R \subset W_{pR}. \quad (5)$$

Also $V_R \subset V_{pR}$, so $W_R \subset W_{pR}$. Burnol's identity $GE = EI$, together with commutation of I and $\mathcal{D}$ on the compact source domain, gives

$$GW_{pR} = W_{pR}.$$

Theorem 4.1 (Three-piece arithmetic source decomposition). *If $R^2 > p$, then*

$$W_{pR} = \overline{W_R + D_p W_R + GD_p W_R}. \quad (6)$$

Since $GW_R = W_R$ and $GD_p = D_{1/p}G$, the last summand may equivalently be written $D_{1/p}W_R$.

Proof. The inclusion from right to left follows from (5), $W_R \subset W_{pR}$, and G -invariance of W_{pR} .

For the reverse inclusion, the three source intervals

$$[1/(pR), R/p], \quad [1/R, R], \quad [p/R, pR]$$

cover $[1/(pR), pR]$ when $R^2 > p$. A smooth partition of unity therefore splits every $\phi \in V_{pR}$ as

$$\phi = \phi_- + \phi_0 + \phi_+,$$

where $\phi_0 \in V_R$, $\phi_+ = \tau_p \psi_+$ for some $\psi_+ \in V_R$, and $\phi_- = I\tau_p \psi_-$ for some $\psi_- \in V_R$. Applying $E\mathcal{D}$, using (4) and $GE = EI$, places the three pieces in W_R , $D_p W_R$, and $GD_p W_R$ respectively. Taking closures proves (6). $\square$

Corollary 4.2 (Arithmetic projection disappears in two adjoint channels). *Assume $R^2 > p$ and let $y \in Q_{pR}$. Then*

$$P_R y = S_R y, \quad P_R D_p y = S_R D_p y, \quad P_R D_{1/p} y = S_R D_{1/p} y. \quad (7)$$

Proof. Since $Q_{pR} \perp W_{pR}$, Theorem 4.1 gives $y \perp W_R$, $y \perp D_p W_R$, and $y \perp D_{1/p} W_R$. These imply respectively that $S_R y$, $S_R D_{1/p} y$, and $S_R D_p y$ are orthogonal to W_R . Each therefore lies in Q_R , yielding (7). $\square$

5 A quotient-native prime operator pencil

Define two contractions from the old quotient into the enlarged quotient,

$$J_{R,p} = P_{pR}|_{Q_R} : Q_R \rightarrow Q_{pR},$$

$$B_{R,p} = P_{pR} D_{1/p}|_{Q_R} : Q_R \rightarrow Q_{pR}.$$

Their adjoints are

$$J_{R,p}^* = P_R|_{Q_{pR}}, \quad B_{R,p}^* = P_R D_p|_{Q_{pR}}.$$

For the zero vector $v_+ = v_\rho^{pR}$, nested evaluation and the corrected boundary calculation give the exact identities

$$J_{R,p}^* v_+ = v, \quad B_{R,p}^* v_+ = \lambda_\rho v. \quad (8)$$

Equivalently,

$$(B_{R,p}^* - \lambda_\rho J_{R,p}^*) v_\rho^{pR} = 0. \quad (9)$$

Thus the arithmetic prime multiplier already appears as an exact generalized eigenvalue of a fixed, zero-independent operator pencil. The unresolved issue is converting this generalized eigenrelation into a single positive fixed-space action without inserting the zero location by hand.

6 Exact defect kernel and a sufficient fixed-space theorem

Define on Q_{pR}

$$\Delta_{R,p} = J_{R,p}J_{R,p}^* - B_{R,p}B_{R,p}^*. \quad (10)$$

This is self-adjoint. From (8), for every nontrivial zero,

$$\langle \Delta_{R,p} v_\rho^{pR}, v_\rho^{pR} \rangle = (1 - |\lambda_\rho|^2) \|v_\rho^R\|^2. \quad (11)$$

More generally, for zeros ρ, σ ,

$$\langle \Delta_{R,p} v_\sigma^{pR}, v_\rho^{pR} \rangle = (1 - \lambda_\sigma \overline{\lambda_\rho}) \langle v_\sigma^R, v_\rho^R \rangle. \quad (12)$$

This is a Pick-type difference-of-Gram kernel on the zero-evaluation system.

Proposition 6.1 (Finite-cutoff span positivity would imply RH). *If, for one fixed $p > 1$ and admissible cutoffs, $\Delta_{R,p}$ were nonnegative on the span of the zero-evaluation vectors v_ρ^{pR} , then every nontrivial zero would satisfy $|\lambda_\rho| \leq 1$. Functional-equation symmetry would then imply RH.*

Proof. Apply nonnegativity to a single vector in (11). Since $v_\rho^R \neq 0$, $1 - |\lambda_\rho|^2 \geq 0$. Hence $p^{1/2-\Re\rho} \leq 1$, or $\Re\rho \geq 1/2$. The functional equation supplies the reflected zero $1 - \rho$, forcing equality. $\square$

The span-positivity hypothesis is substantially stronger than RH and should not be treated as the main target. Indeed, assume RH for the moment. Then every diagonal quantity in (11) is zero. If $\Delta_{R,p}$ were positive semidefinite on the whole zero span, positivity would force each v_ρ^{pR} to lie in the nullspace of the restricted form. Hence for all critical zeros ρ, σ ,

$$(1 - \lambda_\sigma \overline{\lambda_\rho}) \langle v_\sigma^R, v_\rho^R \rangle = 0. \quad (13)$$

Thus distinct prime phases would require the corresponding evaluation vectors to be orthogonal. Burnol's zero systems are studied as complete/minimal systems, not supplied here as an orthogonal basis. Therefore RH by itself does not furnish the extra cross-term vanishing in (13). The safe target is the *diagonal* sign on each zero evaluator, not positivity of the full Pick matrix.

For the same reason, the stronger global operator inequality

$$B_{R,p}B_{R,p}^* \leq J_{R,p}J_{R,p}^* \quad (14)$$

should be regarded only as an over-strong sufficient condition. Were it true, the Douglas factorization lemma would produce a contraction $C_{R,p} : Q_R \rightarrow Q_R$ with $B_{R,p} = J_{R,p}C_{R,p}$ and hence $C_{R,p}^* v_\rho^R = \lambda_\rho v_\rho^{pR}$. Nothing in the present work proves the cross-term conditions required for such a global factorization, even under RH.

7 The defect becomes a pure Sonine comparison

Theorem 4.1 removes the arithmetic projection from the defect form. For $y \in Q_{pR}$ and $R^2 > p$, Corollary 4.2 gives

$$J_{R,p}^* y = S_R y, \quad B_{R,p}^* y = S_R D_p y.$$

Therefore

$$\langle \Delta_{R,p} y, y \rangle = \|S_R y\|^2 - \|S_R D_p y\|^2. \quad (15)$$

The diagonal sign problem has thus been reduced to the following concrete question:

For each zero evaluator $y = v_\rho^{pR}$, can an independent arithmetic estimate prove $\|S_R D_p y\| \leq \|S_R y\|$?

A proof of

$$\|S_R D_p y\| \leq \|S_R y\|$$

on each zero evaluator would prove RH by (11). Positivity on the whole span is a stronger sufficient condition, as explained in Proposition 6.1. A proof on all of Q_{pR} would give the stronger Douglas factorization above.

This is the current fixed-space compatibility target. It is more explicit than the earlier boundary-distance statement because the co-Poisson projection has disappeared from the quadratic form exactly. No sign for (15) has yet been proved.

8 Exact Fredholm reduction to a boundary-shell imbalance

We now make (15) explicit using Burnol's finite-interval Fredholm projection. Put

$$a = R^{-1}, \quad q = Iy \in K_{a/p},$$

where $If(x) = x^{-1}f(1/x)$ and $K_{a/p}$ is the cosine-Sonine space with gap a/p . Let $\mathcal{F}_+$ be the unitary cosine transform, let P_a denote restriction/projection to $(0, a)$, and set

$$F_a = P_a \mathcal{F}_+ P_a, \quad D_a = F_a^2.$$

Under inversion, S_R becomes the orthogonal projection π_a onto K_a , while $ID_p = D_{1/p}I$. Hence

$$\|S_R y\|^2 - \|S_R D_p y\|^2 = \|\pi_a q\|^2 - \|\pi_a D_{1/p} q\|^2. \quad (16)$$

For a general $f \in L^2(0, \infty)$ define

$$u = P_a f, \quad v = P_a \mathcal{F}_+ f.$$

Burnol's projection theorem (equivalently the 2×2 Fredholm inverse in Lemma 2 of [2]) gives the exact lost energy

$$\mathcal{E}_a(f) := \|f - \pi_a f\|^2 \quad (17)$$

$$= \frac{1}{2} \langle u + v, (I + F_a)^{-1}(u + v) \rangle + \frac{1}{2} \langle u - v, (I - F_a)^{-1}(u - v) \rangle. \quad (18)$$

In particular $\mathcal{E}_a(f) \geq 0$ and

$$\|\pi_a f\|^2 = \|f\|^2 - \mathcal{E}_a(f).$$

Since dilation is unitary, (16) becomes

$$\|S_R y\|^2 - \|S_R D_p y\|^2 = \mathcal{E}_a(D_{1/p} q) - \mathcal{E}_a(q). \quad (19)$$

The Sonine condition $q \in K_{a/p}$ now produces a one-channel collapse after dilation. Indeed $\mathcal{F}_+ q$ vanishes on $(0, a/p)$ and

$$\mathcal{F}_+ D_{1/p} = D_p \mathcal{F}_+,$$

so

$$P_a \mathcal{F}_+ D_{1/p} q = 0.$$

Writing

$$u = P_a q, \quad v = P_a \mathcal{F}_+ q, \quad u' = P_a D_{1/p} q,$$

we obtain exactly

$$\mathcal{E}_a(D_{1/p} q) = \langle u', (I - D_a)^{-1} u' \rangle. \quad (20)$$

Equations (18)–(20) are the desired finite- R Fredholm formula for the sign problem.

There is also a transparent shell approximation. The Hilbert–Schmidt estimate gives $\|F_a\| \leq 2a$. Consequently, for $a < 1/2$,

$$\left| \mathcal{E}_a(q) - (\|u\|^2 + \|v\|^2) \right| \leq \frac{2a}{1 - 2a} (\|u\|^2 + \|v\|^2), \quad (21)$$

$$\left| \mathcal{E}_a(D_{1/p} q) - \|u'\|^2 \right| \leq \frac{4a^2}{1 - 4a^2} \|u'\|^2. \quad (22)$$

Because q and $\mathcal{F}_+ q$ both vanish below a/p ,

$$\|u'\|^2 - \|u\|^2 = \int_a^{pa} |q(t)|^2 dt, \quad \|v\|^2 = \int_{a/p}^a |\mathcal{F}_+ q(t)|^2 dt.$$

Thus

$$\boxed{\|S_R y\|^2 - \|S_R D_p y\|^2 = \int_a^{pa} |q(t)|^2 dt - \int_{a/p}^a |\mathcal{F}_+ q(t)|^2 dt + R_{a,p}(q)}, \quad (23)$$

where, explicitly,

$$\begin{aligned} |R_{a,p}(q)| &\leq \frac{4a^2}{1 - 4a^2} \int_{a/p}^{pa} |q(t)|^2 dt \\ &\quad + \frac{2a}{1 - 2a} \left(\int_{a/p}^a |q(t)|^2 dt + \int_{a/p}^a |\mathcal{F}_+ q(t)|^2 dt \right). \end{aligned}$$

The sign problem has therefore been reduced, up to a quantitatively small Fredholm correction, to a signed imbalance between two adjacent multiplicative shells: a physical outer shell and a Fourier inner shell.

The arithmetic decomposition from Theorem 4.1 implies more than mere Sonine membership. For $y \in Q_{pR}$,

$$S_R y, \quad S_R D_p y, \quad S_R D_{1/p} y \in Q_R.$$

Equivalently, the shell pair in (23) comes from a vector orthogonal to the old, prime-expanded, and inverse-prime arithmetic source directions simultaneously. This is the precise additional arithmetic structure available for a future sign theorem.

What this does not prove. There is no generic Sonine inequality forcing the first shell integral in (23) to dominate the second. On a zero evaluator $y = v_\rho^R$ the exact dual identity already gives

$$\|S_R y\|^2 - \|S_R D_p y\|^2 = (1 - |\lambda_\rho|^2) \|v_\rho^R\|^2.$$

Hence positivity on all zero evaluators is itself equivalent to the required one-sided zero exclusion. The Fredholm reduction is useful because it identifies the missing mechanism as a sign-sensitive arithmetic shell theorem; it does not manufacture that theorem from generic uncertainty geometry.

8.1 Signed moments and an unconditional relative error budget

The $O(a)$ correction in (23) contains a signed cross term. It can be retained explicitly, and even resummed with all constant-kernel contributions, before estimating the error.

Lemma 8.1 (Signed Fredholm formula and rank-one resummation). *Let $0 < a < 1/2$ and use u, v, u' as above. Put $A_a = (I - F_a^2)^{-1}$ and*

$$H_a = \|u'\|^2 + \|u\|^2 + \|v\|^2, \quad \mathcal{B}_{a,p}(q) = \int_a^{pa} |q|^2 - \int_{a/p}^a |\mathcal{F}_+ q|^2.$$

For $\mathfrak{d}_a(q) = \mathcal{E}_a(D_{1/p}q) - \mathcal{E}_a(q)$,

$$\mathfrak{d}_a(q) = \langle u', A_a u' \rangle - \langle u, A_a u \rangle - \langle v, A_a v \rangle + 2\Re \langle u, F_a A_a v \rangle. \quad (24)$$

Define $m = \int_0^a u$, $n = \int_0^a v$, and $m' = \int_0^a u'$. Then

$$\mathfrak{d}_a(q) = \mathcal{B}_{a,p}(q) + \mathcal{C}_a(q) + \mathcal{R}_a(q), \quad (25)$$

$$\mathcal{C}_a(q) = \frac{4\Re(m\bar{n}) + 4a(|m'|^2 - |m|^2 - |n|^2)}{1 - 4a^2}, \quad |\mathcal{R}_a(q)| \leq \frac{4\pi^2 a^5}{5(1 - 2a)^2} H_a. \quad (26)$$

In particular, $m' = p^{-1/2} \int_{a/p}^{pa} q(t) dt$. These formulas require Sonine membership, not arithmetic orthogonality.

Proof. The identities $(I \pm F_a)^{-1} = A_a(I \mp F_a)$ in (18) give (24). Self-adjointness and commutation of F_a, A_a make the two mixed terms conjugate; they enter the *difference* with a plus sign.

Let $T_a f = 2 \int_0^a f(t) dt$, the constant-kernel operator on $(0, a)$. The actual cosine kernel gives

$$|2 \cos(2\pi x t) - 2| \leq 4\pi^2 x^2 t^2, \quad \|F_a - T_a\| \leq \|F_a - T_a\|_{\text{HS}} \leq \frac{4\pi^2}{5} a^5.$$

Both $\|F_a\|$ and $\|T_a\|$ are at most $2a$. On $L^2(0, a) \oplus L^2(0, a)$ set

$$K_F = \begin{pmatrix} I & F_a \\ F_a & I \end{pmatrix}, \quad K_T = \begin{pmatrix} I & T_a \\ T_a & I \end{pmatrix}.$$

The resolvent identity yields

$$\|K_F^{-1} - K_T^{-1}\| \leq \frac{4\pi^2 a^5}{5(1 - 2a)^2}.$$

Apply this bound separately to (u, v) and $(u', 0)$ in the lost-energy quadratic forms. With $e = 1_{(0, a)}$ and $e \otimes e : f \mapsto \langle f, e \rangle e$, one has

$$(I - T_a^2)^{-1} = I + \frac{4a}{1 - 4a^2} e \otimes e, \quad T_a(I - T_a^2)^{-1} = \frac{2}{1 - 4a^2} e \otimes e.$$

Substitution proves the stated moment formula and error. The final identity for m' follows from $D_{1/p}q(x) = \sqrt{p}q(px)$ and the gap below a/p . $\square$

Proposition 8.2 (Normalized shell reduction without evaluator asymptotics). *Fix a nontrivial off-line zero ρ , put $\beta = \Re \rho$, $d = |\beta - 1/2| \in (0, 1/2)$, and fix a prime p . For $y_R = v_\rho^{pR}$, $q_R = Iy_R$, $a = R^{-1}$ and $N_R = \|v_\rho^R\|^2$, the following estimates do not use Assumption 2.1:*

$$\frac{|\mathcal{R}_a(q_R)|}{N_R} = O_{\rho,p}(R^{2d-5}), \quad \frac{|\mathcal{C}_a(q_R)|}{N_R} = O_{\rho,p}(R^{2d-1}), \quad (27)$$

$$\frac{\mathcal{B}_{1/R,p}(q_R)}{N_R} = 1 - p^{1-2\beta} + O_{\rho,p}(R^{2d-1}). \quad (28)$$

Constants may also depend on a fixed initial cutoff with nonzero evaluation. In particular all displayed error terms tend to zero.

Proof. For $\Re s = b > 1/2$ and $f \in H_R$, support in $(0, R]$ and Cauchy–Schwarz give

$$|Mf(s)| \leq |c_0(s)| \left(\frac{R^{2b-1}}{2b-1} \right)^{1/2} \|f\|.$$

The Mellin cosine multiplier, with the left-Mellin convention of Section 2, gives

$$M_0(Gf)(s) = \frac{c_0(1-s)}{c_0(s)} M_0 f(1-s), \quad M(Gf)(s) = Mf(1-s).$$

These identities hold first on the Mellin line and extend by the Sonine continuation; see [4, §1, (7)], converting its right-Mellin convention to the present one. Since G is a unitary involution preserving H_R , the same bound on the reflected parameter applies when $b < 1/2$. The norm of the evaluator at $\bar{\rho}$ is thus

$$\|v_\rho^R\|^2 \leq C_\rho R^{2d}, \quad C_\rho = \begin{cases} |c_0(\rho)|^2/(2\beta-1), & \beta > 1/2, \\ |c_0(1-\rho)|^2/(1-2\beta), & \beta < 1/2. \end{cases}$$

Nested evaluation gives $N_R \geq N_{R_0} > 0$ for $R \geq R_0$; this uses only the nonzero evaluator and orthogonal projection, not a lower growth asymptotic. Also $H_a \leq 3 \|q_R\|^2 = 3 \|v_\rho^{pR}\|^2 \ll_{\rho,p} R^{2d}$. The remainder estimate follows. By Cauchy–Schwarz, $|m|^2 \leq a \|u\|^2$, and likewise for n, m' , whence

$$|\mathcal{C}_a(q_R)| \leq \frac{2a(\|u\|^2 + \|v\|^2) + 4a^2 H_a}{1 - 4a^2}.$$

This proves the second error estimate. Finally use the exact arithmetic diagonal identity (11) in (25). $\square$

The remaining arithmetic obligation, now with no Fredholm ambiguity. It suffices to fix $p = 2$ and prove independently, for each hypothetical zero with $\Re \rho < 1/2$, that

$$\liminf_{R \rightarrow \infty} \frac{\int_{1/R}^{2/R} |Iv_\rho^{2R}|^2 - \int_{1/(2R)}^{1/R} |\mathcal{F}_+ Iv_\rho^{2R}|^2}{\|v_\rho^R\|^2} \geq 0. \quad (29)$$

Indeed (28) would instead give a strictly negative limit, and reflection then excludes right-hand zeros too. Multiplicity does not alter this criterion, since the value evaluator already exists at every zero. No uniformity in ρ , no full-span positivity, and no matching evaluator lower asymptotic is

required. The rates are not uniform as the parameter approaches the critical line or strip edges. The three arithmetic source orthogonalities remain available, but neither they nor the signed moment formula prove (29). In particular the explicit cross term is asymptotically negligible on each fixed off-line evaluator; controlling it alone cannot settle the leading shell sign. This closes a normalization prerequisite of this route, not the signed G2 estimate or RH.

9 Projection-angle form of the same obstruction

There is a second exact reformulation that removes the Fredholm coordinates altogether. For any closed subspace $M \subset K$ and unitary U , the projection onto $U^{-1}M$ is $U^{-1}P_M U$. Hence

$$D_{1/p} S_R D_p$$

is the orthogonal projection onto $D_{1/p} H_R$. If $y \in Q_{pR}$, Theorem 4.1 gives $y \perp W_R$ and $y \perp D_{1/p} W_R$. Therefore

$$S_R y \in Q_R, \quad D_{1/p} S_R D_p y \in D_{1/p} Q_R.$$

The two vectors are in fact the orthogonal projections of y onto the indicated arithmetic quotient spaces. Thus

$$\boxed{\|S_R y\|^2 - \|S_R D_p y\|^2 = \|P_{Q_R} y\|^2 - \|P_{D_{1/p} Q_R} y\|^2.} \quad (30)$$

Here the second P denotes the ordinary Hilbert projection onto the closed subspace $D_{1/p} Q_R$.

Equation (30) is useful conceptually. The two comparison spaces Q_R and $D_{1/p} Q_R$ are unitarily congruent; neither is generically contained in the other. The missing sign is therefore not a consequence of dimension, positivity, or ordinary projection monotonicity. It asks for a genuinely prime-oriented arithmetic preference inside Q_{pR} .

For a zero evaluator $y = v_\rho^{pR}$ the two projections are explicit:

$$P_{Q_R} y = v_\rho^R, \quad P_{D_{1/p} Q_R} y = \lambda_\rho D_{1/p} v_\rho^R,$$

so (30) reduces again to

$$(1 - |\lambda_\rho|^2) \|v_\rho^R\|^2.$$

Thus any proof based only on the abstract fact that the two spaces are unitary images of one another cannot decide the sign. A successful argument must exploit how the co-Poisson arithmetic range sits relative to the dilation direction, not merely the ambient Sonine geometry.

The next concrete arithmetic object

Theorem 4.1 identifies the enlarged arithmetic space with the closure of the range of the synthesis map

$$\mathcal{A}_{R,p} : W_R \oplus W_R \oplus W_R \longrightarrow H_{pR}, \quad \mathcal{A}_{R,p}(w_0, w_+, w_-) = w_0 + D_p w_+ + D_{1/p} w_-.$$

Thus $W_{pR} = \overline{\text{Ran } \mathcal{A}_{R,p}}$; closed range is not asserted. Its Gram operator is the 3×3 Toeplitz-type block matrix

$$\mathcal{A}_{R,p}^* \mathcal{A}_{R,p} = \begin{pmatrix} I & C_p & C_{1/p} \\ C_p^* & I & C_{1/p^2} \\ C_{1/p}^* & C_{1/p^2}^* & I \end{pmatrix},$$

where, for example,

$$C_p = P_{W_R} D_p|_{W_R}.$$

The exact placement of adjoints depends only on the chosen first-slot convention. Positivity of this Gram matrix is automatic because it is a Gram operator; what is not automatic is the Schur-complement inequality needed to orient (30). This identifies a new, genuinely arithmetic target: obtain sign-sensitive control of the cross-scale correlation operators C_p and C_{p^2} from the co-Poisson source formula. Generic Hilbert-space positivity alone cannot provide it.

9.1 A convergent arithmetic adjoint for the shell constraints

To use the source ranges in a signed estimate, their infinite sums must first be paired legitimately with arbitrary L^2 vectors. The formal sum $\sum_{n \geq 1} y(t/n)/n$ need not converge. The following cutoff supplies an explicit renormalization in the conventions of Section 2. The integer M here is a co-Poisson summation cutoff, not a Fourier cut or the Hardy smoothing order used in later sections.

Lemma 9.1 (Integral-corrected source cutoff). *Let $h \in C_c^\infty((\alpha, b))$, $0 < \alpha < b$, put $V(h) = \int |h'|$, and define*

$$E_M h(x) = \sum_{n=1}^M h(nx) - \frac{1}{x} \int_0^{Mx} h(t) dt.$$

Then $E_M h = Eh$ for $x \geq b/M$, and

$$\|E_M h - Eh\|_2 \leq V(h) \sqrt{b/M}. \quad (31)$$

If $h \neq 0$ and $\int h = 0$, put $H(t) = \int_0^t h$. The uncorrected sum $U_M h = \sum_{n \leq M} h(nx)$ instead satisfies

$$\|U_M h - Eh\|_2 = \sqrt{M} \left(\int_\alpha^b \frac{|H(t)|^2}{t^2} dt \right)^{1/2} + O_h(M^{-1/2}). \quad (32)$$

The leading constant is strictly positive.

Proof. Subtract the defining co-Poisson counterterm to obtain

$$E_M h - Eh = - \sum_{n > M} h(nx) + \frac{1}{x} \int_{Mx}^\infty h(t) dt.$$

On each interval $((n-1)x, nx)$, the error between the right endpoint value and the integral average is at most the variation of h on that interval. Thus the displayed difference is bounded by $\int_{Mx}^\infty |h'| \leq V(h)$ and vanishes above b/M . This proves (31). The same quadrature bound shows Eh bounded near zero; above b it is $-x^{-1} \int h$, so $Eh \in L^2$. If $\int h = 0$, then H is supported in $[\alpha, b]$ and $U_M h - Eh = (E_M h - Eh) + H(Mx)/x$. Changing variables gives $\|H(Mx)/x\|_2 = \sqrt{M} \|H(t)/t\|_2$. The triangle inequality proves (32); its constant cannot vanish unless $h = H' = 0$. $\square$

Proposition 9.2 (Regularized adjoint and arithmetic source equation). *For $y \in L^2(0, \infty)$ set*

$$(\mathcal{A}_M y)(t) = \sum_{n=1}^M \frac{y(t/n)}{n} - \int_{t/M}^\infty \frac{y(x)}{x} dx. \quad (33)$$

There is a distribution $\mathcal{A}y \in H_{\text{loc}}^{-1}(0, \infty)$ such that $\mathcal{A}_M y \rightarrow \mathcal{A}y$ locally in H^{-1} , and

$$\langle E_M h, y \rangle = \int h(t) \overline{\mathcal{A}_M y(t)} dt, \quad \langle E h, y \rangle = \langle h, \mathcal{A}y \rangle, \quad (34)$$

$$|\langle h, (\mathcal{A}_M - \mathcal{A})y \rangle| \leq V(h) \sqrt{b/M} \|y\|_2. \quad (35)$$

The distribution pairing in the second slot extends the first-slot-linear Hermitian convention. For $y \in H_R$, one has the exact characterization

$$y \in Q_R \iff \mathcal{A}y(t) = A_y + B_y/t \quad \text{on } (1/R, R) \quad (36)$$

as distributions, for some constants A_y, B_y . For every dilation factor $c > 0$, $\mathcal{A}D_c = D_c \mathcal{A}$.

Proof. For finite M all sums may be integrated term by term. In the integral term, Fubini is justified by

$$\int_{\alpha/M}^{\infty} \frac{|y(x)|}{x} dx \leq \sqrt{M/\alpha} \|y\|_2.$$

Changing $t = nx$ in each summand and reversing the counterterm integral gives (34). Lemma 9.1 gives (35). Since $V(h) \leq \sqrt{b - \alpha} \|h'\|_2$, the differences are Cauchy in the dual of $H_0^1(\alpha, b)$, using its derivative norm. These local limits agree on overlaps and define $\mathcal{A}y$.

By the definition of W_R , arithmetic orthogonality is exactly $\langle \mathcal{D}\phi, \mathcal{A}y \rangle = 0$ for all $\phi \in C_c^\infty((1/R, R))$. The operator $\mathcal{D}\phi = (t^2 \phi')'$ is formally self-adjoint. Hence this is $(t^2(\mathcal{A}y)')' = 0$ in distributions. Integrating twice on the connected interval gives precisely $A_y + B_y/t$. The converse follows by integration by parts and passage to the closure defining W_R . Finally a change of variables in (33) gives $\mathcal{A}_M D_c = D_c \mathcal{A}_M$ exactly. Distributional passage to the limit proves the covariance assertion. $\square$

For the three-scale problem this becomes $\mathcal{A}y = A_y + B_y/t$ on $(1/(pR), pR)$, in addition to $y \in H_{pR}$. This makes all three source constraints explicit; it does not add three independent scalar conditions to that one local equation. In particular the two coefficients need not vanish. No pointwise or local L^2 convergence of $\mathcal{A}_M y$ is claimed for arbitrary y . There is also an explicit growing-interval budget. With $I_R = (1/(pR), pR)$ and the derivative norm on $H_0^1(I_R)$, (35) gives

$$\|(\mathcal{A}_M - \mathcal{A})y\|_{(H_0^1(I_R))'} \leq \sqrt{pR(pR - 1/(pR))/M} \|y\|_2.$$

For the fixed off-line evaluator $y = v_\rho^{pR}$, Proposition 8.2 therefore bounds this error divided by $\|v_\rho^R\|$ by $O_{\rho,p}(R^{1+d}/\sqrt{M})$. Choosing $M = \lceil R^4 \rceil$ makes it tend to zero. This controls source testing in the stated dual norm, not shell energies or a source Gram inverse; it does not change any later Fourier cutoff.

Corollary 9.3 (Controlled three-channel source Gram limits). *Fix finitely many smooth sources h_i supported in (α, b) , and write $w_i = E h_i$, $w_i^{(M)} = E_M h_i$ and $\epsilon_i = V(h_i) \sqrt{b/M}$. For $c, d \in \{1, p, 1/p\}$,*

$$\begin{aligned} & \left| \langle D_c w_i^{(M)}, D_d w_j^{(M)} \rangle - \langle D_c w_i, D_d w_j \rangle \right| \\ & \leq \epsilon_i \|w_j\| + \epsilon_j \|w_i\| + \epsilon_i \epsilon_j. \end{aligned}$$

Thus every fixed finite source-Gram matrix has a convergent arithmetic quadrature representation. No convergence of the associated inverse or uniform lower singular-value bound is implied.

Proof. Expand the difference into its two single-error terms and its double-error term, use unitarity of both dilations and apply Cauchy–Schwarz. $\square$

A check against a circular sign argument. For $1/2 < \Re w < 1$ and $y(x) = x^{w-1}1_{(0,B)}(x)$, which is in L^2 , the finite formula on $0 < t < B$ gives

$$\mathcal{A}_M y(t) = t^{w-1} \left(\sum_{n \leq M} n^{-w} - \frac{M^{1-w}}{1-w} \right) + \frac{B^{w-1}}{1-w}.$$

The elementary Euler-summation continuation of ζ therefore gives

$$\mathcal{A} y(t) = \zeta(w) t^{w-1} + \frac{B^{w-1}}{1-w}. \quad (37)$$

Indeed, comparison of n^{-w} with its integral over $(n, n+1)$ gives a locally uniform tail $O_w(M^{-\Re w})$ on $\Re w > 0$, $w \neq 1$; the limiting analytic function agrees with ζ on $\Re w > 1$ and hence throughout that domain. This is consistent with the Mellin co-Poisson identities in [4, 3]. At a hypothetical zero the nonconstant power disappears, but this only re-encodes the zero condition. The raw power is not asserted to satisfy the second Sonine support condition. Thus neither this scalar calculation nor positivity of the source Gram matrix proves the signed shell target (29). The new adjoint supplies a justified arithmetic constraint for a future quadratic estimate, not that estimate.

10 A local-prime sieve identity inside the three-scale range

The three-piece decomposition contains more arithmetic information than support covariance alone. Let $h = \mathcal{D}\phi$ with $\phi \in V_R$. Then $\int h = 0$, so the co-Poisson counterterm vanishes and

$$E(h)(x) = \sum_{n \geq 1} h(nx).$$

Since

$$p^{-1/2} D_{1/p} f(x) = f(px),$$

we obtain the exact identity

$$\boxed{\left(I - p^{-1/2} D_{1/p} \right) E(h)(x) = \sum_{\substack{n \geq 1 \\ p \nmid n}} h(nx).} \quad (38)$$

Thus the difference between the old arithmetic direction and its inverse-prime dilation is precisely the co-Poisson sum with the p -divisible lattice points removed.

On Mellin transforms, (38) is the familiar local Euler-factor cancellation

$$\widehat{E(h)}(s) \mapsto (1 - p^{-s}) \widehat{E(h)}(s) = \zeta(s)(1 - p^{-s}) \widehat{h}(s), \quad (39)$$

within the strip where the co-Poisson/Mellin identity is interpreted. The factor $\zeta(s)(1 - p^{-s})$ is the zeta Euler product with the local factor at p removed.

Applying the functional-equation involution G gives the companion direction

$$G \left(I - p^{-1/2} D_{1/p} \right) G = I - p^{-1/2} D_p,$$

whose Mellin factor is $1 - p^{s-1}$. On the critical line these two local factors are complex conjugates and have equal modulus; off the line their radial behavior differs. This is a genuine prime-local structure absent from a generic pair of Sonine subspaces.

The identity does not by itself prove the shell sign: $y \in Q_{pR}$ is already orthogonal to both W_R and $D_{1/p}W_R$, hence also to their p -sieved difference. Its value is that it identifies a concrete arithmetic correlation operator to study. If

$$C_c = P_{W_R} D_c|_{W_R},$$

then norms of the p -sieved vectors determine the real part of $C_{1/p}$ through

$$\left\| w - p^{-1/2} D_{1/p} w \right\|^2 = (1 + p^{-1}) \|w\|^2 - 2p^{-1/2} \Re \langle w, D_{1/p} w \rangle.$$

Consequently a quantitative lower bound for the p -sieved co-Poisson norm, uniform over the relevant source class and strong enough to survive the three-space Schur complement, would give new control of the cross-scale Gram operator appearing above. This is a narrower arithmetic target than an unrestricted positivity claim.

11 Why a finite Euler-product sieve fails in the Hilbert norm

The one-prime identity (38) suggests iterating over all primes up to a cutoff z . Put

$$\mathfrak{P}(z) = \prod_{p \leq z} p, \quad \mathcal{S}_z = \prod_{p \leq z} (I - p^{-1/2} D_{1/p}).$$

For an admissible compact source $h = \mathcal{D}\phi$ one obtains exactly

$$\mathcal{S}_z E(h)(x) = \sum_{(n, \mathfrak{P}(z))=1} h(nx). \quad (40)$$

For each fixed $x > 0$ the right side eventually equals $h(x)$, because only finitely many integers occur and every fixed $n > 1$ eventually has a prime divisor at most z . Thus the finite Euler-product sieve converges pointwise to the bare source. The Hilbert norm behaves very differently.

Let

$$\Psi(s) = \int_0^\infty h(t) t^{s-1} dt.$$

Admissibility gives $\Psi(1) = 0$ (and also $\Psi(0) = 0$). If $h \neq 0$, analyticity implies that there is a least integer $m \geq 1$ such that $\Psi^{(m)}(1) \neq 0$. Define

$$R_z(x) = \mathcal{S}_z E(h)(x) - h(x).$$

Choose a compact interval $U \Subset (0, \inf \text{supp } h)$. For $x = u/z$, $u \in U$, and z sufficiently large, every integer contributing to $R_z(u/z)$ lies between z and $C_U z < z^2$. Hence an integer coprime to $\mathfrak{P}(z)$ in this range is necessarily prime. The prime number theorem with a smooth weight then gives, uniformly for $u \in U$,

$$R_z(u/z) = \frac{(-1)^m \Psi^{(m)}(1)}{u} \frac{z}{(\log z)^{m+1}} + O_{h,U} \left(\frac{z}{(\log z)^{m+2}} \right). \quad (41)$$

Indeed, the main term is

$$\frac{z}{u} \int \frac{h(t)}{\log(z t/u)} dt,$$

and expansion of the denominator in powers of $1/\log z$ kills precisely the first m logarithmic moments. Consequently

$$\int_{U/z} |R_z(x)|^2 dx \sim |\Psi^{(m)}(1)|^2 \left(\int_U \frac{du}{u^2} \right) \frac{z}{(\log z)^{2m+2}}, \quad (42)$$

which tends to $+\infty$.

Thus the sharp finite Euler-product inversion has the paradoxical behavior

$$\mathcal{S}_z E(h)(x) \rightarrow h(x) \quad \text{for every fixed } x > 0, \quad \|\mathcal{S}_z E(h) - h\|_2 \not\rightarrow 0;$$

in fact a fixed nonzero admissible source has a diverging prime-shell contribution. This is a scoped no-go theorem for the raw finite-prime sieve, not for all smoothed or nonlocal arithmetic inverses. It also explains why imposing any *fixed* finite number of additional logarithmic moments cannot repair the mechanism: powers of $\log z$ cannot beat the factor $z^{1/2}$ in the L^2 norm. A successful Euler-factor construction would need a genuinely nonlocal tail or a number of matched conditions growing with the scale.

12 The arithmetic-reflection defect as a dual zero channel

Let w be a hypothetical off-line zero and put $w^\sharp = 1 - \bar{w}$. The ambient de Branges reflection multiplier

$$B_w(s) = \frac{s - w^\sharp}{s - w}$$

has modulus one on the critical line. The compact-division argument below shows that, at an off-line zero of multiplicity m , the finite-cutoff leakage

$$K_{w,R} = P_{Q_R} B_w|_{W_R}$$

has rank one and factors as

$$K_{w,R} f = \ell_{w,R}(f) q_{w,R}, \quad \ell_{w,R}(E(\psi)) = \Psi(w).$$

At a simple zero the vector $q_{w,R}$ is characterized by vanishing completed Mellin values at all other zero constraints and a fixed nonzero value at w . This is the finite-cutoff analogue of the biorthogonal zero direction appearing in Burnol's complete/minimal systems; at the critical cutoff $a = 1$ the dual system is represented, up to triangular changes within a multiple zero, by the functions $\zeta(s)/(s - \rho)^k$ [4].

The rank-one channel has an exact prime transport law. Dilation covariance of the source gives

$$\ell_{w,pR}(D_p f) = p^{w-1/2} \ell_{w,R}(f), \quad \ell_{w,pR}(D_{1/p} f) = p^{1/2-w} \ell_{w,R}(f).$$

Since B_w commutes with dilation and $D_{\pm 1/p} W_R \subset W_{pR}$, comparison of the two rank-one factorizations yields

$$\boxed{P_{Q_{pR}} D_p q_{w,R} = p^{w-1/2} q_{w,pR}}, \quad \boxed{P_{Q_{pR}} D_{1/p} q_{w,R} = p^{1/2-w} q_{w,pR}}. \quad (43)$$

Writing $d = |\Re w - 1/2|$, contractivity gives

$$p^d \|q_{w,pR}\| \leq \|q_{w,R}\|. \quad (44)$$

For a simple zero, $M(q_{w,R})(w)$ is the fixed nonzero number

$$(w - w^\sharp)c_0(w)\zeta'(w),$$

when the normalizing source has $\Psi(w) = 1$. Cauchy–Schwarz against the zero-evaluation Riesz vector, together with the off-line evaluator growth in Assumption 2.1 $\|v_w^R\| \asymp_w R^d$, therefore gives

$$\|q_{w,R}\| \gg_w R^{-d}. \quad (45)$$

Iterating (44) supplies the matching upper estimate along geometric cutoffs $R = p^n R_0$. Hence

$$\|q_{w,p^n R_0}\| \asymp_{w,p,R_0} (p^n R_0)^{-d}. \quad (46)$$

Let $r_{w,R} \in W_R$ be the Riesz vector of the source-moment functional $\ell_{w,R}$. The same two dilation inclusions imply

$$\|r_{w,pR}\| \geq p^d \|r_{w,R}\|.$$

Since B_w is unitary on its ambient domain,

$$\|K_{w,R}\| = \|r_{w,R}\| \|q_{w,R}\| \leq 1.$$

Combining these estimates with (45) yields, for a hypothetical simple off-line zero,

$$0 < c_{w,p,R_0} \leq \|K_{w,p^n R_0}\| \leq 1 \quad (n \geq 0). \quad (47)$$

Thus the arithmetic-reflection defect does not disappear in operator norm even though every fixed source can be repaired asymptotically in the enlarging arithmetic spaces. The source-moment Riesz vector grows like R^d , the dual zero channel shrinks like R^{-d} , and the product retains order-one strength. This gives a precise moving-boundary explanation for why fixed-input convergence is insufficient.

12.1 Full multiplicity channel: exact jet transport and persistent leakage volume

The rank-one calculation above uses one division by $s - w$ and therefore exposes only the first source moment, even when the zero w has higher multiplicity. To treat a zero of multiplicity m without discarding its jet structure, use the m -fold ambient reflection

$$\mathcal{B}_w^{(m)}(s) = \left(\frac{s - w^\sharp}{s - w} \right)^m, \quad w^\sharp = 1 - \bar{w},$$

and define

$$\mathcal{K}_{w,R}^{(m)} = P_{Q_R} \mathcal{B}_w^{(m)}|_{W_R}.$$

For $f = E(\psi)$ write $\Psi = M_0\psi$ and introduce the factorially normalized source jet

$$\mathcal{J}_{w,R}f = \left(\Psi(w), \Psi'(w), \dots, \frac{\Psi^{(m-1)}(w)}{(m-1)!} \right)^T \in \mathbb{C}^m.$$

Because ζ has order exactly m at w , these functionals extend continuously from the source core to W_R : they are fixed triangular linear combinations of the completed-Mellin output derivatives of orders $m, \dots, 2m - 1$.

Theorem 12.1 (Exact multiplicity- m leakage factorization). *Conditionally suppose that w is a nontrivial zero of multiplicity m with $\Re w \neq 1/2$. Then there is an injective map*

$$\mathcal{Q}_{w,R} : \mathbb{C}^m \longrightarrow Q_R$$

such that

$$\mathcal{K}_{w,R}^{(m)} = \mathcal{Q}_{w,R} \mathcal{J}_{w,R}, \quad \text{rank } \mathcal{K}_{w,R}^{(m)} = m. \quad (48)$$

If $h = \log p$ and N is the nilpotent shift on factorial jet coordinates, set

$$M_p = p^{w-1/2} e^{hN}.$$

Then the leakage basis obeys the exact prime transport laws

$$P_{Q_{pR}} D_p \mathcal{Q}_{w,R} = \mathcal{Q}_{w,pR} M_p, \quad P_{Q_{pR}} D_{1/p} \mathcal{Q}_{w,R} = \mathcal{Q}_{w,pR} M_p^{-1}. \quad (49)$$

Proof. Write locally

$$\zeta(s) = (s - w)^m a_w(s), \quad a_w(w) \neq 0.$$

For a core vector $f = E(\psi)$,

$$M(\mathcal{B}_w^{(m)} f)(s) = c_0(s) a_w(s) (s - w^\sharp)^m \Psi(s).$$

The first m completed-Mellin derivatives at w are therefore an invertible lower-triangular transformation of the source jet $\mathcal{J}_{w,R} f$. Every element of W_R has those first m derivatives equal to zero. This alone does not prove factorization. If the source jet vanishes, Lemma 2.2 shows that $\Psi(s)/(s - w)^m$ is again the transform of an admissible compact source. Multiplication by $(s - w^\sharp)^m$ preserves that class, so $\mathcal{B}_w^{(m)} f \in W_R$. The kernel inclusion extends to W_R by Lemma 2.3, using continuity of the jet and of the ambient unitary multiplier. Therefore the leakage factors through the source jet. The jet map is onto by Lemma 2.3. The displayed triangular output map is invertible because $w \neq w^\sharp$, so its factor into Q_R is injective and the rank is exactly m . On the critical line the multiplier is the identity and the leakage is zero; that case is excluded.

Dilation gives

$$\mathcal{J}_{w,pR}(D_p f) = p^{w-1/2} e^{hN} \mathcal{J}_{w,R} f = M_p \mathcal{J}_{w,R} f.$$

The multiplier $\mathcal{B}_w^{(m)}$ commutes with dilations, and $D_p W_R, D_{1/p} W_R \subset W_{pR}$. Projecting the identity $D_{\pm p} \mathcal{B}_w^{(m)} f = \mathcal{B}_w^{(m)} D_{\pm p} f$ to Q_{pR} gives (49). Notice that $\det e^{hN} = 1$, hence

$$|\det M_p| = p^{m(\Re w - 1/2)}.$$

□

The theorem has a basis-independent consequence. Let

$$G_R = \mathcal{Q}_{w,R}^* \mathcal{Q}_{w,R}, \quad H_R = \mathcal{J}_{w,R} \mathcal{J}_{w,R}^*.$$

Thus G_R is the Gram matrix of the dual leakage directions and H_R is the Gram matrix of the Riesz representers of the source-jet functionals. Contractivity of the two projected dilation maps in (49) gives

$$M_p^* G_{pR} M_p \preceq G_R, \quad M_p^{-*} G_{pR} M_p^{-1} \preceq G_R, \quad (50)$$

$$H_{pR} \succeq M_p H_R M_p^*, \quad H_{pR} \succeq M_p^{-1} H_R M_p^{-*}. \quad (51)$$

Writing

$$d = \left| \Re w - \frac{1}{2} \right|,$$

taking determinants yields

$$\det G_{pR} \leq p^{-2md} \det G_R, \quad \det H_{pR} \geq p^{2md} \det H_R. \quad (52)$$

The remaining quantitative leakage conclusions in this section assume Assumption 2.1. Only a one-sided ambient jet bound is needed. Let $V_{w,R}$ be the Gram matrix of the first m completed-Mellin evaluation jets at w on the ambient Sonine space H_R . For the raw representatives specified in Assumption 2.1, the fixed finite raw jet family has the exact cutoff scaling

$$R^{1/2-w} e^{(\log R)N} (1_{t \geq 1} t^{w-1}, 1_{t \geq 1} t^{w-1} \log t, \dots, 1_{t \geq 1} t^{w-1} (\log t)^{m-1})$$

for $\Re w < 1/2$; for $\Re w > 1/2$ the Fourier involution gives the reflected model. The nilpotent factor has determinant one, and the fixed raw Gram matrix is finite. Orthogonal Sonine projection can only decrease the Gram matrix in Loewner order. Therefore

$$\boxed{\det V_{w,R} \ll_{w,m} R^{2md}.} \quad (53)$$

No relative $o(1)$ estimate for the whole projected jet family is used here.

Let A_w be the fixed invertible triangular matrix taking the source jet to the first m output jets in the proof of Theorem 12.1. The Gram determinant inequality applied to

$$\Gamma_{w,R} \mathcal{Q}_{w,R} = A_w$$

gives

$$|\det A_w|^2 \leq \det V_{w,R} \det G_R.$$

Combining this with (53) and the upper bound obtained by iterating (52), along every geometric sequence $R = p^n R_0$,

$$\boxed{\det G_R \asymp_{w,m,p,R_0} R^{-2md}.} \quad (54)$$

The nonzero singular values of $\mathcal{K}_{w,R}^{(m)} = \mathcal{Q}_{w,R} \mathcal{J}_{w,R}$ satisfy

$$\prod_{j=1}^m \sigma_j(\mathcal{K}_{w,R}^{(m)})^2 = \det G_R \det H_R. \quad (55)$$

Since $\mathcal{B}_w^{(m)}$ is unitary on the ambient critical-line space and P_{Q_R} is contractive, every $\sigma_j \leq 1$. Equations (51) and (54) imply

$$\det H_R \gg R^{2md},$$

whereas (55), $\sigma_j \leq 1$, and the lower bound for $\det G_R$ give

$$\det H_R \ll R^{2md}.$$

Thus

$$\boxed{\det H_R \asymp_{w,m,p,R_0} R^{2md}} \quad (56)$$

and, most importantly,

$$\boxed{0 < c_{w,m,p,R_0} \leq \prod_{j=1}^m \sigma_j(\mathcal{K}_{w,R}^{(m)}) \leq 1} \quad (R = p^n R_0). \quad (57)$$

Because each singular value is at most one, (57) also bounds the smallest active singular value away from zero. Hence the entire multiplicity- m arithmetic-reflection leakage channel remains uniformly nondegenerate along geometric cutoffs.

Remark 12.2. This is the multiplicity analogue of the simple-zero compensation law. The dual leakage volume decays like R^{-md} , the source-jet volume grows like R^{md} , and the nilpotent logarithmic factors cancel from the determinant. Higher multiplicity therefore does not make the off-line defect disappear; it converts the scalar renormalization into a Jordan-coupled m -channel renormalization.

Remark 12.3 (What this still does not prove). The persistent leakage theorem is conditional on the existence of an off-line zero and describes what the arithmetic reflection channel would then do. It is not itself a contradiction: a unitary ambient reflection can have a uniformly nontrivial finite-dimensional leakage into Q_R . To deduce RH one still needs an independent arithmetic or geometric principle forcing this leakage channel to vanish, become asymptotically singular, or conflict with a positive fixed-space prime action. No such principle is proved here.

13 Bridge to the prime-compression defect

The persistent reflection leakage is not merely an auxiliary finite-dimensional construction. It is itself the defect of a canonical compression of the unitary zero-reflection multiplier, and it intertwines exactly with the cross-scale prime compression.

Let R_R denote orthogonal projection onto W_R , and for a fixed off-line parameter w define

$$A_{w,R} = R_R B_w|_{W_R}, \quad K_{w,R} = P_R B_w|_{W_R}.$$

At a zeta zero w , every $f \in W_R$ has completed Mellin value zero at w , so Burnol's division theorem places $B_w f$ back in $H_R = W_R \oplus Q_R$. Since B_w is unitary on the ambient critical-line L^2 space,

$$\boxed{I_{W_R} - A_{w,R}^* A_{w,R} = K_{w,R}^* K_{w,R}} \quad (58)$$

Thus the rank-one leakage for one reflection, and the multiplicity- m leakage for B_w^m , are canonical positive defect operators of the arithmetic reflection compression.

Now define the cross-scale prime compression

$$C_{R,p} = P_{pR} D_p|_{Q_R} : Q_R \longrightarrow Q_{pR}.$$

Because $D_p W_R \subset W_{pR}$ and $B_w D_p = D_p B_w$, for every $f \in W_R$ one has

$$\boxed{K_{w,pR} D_p f = C_{R,p} K_{w,R} f.} \quad (59)$$

The same identity holds for the multiplicity- m leakage operator $\mathcal{K}_{w,R}^{(m)}$. In the jet basis of Theorem 12.1,

$$C_{R,p} \mathcal{Q}_{w,R} = \mathcal{Q}_{w,pR} M_p, \quad M_p = p^{w-1/2} e^{(\log p)N}.$$

Consequently the cross-scale defect restricted to the leakage channel has the exact Gram form

$$\boxed{\mathcal{Q}_{w,R}^*(I - C_{R,p}^* C_{R,p}) \mathcal{Q}_{w,R} = G_R - M_p^* G_{pR} M_p \succeq 0.} \quad (60)$$

This is the first exact positive operator identity in the project in which the persistent arithmetic-reflection channel and the prime compression appear in the same formula.

The fixed-space compression is nevertheless different. Since $D_p Q_R \subset H_{pR}$, write

$$F_{R,p} = R_{pR} D_p|_{Q_R}.$$

Then

$$\boxed{I - C_{R,p}^* C_{R,p} = F_{R,p}^* F_{R,p}.} \quad (61)$$

Let

$$R_{R,p} = P_R|_{Q_{pR}} : Q_{pR} \rightarrow Q_R, \quad E_{R,p} = P_R R_{pR} D_p|_{Q_R} : Q_R \rightarrow Q_R.$$

Decomposing $D_p q$ into its Q_{pR} and W_{pR} components and then projecting to Q_R gives the exact return formula

$$\boxed{T_{R,p} = R_{R,p} C_{R,p} + E_{R,p}.} \quad (62)$$

Moreover $E_{R,p} = P_R F_{R,p}$, hence

$$\boxed{E_{R,p}^* E_{R,p} \preceq I - C_{R,p}^* C_{R,p}.} \quad (63)$$

The term $E_{R,p}$ is the arithmetic *aliasing* created by projecting the newly available relations in W_{pR} back into the older quotient Q_R . It vanishes if the quotients are nested in the required direction, but no such nesting is available here.

Equations (60)–(63) isolate the precise missing bridge. Reflection leakage controls the cross-scale prime defect, while the RH-sensitive fixed-space operator also depends on the angle map $R_{R,p}$ and the aliasing $E_{R,p}$. Hilbert-space positivity alone does not compare the two defects.

Proposition 13.1 (No generic domination between cross-scale and fixed-space defects). *Even with nested ambient spaces $H_R \subset H_{pR}$, nested arithmetic spaces $W_R \subset W_{pR}$, and a unitary U satisfying $UW_R \subset W_{pR}$, there is no universal operator inequality in either direction between the defects of*

$$P_{Q_{pR}} U|_{Q_R} \quad \text{and} \quad P_{Q_R} U|_{Q_R}.$$

Proof. Take an orthonormal basis e_1, e_2, e_3 , set

$$H_R = \text{span}\{e_1, e_2\}, \quad W_R = \text{span}\{e_2\}, \quad Q_R = \text{span}\{e_1\},$$

and

$$H_{pR} = \mathbb{C}^3, \quad W_{pR} = \text{span}\{e_2, \cos \theta e_1 + \sin \theta e_3\}.$$

Then

$$Q_{pR} = \text{span}\{-\sin \theta e_1 + \cos \theta e_3\}.$$

For $U = I$, one has $UW_R \subset W_{pR}$, the fixed-space compression on Q_R is the identity, while the cross-scale compression has norm $|\sin \theta|$. Thus the fixed-space defect is zero while the cross-scale defect is $\cos^2 \theta$.

Conversely choose a unitary U fixing e_2 and sending e_1 to the unit generator of Q_{pR} . Again $UW_R \subset W_{pR}$, but now the cross-scale compression is isometric while the fixed-space compression has norm $|\sin \theta|$. The two defects therefore reverse their sizes. This rules out a comparison based only on positivity, nesting, and unitary transport. $\square$

The proposition does not use the additional arithmetic structure of Burnol's spaces, so it is not a no-go theorem for the RH program. Its role is diagnostic: any successful passage from (60) to the fixed-space defect must exploit a genuinely arithmetic estimate on $R_{R,p}$ or $E_{R,p}$, not a generic projection inequality.

14 Infinitesimal dilation and the exact source-range defect

The finite-prime picture has an infinitesimal counterpart which unifies the source-moment obstruction, the reflection leakage, and the Jordan transport law. Let

$$(\mathbf{A}f)(x) = -\frac{1}{2}f(x) - xf'(x)$$

be the skew-adjoint generator of the ambient unitary dilation group $D_{e^t} = e^{t\mathbf{A}}$. On the uncompleted Mellin transform,

$$M_0(\mathbf{A}f)(s) = (s - \frac{1}{2})M_0f(s). \quad (64)$$

Let $\mathcal{A}_R = \mathcal{D}V_R$ be the admissible compact source class and let

$$\mathfrak{C}_R = E(\mathcal{A}_R),$$

which is dense in W_R by definition. Dilation covariance implies

$$\mathbf{A}E(\psi) = E(\mathbf{A}\psi), \quad \mathbf{A}\mathcal{D} = \mathcal{D}\mathbf{A}.$$

Moreover, if $\Psi(0) = \Psi(1) = 0$, then $(s - \frac{1}{2})\Psi(s)$ has the same two zero moments. Hence

$$\boxed{\mathbf{A}\mathfrak{C}_R \subset \mathfrak{C}_R}. \quad (65)$$

The restriction $\mathbf{A}_{W,R} = \mathbf{A}|_{\mathfrak{C}_R}$ is therefore a densely defined skew-symmetric operator in W_R .

Fix w in the open strip, put $\lambda_w = w - \frac{1}{2}$, and let $r \geq 1$. For $f = E(\theta)$ with $\Theta = M_0\theta$, equation (64) gives

$$(\mathbf{A}_{W,R} - \lambda_w)^r E(\eta) = E(\theta) \iff (s - w)^r H(s) = \Theta(s),$$

where $H = M_0\eta$. Lemma 2.2 shows that $H = \Theta/(s - w)^r$ is again the Mellin transform of an admissible source in the same cutoff exactly when

$$\Theta(w) = \Theta'(w) = \dots = \Theta^{(r-1)}(w) = 0.$$

Thus the source-core range is exactly

$$\text{Ran}(\mathbf{A}_{W,R} - \lambda_w)^r = \left\{ E(\theta) : \theta \in \mathcal{A}_R, \Theta^{(j)}(w) = 0 \ (0 \leq j < r) \right\}. \quad (66)$$

At a zeta zero w of multiplicity m , the source-jet functionals

$$\ell_j(E(\theta)) = \frac{\Theta^{(j)}(w)}{j!}, \quad 0 \leq j < m,$$

extend continuously to W_R . Indeed they are triangular linear combinations of completed-Mellin output derivatives of orders $m, \dots, 2m - 1$, with nonzero leading coefficient determined by $\zeta^{(m)}(w)$. They are linearly independent: choose one interior source η with $H(w) \neq 0$ and several sufficiently small dilations $D_{a_k}\eta$; the jet matrix is a nonzero triangular factor times a Vandermonde matrix in $\log a_k$. Therefore

Theorem 14.1 (Exact generator range defect). *Conditionally suppose that w is a zeta zero of multiplicity m . Then*

$$\overline{\text{Ran}(\mathbf{A}_{W,R} - \lambda_w)^m} = \bigcap_{j=0}^{m-1} \ker \ell_j, \quad \text{codim}_{W_R} = m. \quad (67)$$

The codimension is exact at every finite cutoff for which the source class is nontrivial.

Proof. The range on the left is the range of the algebraic power on the invariant source core, not an asserted power of a closed operator. Compact division gives (66). The continuous independent source jets have a surjective finite-dimensional joint map. Lemma 2.3 identifies the closure of its core kernel with its full kernel, which has codimension m . $\square$

This theorem identifies the multiplicity- m leakage channel as a higher-order range defect rather than an ad hoc correction. The basic deficiency index is one; multiplicity m appears through a length- m generalized deficiency chain, as made precise below. Indeed on the Mellin line

$$B_w(s) = \frac{s - w^\sharp}{s - w} = 1 + \frac{w - w^\sharp}{s - w}, \quad w^\sharp = 1 - \overline{w}, \quad (68)$$

so B_w is the Cayley–Blaschke transform of the dilation generator at the spectral parameter λ_w . The condition $\ell_0 = 0$ is precisely the condition under which the resolvent term in (68) remains inside the same compact arithmetic source space. At a zeta zero, the output factor $\zeta(w) = 0$ guarantees that $B_w W_R \subset H_R$, and the unresolved source coordinate is therefore forced to appear internally as the leakage map $K_{w,R} : W_R \rightarrow Q_R$. Repeating the division m times produces the full jet channel of Theorem 12.1.

The prime transport matrix from that theorem is exactly the exponential of the jet generator:

$$M_p = p^{w-1/2} e^{(\log p)N} = \exp\left((\log p)\left((w - \tfrac{1}{2})I + N\right)\right). \quad (69)$$

Thus the persistent finite-cutoff leakage, its Jordan factors, and its prime scaling are the finite-time form of the same infinitesimal range defect.

This also rules out a tempting shortcut. The ambient generator $\mathbf{A}$ is skew-adjoint, but its arithmetic restriction on W_R is not automatically maximal skew-symmetric: equation (67) exhibits a finite-codimensional resolvent obstruction. Merely invoking the ambient unitary dilation group therefore cannot yield a Hilbert–Polya proof. One would have to show that the relevant deficiency channel is eliminated by an additional arithmetic boundary condition while retaining every zero mode. That is precisely the compatibility theorem still missing from the fixed-space prime construction.

15 Boundary-triplet audit and an exact zero-incidence criterion

The infinitesimal formulation permits a sharper operator-theoretic audit. Let

$$S_R = \overline{\mathbf{A}_{W,R}}$$

be the closure in W_R of the skew-symmetric arithmetic dilation generator on the compact source core. The first conclusion is that the basic deficiency dimension is one, not the multiplicity of a zeta zero.

For w with $0 < \Re w < 1$ and $\zeta(w) \neq 0$, define the source-evaluation functional on W_R by

$$\ell_w(f) = \frac{\mathcal{M}f(w)}{c_0(w)\zeta(w)}.$$

It is continuous because completed Mellin evaluation is continuous on H_R . At a zeta zero the same functional extends by the derivative formula already used in the source-moment audit. On the source core,

$$\text{Ran}(\mathbf{A}_{W,R} - (w - \tfrac{1}{2})) = \ker \ell_w \cap \mathfrak{C}_R.$$

Since one may correct a core approximant by a fixed core vector with source moment one, closure gives

$$\overline{\text{Ran}(\mathbf{S}_R - (w - \tfrac{1}{2}))} = \ker \ell_w. \quad (70)$$

Choose $0 < a < 1/2$ and the real points $w_{\pm} = 1/2 \pm a$, where $\zeta(w_{\pm}) \neq 0$. Because $\mathbf{S}_R$ is skew-symmetric,

$$\|(\mathbf{S}_R \mp a)f\|^2 = \|\mathbf{S}_R f\|^2 + a^2 \|f\|^2,$$

so these ranges are closed. Equation (70) therefore implies

$$\dim \ker(\mathbf{S}_R^* - a) = \dim \ker(\mathbf{S}_R^* + a) = 1.$$

Thus $\mathbf{S}_R$ has deficiency indices $(1, 1)$. If $r_w \in W_R$ denotes the Riesz vector of ℓ_w , then

$$r_w \in \ker(\mathbf{S}_R^* - (\overline{w} - \tfrac{1}{2})). \quad (71)$$

The involution G preserves W_R and anti-commutes with the dilation generator. On source moments it sends $\Psi(s)$ to $\Psi(1-s)$, hence

$$Gr_w = r_{1-w}. \quad (72)$$

The deficiency vectors are total: if $f \in W_R$ is orthogonal to r_w on any open set of parameters avoiding zeta zeros, then the entire completed Mellin transform of f vanishes on an open set and therefore vanishes identically. Hence $\mathbf{S}_R$ is simple in the standard symmetric-operator sense.

Standard Livšic/von Neumann theory therefore supplies a one-parameter family of maximal skew-adjoint extensions of $\mathbf{S}_R$. If $e_{\pm}$ are normalized generators of $\ker(\mathbf{S}_R^* \mp a)$ and

$$f = f_0 + c_+ e_+ + c_- e_-, \quad f_0 \in \mathcal{D}(\mathbf{S}_R),$$

then the boundary form is

$$\langle \mathbf{S}_R^* f, g \rangle + \langle f, \mathbf{S}_R^* g \rangle = 2a (c_+ \overline{d_+} - c_- \overline{d_-}). \quad (73)$$

A maximal skew-adjoint extension is the restriction of $-\mathbf{S}_R^*$ (not $\mathbf{S}_R^*$) to the domain specified by one fixed unitary boundary relation $c_- = e^{i\vartheta} c_+$. This statement is ordinary extension theory; it does not identify any such extension with the zeta zeros.

The zeta condition enters through an additional ambient incidence statement. Fix an off-critical parameter w and choose an admissible source $\eta_w \in \mathcal{A}_R$ with Mellin transform H_w satisfying $H_w(w) \neq 0$; the bump construction from the source-moment repair gives such a source. Put

$$f_w = E(\eta_w) \in W_R, \quad B_w(s) = \frac{s - (1 - \overline{w})}{s - w}.$$

Then

Theorem 15.1 (Exact zero-incidence criterion). *For $0 < \Re w < 1$, $\Re w \neq 1/2$,*

$$\boxed{\zeta(w) = 0 \iff B_w f_w \in H_R.} \quad (74)$$

Equivalently, if $S_R^{(H)}$ denotes orthogonal projection onto H_R ,

$$\boxed{\zeta(w) = 0 \iff \|(I - S_R^{(H)})B_w f_w\| = 0.} \quad (75)$$

At a zero the non-arithmetic component therefore lies entirely in Q_R ; away from a zero it has a nonzero $H_R^\perp$ component.

Proof. The completed Mellin transform of f_w is

$$\mathcal{M}f_w(s) = c_0(s)\zeta(s)H_w(s).$$

If $\zeta(w) = 0$, then $\mathcal{M}f_w(w) = 0$, and Burnol's ambient division theorem places $B_w f_w$ in H_R . Conversely, if $B_w f_w \in H_R$, its completed Mellin transform is entire. On the critical line it agrees with

$$\frac{s - (1 - \bar{w})}{s - w} c_0(s)\zeta(s)H_w(s),$$

and hence with its meromorphic continuation. Since $w \neq 1 - \bar{w}$ and $c_0(w)H_w(w) \neq 0$, removability of the apparent pole at $s = w$ forces $\zeta(w) = 0$. $\square$

The theorem clarifies both the promise and the limitation of the boundary-triplet route. The source generator S_R itself has the familiar deficiency pattern $(1, 1)$ and therefore admits fixed maximal skew-adjoint boundary conditions. But the zeta zeros are not presently characterized as eigenvalues of one of those extensions. They are characterized by the external incidence condition (74): the Cayley–Blaschke defect generated from W_R must land in the larger Sonine space H_R , and at a zero its residual part is precisely the Q_R leakage channel studied above. The zero-evaluation vectors themselves live in Q_R , not in W_R .

Consequently a genuine Hilbert–Pólya conclusion from this picture still requires one new theorem: identify the incidence condition (74), or an equivalent condition on the Q_R leakage, with a *single zero-independent* maximal skew-adjoint boundary condition or with the characteristic function of one fixed conservative coupling. No such identification is proved here. In particular, choosing a boundary condition separately as a function of w would simply repack the zero problem and would be circular.

The correction above also explains the role of multiplicity. The operator S_R has one-dimensional basic deficiency spaces. A zero of multiplicity m produces the codimension- m closure of the algebraic source-core range of $(A_{W,R} - (w - 1/2))^m$ and the length- m Jordan/source-jet chain of Theorem 12.1; it does not change the basic deficiency indices from $(1, 1)$.

16 A nested rank-one defect ladder and defect transfer

The zero-incidence theorem admits a sharper quantitative form. There are two nested compression defects associated with the same ambient Blaschke multiplier B_w : one for the Sonine space H_R , and one for its arithmetic subspace W_R .

Let $S_R^{(H)}$ and R_R denote the orthogonal projections onto H_R and W_R , respectively, and define

$$\Delta_{w,R}^H = (I - S_R^{(H)})B_w|_{H_R}, \quad \Delta_{w,R}^W = (I - R_R)B_w|_{W_R}.$$

Burnol's division property implies that $\Delta_{w,R}^H$ vanishes on the codimension-one evaluation kernel

$$\{f \in H_R : \mathcal{M}f(w) = 0\}.$$

It is not the zero operator: if k_w^R is the Riesz vector for completed Mellin evaluation, then $B_w k_w^R$ has an uncanceled pole at w and therefore cannot belong to H_R . Hence $\Delta_{w,R}^H$ has rank exactly one. Put

$$d_{w,R}^H = \frac{\Delta_{w,R}^H k_w^R}{\|k_w^R\|^2}.$$

Then

$$\boxed{\Delta_{w,R}^H f = \mathcal{M}f(w) d_{w,R}^H \quad (f \in H_R).} \quad (76)$$

Likewise the exact source-division identity shows that $\Delta_{w,R}^W$ vanishes on $\ker \ell_w$. Thus it also has rank one. If r_w^R is the Riesz vector of the source-evaluation functional ℓ_w on W_R , set

$$d_{w,R}^W = \frac{\Delta_{w,R}^W r_w^R}{\|r_w^R\|^2}.$$

Then

$$\boxed{\Delta_{w,R}^W f = \ell_w(f) d_{w,R}^W \quad (f \in W_R).} \quad (77)$$

For $f \in W_R$ the arithmetic Mellin factorization gives

$$\mathcal{M}f(w) = c_0(w)\zeta(w)\ell_w(f).$$

Projecting (77) further onto $H_R^\perp$ and comparing with (76) therefore gives the exact vector identity

$$\boxed{d_{w,R}^W = q_{w,R} + c_0(w)\zeta(w)d_{w,R}^H, \quad q_{w,R} = P_{Q_R} d_{w,R}^W \in Q_R.} \quad (78)$$

The two summands are orthogonal because

$$W_R^\perp = Q_R \oplus H_R^\perp.$$

Consequently

$$\boxed{\|d_{w,R}^W\|^2 = \|q_{w,R}\|^2 + |c_0(w)\zeta(w)|^2 \|d_{w,R}^H\|^2.} \quad (79)$$

Equation (79) is a defect-conservation law. The source generator always has one nontrivial Blaschke/Cayley defect direction off the critical line. At a zeta zero its $H_R^\perp$ component vanishes and the entire defect lies internally in the arithmetic quotient Q_R ; away from a zero a nonzero portion necessarily escapes the Sonine space. Thus a zero does not annihilate the defect. It transfers the defect from the outer Sonine channel to the inner arithmetic channel.

Theorem 15.1 can therefore be restated as the angle condition

$$\zeta(w) = 0 \quad \Longleftrightarrow \quad d_{w,R}^W \in Q_R \quad \Longleftrightarrow \quad \frac{|c_0(w)\zeta(w)|^2 \|d_{w,R}^H\|^2}{\|d_{w,R}^W\|^2} = 0. \quad (80)$$

This resembles a scalar characteristic or transmission coefficient for the nested pair $W_R \subset H_R$, but no identification with the Livšic characteristic function of $\mathbf{S}_R$ is asserted here. Establishing such an identification, with a zero-independent conservative colligation, would be a genuinely new theorem rather than a change of notation.

17 Exact same-label returns are overstrong; the asymptotic regime is calibrated

The complete/minimal zero systems suggest a formally natural same-label return between two scale fibers. A necessary caution is that *exact contractivity* of such a return is stronger than RH, whereas the asymptotic compression used earlier is correctly calibrated.

Proposition 17.1 (Exact diagonal contractions force orthogonality). *Let $\mathcal{H}$ be a Hilbert space and let $(x_j)_{j \in J}$ have dense linear span. Suppose a contraction A satisfies*

$$Ax_j = \lambda_j x_j, \quad |\lambda_j| = 1$$

for every j . Then A is unitary. In particular,

$$\lambda_i \neq \lambda_j \implies \langle x_i, x_j \rangle = 0.$$

Proof. For every j , equality of norms gives

$$0 = \|x_j\|^2 - \|Ax_j\|^2 = \langle (I - A^*A)x_j, x_j \rangle.$$

Since $I - A^*A \succeq 0$, this implies $(I - A^*A)x_j = 0$. Density and continuity give $A^*A = I$, so A is an isometry. Every x_j belongs to $\text{Ran } A$ because $x_j = \lambda_j^{-1}Ax_j$; hence the range is dense. The range of an isometry is closed, so A is onto and therefore unitary. Distinct eigenspaces of a unitary operator are orthogonal. $\square$

Under RH the desired prime multipliers $\lambda_p = p^{1/2-\rho}$ all have modulus one. Thus an *exact* contractive fixed-space operator diagonal on a complete zero system would force exact orthogonality between all zero vectors carrying distinct prime phases. That orthogonality is an additional requirement not contained in RH. Consequently exact same-label contractivity is not the correct target for the Burnol zero systems.

This is consistent with Burnol's later study of the biorthogonal family $\zeta(s)/(s-\rho)^k$: the system and its dual are complete and minimal, but bounded unconditional reconstruction is not supplied by completeness/minimality alone. The hereditary-completeness problem is genuinely subtler; in the admissible mixed systems studied there the possible defect is controlled only up to codimension one. Thus a formal biorthogonal return must not be treated as a bounded, let alone contractive, operator without a separate theorem.

Under Assumption 2.1, the moving-cutoff construction avoids the preceding overconstraint because fixed critical-line zero kernels become asymptotically orthogonal. The following calculation records the consequence of those quantitative inputs, not a new proof of them. For two fixed distinct critical-line parameters

$$w = \frac{1}{2} + i\tau, \quad z = \frac{1}{2} + i\sigma, \quad \tau \neq \sigma,$$

let Z_w^R, Z_z^R be Burnol's completed-Mellin evaluation vectors. The raw Sonine representatives satisfy

$$C_1(v, w) = \gamma_+(w)v^{-w} + O_w(1), \quad C_1(v, z) = \gamma_+(z)v^{-z} + O_z(1)$$

near zero. Hence their cross term has leading integral

$$\gamma_+(w)\overline{\gamma_+(z)} \int_{R^{-2}}^1 v^{-1-i(\tau-\sigma)} dv = O_{w,z}(1),$$

while all remaining local and tail terms are bounded. The Sonine projection changes each raw evaluator by $O_w(R^{-1})$, so

$$\langle Z_w^R, Z_z^R \rangle = O_{w,z}(1). \quad (81)$$

On the other hand

$$\|Z_w^R\|^2 = C_w \log R + O_w(1), \quad \|Z_z^R\|^2 = C_z \log R + O_z(1), \quad C_w, C_z > 0.$$

Therefore

$$\boxed{\frac{\langle Z_w^R, Z_z^R \rangle}{\|Z_w^R\| \|Z_z^R\|} = O_{w,z}\left(\frac{1}{\log R}\right)}. \quad (82)$$

For any fixed finite set of distinct critical-line parameters, the normalized Gram matrix therefore tends to the identity.

Corollary 17.2 (Finite-mode asymptotic calibration). *Assume Assumption 2.1. Fix a finite set F of distinct critical-line parameters and unit-modulus numbers λ_w . On*

$$\mathcal{V}_R(F) = \text{span}\{Z_w^R : w \in F\}$$

define the diagonal operator $A_R(F)Z_w^R = \lambda_w Z_w^R$. Then

$$\|A_R(F)\| = 1 + O_F\left(\frac{1}{\log R}\right).$$

In contrast, if a finite set contains a parameter with $|\lambda_w| > 1$, then every operator having that vector as an exact eigenvector satisfies $\|A\| \geq |\lambda_w| > 1$.

Proof. After normalizing the basis vectors, equation (82) gives a Gram matrix $G_R = I + O_F(1/\log R)$. The squared norm of the diagonal phase operator is the largest generalized eigenvalue of the pencil $(D^* G_R D, G_R)$. Here $D = \text{diag}(\lambda_w)$, so this eigenvalue equals $1 + O_F(1/\log R)$. The second assertion is immediate from the norm of an eigenvector. $\square$

The correct fixed-space target is therefore *asymptotic* rather than exact contractivity. A useful non-circular formulation would construct, from the arithmetic source geometry and not from the zero list itself, operators $R_{R,p,T}$ on a fixed finite spectral window such that

$$\limsup_{R \rightarrow \infty} \|R_{R,p,T}\| \leq 1$$

for every fixed T . Any hypothetical off-line zero inside the window would force a fixed eigenvalue modulus $p^{1/2 - \Re \rho} > 1$ and contradict this bound, while under RH the finite-mode Gram calibration above is compatible with norm $1 + o(1)$. The remaining difficulty is to define such a return by a zero-independent co-Poisson/prime formula. Defining it spectrally from the zero vectors would merely encode the desired conclusion.

18 Off-line resolvent localization and why the determinant shortcut is tautological

The fixed Hilbert-space resolvent construction admits a useful extension away from the critical line. This gives a zero-independent spectral window, but it also shows that determinant or spectral-radius inequalities on that window are endpoints equivalent to zero exclusion rather than independent mechanisms.

Assumption 18.1 (Fixed closed arithmetic realization). A fixed Hilbert space $\mathcal{H}_B$ of functions holomorphic in the nontrivial strip is specified, with continuous off-line evaluations and an isometric critical-line realization. A closed densely defined operator A has graph formula

$$AF(s) = sF(s) - \eta_B(F)\zeta(s)$$

on its stated domain. The finite span of $F_{\rho,j} = \zeta(s)/(s-\rho)^j$ is a graph core, with $AF_{\rho,j} = \rho F_{\rho,j} + F_{\rho,j-1}$ for $j > 1$ and $AF_{\rho,1} = \rho F_{\rho,1}$. The bounded divided-difference resolvents below preserve the realization, and the resulting off-line meromorphic resolvent has no additional spectral subspaces beyond the indicated finite root blocks.

This is an explicit realization hypothesis, not a completed operator construction in this manuscript. The inherited critical-line norm, the domain, closure, and graph-core theorem must be supplied together before using the spectral claims that follow. Completeness and minimality of the root system do not imply the graph-core assertion. Later finite root-space sampling identities can be defined algebraically without this assumption; calling their source space a Riesz window requires it.

Let $\mathcal{H}_B$ denote the fixed right-Mellin Hilbert realization in Assumption 18.1, let A be the closed operator defined there, and let

$$F_{\rho,j}(s) = \frac{\zeta(s)}{(s-\rho)^j}, \quad 1 \leq j \leq m_\rho,$$

be its complete minimal zero/jet system. For a point z satisfying

$$\Re z \neq \frac{1}{2}, \quad \zeta(z) \neq 0,$$

define

$$(R_z F)(s) = \frac{F(s) - F(z)\zeta(s)/\zeta(z)}{s - z}. \quad (83)$$

Point evaluation at z is continuous on $\mathcal{H}_B$. Moreover, if $\delta_z = |\Re z - 1/2|$, then multiplication by $(s-z)^{-1}$ on the critical line has norm at most δ_z^{-1} . Since $\zeta(s)/s$ belongs to the ambient Mellin L^2 space and $s/(s-z)$ is bounded on the critical line, one also has

$$q_z(s) := \frac{\zeta(s)}{s-z} \in L^2.$$

Consequently R_z is bounded in the ambient norm, with the explicit coarse estimate

$$\|R_z F\| \leq \left(\delta_z^{-1} + \frac{\|k_z\| \|q_z\|}{|\zeta(z)|} \right) \|F\|, \quad (84)$$

where k_z is the Riesz vector for evaluation at z .

On the finite zero/jet core the numerator in (83) factors algebraically, so R_z maps the core to itself and is the triangular Jordan resolvent. Boundedness therefore extends R_z to $\mathcal{H}_B$. The separately assumed graph-core property of the zero/jet system then upgrades the algebraic identities to

$$(A - zI)R_z = I, \quad R_z(A - zI) = I$$

on their natural domains. Thus:

Conditional result 18.2 (Off-line spectrum under a closed-operator realization). Assume Assumption 18.1. In the nontrivial strip away from the critical line,

$$\sigma(A) \cap \{0 < \Re z < 1 : \Re z \neq \tfrac{1}{2}\} = \{\rho : \zeta(\rho) = 0, \Re \rho \neq \tfrac{1}{2}\},$$

with algebraic multiplicities matching the zeta multiplicities.

Proof. If $\zeta(z) \neq 0$, equations (83)–(84) and the graph-core argument above show that $z \in \rho(A)$. If $z = \rho$ is a nontrivial zeta zero of order m_ρ , $F_{\rho,1}$ is an eigenvector of A and $F_{\rho,1}, \dots, F_{\rho,m_\rho}$ form the displayed Jordan chain. The chain cannot be extended: if $G \in D(A)$ satisfied $(A - \rho I)G = F_{\rho,m_\rho}$, then the graph formula for A would force

$$G(s) = \frac{\zeta(s)}{(s - \rho)^{m_\rho+1}} + c \frac{\zeta(s)}{s - \rho}$$

for some scalar c . The first term has a nonremovable pole at the nontrivial zero ρ , whereas elements of $\mathcal{H}_B$ have no such pole. Thus the root space has exactly the zeta multiplicity. No assertion about spectrum exactly on the critical line is needed for this theorem. $\square$

Under the same realization assumption, a compact contour $\Gamma = \partial\Omega$ lying strictly to one side of the critical line and avoiding zeta zeros defines a genuinely zero-independent Riesz projection

$$\Pi_\Omega = \frac{1}{2\pi i} \oint_\Gamma (zI - A)^{-1} dz. \quad (85)$$

Its range is precisely the finite-dimensional sum of the zero/jet blocks inside Ω , without listing those zeros in the definition.

For a prime p define on this window

$$U_{p,\Omega} = \frac{1}{2\pi i} \oint_\Gamma p^{1/2-z} (zI - A)^{-1} dz. \quad (86)$$

On a zero block at ρ this has the exact action

$$p^{1/2-\rho} \exp[-(\log p)N].$$

Thus (86) is an exact fixed-space, zero-independent finite-window prime action.

It is tempting to seek $r(U_{p,\Omega}) \leq 1$ or $|\det U_{p,\Omega}| \leq 1$ for a window strictly left of the critical line. The determinant calculation shows why this is not by itself a new mechanism. Holomorphic functional calculus gives

$$\det(U_{p,\Omega}|_{\text{Ran } \Pi_\Omega}) = \prod_{\rho \in \Omega} p^{m_\rho(1/2-\rho)}, \quad (87)$$

so

$$\log |\det U_{p,\Omega}| = (\log p) \sum_{\rho \in \Omega} m_\rho(1/2 - \Re \rho). \quad (88)$$

Equivalently, when the contour avoids the pole and trivial zeros,

$$\log \det U_{p,\Omega} = \frac{\log p}{2\pi i} \oint_\Gamma \left(\frac{1}{2} - z\right) \frac{\zeta'(z)}{\zeta(z)} dz, \quad (89)$$

with a consistent branch on the finite-dimensional determinant. For a left window every summand in (88) is strictly positive. Therefore the inequality $|\det U_{p,\Omega}| \leq 1$ is equivalent to the emptiness of that window. The spectral-radius inequality has the same issue. These quantities are useful diagnostics and exact spectral selectors, but a proof still requires an independent arithmetic estimate before taking the determinant.

A global hyponormality shortcut is impossible

Let $T_{R,p} = P_R D_p|_{Q_R}$ and use Burnol's involution G , which preserves Q_R and satisfies $GT_{R,p}G = T_{R,p}^*$. Its self-commutator

$$C_{R,p} = T_{R,p}^* T_{R,p} - T_{R,p} T_{R,p}^*$$

therefore obeys the exact anti-symmetry

$$\boxed{GC_{R,p}G = -C_{R,p}} \quad (90)$$

Hence $C_{R,p} \succeq 0$ or $C_{R,p} \preceq 0$ on all of Q_R would force $C_{R,p} = 0$, i.e. normality of the compression. Thus global hyponormality/cohyponormality cannot supply the missing one-sided zero exclusion unless one first proves the much stronger normality statement. We do not claim a side-dependent asymptotic value for the diagonal of $C_{R,p}$: the existing inward-mode audit determines one of the two norm limits in each off-line half-plane, but not the complementary outward norm required for such a formula.

19 Comparison with the finite-prime Weil-form program

A recent and independently developed spectral program of Connes–Consani–Moscovici constructs, from the Weil quadratic form on $[\lambda^{-1}, \lambda]$, self-adjoint finite-prime operators whose regularized determinants have only critical-line zeros. Their finite form on this interval involves only prime-power terms $n \leq \lambda^2$, the same multiplicative scale ratio that appears naturally in the Burnol source window $[1/R, R]$. Their theorem produces a self-adjoint rank-one perturbation of the logarithmic scaling operator in a positive inner product obtained from the truncated Weil form, and hence an entire determinant with real zeros. The stated missing step is convergence of those finite-prime determinants/spectra toward the Riemann Ξ function; a rigorous proof of that convergence would establish RH [7].

A conditional obstruction to norm-resolvent convergence

A unitary norm-resolvent comparison has the following necessary condition in the inherited Burnol norm. Set

$$B = A - \frac{1}{2}I.$$

On a critical-line zero block its eigenvalues are $i\gamma$. If D_n are self-adjoint finite-prime operators and U_n are unitary identifications into one fixed Hilbert space, then $iU_n D_n U_n^*$ are skew-adjoint. A norm-resolvent limit of skew-adjoint operators is again skew-adjoint. But the exact boundary form from the resolvent audit gives

$$\langle BF_\rho, F_\sigma \rangle + \langle F_\rho, BF_\sigma \rangle = i(\gamma_\rho - \gamma_\sigma) \langle F_\rho, F_\sigma \rangle$$

for critical-line zero vectors. If a pair of distinct critical ordinates satisfies $\langle F_\rho, F_\sigma \rangle \neq 0$, this identity rules out skew-symmetry of B and hence such a norm-resolvent limit to B (equivalently a shifted limit to A). The earlier draft referred to a certified two-zero calculation but supplied no reproducible certificate. This revision does not assert that numerical premise. The obstruction is conditional on verifying it in the specified Hilbert realization. Finite-dimensional approximations also require a precise common-space realization before norm-resolvent convergence is a well-typed statement.

This does not rule out a comparison through a changing positive metric. A nonnormal operator may be similar to a skew-adjoint one in an equivalent Hilbert norm, and the finite-prime Weil construction naturally provides such positive forms at each cutoff.

A finite-window Lyapunov metric criterion

The preceding observation suggests a weaker and correctly typed bridge. Let Ω be a compact Riesz window contained strictly in one half of the critical strip, let Π_Ω be (85), and keep $B = A - \frac{1}{2}I$.

Theorem 19.1 (Uniform metric defect excludes an off-line window). *Assume the Riesz realization in Assumption 18.1. Suppose there are bounded positive operators G_n on $\mathcal{H}_B$ and numbers $c_\Omega > 0$, $\varepsilon_n \rightarrow 0$ such that for all sufficiently large n ,*

$$\langle G_n f, f \rangle \geq c_\Omega \|f\|^2, \quad f \in \text{Ran } \Pi_\Omega, \quad (91)$$

and, on this finite-dimensional spectral subspace,

$$|\langle G_n B f, f \rangle + \langle G_n f, B f \rangle| \leq \varepsilon_n \|f\|^2. \quad (92)$$

Then $\text{Ran } \Pi_\Omega = \{0\}$.

Proof. If $\rho \in \Omega$ is a zero and F_ρ is a value eigenvector, then $B F_\rho = (\rho - \frac{1}{2})F_\rho$. Substituting $f = F_\rho$ in (92) gives

$$2 \left| \Re \rho - \frac{1}{2} \right| \langle G_n F_\rho, F_\rho \rangle \leq \varepsilon_n \|F_\rho\|^2.$$

By (91),

$$2c_\Omega \left| \Re \rho - \frac{1}{2} \right| \leq \varepsilon_n,$$

which is impossible for large n because Ω stays a positive distance from the critical line. Thus no zero block occurs in Ω , and Theorem 18.2 gives $\Pi_\Omega = 0$. $\square$

The theorem is deliberately finite-window. A single globally bounded positive metric making B skew-adjoint would be substantially stronger and would impose global basis geometry on Burnol's complete/minimal zero system. For RH it is enough to obtain (91)–(92) separately on each compact off-line window.

The finite-prime Weil forms make this target concrete. On the Connes–Consani–Moscovici finite-dimensional space, under their stated hypotheses, including the required ground-state and boundary properties, subtracting the lowest form eigenvalue and quotienting its null direction makes the corrected logarithmic operator self-adjoint in the resulting positive form [7]. To turn that finite-prime positivity into Theorem 19.1, one would need a zero-independent transport of those forms into the Burnol fixed space for which two properties survive simultaneously: a cutoff-uniform coercive lower bound on $\text{Ran } \Pi_\Omega$, and a Lyapunov defect tending to zero. This is weaker than direct norm-resolvent convergence and avoids the exact-orthogonality obstruction, but no such transport estimate is proved here.

This formulation also identifies what would count as genuine progress from the finite-prime side. Mere convergence of eigenvalue lists, or convergence of the determinant after zeros are already matched, is not enough. The required new input is an operator/form comparison that remains positive before any zero locations are inserted.

A finite-core criterion without a closed-operator assumption

For zero exclusion the Riesz realization can be bypassed. Let $\mathcal{V}$ be the algebraic span of finitely many genuine zero chains $F_{\rho,j} = \zeta(s)/(s - \rho)^j$, $1 \leq j \leq m_\rho$. Distinct partial fractions are linearly independent, and the formula

$$B F_{\rho,1} = (\rho - \tfrac{1}{2})F_{\rho,1}, \quad B F_{\rho,j} = (\rho - \tfrac{1}{2})F_{\rho,j} + F_{\rho,j-1} \quad (j > 1)$$

defines an operator on this finite-dimensional space without a global domain or graph-core theorem. Equip $\mathcal{V}$ with any fixed positive Hilbert norm. The source functional η_B is one on each $F_{\rho,1}$ and zero on the higher members of each canonical chain.

Theorem 19.2 (Finite-core metric criterion and exact transport budget). *Let $J_n : \mathcal{V} \rightarrow E_n$, let $D_n = D_n^*$, and let $S_n = S_n^*$ be finite matrices. Define*

$$Y_n = iD_n J_n - J_n B, \quad G_n = J_n^* S_n J_n, \quad \mathcal{L}_n = G_n B + B^* G_n.$$

If $G_n \succeq c_n I$ for some $c_n > 0$ and

$$\frac{\|\mathcal{L}_n\|}{c_n} \longrightarrow 0, \quad (93)$$

then every eigenvalue of $B|_{\mathcal{V}}$ has real part zero. In particular a space consisting of off-line chains must be zero. No global closed-operator or quantitative evaluator assumption is used. The exact finite identity is

$$\boxed{\mathcal{L}_n = -iJ_n^*[D_n, S_n]J_n - J_n^* S_n Y_n - Y_n^* S_n J_n.} \quad (94)$$

Thus a sufficient condition for (93) is

$$\|J_n^*[D_n, S_n]J_n\| + 2\|J_n^* S_n Y_n\| = o(c_n).$$

Estimating the combined signed expression in (94) can be weaker than estimating its terms separately.

Proof. For a genuine eigenvector $Bf = \mu f$, first-slot linearity gives

$$\langle \mathcal{L}_n f, f \rangle = 2\Re \mu \langle G_n f, f \rangle.$$

Consequently $2|\Re \mu|c_n \leq \|\mathcal{L}_n\|$. Every zero chain contains its first, genuine eigenvector, so multiplicity creates no gap in this argument. To obtain the budget, substitute $J_n B = iD_n J_n - Y_n$ into both terms of $G_n B + B^* G_n$ and use $D_n = D_n^*$ and $S_n = S_n^*$. This proves the displayed identity with the indicated sign and adjoints. The norm bound is the triangle inequality. $\square$

The criterion is invariant under multiplication of S_n by a positive scalar. A vanishing unnormalized defect obtained by collapsing the metric does not suffice. On the corrected sampler of (329), its total source is exactly

$$Y_n = z_{L,2K}^c \eta_B + \mathcal{R}_{L,K}. \quad (95)$$

Both terms must enter (94). Weak G1 concerns particular Weil residual pairings and does not bound this budget for an arbitrary positive S_n . Applying the criterion to every hypothetical off-line zero would prove RH, but the required small signed budget remains an independent arithmetic obligation. The old Riesz theorem is still useful for spectral projections; its realization assumption has not been proved.

A positive metric that annihilates the source and shell

There is a concrete zero-list-independent benchmark for this criterion. Use the physical scaling $t_n = 2\pi n/L$, $D = D_L$, the centered output $H_F = \kappa F$ of (308), and the corrected sampler $J = \tilde{J}_{L,K}$ on E_{2K} . Fix $T > 0$, write $P = P_{T,L}$ for the orthogonal projection to $|t_n| \leq T$, and choose $K \geq \lfloor TL/(2\pi) \rfloor$ so the endpoint shell lies outside this band. Put

$$z = z_{L,2K}^c, \quad z_T = Pz, \quad w = \frac{z_T}{\|z_T\|}, \quad S_{T,L} = P - |w\rangle\langle w|.$$

For all sufficiently large L , $z_T \neq 0$. Here the rank-one map is $x \mapsto w \langle x, w \rangle$, in the fixed first-slot-linear convention. The construction uses values of $\kappa\zeta$ on a physical band, not a list of nontrivial zeros. It is an analytic benchmark, not a newly proved finite-prime Weil metric or positivity transfer.

Proposition 19.3 (Source-annihilating metric with fixed-core coercivity). *The matrix $S_{T,L}$ is an orthogonal projection and satisfies exactly*

$$S_{T,L}z = 0, \quad S_{T,L}\mathcal{R}_{L,K} = 0, \quad S_{T,L}J = S_{T,L}J_{L,2K}^c. \quad (96)$$

For every fixed finite root space $\mathcal{V}$ as above, there is $c_{T,\mathcal{V}} > 0$ such that, for sufficiently large L ,

$$\langle S_{T,L}JF, JF \rangle \geq c_{T,\mathcal{V}}\|F\|^2 \quad (F \in \mathcal{V}). \quad (97)$$

These assertions are uniform in the ambient cutoff once its shell lies outside the fixed band. The exact remaining Lyapunov defect is

$$G_L B + B^* G_L = -iJ^*[D, S_{T,L}]J, \quad G_L = J^* S_{T,L}J. \quad (98)$$

Proof. Since $w \in \text{Ran } P$ is a unit vector, $S_{T,L}$ is the orthogonal projection onto $\text{Ran } P \cap w^\perp$. It annihilates z . Both $c_{L,K}$ and $D_L c_{L,K}$ are supported on $K < |n| \leq 2K$, so it also annihilates the two summands of $\mathcal{R}_{L,K}F$ and the endpoint correction itself. This proves (96), without any estimate on their sizes.

For $U, V \in \mathcal{V} + \mathbb{C}\zeta$, set

$$[U, V]_T = \frac{1}{2\pi} \int_{-T}^T H_U(\tfrac{1}{2} + it) \overline{H_V(\tfrac{1}{2} + it)} dt.$$

The centered factors $(-1)^n$ cancel in ordinary Gram pairings. Riemann sums with coefficient $L^{-1/2}$ therefore give, uniformly on the fixed finite-dimensional unit sphere,

$$\langle S_{T,L}JF, JG \rangle \longrightarrow [F, G]_T - \frac{[F, \zeta]_T [\zeta, G]_T}{[\zeta, \zeta]_T}. \quad (99)$$

The denominator is positive because H_ζ is not identically zero on an interval. Equality in the Cauchy–Schwarz inequality on the right would give $H_F = aH_\zeta$ there. Since $F = \zeta R_F$ for a proper rational function R_F , analytic continuation away from isolated zeros forces $R_F \equiv a$. Its decay at infinity gives $a = 0$ and $F = 0$. Thus the limiting Gram form is positive definite. Finite-dimensional compactness proves (97). Finally use (94) and (96). $\square$

This disposes of the original source and shell terms for this particular metric, not of the whole compatibility problem. The remaining commutator can be computed exactly.

Proposition 19.4 (Variance of the remaining commutator). *Put $m_{T,L} = \langle Dw, w \rangle \in \mathbb{R}$, $h = (D - m_{T,L})w$, and $\varsigma_{T,L} = \|h\|$. Then*

$$[D, S_{T,L}] = -|h\rangle\langle w| + |w\rangle\langle h|, \quad \|[D, S_{T,L}]\| = \varsigma_{T,L}. \quad (100)$$

At each fixed $T > 0$,

$$\varsigma_{T,L}^2 \longrightarrow \frac{\int_{-T}^T (t - m_T)^2 |H_\zeta(\tfrac{1}{2} + it)|^2 dt}{\int_{-T}^T |H_\zeta(\tfrac{1}{2} + it)|^2 dt} > 0, \quad m_T = \frac{\int_{-T}^T t |H_\zeta(\tfrac{1}{2} + it)|^2 dt}{\int_{-T}^T |H_\zeta(\tfrac{1}{2} + it)|^2 dt}. \quad (101)$$

Moreover the self-adjoint compression $\widehat{D} = S_{T,L} D S_{T,L}$ on $\text{Ran } S_{T,L}$, with $\widehat{J} = S_{T,L} J$, has the exact new graph source

$$i\widehat{D}\widehat{J}F - \widehat{J}BF = -ih \langle JF, w \rangle. \quad (102)$$

Proof. D commutes with P and $h \perp w$. Expanding the commutator gives (100); on the span of w and h its two singular values are $\|h\|$, and it vanishes on the orthogonal complement. The Riemann-sum limits for the first two moments give (101). The variance is strictly positive since the nonzero continuous density is not supported at a single frequency. For (102), first use $S_{T,L}Y = 0$, then

$$i\hat{D}\hat{J} - \hat{J}B = -iS_{T,L}D(I - S_{T,L})J.$$

The part outside P is killed by $S_{T,L}D$, and the part along w gives $S_{T,L}Dw = h$, proving the formula. $\square$

Thus the elementary operator-norm bound for this commutator does not tend to zero. This does not by itself decide a restricted signed pairing. For a hypothetical eigenvector $BF = \mu F$, however, the exact restriction is

$$-i \langle [D, S_{T,L}]JF, JF \rangle = 2\Re \mu \langle S_{T,L}JF, JF \rangle.$$

It has the required nonzero size under an off-line hypothesis; it is not an independent upper estimate contradicting that hypothesis. Positivity and source annihilation alone have simply moved the defect into the commutator, or equivalently into (102).

Why removing successive source derivatives loses coercivity

Projecting away $Dz, D^2z, \dots$ is a natural proposed repair of the last commutator. On a fixed physical band it encounters an explicit obstruction.

Proposition 19.5 (Polynomial source annihilation suppresses zero vectors). *Fix $T > 0$ and a hypothetical off-line zero $\rho = 1/2 + \mu$, where $\mu = d + i\gamma$ and $d \neq 0$. Let $Q_{m,L}$ be contractions supported on $\text{Ran } P_{T,L}$ such that*

$$Q_{m,L}D^j z_T = 0 \quad (0 \leq j \leq 2m+1).$$

Set $M^2 = d^2 + (T + |\gamma|)^2$ and $r = 1 - d^2/M^2 \in (0, 1)$. For the genuine eigenvector $F_{\rho,1} = \zeta/(s - \rho)$ one has

$$\|Q_{m,L}JF_{\rho,1}\| \leq \frac{r^{m+1}}{|d|} \|z_T\|. \quad (103)$$

Consequently bounded projection metrics obtained by removing an increasing number of these source derivatives cannot retain a uniform positive lower bound on this vector when $m \rightarrow \infty$ at fixed T .

Proof. On $[-T, T]$ define $\xi(t) = 1 - (d^2 + (t - \gamma)^2)/M^2 \in [0, r]$. The polynomial

$$p_m(t) = \frac{-d - i(t - \gamma)}{M^2} \sum_{j=0}^m \xi(t)^j$$

has degree $2m + 1$. A finite geometric sum gives the exact remainder

$$\frac{1}{it - \mu} - p_m(t) = \frac{\xi(t)^{m+1}}{it - \mu}, \quad \sup_{|t| \leq T} \left| \frac{1}{it - \mu} - p_m(t) \right| \leq |d|^{-1} r^{m+1}.$$

The centered source and root samples have the same half-period phase, so $P_{T,L}JF_{\rho,1} = (iD - \mu)^{-1}z_T$ exactly; the high shell does not meet the band. Applying $Q_{m,L}$ removes $p_m(D)z_T$. The diagonal remainder estimate proves (103). Finally $\|z_T\|$ stays bounded by its Riemann-sum limit. $\square$

This is a limited obstruction to bounded polynomial-source projection metrics, not to all positive metrics, changing bands, or signed arithmetic estimates. Rescaling a collapsing metric must be assessed by the ratio in (93); it cannot be judged by an unnormalized commutator alone. Section 19 therefore now has a domain-free finite-core criterion and a coercive positive benchmark with exact source removal, but no independent small Lyapunov defect. G2 and RH remain open.

Why a sharp band projection does not inherit strip transfer

The ordinary Gram convergence in Proposition 19.3 does not authorize applying the holomorphic Weil-transfer theorem after the physical band projection. The following calculation identifies the new cut terms on exactly that finite object.

Lemma 19.6 (Transform of a sharp physical band). *Fix $T > 0$, set $\Delta = 2\pi/L$, $N = \lfloor T/\Delta \rfloor$ and $T_L = N\Delta$. Let a_L be functions with uniformly bounded C^2 norm on $[-T, T]$. In centered physical coordinates define*

$$f_L(y) = \frac{1}{L} \sum_{n=-N}^N a_L(t_n) e^{it_n y}, \quad |y| \leq L/2, \quad t_n = n\Delta,$$

and extend by zero. For fixed $v \in \mathbb{C}$ with $\Re v \neq 0$, put $\mathcal{H}_L(v) = \int f_L(y) e^{-vy} dy$. Then exactly

$$\mathcal{H}_L(v) = -\frac{2 \sinh(vL/2)}{L} \sum_{n=-N}^N \frac{(-1)^n a_L(t_n)}{it_n - v}. \quad (104)$$

Moreover, uniformly when v ranges over a compact subset of $\{\Re v \neq 0\}$,

$$\mathcal{H}_L(v) = -\frac{(-1)^N \sinh(vL/2)}{L} \left\{ \frac{a_L(T_L)}{iT_L - v} + \frac{a_L(-T_L)}{-iT_L - v} + O(L^{-1}) \right\}. \quad (105)$$

If $a_L = a + O(L^{-1})$ in C^2 , define

$$C_a(v) = \frac{a(T)}{iT - v} + \frac{a(-T)}{-iT - v}.$$

Whenever $C_a(v) \neq 0$,

$$|\mathcal{H}_L(v)| \sim \frac{|C_a(v)|}{2L} e^{|\Re v|L/2}. \quad (106)$$

This is compatible with a bounded ordinary norm, since $\|f_L\|_2^2 = L^{-1} \sum_{|n| \leq N} |a_L(t_n)|^2 = O(1)$.

Proof. Integrate each exponential over $[-L/2, L/2]$; both endpoint phases are $(-1)^n$, which proves (104) with its sign and factor two. Set $q(t) = a_L(t)/(it - v)$ and $q_j = q(-T_L + j\Delta)$. An exact telescoping identity is

$$2 \sum_{j=0}^{2N} (-1)^j q_j - q_0 - q_{2N} = \sum_{k=0}^{N-1} (q_{2k} - 2q_{2k+1} + q_{2k+2}).$$

The right side has absolute value at most $N\Delta^2 \|q''\|_\infty = T_L \Delta \|q''\|_\infty$. The denominators are uniformly separated from zero on the stated compact sets of v , giving (105). Now $T_L = T + O(L^{-1})$ and the assumed C^2 convergence replace the braces by $C_a(v) + O(L^{-1})$. Finally $|\sinh(vL/2)| \sim \frac{1}{2} e^{|\Re v|L/2}$. Parseval on the finite interval proves the ordinary norm assertion. $\square$

Proposition 19.7 (Source removal does not remove the sharp-cut growth). *Use the corrected sampler and the projection $S_{T,L}$ of Proposition 19.3, with its endpoint shell outside the fixed band. Suppose $\rho = \frac{1}{2} + \mu$ is a hypothetical off-line zero, take its genuine first eigenvector $F = \zeta/(s - \rho)$, and choose $T > 0$ such that $H_\zeta(\frac{1}{2} \pm iT) \neq 0$. Such bands exist because the excluded endpoint zeros are isolated. Put*

$$Z(t) = H_\zeta(\tfrac{1}{2} + it), \quad R(t) = \frac{1}{it - \mu}, \quad \alpha_T = \frac{\int_{-T}^T |Z(t)|^2 R(t) dt}{\int_{-T}^T |Z(t)|^2 dt}.$$

The centered physical representative of $S_{T,L} JF$ has the coefficients of Lemma 19.6 with

$$a_L(t) = Z(t)\{R(t) - \alpha_L\}, \quad \alpha_L = \frac{\sum_{|n| \leq N} |Z(t_n)|^2 R(t_n)}{\sum_{|n| \leq N} |Z(t_n)|^2} = \alpha_T + O(L^{-1}).$$

Let $a(t) = Z(t)\{R(t) - \alpha_T\}$. For every fixed real $b > 0$, at least one of $C_a(b)$ and $C_a(-b)$ is nonzero. Thus at least one of the two transforms at $s = \frac{1}{2} \pm b$ grows as a nonzero constant times $e^{bL/2}/L$. In particular these source-projected vectors have no bounded full-strip transform limit, even on a compact substrip containing those two points when $0 < b < 1/2$.

Proof. The projection coefficient is the first-slot-linear ratio $\langle PJF, z_T \rangle / \|z_T\|^2$, giving the displayed α_L ; the half-period factors cancel in this ratio. The shell is absent from the band. Since $\Re \mu \neq 0$, R is smooth there. Ordinary Riemann-sum estimates give the $O(L^{-1})$ rate, including the positive limiting denominator. Consequently $a_L = a + O(L^{-1})$ in C^2 .

Write $A = a(T)$ and $B = a(-T)$. The two equations $C_a(b) = C_a(-b) = 0$ form a nonsingular two-by-two system in A, B : its determinant is

$$-\frac{4ibT}{(T^2 + b^2)^2} \neq 0.$$

They would force $A = B = 0$. Since $Z(\pm T) \neq 0$, this would imply $R(T) = \alpha_T = R(-T)$, which is impossible for $T > 0$. Apply (106) to a nonzero coefficient. $\square$

The functional-equation partner does not, by itself, cancel this edge mechanism. The next statement concerns one reflected pair of summands in the zero-side formula, not the complete Weil form.

Proposition 19.8 (Reflected-pair edge contribution). *In Lemma 19.6, assume $a_L = a + O(L^{-1})$ in $C^2[-T, T]$. Fix $v = d + i\gamma$ with $d > 0$, put $v^\sharp = -\bar{v}$, and suppose the zeros $\frac{1}{2} + v$ and $\frac{1}{2} + v^\sharp$ have common multiplicity m . The contribution of this reflected pair to the value-level zero formula is*

$$\mathcal{P}_{v,L} = m \left\{ \mathcal{H}_L(v) \overline{\mathcal{H}_L(v^\sharp)} + \mathcal{H}_L(v^\sharp) \overline{\mathcal{H}_L(v)} \right\}.$$

Writing

$$A_{a,T}(v) = C_a(v) \overline{C_a(v^\sharp)},$$

one has

$$\boxed{\mathcal{P}_{v,L} = -\frac{me^{dL}}{2L^2} \Re(e^{i\gamma L} A_{a,T}(v)) + O\left(\frac{e^{dL}}{L^3}\right)}. \quad (107)$$

Thus reflection converts the two growing transforms into a real oscillatory term; it does not cancel them identically. If $A_{a,T}(v) \neq 0$ and $\gamma \neq 0$, this pair contribution has subsequences of both signs and magnitude comparable to e^{dL}/L^2 . For the source-projected root vector of Proposition 19.7, the coefficient is the explicit product formed from $a(t) = Z(t)\{(it - \mu)^{-1} - \alpha_T\}$. No assertion is made when one of these two edge coefficients vanishes.

Proof. Lemma 19.6 gives, with $\epsilon_L = (-1)^N$,

$$\mathcal{H}_L(v) = -\frac{\epsilon_L \sinh(vL/2)}{L} \{C_a(v) + O(L^{-1})\}.$$

Since $v^\sharp = -\bar{v}$,

$$\overline{\mathcal{H}_L(v^\sharp)} = \frac{\epsilon_L \sinh(vL/2)}{L} \{\overline{C_a(v^\sharp)} + O(L^{-1})\}.$$

Their product therefore equals

$$-\frac{\sinh^2(vL/2)}{L^2} \{A_{a,T}(v) + O(L^{-1})\}.$$

The other reflected summand is its complex conjugate. Finally $\sinh^2(vL/2) = \frac{1}{4}e^{vL}\{1 + O(e^{-dL})\}$, which proves (107). Choosing the phase of γL near $-\arg A_{a,T}(v)$ and near $\pi - \arg A_{a,T}(v)$ gives the two subsequences. $\square$

This does not prove divergence or negativity of the complete Weil form: other zero summands may cancel the displayed contribution. The combined prime, pole and archimedean expression is an equivalent representation of that same form, not an additional contribution. Nor does this rule out a new signed product estimate just on the zero set. It proves that the existing bounded holomorphic-strip transfer cannot simply be inherited by these sharply projected vectors. There is no conflict with weak G1, which estimates a residual against band tests without asserting that those tests retain the original Hardy output's strip bounds.

For a proposed correction $M = S_{T,L}(W + a_0 I)S_{T,L}$, source and shell annihilation still hold, but all three terms in

$$[D, M] = [D, S_{T,L}](W + a_0 I)S_{T,L} + S_{T,L}[D, W]S_{T,L} + S_{T,L}(W + a_0 I)[D, S_{T,L}]$$

remain in the metric budget. Neither positivity of this compression nor a small signed budget is proved. A valid next step must control its actual finite prime, pole and archimedean terms together, or introduce a different filter with independently justified product estimates.

Full-space projection with a proved strip-product bound

The preceding obstruction concerns a fixed physical band. It can be avoided without discarding the graph defect. Keep the original ambient space E_{2K} , the physical D , the source $z = J_{L,2K}^c \zeta$, the shell $c = c_{L,K}$ and the corrected $J = \tilde{J}_{L,K}$. Define

$$e_0 = c/\|c\|, \quad e_1 = Dc/\|Dc\|, \quad P_{\text{sh}} = |e_0\rangle\langle e_0| + |e_1\rangle\langle e_1|, \quad Q_0 = I - P_{\text{sh}}.$$

Parity makes e_0, e_1 orthonormal. Put

$$u = Q_0 z, \quad S_{L,K} = Q_0 - \frac{|u\rangle\langle u|}{\|u\|^2}, \quad \alpha_{L,K}(F) = \frac{\langle Q_0 J_{L,2K}^c F, u \rangle}{\|u\|^2}. \quad (108)$$

The nonzero denominator for large L is proved below. This construction uses analytic samples of $\kappa\zeta$ and the two known shell vectors; it does not use a zero list. It changes the metric, not J , its physical endpoint, or the finite Weil matrix.

Proposition 19.9 (Full-space source removal and Weil transfer). *Let $\mathcal{V}$ be a fixed finite span of genuine root chains, on either side of or on the critical line. Choose the fixed Hardy smoothing so that, on $\mathcal{U} = \mathcal{V} + \mathbb{C}\zeta$,*

$$|H_U(\tfrac{1}{2} + it)| \leq C\|U\|(1 + |t|)^{-R}, \quad R \geq 3, \quad |g_U^{(j)}(y)| \leq C\|U\|e^{-p|y|}, \quad j \leq 2,$$

with $p > 1/2$, as in (309). Let $K \geq \lceil e^{2L} L^4 \rceil$. Then $S = S_{L,K}$ is an orthogonal projection and, exactly,

$$Sz = Sc = SDc = 0, \quad S\mathcal{R} = 0, \quad SJF = Q_0 J_{L,2K}^c (F - \alpha_{L,K}(F)\zeta). \quad (109)$$

If $\mathcal{H}_{L,K,F}$ is the transform of SJF translated back to $[-L/2, L/2]$ and zero extended, then throughout $0 \leq \Re s \leq 1$,

$$|\mathcal{H}_{L,K,F}(s) - H_F(s) + \alpha_{L,K}(F)H_\zeta(s)| \leq \frac{C\varepsilon_{L,K}\|F\|}{1 + |\Im s|}, \quad (110)$$

where, uniformly over these cutoffs,

$$\varepsilon_{L,K} = e^{-(p-1/2)L/2} + Le^{L/4}(1 + K/L)^{2-R} \longrightarrow 0.$$

Write $[U, V]_\infty = (2\pi)^{-1} \int_{\mathbb{R}} H_U(1/2 + it) \overline{H_V(1/2 + it)} dt$. Uniformly on bounded subsets of $\mathcal{V}$,

$$\langle SJF, SJG \rangle \longrightarrow [F, G]_\infty - \frac{[F, \zeta]_\infty [\zeta, G]_\infty}{[\zeta, \zeta]_\infty}, \quad (111)$$

$$\langle WSJF, SJG \rangle \longrightarrow \mathcal{Q}(g_F, g_G). \quad (112)$$

The first limiting form is positive definite on $\mathcal{V}$. The second limit has error $O(\varepsilon_{L,K} + \varepsilon_{L,K}^2)$.

Proof. Let $X = 1 + K/L$. A centered sample vector $a = J_{L,2K}^c U$ obeys

$$\|1_{K < |n| \leq 2K} a\|_2 \leq C\|U\|X^{1/2-R}, \quad \sum_n (1 + |t_n|) |(P_{\text{sh}} a)_n| \leq C\sqrt{L}\|U\|X^{2-R}.$$

For the second inequality use Cauchy–Schwarz on the $2K$ shell coordinates, $\|P_{\text{sh}} a\| \leq \|1_{\text{sh}} a\|$ and $1 + |t_n| \leq CX$ there. The omitted coefficients of the infinite centered Fourier series satisfy the same weighted sum bound, since $R > 2$. Riemann sums on the line and these tail bounds give

$$\|u\|^2 \longrightarrow [\zeta, \zeta]_\infty > 0, \quad \alpha_{L,K}(F) \longrightarrow [F, \zeta]_\infty / [\zeta, \zeta]_\infty.$$

In particular $\alpha_{L,K}$ is uniformly bounded on the fixed space. The projection and annihilation identities now follow directly from (108); $\mathcal{R}F$ lies in $\text{span}\{c, Dc\}$ exactly. Since $Sc = 0$, the endpoint correction in J is annihilated, which proves the last identity in (109).

Here is the strip estimate, including the physical endpoint jumps. Set $U = F - \alpha_{L,K}(F)\zeta$. Compare the centered representative of $Q_0 J_{L,2K}^c U$ with $1_{[-L/2, L/2]} P_L g_U$. Their difference b consists of the omitted Fourier tail and the negative shell projection. The weighted coefficient bounds above, with the physical basis factor $L^{-1/2}$, give on this interval

$$\|b\|_\infty + \|b'\|_\infty \leq C\|F\|X^{2-R}.$$

For $|d| \leq 1/2$, zero extension therefore has

$$\|e^{-d \cdot} b\|_1 + \text{Var}(e^{-d \cdot} b) \leq CL e^{L/4} \|F\| X^{2-R}.$$

The variation includes both cut atoms. The L^1 estimate at bounded ordinates and one integration by parts against this finite derivative measure at large ordinates give the second term of (110). Lemma 24.1 supplies the first term. At $R = 3$ the second term is at most $Ce^{-7L/4}L^{-2}$ for the stated cutoffs. No fixed-band edge asymptotic is used.

Ordinary Gram convergence and the projection formula prove (111). Equality in its Cauchy–Schwarz bound would give $H_F = aH_\zeta$ almost everywhere on the line. Analyticity then forces the proper rational function F/ζ to be constant, hence zero. This proves positive definiteness and thus a uniform lower bound on the fixed finite-dimensional unit sphere.

For the Weil limit, $H_\zeta(\rho) = 0$ at every nontrivial zero; the pole at 1 has already been canceled by κ . Thus subtracting $\alpha_{L,K}(F)H_\zeta$ changes none of the global zero-form values. Equation (110) and the bounded holomorphic transforms bound each product error by

$$C\|F\|\|G\|(\varepsilon_{L,K} + \varepsilon_{L,K}^2)(1 + |\Im \rho|)^{-2}.$$

This is summable over all zeros, without RH. Apply (310) and translate back to $[0, L]$; reflected translation factors cancel in each product. This proves (112) on the actual matrix $W_{L,2K}$. $\square$

The projection above resolves a transfer issue for a different metric; it does not repair the fixed-band vectors of Propositions 19.7–19.8. Nor does it make the original graph defect small: it annihilates that defect inside a particular metric. Its remaining signed budget is exactly

$$G_L B + B^* G_L = -iJ^*[D, S_{L,K}]J, \quad G_L = J^* S_{L,K} J. \quad (113)$$

In particular Q_0 must not be commuted through D ; removing the two shell directions does not make their orthogonal complement invariant.

Corollary 19.10 (A fixed scalar shift does not close the repaired metric). *Let $\mathcal{V}$ be a nonempty hypothetical one-sided off-line root space and fix $a_0 > 0$. With $S = S_{L,K}$ as above, put*

$$M_L = S(W + a_0 I)S, \quad A_L = J^* S W S J, \quad G_L = J^* S J, \quad H_L = J^* M_L J.$$

Then $A_L \rightarrow 0$, $G_L \rightarrow G_\infty \succ 0$ and $H_L \rightarrow a_0 G_\infty \succ 0$, but

$$H_L B + B^* H_L \longrightarrow a_0(G_\infty B + B^* G_\infty) \neq 0. \quad (114)$$

Consequently this candidate does not satisfy the relative defect criterion of Theorem 19.2.

Proof. The global Weil form vanishes on a one-sided root space, so (112) gives $A_L \rightarrow 0$ in its finite matrix norm. Since B is fixed, $A_L B + B^* A_L \rightarrow 0$ as well. The identity $H_L = A_L + a_0 G_L$ proves the limits. On a first root vector f , $Bf = \mu f$ with $\Re \mu \neq 0$, the limiting signed pairing is $2a_0 \Re \mu \langle G_\infty f, f \rangle \neq 0$. Its ratio to the limiting metric pairing is $2\Re \mu$. The pullback metrics are bounded above as well as below, so their positive lower bounds cannot turn this nonzero limit into a vanishing relative defect. $\square$

This is restricted coercivity, not positivity of M_L on the entire ambient space. The exact budget for M_L , whose source terms also vanish, still contains all three terms

$$[D, M_L] = [D, S](W + a_0 I)S + S[D, W]S + S(W + a_0 I)[D, S].$$

The new transfer theorem controls their combined pulled-back value via (114); it supplies no independent small signed estimate. A successful G2 metric must do more than preserve the one-sided zero Weil limit and add a fixed ordinary positive shift.

Norm-level Weil action and the squared-metric candidate

Squaring the finite Weil matrix supplies positivity without a scalar shift. Its behavior cannot be inferred from form convergence alone. We first prove the needed norm estimate. The classical unconditional zero count gives, with multiplicities,

$$\#\{\rho : j \leq \Im \rho < j+1\} \leq C \log(2+|j|), \quad j \in \mathbb{Z}. \quad (115)$$

For example, this follows by subtracting the estimates for $N(T)$ at the two endpoints in Trudgian [17, Corollary 1]; finitely many small ordinates are absorbed into C . Neither simplicity nor RH is used.

Lemma 19.11 (Local synthesis from decaying zero coefficients). *Suppose $|a_\rho| \leq \epsilon/(1+|\Im \rho|)$ on the multiset of nontrivial zeros. Then, for $A \geq 1$, the series*

$$r(y) = \sum_{\rho \in \mathcal{Z}} a_\rho e^{(\rho-1/2)y}$$

converges in $L^2(-A, A)$, and

$$\|r\|_{L^2(-A, A)} \leq C\epsilon\sqrt{A+1}e^{A/2}. \quad (116)$$

The constant is independent of A , ϵ and the coefficients.

Proof. Choose a nonnegative smooth function χ_A equal to one on $[-A, A]$, zero outside $[-A-1, A+1]$, with its first two derivatives bounded independently of A . Write $\rho = 1/2 + d_\rho + i\gamma_\rho$. Since $|d_\rho| < 1/2$, two integrations by parts, together with the direct integral bound, give

$$\left| \int \chi_A(y) e^{(d_\rho + d_\tau)y} e^{i(\gamma_\rho - \gamma_\tau)y} dy \right| \leq \frac{C(A+1)e^A}{1+|\gamma_\rho - \gamma_\tau|^2}.$$

For a finite partial sum put $b_j = \sum_{j \leq \gamma_\rho < j+1} |a_\rho|$. Grouping both indices by these unit intervals bounds its squared norm by

$$C(A+1)e^A \sum_{j,k} \frac{b_j b_k}{1+|j-k|^2} \leq C(A+1)e^A \sum_j b_j^2.$$

The last inequality is the $\ell^1 * \ell^2$ convolution bound. Equation (115) gives $b_j \leq C\epsilon \log(2+|j|)/(1+|j|)$, whose square is summable. The same bound on omitted indices makes the partial sums Cauchy in $L^2(-A, A)$ and proves (116). It also permits arbitrary finite exhaustions of the zero multiset. $\square$

Let $\mathcal{T}_L$ denote the unitary translation from the physical interval $[0, L]$ to $[-L/2, L/2]$; on coefficient vectors it represents x as $L^{-1/2} \sum x_n (-1)^n e^{it_n y}$. Let $\Pi_{L,2K}$ be the ordinary orthogonal projection onto the corresponding centered Fourier space. These operations do not change the finite matrix W .

Proposition 19.12 (Weil-action transfer on a fixed root family). *Use the objects and cutoffs of Proposition 19.9, but choose the fixed smoothing with tail rate $p > 1$. Define the finite sum*

$$q_F(y) = \sum_{\rho \in \mathcal{Z}} H_F(\rho) e^{(\rho-1/2)y},$$

where zeros are counted with multiplicity; only the finitely many selected root locations can contribute. Then uniformly on the fixed root family,

$$\|\mathcal{T}_L W S J F - \Pi_{L,2K} q_F\|_2 \leq C \Delta_{L,K} \|F\|, \quad \Delta_{L,K} = \sqrt{L+1} e^{L/4} \varepsilon_{L,K} \longrightarrow 0. \quad (117)$$

In particular the first part of $\Delta_{L,K}$ is $\sqrt{L+1} e^{-(p-1)L/2}$. The original condition $p > 1/2$ alone does not make this bound tend to zero.

Proof. Let $x = SJF$ and let $\mathcal{H}_x$ be the transform of $\mathcal{T}_L x$. At a zero, Proposition 19.9 and $H_\zeta(\rho) = 0$ give

$$|\mathcal{H}_x(\rho) - H_F(\rho)| \leq C\varepsilon_{L,K}\|F\|/(1 + |\Im\rho|).$$

Lemma 19.11 synthesizes these errors into a function r_F of norm at most the right side of (117). For each centered finite Fourier test v , the first-slot-linear zero formula gives exactly

$$\mathcal{Q}(\mathcal{T}_L x, v) = \sum_{\rho} \mathcal{H}_x(\rho) \overline{H_v(\rho^\sharp)} = \langle q_F + r_F, v \rangle.$$

Indeed $\overline{\rho^\sharp} - 1/2 = -(\rho - 1/2)$, so the exponential in the first slot is $e^{(\rho-1/2)y}$ with a positive sign. The product series is absolutely convergent for these tests; the synthesized series also converges in L^2 . Translation invariance of the paired form identifies the left side with $\langle \mathcal{T}_L W x, v \rangle$. Thus $\mathcal{T}_L W x = \Pi_{L,2K}(q_F + r_F)$, and projection contractivity proves the bound. At $R = 3$ the cutoff contribution to $\Delta_{L,K}$ is $O(e^{-3L/2}L^{-3/2})$. The other contribution tends to zero because $p > 1$. $\square$

The stronger tail margin is available, for example with fixed $\sigma = 2$, $a = 4$ and sufficient M , taking $1 < p \leq 3/2$ in (309). Critical-line root chains are allowed here and in Proposition 19.9: the zeros remove their apparent transform poles, and the proper-rational independence proof works away from the isolated zero locations. This extension assumes genuine root functions, not inserted meromorphic poles.

Corollary 19.13 (Mass of the squared-Weil metric). *On the same finite space set $M_L^\square = SW^2S \succeq 0$. For a zero $\rho = 1/2 + \mu$ of multiplicity m , write $F_{\rho,j} = \zeta/(s - \rho)^j$, $1 \leq j \leq m$, and*

$$C_\rho = m\kappa(\rho) \frac{\zeta^{(m)}(\rho)}{m!} \neq 0.$$

If $d = \Re\mu \neq 0$, then

$$\langle M_L^\square JF_{\rho,m}, JF_{\rho,m} \rangle = \|WSJF_{\rho,m}\|^2 \sim |C_\rho|^2 \frac{\sinh(|d|L)}{|d|}. \quad (118)$$

For a critical-line zero the corresponding asymptotic is $|C_\rho|^2 L$. For every lower jet $j < m$,

$$\langle M_L^\square JF_{\rho,j}, JF_{\rho,j} \rangle = O(\Delta_{L,K}^2) \longrightarrow 0. \quad (119)$$

These assertions concern the displayed normalization; a rescaled metric must still satisfy the relative criterion, not merely an absolute bound.

Proof. Self-adjointness of W gives positivity and the squared norm identity. For the top root, $q_F = C_\rho e^{\mu y}$; for lower jets $q_F = 0$, because all their zero values vanish. Moreover

$$\|e^{\mu y}\|_{L^2(-L/2, L/2)}^2 = \begin{cases} \sinh(|d|L)/|d|, & d \neq 0, \\ L, & d = 0. \end{cases}$$

Its centered Fourier coefficient at index n is $2(-1)^n \sinh(\mu L/2)/(\sqrt{L}(\mu - it_n))$, with the removable value used when $\mu = it_n$. For $|n| > 2K$ this gives

$$\|(I - \Pi_{L,2K})e^{\mu \cdot}\|_2 \leq C_\mu e^{|d|L/2} \sqrt{L/K} \longrightarrow 0.$$

The cutoff dominates this exponential since $|d| < 1/2$. Apply Proposition 19.12 to obtain both asymptotics and the lower-jet bound. $\square$

This candidate is globally positive and annihilates the exact total source, since $S(z\eta_B + \mathcal{R}) = 0$. Nevertheless its full commutator is

$$[D, M_L^\square] = [D, S]W^2S + S([D, W]W + W[D, W])S + SW^2[D, S].$$

All terms belong to the signed G2 budget. At a hypothetical simple off-line root $Bf = \mu f$, putting $G_L^\square = J^*M_L^\square J$ gives the exact quotient

$$\frac{\langle (G_L^\square B + B^*G_L^\square)f, f \rangle}{\langle G_L^\square f, f \rangle} = 2\Re\mu$$

whenever its denominator is nonzero. At a multiple zero, the lower jets additionally lose an unscaled coercive lower bound by (119). Squaring has not supplied a small relative defect. The norm-action theorem is a stronger transfer result, not a proof of the independent signed arithmetic estimate or RH.

20 Finite Weil residuals: compact-source radicality and G1 transfer

The finite-prime comparison can be narrowed further. The following subsection isolates the part of the Connes–Consani–Moscovici comparison that is controlled by classical prolate asymptotics and the prime number theorem, separately from the still-open spectral-ordering problem.

A rank-two symmetry defect for an arbitrary even candidate

Let $W_{\lambda,N}$ denote the finite Weil matrix, let D_N be the Fourier-index matrix, and identify the equal-coefficient boundary covector with its Riesz vector η_N , writing $\eta_N(x) = \langle x, \eta_N \rangle$ in the first-slot-linear convention. The exact CCM commutator has the form

$$D_N W_{\lambda,N} - W_{\lambda,N} D_N = |\beta_N\rangle\langle\eta_N| - |\eta_N\rangle\langle\beta_N|, \quad (120)$$

where β_N is odd and η_N is even. Let k_N be any real even vector satisfying $\eta_N(k_N) = 1$, and define

$$\mathcal{D}_{k,N} = D_N - |D_N k_N\rangle\langle\eta_N|.$$

Then parity gives $\langle k_N, \beta_N \rangle = 0$, and applying (120) to k_N gives

$$W_{\lambda,N} D_N k_N = D_N W_{\lambda,N} k_N - \beta_N.$$

Substitution yields the exact identity

$$\boxed{W_{\lambda,N} \mathcal{D}_{k,N} - \mathcal{D}_{k,N}^* W_{\lambda,N} = -|D_N W_{\lambda,N} k_N\rangle\langle\eta_N| + |\eta_N\rangle\langle D_N W_{\lambda,N} k_N|.} \quad (121)$$

Thus no ground-state vector is needed to identify the metric-symmetry defect: for an explicit even candidate it is rank at most two and is controlled by one linear residual.

It is useful to state the scale invariant version with the physical logarithmic generator $D_{\log} = (2\pi/L)D_N$, $L = 2\log\lambda$, and the actual boundary-evaluation covector $\delta_N = L^{-1/2}\eta_N$. On the fixed physical test band $|D_{\log}| \leq T$, the restriction of the boundary-evaluation covector δ_N has norm bounded by a constant depending only on T , independently of how large the ambient section E_N is. Thus one may later choose a much larger diagonal section $N = N(\lambda)$ to resolve the prolate

boundary layer without changing the fixed-window estimate. Hence the relevant projective quantity is

$$\frac{\|P_T D_{\log} W_{\lambda, N} k_N\|}{|\delta_N(k_N)|}, \quad (122)$$

where P_T is projection to the physical-frequency window $[-T, T]$. This formulation avoids the factor-of- $\sqrt{L}$ ambiguity that results from simultaneously rescaling the covector and the candidate.

The compact-source radical passage

The classical radical theorem applies to the even Schwartz class with $h(0) = \hat{h}(0) = 0$ [13, Section 3]. The following extension supplies the domain passage needed for the compact prolate source. It is a statement about explicit-formula pairings and the closed *semilocal* form, not a claim of a positive global closed form.

Theorem 20.1 (Radicality with a nonzero compact-source endpoint). *Fix $b > 0$. Let h be even and smooth on $[-b, b]$, extended by zero outside that interval, and suppose*

$$h(0) = 0, \quad \int_0^b h(u) du = 0.$$

No vanishing of the interior trace $h(b^-)$ is required. In logarithmic coordinates put

$$k(x) = \mathcal{E}(h)(e^x) = e^{x/2} \sum_{n \geq 1} h(ne^x).$$

For every v obtained by zero-extending an H^1 function on a compact interval J , all terms of the polarized explicit formula $QW(k, v)$ are finite and $QW(k, v) = 0$. In particular, for a compact interval I containing the support of v , set $k_{\text{in}} = 1_I k$ and $r = k - k_{\text{in}}$. Then k_{in} and v belong to the closed semilocal form domain and

$$QW_I(k_{\text{in}}, v) = -QW(r, v).$$

The assertion holds at each fixed b ; its approximation constants need not be uniform as b varies.

Proof. A primitive gains one power. Write $\beta(t) = \frac{1}{2} - \{t\}$ away from integers. Euler summation, with $N = \lfloor b/u \rfloor$, gives for almost every sufficiently small u

$$\sum_{n=1}^N h(nu) = u^{-1} \int_0^{Nu} h(t) dt + \frac{h(Nu) - h(0)}{2} + O_h(u) = h(b^-) \beta(b/u) + O_h(u).$$

For example, the composite trapezoid remainder is bounded by $Cbu \|h''\|_\infty$ in the first equality; Taylor expansion of the missing interval $[Nu, b]$ gives the second. Values at the arithmetic cuts affect neither an integral nor the correlations below. Consequently

$$\mathcal{E}(h)(u) = h(b^-) u^{1/2} \beta(b/u) + O_h(u^{3/2}). \quad (123)$$

In particular $k \in L^2(\mathbb{R})$, it vanishes for $x > \log b$, and it is of bounded variation on every compact logarithmic interval. Let $K(x) = \int_{-\infty}^x k(t) dt$. Since β has mean zero and a bounded primitive,

$$\int_0^u v^{-1/2} \beta(b/v) dv = \sqrt{b} \int_{b/u}^\infty \beta(w) w^{-3/2} dw = O_b(u^{3/2}).$$

The last bound follows by one integration by parts against that bounded primitive. Integrating the remainder in (123) yields

$$K(x) = O_h(e^{3x/2}) \quad (x \rightarrow -\infty). \quad (124)$$

Both moments matter: without $h(0) = 0$ an additional term $-\frac{1}{2}h(0)u^{1/2}$ generally prevents this gain.

Absolute convergence of the geometric side. The zero extension v_0 of v has bounded variation, including its two endpoint jumps. In the first-slot-linear convention its correlation is

$$C_{k,v}(y) = \int k(x) \overline{v_0(x-y)} dx = - \int K(x) \overline{dv_0(x-y)}.$$

Thus (124) gives $C_{k,v}(y) = O_{h,v}(e^{3y/2})$ as $y \rightarrow -\infty$; it is zero for all sufficiently large positive y . Its reflected correlation has the opposite support orientation. The even combinations therefore decay as $O_{h,v}(e^{-3y/2})$ for $y \rightarrow +\infty$. Each growing pole integral is absolutely convergent, and so is the prime series, since its tail is bounded by a constant times $\sum_{n \geq 2} (\log n) n^{-2}$.

At the archimedean singularity use, for $0 < t \leq 1$,

$$\|v_0(\cdot - t) - v_0\|_2 \leq C \left[t \|v'\|_2 + \sqrt{t} \{|v(\inf J)| + |v(\sup J)|\} \right]. \quad (125)$$

This follows from the fundamental theorem of calculus on the overlap and the two endpoint strips. Cauchy–Schwarz makes the subtracted archimedean numerator $O(t^{1/2})$, which is integrable against its $O(t^{-1})$ kernel. Its large- t part is integrable by L^2 bounds and exponential kernel decay.

Moment-preserving smoothing. Choose $\phi_\epsilon \geq 0$ in $C_c^\infty((\epsilon, 2\epsilon))$ with integral one, and define

$$h_\epsilon(u) = \int \phi_\epsilon(t) e^{t/2} h(e^t u) dt, \quad k_\epsilon(x) = \int \phi_\epsilon(t) k(x+t) dt. \quad (126)$$

Multiplicative convolution smooths the endpoint; near zero it preserves the smooth even germ. Hence h_ϵ is even and compactly supported smooth, with support in $[-be^{-\epsilon}, be^{-\epsilon}]$. Both moments remain exactly zero: dilation multiplies the integral by $e^{-t/2}$ and the central value is identically zero. Direct substitution in the finite source sum proves $\mathcal{E}(h_\epsilon)(e^x) = k_\epsilon(x)$, with all scale factors as in (126). This is the scaling covariance used in [13, Lemma 6.2]. Translation continuity gives $k_\epsilon \rightarrow k$ in $L^2(\mathbb{R})$, while

$$K_\epsilon(x) = \int \phi_\epsilon(t) K(x+t) dt = O_h(e^{3x/2})$$

uniformly for sufficiently small ϵ . In particular none of the estimates needed for the passage uses derivatives of h_ϵ that might blow up near its moving endpoint.

For a smooth compact logarithmic test w , the classical Schwartz radical theorem gives $QW(k_\epsilon, w) = 0$. To allow the stated v , convolve its zero extension with a nonnegative smooth additive approximate identity. These tests converge in L^2 , have uniformly bounded variation and support, and satisfy the same translation bound (125) by convolution contraction. The preceding correlation bounds therefore dominate separately the prime series, both pole integrals and the subtracted archimedean integral. Dominated convergence gives $QW(k_\epsilon, v) = 0$. The same argument, now using the uniform primitive bound for K_ϵ and L^2 convergence, allows $\epsilon \downarrow 0$ and proves $QW(k, v) = 0$.

The semilocal domain and truncation. On any fixed compact interval k is a finite sum of smooth pieces with finitely many jumps, so k_{in} is compactly supported BV. Such a function has Fourier transform $O((1 + |\xi|)^{-1})$, and hence

$$\int_{\mathbb{R}} (1 + \log(2 + |\xi|)) |\widehat{k_{\text{in}}}(\xi)|^2 d\xi < \infty.$$

The same holds for v_0 . On a fixed compact support the archimedean multiplier has logarithmic growth; the finitely many prime translations and pole terms are bounded forms. This is the closed semilocal form domain of the explicit formula [13]. The smooth approximations converge there as well: on a fixed enlarged interval they have uniform BV bounds and L^2 convergence, so a Fourier tail split with $\log(2 + |\xi|)/(1 + |\xi|)^2$ proves convergence in the logarithmic form norm. Compact truncation leaves the far lower primitive unchanged, so $QW(r, v)$ is also finite. Linearity now gives the asserted truncation identity, with the semilocal term equal to its closed-form realization. $\square$

The endpoint of the approximating sources need not equal that of h . The theorem is proved at fixed b *before* normalizing by the original source's separately evaluated one-sided endpoint. No endpoint trace is inferred from form-norm convergence. This corrects the stronger, unnecessary endpoint-preserving approximation requirement in version 1.3.

The focused PSWF cross-tail estimates

The fixed-order radial estimates, the two-sided exterior-energy argument, and Fuchs's concentration asymptotic below establish the source scale. Together with Theorem 20.1, they close the fixed-band and Sobolev-pairing G1 statements of this section for the explicit repaired source. They do not give a uniform operator-norm bound on a growing deep prolate block, spectral ordering, or the G2 graph/form comparison.

CCM's proposed source is the linear combination of the two fixed-order prolate modes $h_{0,\lambda}, h_{4,\lambda}$ with vanishing integral. The order-zero radial normalization can be matched exactly to the endpoint amplitude. In Dunster's notation, for $m = 0$ his matching constant satisfies

$$c_n^0(\gamma) = K_n^0(\gamma),$$

where $K_n^0(\gamma) = Ps_n^0(1, \gamma^2)$. His Bessel approximation is uniform for the full radial interval $1 < x < \infty$.

Proposition 20.2 (Endpoint-normalized radial bound). *Fix $n \in \{0, 4\}$. For all sufficiently large γ there is a constant C_n , independent of γ , such that*

$$|Ps_n^0(x, \gamma^2)| \leq C_n \frac{|Ps_n^0(1, \gamma^2)|}{x}, \quad x \geq 1.$$

Proof. Dunster's formula (3.5) gives, uniformly for $1 < x < \infty$,

$$Ps_n^0(x, \gamma^2) = c_n^0(\gamma) \left\{ \frac{\eta}{(x^2 - 1)(x^2 - \sigma^2)} \right\}^{1/4} \left[J_0(\gamma\eta^{1/2}) + O(\gamma^{-1}) \text{env } J_0(\gamma\eta^{1/2}) \right],$$

with $\eta = \xi^2$. His local relation

$$\eta = 2(1 - \sigma^2)(x - 1) + O((x - 1)^2)$$

shows that the geometric prefactor tends to one as $x \downarrow 1$; since $J_0(0) = 1$ and $c_n^0(\gamma) = Ps_n^0(1, \gamma^2)$, the normalization is exactly the endpoint scale up to the uniform $O(\gamma^{-1})$ error. For fixed n , Dunster also gives $\sigma^2 = O_n(\gamma^{-1})$.

Use the standard envelope bound

$$\text{env } J_0(y) \ll \min(1, y^{-1/2}).$$

If $\gamma\xi \leq 1$, then $x - 1 = O_n(\gamma^{-2})$, so the geometric prefactor is uniformly bounded and $x^{-1} \asymp 1$. If $\gamma\xi \geq 1$, the Bessel-envelope bound gives

$$\left\{ \frac{\xi^2}{(x^2 - 1)(x^2 - \sigma^2)} \right\}^{1/4} \text{env } J_0(\gamma\xi) \ll_n \frac{\gamma^{-1/2}}{\{(x^2 - 1)(x^2 - \sigma^2)\}^{1/4}}.$$

On the boundary layer where $x - 1 = O(1)$ this is $O_n(1)$, while for x bounded away from 1 it is $O_n(x^{-1})$. Enlarging the constant across the compact transition interval proves the claim. $\square$

The preceding endpoint estimate can be sharpened away from the radial turning point, and the resulting gain is the key to summing the co-Poisson lattice.

Lemma 20.3 (Away-boundary and exterior-energy bounds). *Fix $n \in \{0, 4\}$ and write*

$$\Psi_{n,c}(x) = Ps_n^0(x, c^2), \quad B_{n,c} = |\Psi_{n,c}(1)|.$$

For all sufficiently large c ,

$$|\Psi_{n,c}(x)| \ll_n \frac{B_{n,c}}{\sqrt{c}x}, \quad x \geq 2, \quad (127)$$

$$\int_1^\infty |\Psi_{n,c}(x)|^2 dx \ll_n \frac{B_{n,c}^2}{c}. \quad (128)$$

Proof. In Dunster's uniform formula (3.5), for $x \geq 2$ one has $\xi(x) \asymp_n x$ and $\{(x^2 - 1)(x^2 - \sigma^2)\}^{1/4} \asymp_n x$. The Bessel-envelope estimate therefore gives

$$\left\{ \frac{\xi^2}{(x^2 - 1)(x^2 - \sigma^2)} \right\}^{1/4} \text{env } J_0(c\xi) \ll_n \frac{1}{\sqrt{c}x},$$

which proves (127).

For $1 < x < 2$, use $d\xi/dx = ((x^2 - \sigma^2)/(x^2 - 1))^{1/2}$. Squaring Dunster's formula and changing variables from x to ξ reduces the main term, up to bounded fixed- n factors, to

$$B_{n,c}^2 \int_0^{\xi(2)} \xi \text{env } J_0(c\xi)^2 d\xi.$$

The bound $\text{env } J_0(t)^2 \ll \min(1, t^{-1})$ makes this $O_n(B_{n,c}^2/c)$; the uniform $O(c^{-1})$ remainder in Dunster's formula is smaller. On $[2, \infty)$, (127) is square integrable and contributes another $O_n(B_{n,c}^2/c)$. This proves (128). $\square$

Normalized source coefficients and endpoint noncancellation

We now settle the coefficient and repair-size parts of the former source assumption. These estimates concern the actual two-mode source, not a choice of independently rescaled endpoint values. The form-domain issue is treated separately from the elementary correlation consequences below.

Let $\psi_{n,c}$ be the real angular PSWF with $\|\psi_{n,c}\|_{L^2(-1,1)} = 1$, extended to the real line by its finite Fourier eigenrelation. For $n = 0, 4$ choose its sign so that $\psi_{n,c}(0) > 0$. Write $\vartheta_n(c) = \chi_n(c)^2$ for its concentration eigenvalue, and put $b_n(c) = |\psi_{n,c}(1)|$. These $\psi_{n,c}$ are scalar multiples of the preceding Dunster normalization; its homogeneous radial estimates are unchanged.

Lemma 20.4 (Endpoint size from exterior energy). *For each $n \in \{0, 4\}$, as $c \rightarrow \infty$,*

$$b_n(c)^2 \asymp_n c\{1 - \vartheta_n(c)\}, \quad b_n(c) \asymp_n c^{n/2+3/4}e^{-c}. \quad (129)$$

In particular $b_4(c)/b_0(c) \asymp c^2$.

Proof. For clarity, the eigenrelation on $[-1, 1]$ is

$$\int_{-1}^1 e^{-icxy} \psi_{n,c}(y) dy = \mu_n(c) \psi_{n,c}(x), \quad |\mu_n(c)|^2 = \frac{2\pi}{c} \vartheta_n(c).$$

Using the same formula to extend the function to all real x , Plancherel gives the exact identity

$$\int_1^\infty |\psi_{n,c}(x)|^2 dx = \frac{1 - \vartheta_n(c)}{2\vartheta_n(c)}. \quad (130)$$

The upper bound for this integral by $C_n b_n(c)^2/c$ was proved in Lemma 20.3. We also need a lower bound; an upper bound alone would not justify an endpoint comparison.

On the fixed interval $2 \leq x \leq 3$, Dunster's radial formula [9, eqs. (3.5)–(3.7)] and the large-argument expansion of J_0 give, with a real choice of the endpoint sign,

$$\psi_{n,c}(x) = \psi_{n,c}(1) \sqrt{\frac{2}{\pi c}} w_c(x) \left\{ \cos(c\xi_c(x) - \pi/4) + O_n(c^{-1}) \right\},$$

where $w_c(x) = \{(x^2 - 1)(x^2 - \sigma_n(c)^2)\}^{-1/4}$ and $\xi'_c(x) = \{(x^2 - \sigma_n(c)^2)/(x^2 - 1)\}^{1/2}$. Here $\sigma_n(c)^2 = O_n(c^{-1})$. On this interval w_c and ξ'_c have uniform positive lower bounds and bounded derivatives. One integration by parts therefore yields

$$\int_2^3 w_c(x)^2 \cos^2(c\xi_c(x) - \pi/4) dx = \frac{1}{2} \int_2^3 w_c(x)^2 dx + O_n(c^{-1}) \gg_n 1.$$

The integrated cross and squared remainder terms are smaller by an additional factor c^{-1} . Hence the exterior energy is also at least $c_n b_n(c)^2/c$ for some $c_n > 0$. Equation (130) and $\vartheta_n(c) \rightarrow 1$ prove the first comparison in (129).

Fuchs's fixed-index asymptotic [10, Theorem 1], in this concentration normalization, is

$$1 - \vartheta_n(c) \sim \frac{4\sqrt{\pi} 8^n}{n!} c^{n+1/2} e^{-2c}. \quad (131)$$

Combining it with the two-sided energy comparison proves the second comparison and the ratio assertion. No differentiation of an asymptotic remainder is used. $\square$

Proposition 20.5 (An explicit normalized two-mode source). *Set $c = 2\pi\lambda^2$ and $h_{n,\lambda}(u) = \lambda^{-1/2} \psi_{n,c}(u/\lambda)$ on $[-\lambda, \lambda]$, zero-extended for Fourier transformation. Endpoint values denote interior traces. Define*

$$\begin{aligned} a_{4,\lambda} &= \chi_0(c) h_{0,\lambda}(0), & a_{0,\lambda} &= -\chi_4(c) h_{4,\lambda}(0), \\ h_\lambda^\circ &= a_{0,\lambda} h_{0,\lambda} + a_{4,\lambda} h_{4,\lambda}, & B_\lambda^+ &= \sqrt{\lambda} h_\lambda^\circ(\lambda). \end{aligned} \quad (132)$$

The source has exactly zero integral. Both coefficients have finite nonzero limits. For all sufficiently large λ ,

$$|B_\lambda^+| \asymp c^{11/4} e^{-c}, \quad (133)$$

$$\frac{\sum_{n=0,4} |a_{n,\lambda}| \sqrt{\lambda} |h_{n,\lambda}(\lambda)|}{|B_\lambda^+|} = 1 + O(c^{-2}), \quad (134)$$

$$\frac{|h_\lambda^\circ(0)|}{|B_\lambda^+|} = O(c^{7/4} e^{-c}) = O_A(\lambda^{-A}) \quad \text{for every fixed } A. \quad (135)$$

The last two conclusions are invariant under a common nonzero scalar rescaling of the source.

Proof. The compressed physical Fourier eigenvalue is $\chi_n(c) > 0$ for $n = 0, 4$, so evaluation at zero gives exactly

$$\int_{-\lambda}^{\lambda} h_{n,\lambda}(u) du = \chi_n(c) h_{n,\lambda}(0).$$

Thus (132) cancels the integral. The fixed-order Hermite limit, with the unit-norm scaling just specified, gives

$$h_{0,\lambda}(0) \longrightarrow 2^{1/4}, \quad h_{4,\lambda}(0) \longrightarrow \frac{\sqrt{3}}{2^{1/4}}.$$

This central-value normalization follows, for example, from the fixed-order limit used in [7, Lemma 7.2]; the uniform error there also makes the physical L^2 norm tend to one, so unit-norm renormalization preserves these limits. Since $\chi_n \rightarrow 1$, both coefficients converge to the stated nonzero central constants, with a minus sign for $a_{0,\lambda}$.

By Lemma 20.4, the magnitude of the order-zero endpoint contribution divided by that of order four is $O(c^{-2})$. The triangle and reverse triangle inequalities prove (133) and (134), without needing a claim about their relative signs. Finally,

$$h_{\lambda}^{\circ}(0) = \{\chi_0(c) - \chi_4(c)\} h_{0,\lambda}(0) h_{4,\lambda}(0).$$

Equation (131) and $1 - \chi_n = (1 - \vartheta_n)/(1 + \chi_n)$ give $|h_{\lambda}^{\circ}(0)| = O(c^{9/2} e^{-2c})$. Division by (133) gives (135). $\square$

Corollary 20.6 (Size of the exact value-and-integral repair). *Fix an even $\psi \in C_c^{\infty}((-1, 1))$ with $\psi(0) = 1$ and $\int \psi = 0$. For the source in Proposition 20.5,*

$$\tilde{h}_{\lambda} = h_{\lambda}^{\circ} - h_{\lambda}^{\circ}(0) \psi$$

has zero value at zero, zero integral and the same endpoint as h_{λ}° . In lower-tail coordinates $u = \lambda^{-1} e^{-s}$, for every fixed A and every $S \geq 0$ its co-Poisson correction obeys

$$\left\| 1_{s \geq S} h_{\lambda}^{\circ}(0) \mathcal{E}(\psi)(\lambda^{-1} e^{-s}) \right\|_2 \leq C_A |B_{\lambda}^+| \lambda^{-A} e^{-S/2}. \quad (136)$$

This proves the repair's lower-tail size. Its repaired compact source also meets Theorem 20.1, which supplies the radical and semilocal form-domain passage used below.

Proof. The moment and endpoint statements follow directly from the fixed bump. Poisson summation, retaining its zero-value term, gives

$$\mathcal{E}(\psi)(u) = -\frac{1}{2} u^{1/2} + u^{-1/2} \sum_{m \geq 1} \widehat{\psi}(m/u) = O(u^{1/2}) \quad (0 < u \leq 1).$$

The last bound follows from Schwartz decay, taking any fixed power greater than one in the sum. The L^2 tail of $u^{1/2}$ in the s coordinate equals $\lambda^{-1/2} e^{-S/2}$. Apply (135), increasing its arbitrary power if needed. $\square$

The coefficient noncancellation and repair-size bounds are therefore no longer assumptions for this explicitly normalized source. The tail-to-form-correlation consequences and translated endpoint terms are proved below; the global radical/form-domain passage is supplied by Theorem 20.1.

The possible lattice resonance can be isolated entirely in a range where absolute summation is cheap. Beyond that range the radial differential equation itself supplies a genuinely linear phase with a summable remainder.

Lemma 20.7 (Uniform far field with controlled c -dependence). *For fixed $n \in \{0, 4\}$ there is a scalar $A_{n,c}$ such that, uniformly for $x \geq c^2$,*

$$\Psi_{n,c}(x) = A_{n,c} \frac{\sin(cx - n\pi/2)}{x} + O_n\left(\frac{B_{n,c}\sqrt{c}}{x^2}\right), \quad |A_{n,c}| \ll_n \frac{B_{n,c}}{\sqrt{c}}. \quad (137)$$

Proof. Dunster's radial equation, with $m = 0$, becomes after the standard substitution $w(x) = (x^2 - 1)^{1/2} \Psi_{n,c}(x)$,

$$w'' + c^2 w = V_{n,c}(x)w, \quad V_{n,c}(x) = \frac{\lambda_n^0(c^2)}{x^2 - 1} - \frac{1}{(x^2 - 1)^2}.$$

His fixed-order relation $\sigma^2 = 2(n + \frac{1}{2})c^{-1} + O_n(c^{-2})$, together with $\sigma^2 = 1 + c^{-2}\lambda_n^0(c^2)$, gives

$$\lambda_n^0(c^2) = -c^2 + (2n + 1)c + O_n(1).$$

Hence $|V_{n,c}(x)| \ll_n c^2 x^{-2}$ for $x \geq 2$ and

$$\int_{c^2}^{\infty} \frac{|V_{n,c}(t)|}{c} dt \ll_n c^{-1}.$$

By Lemma 20.3, $|w(x)| \ll_n B_{n,c}/\sqrt{c}$ for $x \geq 2$. On an interval of length c^{-1} about $x_0 = c^2$, the differential equation then gives $|w''| \ll_n B_{n,c}c^{3/2}$; Taylor's formula yields $|w'(x_0)| \ll_n B_{n,c}\sqrt{c}$. Variation of constants for the perturbed oscillator and Gronwall therefore give

$$|w(x)| + c^{-1}|w'(x)| \ll_n B_{n,c}/\sqrt{c}, \quad x \geq c^2.$$

The usual incoming/outgoing amplitudes have derivatives bounded by $|V_{n,c}(x)w(x)|/c$, which is integrable. They therefore converge at infinity, and integration from x to infinity gives an error

$$\ll_n \frac{1}{c} \int_x^{\infty} |V_{n,c}(t)w(t)| dt \ll_n \frac{B_{n,c}\sqrt{c}}{x}.$$

After division by $(x^2 - 1)^{1/2} \asymp x$, this is the remainder in (137). Finally Dunster's exact radial phase at infinity, equation (1.24), identifies the limiting real combination with $\sin(cx - n\pi/2)$, proving the stated form and amplitude bound. $\square$

Theorem 20.8 (Endpoint-normalized radial lattice bound). *Put $c = 2\pi\lambda^2$. For fixed $n \in \{0, 4\}$ and all $s \geq 0$,*

$$\left| \sum_{m \geq 2} \Psi_{n,c}(me^s) \right| \ll_n \frac{B_{n,c}}{\sqrt{c}e^s} (1 + \log c). \quad (138)$$

The estimate is uniform in s and hence contains no unproved nonresonance hypothesis.

Proof. Set $x = e^s$ and

$$M = \max\left(2, \left\lceil \frac{c^2}{x} \right\rceil\right).$$

For $2 \leq m < M$, Lemma 20.3 gives, by absolute summation,

$$\sum_{2 \leq m < M} |\Psi_{n,c}(mx)| \ll_n \frac{B_{n,c}}{\sqrt{c}x} \sum_{m < M} \frac{1}{m} \ll_n \frac{B_{n,c}(1 + \log c)}{\sqrt{c}x}.$$

For $m \geq M$, Lemma 20.7 applies. Since $n = 0$ or 4 , the fixed phase $n\pi/2$ is an integral multiple of 2π . The elementary harmonic-sine bound

$$\sup_{\theta \in \mathbb{R}} \sup_{M \geq 1} \left| \sum_{m=M}^{\infty} \frac{\sin(m\theta)}{m} \right| < \infty$$

then gives

$$\left| \frac{A_{n,c}}{x} \sum_{m \geq M} \frac{\sin(cm x - n\pi/2)}{m} \right| \ll_n \frac{B_{n,c}}{\sqrt{c} x}.$$

(The displayed uniform bound follows, for example, by splitting at $m \asymp |\theta|^{-1}$ and applying Dirichlet summation to the remaining tail.) The remainder in (137) contributes

$$\ll_n \frac{B_{n,c} \sqrt{c}}{x^2} \sum_{m \geq M} m^{-2}.$$

If $M \asymp c^2/x$ this is $O_n(B_{n,c}/(c^{3/2}x))$; if $M = 2$, then $x \gg c^2$ and it is even smaller than $B_{n,c}/(\sqrt{c}x)$. Combining the two ranges proves (138). $\square$

The $m = 1$ Poisson sample is a boundary-layer term rather than a lattice-summation problem. The finite-Fourier normalization introduces no additional power of λ : in the standard PSWF convention

$$\mathcal{F}_c \Psi_{n,c} = \mu_n(c) \Psi_{n,c}, \quad \frac{c}{2\pi} |\mu_n(c)|^2 = \chi_n(c)^2,$$

where χ_n^2 is the concentration eigenvalue. Since $c = 2\pi\lambda^2$, $|\mu_n(c)| = \chi_n(c)/\lambda$. After scaling from $[-1, 1]$ to $[-\lambda, \lambda]$, if $u = \lambda^{-1}e^{-s}$ then the Fourier-lattice part of the lower tail is, up to the harmless fixed even-mode Fourier phase,

$$r_{n,\lambda}^{\text{lat}}(s) = \chi_n(c) e^{s/2} \sum_{m \geq 1} \Psi_{n,c}(m e^s). \quad (139)$$

For an even source with $\int h = 0$ but $h(0) \neq 0$, Poisson summation has in addition the elementary term

$$\mathcal{E}(h)(u) = -\frac{1}{2} u^{1/2} h(0) + u^{-1/2} \sum_{m \geq 1} \hat{h}(m/u). \quad (140)$$

Thus the raw CCM two-mode source has a small zero-value defect in addition to the radial lattice term. It will be estimated explicitly below rather than silently omitted. In particular, at $s = 0$ the $m = 1$ lattice term is $\chi_n(c)$ times the upper-window boundary amplitude.

Corollary 20.9 (Fixed-order co-Poisson tail and endpoint comparison). *Let h_λ° be the explicitly normalized two-mode source of Proposition 20.5, and let $B_\lambda^+ = \sqrt{\lambda} h_\lambda^\circ(\lambda)$ be its upper-window co-Poisson boundary amplitude. Then the lower logarithmic tail r_λ° satisfies, for every $S \geq 0$,*

$$\left\| 1_{\{s \geq S\}} r_\lambda^\circ \right\|_2 \ll \frac{|B_\lambda^+|}{\lambda} (1 + \log \lambda) e^{-S/2}. \quad (141)$$

The same estimate holds for the exact value-and-integral repair. If B_λ is the common one-sided endpoint value of the inversion-even finite-window proxy, then

$$\frac{B_\lambda}{B_\lambda^+} = 1 + O\left(\frac{1 + \log c}{\sqrt{c}}\right), \quad c = 2\pi\lambda^2. \quad (142)$$

In particular $B_\lambda \asymp B_\lambda^+$ and it is nonzero for all sufficiently large λ .

Proof. For the $m = 1$ term in (139), Lemma 20.3 gives the exact log-coordinate estimate

$$\int_0^\infty \left| \chi_n(c) e^{s/2} \Psi_{n,c}(e^s) \right|^2 ds = \chi_n(c)^2 \int_1^\infty |\Psi_{n,c}(x)|^2 dx \ll_n \frac{B_{n,c}^2}{c}.$$

For $s \geq \log 2$, (127) also gives the pointwise bound $O_n(B_{n,c} c^{-1/2} e^{-s/2})$. Thus its L^2 tail beyond depth S is $O_n(B_{n,c} c^{-1/2} e^{-S/2})$ for all $S \geq 0$, after enlarging the constant on $0 \leq S \leq \log 2$.

For $m \geq 2$, Theorem 20.8 and (139) give directly

$$|r_{n,\lambda}^{(\geq 2)}(s)| \ll_n \frac{B_{n,c}(1 + \log c)}{\sqrt{c}} e^{-s/2}.$$

Since $\sqrt{c} \asymp \lambda$, this proves (141) mode by mode. The endpoint noncancellation bound in Proposition 20.5 controls the sum of the individual endpoint magnitudes by $|B_\lambda^+|$, so the modewise estimate passes to the actual normalized combination. This part no longer uses an unproved coefficient estimate.

For completeness, the exact radical source condition can be imposed without changing this scale. Choose a fixed even $\psi \in C_c^\infty((-1, 1))$ with $\psi(0) = 1$ and $\int \psi = 0$, and put

$$\tilde{h}_\lambda = h_\lambda^\circ - h_\lambda^\circ(0)\psi.$$

Then $\tilde{h}_\lambda(0) = 0$ and $\int \tilde{h}_\lambda = 0$, while its endpoint at $\pm\lambda$ is unchanged. Proposition 20.5 proves $|h_\lambda^\circ(0)|/|B_\lambda^+| = O_A(\lambda^{-A})$ for every fixed A , and Corollary 20.6 supplies the lower-tail repair bound. The repaired compact source satisfies Theorem 20.1, so its global radical and semilocal form-domain use is justified. For the fixed bump one must retain its zero-value term: since $\hat{\psi}(0) = \int \psi = 0$ and $\psi(0) = 1$, Poisson summation gives

$$\mathcal{E}(\psi)(u) = -\frac{1}{2}u^{1/2} + u^{-1/2} \sum_{m \geq 1} \hat{\psi}(m/u).$$

The second term is rapidly decreasing as $u \downarrow 0$, while the first is $O(u^{1/2})$. Because $|h_\lambda^\circ(0)|/|B_\lambda^+| = O_A(\lambda^{-A})$ for every fixed A , the whole correction is still $o(|B_\lambda^+| \lambda^{-1} e^{-s/2})$ at $u = \lambda^{-1} e^{-s}$ and is negligible in (141). The same estimate also absorbs the raw zero-defect term in (140).

It remains to identify the endpoint after inversion. Put $f_\lambda = \mathcal{E}(\tilde{h}_\lambda)$. At the upper endpoint only the $m = 1$ source summand remains, so $f_\lambda(\lambda^-) = B_\lambda^+$. At the lower endpoint, Poisson summation and (139) give

$$f_\lambda((\lambda^{-1})^+) = \sum_{n=0,4} a_{n,\lambda} \chi_n(c) \sqrt{\lambda} h_{n,\lambda}(\lambda) + O\left(\frac{1 + \log c}{\sqrt{c}} \sum_{n=0,4} |a_{n,\lambda}| \sqrt{\lambda} |h_{n,\lambda}(\lambda)|\right) + o(|B_\lambda^+|).$$

The last term includes the fixed-bump repair. If λ^2 is an integer, taking the indicated interior trace can also remove the single source summand at the cut; its size is $O(|B_\lambda^+|/\lambda)$ and is absorbed above. Since $\chi_n(c) = 1 + o(1)$ and (134) controls the sum of the individual endpoint magnitudes by $|B_\lambda^+|$, the displayed expression equals $B_\lambda^+ \{1 + O((1 + \log c)/\sqrt{c})\}$. Finally $p_\lambda = (f_\lambda + Jf_\lambda)/2$, so its two common one-sided endpoint values are the average of the upper and lower traces. This proves (142). $\square$

Lemma 20.10 (Uniform zero-extension correlations and the local term). *Let $L = 2 \log \lambda$, let r_λ be either oriented logarithmic tail of the repaired source, and zero-extend $v \in H^1([-L, 0])$. For*

$$C_{r,v}(y) = \int_{\mathbb{R}} r_\lambda(t) \overline{v(t-y)} dt$$

and for each reflected correlation occurring in the polarized even Weil kernel, the interior-test derivative contribution satisfies

$$|C'_{r,v,\text{bulk}}(y)| \leq C \frac{|B_\lambda|}{\lambda} (1 + \log \lambda) \|v'\|_2 \begin{cases} 1, & 0 \leq y \leq L, \\ e^{-(y-L)/2}, & y > L. \end{cases} \quad (143)$$

Here $C'_{r,v,\text{bulk}}(y) = -\int r_\lambda(t) \overline{v'(t-y)} 1_{(-L,0)}(t-y) dt$. The full distributional derivative has, in addition, precisely the two terms

$$-\overline{v(-L)} r_\lambda(y-L) + \overline{v(0)} r_\lambda(y), \quad (144)$$

up to the harmless sign and reflection dictated by the chosen orientation; a term with a negative tail argument is zero. These translated-tail terms are themselves locally integrable densities, not singular atoms in the derivative of the correlation. They must not be omitted from its absolutely continuous part. Moreover the local archimedean functional of the even polarized sum q_λ obeys

$$|A_\infty(q_\lambda)| \leq C \frac{|B_\lambda|}{\lambda} (1 + \log \lambda) (\|v\|_2 + \|v'\|_2 + |v(-L)| + |v(0)|). \quad (145)$$

These estimates are uniform in v and require no form-domain assumption.

Proof. Inside the support interval, differentiation in y puts one derivative on v and never on the prolate tail. The overlap uses tail depths $t \geq (y-L)_+$. Cauchy–Schwarz and (141), followed by (142), prove (143). In distributions,

$$D(1_{[-L,0]}v) = 1_{(-L,0)}v' + v(-L)\delta_{-L} - v(0)\delta_0.$$

Substitution into the correlation, with the minus sign from differentiating $v(t-y)$, gives (144). Reflection gives the other orientation and changes none of the bounds.

For the singularity at the origin use the elementary zero-extension translation estimate, valid for $0 < x \leq 1$,

$$\|v(\cdot - x) - v\|_{L^2(\mathbb{R})} \leq C [x\|v'\|_2 + \sqrt{x}\{|v(-L)| + |v(0)|\}]. \quad (146)$$

It follows by applying the fundamental theorem of calculus on the overlap and estimating the two boundary strips separately. Thus, with $R_\lambda = |B_\lambda|(1 + \log \lambda)/\lambda$,

$$|q_\lambda(x) + q_\lambda(-x) - 2e^{-x/2}q_\lambda(0)| \leq CR_\lambda [\sqrt{x}\{|v(-L)| + |v(0)|\} + x\{\|v'\|_2 + \|v\|_2\}].$$

The archimedean kernel is $O(x^{-1})$ on $(0, 1]$, so this is integrable. For $x \geq 1$ the kernel is $O(e^{-x/2})$, while Cauchy–Schwarz and (141) bound every correlation by $CR_\lambda\|v\|_2$. Integration proves (145). $\square$

The truncated-Mellin identity remains a useful independent check on the powers and normalizations. For an exact two-moment source,

$$\mathcal{M}_\lambda^-(\tau) = \int_0^{1/\lambda} \mathcal{E}(h_\lambda)(u) u^{-i\tau} \frac{du}{u}, \quad (147)$$

and Poisson summation gives

$$\mathcal{M}_\lambda^-(\tau) = \sum_{m \geq 1} m^{-1/2-i\tau} \int_{m\lambda}^\infty \widehat{h}_\lambda(x) x^{-1/2+i\tau} dx. \quad (148)$$

The preceding argument unconditionally supplies the physical/logarithmic tail and correlation estimates needed below, given the cited classical fixed-order PSWF inputs. Theorem 20.1 justifies passing the global radical identity to the closed semilocal form at each fixed λ ; the quantitative estimates, not its approximation constants, provide the required uniformity as $\lambda \rightarrow \infty$.

Lemma 20.11 (Quantitative endpoint-jump bound). *Let $E_{\lambda,T}$ be the periodic trigonometric space on $[-a, a]$, $a = \log \lambda$, with physical logarithmic frequencies in $[-T, T]$. Extend $v \in E_{\lambda,T}$ by zero outside $[-a, a]$, and let q_λ be its logarithmic correlation with the lower tail r_λ . In the distributional derivative of q_λ , the two endpoint jumps contribute translates of r_λ multiplied by the common periodic endpoint value $v(a) = v(-a)$. There is a constant $c_1 > 0$, independent of λ , such that uniformly for $\|v\|_2 = 1$ their total contribution to the centered pole–prime Stieltjes pairing is*

$$\ll_T |B_\lambda| \left((1+L)e^{-L/2} + (1+L)^{3/2}e^{-c_1\sqrt{L}} \right), \quad L = 2 \log \lambda. \quad (149)$$

In particular the endpoint-jump contribution is $o_T(|B_\lambda|)$.

Proof. The endpoint evaluation functional on $E_{\lambda,T}$ has norm $O_T(1)$ because the normalized Fourier basis has $O_T(L)$ modes and each basis vector has endpoint magnitude $L^{-1/2}$. For the jump meeting the lower tail at depth $s = y$, split $0 \leq s \leq \log 2$ from $s \geq \log 2$. On the first interval, $V(s) = O(s)$ and Cauchy–Schwarz with (141) gives

$$O_T \left(|B_\lambda| \frac{1 + \log \lambda}{\lambda} \right) = O_T \left(|B_\lambda| (1+L)e^{-L/2} \right).$$

On the second, Theorem 20.8 together with the $m = 1$ away-bound gives

$$|r_\lambda(s)| \ll |B_\lambda| \frac{1 + \log \lambda}{\lambda} e^{-s/2}, \quad s \geq \log 2,$$

and (151) below supplies an integrable factor $e^{-c\sqrt{s}}$. This contributes the same $O_T(|B_\lambda|(1+L)e^{-L/2})$ scale.

The second endpoint enters only after a shift $y = L + s$. There

$$|V(L + s)| \ll \lambda e^{s/2} e^{-c\sqrt{L+s}}.$$

For $0 \leq s \leq \log 2$, Cauchy–Schwarz and (141) cancel the factor λ and give $O_T(|B_\lambda|(1+L)e^{-c\sqrt{L}})$. For $s \geq \log 2$, the preceding pointwise tail estimate cancels $e^{s/2}$ and leaves

$$O_T \left(|B_\lambda| (1+L) \int_0^\infty e^{-c\sqrt{L+s}} ds \right).$$

The integral is explicit:

$$\int_0^\infty e^{-c\sqrt{L+s}} ds = 2e^{-c\sqrt{L}} \left(\frac{\sqrt{L}}{c} + \frac{1}{c^2} \right).$$

After decreasing the exponential constant if necessary, these estimates give (149). $\square$

Pole–prime reduction and closure of the weak G1 estimate

Let k_λ be the exact-radical co-Poisson vector obtained from the corrected source, and write

$$k_\lambda = k_{\lambda,\text{in}} + r_\lambda, \quad k_{\lambda,\text{in}} = 1_{I_\lambda} k_\lambda.$$

For every zero-extended H^1 test vector v supported in I_λ , Theorem 20.1 gives

$$QW(k_{\lambda,\text{in}}, v) = -QW(r_\lambda, v). \quad (150)$$

The dominant point-mass part and the prime part of the explicit formula pair the logarithmic correlation q_λ with

$$e^{y/2} dy \quad \text{and} \quad \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} \delta_{\log n},$$

respectively; the additional $e^{-y/2}$ part is absolutely integrable. Put

$$V(Y) = 2(e^{Y/2} - 1) - \sum_{2 \leq n \leq e^Y} \frac{\Lambda(n)}{\sqrt{n}}.$$

Partial summation together with the classical de la Vallée-Poussin prime-number-theorem remainder gives constants $c, C > 0$ such that

$$|V(Y)| \leq C e^{Y/2} e^{-c\sqrt{Y}} \quad (Y \geq 1). \quad (151)$$

Stieltjes integration by parts can be made quantitative. Put

$$A_{\lambda,T} = C_T |B_\lambda| (1+L) e^{-L/2},$$

so that (143) bounds the interior-test derivative part by $A_{\lambda,T}$ on $[0, L]$ and by $A_{\lambda,T} e^{-(y-L)/2}$ on $[L, \infty)$. On $1 \leq y \leq L$ split at $L/4$. The first part is $O_T(|B_\lambda|(1+L)L e^{-3L/8})$, while on $[L/4, L]$ the PNT factor gives $e^{-c\sqrt{y}} \leq e^{-c\sqrt{L}/2}$ and the $e^{y/2}$ integral cancels the factor $e^{-L/2}$. Beyond L one obtains

$$A_{\lambda,T} e^{L/2} \int_L^\infty e^{-c\sqrt{y}} dy = 2A_{\lambda,T} e^{L/2-c\sqrt{L}} \left(\frac{\sqrt{L}}{c} + \frac{1}{c^2} \right).$$

Together with Lemma 20.11, this yields constants $c_1, C_T > 0$ such that

$$\left| \int_0^\infty q_\lambda(y) e^{y/2} dy - \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} q_\lambda(\log n) \right| \leq C_T |B_\lambda| \left((1+L)^{3/2} e^{-c_1\sqrt{L}} + (1+L) e^{-L/2} \right). \quad (152)$$

The constants use only the classical unconditional zero-free-region/PNT remainder; no RH assumption or information about individual nontrivial zeros is inserted.

For completeness, the archimedean local term may be estimated directly rather than hidden inside a multiplier statement. In logarithmic coordinates the standard Weil formula contributes, after polarization,

$$\mathcal{A}_\infty(q) = -(\log 4\pi + \gamma_0) q(0) - \int_0^\infty \{q(x) + q(-x) - 2e^{-x/2} q(0)\} \frac{e^{x/2}}{e^x - e^{-x}} dx.$$

The kernel is $O(e^{-x/2})$ for $x \geq 1$ and is $O(x^{-1})$ at the origin. For zero-extended H^1 tests the numerator need only be $O(x^{1/2})$ because of the endpoint strips; this is still integrable. Lemma 20.10 gives precisely the required uniform estimate:

$$|q(0)| \ll |B_\lambda| e^{-L/2} (1 + \log \lambda), \quad |\mathcal{A}_\infty(q_\lambda)| \ll_T |B_\lambda| e^{-L/2} (1 + \log \lambda) = O_T \left(|B_\lambda| \frac{1 + \log \lambda}{\lambda} \right).$$

The remaining $e^{-y/2}$ pole contribution is absolutely integrable and obeys the same bound.

Define the explicit zero-independent envelope

$$\mathfrak{r}(\lambda) := (1+L)^{3/2} e^{-c_1\sqrt{L}} + (1+L) e^{-L/2}, \quad L = 2 \log \lambda. \quad (153)$$

Combining the pole–prime, archimedean, and decaying-pole pieces with global radicality gives the quantitative continuum G1 estimate

$$\boxed{\mathcal{E}_T(\lambda) := \sup_{\substack{\|v\|_2=1 \\ v \in E_{\lambda,T}}} \frac{|QW(k_{\lambda,\text{in}}, v)|}{|B_\lambda|} \leq C_T \mathfrak{r}(\lambda) \longrightarrow 0.} \quad (154)$$

The quantitative estimates use only fixed-order PSWF asymptotics, the radial differential equation, Poisson summation, and the classical unconditional prime-number-theorem remainder. The global radical and semilocal domain passage is now Theorem 20.1; hence (154) is proved for the explicitly repaired source, with no remaining source assumption. The construction and all displayed cutoff rules are independent of a nontrivial-zero list. This is the weak G1 conclusion for a fixed physical band, not a uniform bound on growing prolate blocks or an RH implication by itself.

Lemma 20.12 (Inversion symmetrization preserves the G1 envelope). *Let*

$$p_\lambda = \tfrac{1}{2}(I + J)k_{\lambda,\text{in}}, \quad Jf(u) = f(u^{-1}).$$

Then

$$\sup_{\substack{\|v\|_2=1 \\ v \in E_{\lambda,T}}} \frac{|QW(p_\lambda, v)|}{|B_\lambda|} \leq \mathcal{E}_T(\lambda) \leq C_T \mathfrak{r}(\lambda). \quad (155)$$

Proof. The Weil form is invariant under simultaneous inversion, $QW(Jf, Jg) = QW(f, g)$, and J is a unitary involution preserving the physical frequency window $E_{\lambda,T}$; this is the semilocal form of the inversion symmetry in Connes–Consani–Moscovici [7]. Hence

$$QW(Jk_{\lambda,\text{in}}, v) = QW(k_{\lambda,\text{in}}, Jv).$$

Taking the average with $QW(k_{\lambda,\text{in}}, v)$ and then the supremum proves (155). $\square$

Proposition 20.13 (Sobolev form of the continuum G1 estimate). *Use the explicit repaired source above. Let u be a periodic H^1 function on the logarithmic window $[-a, a]$, $a = \log \lambda$, and interpret it by zero extension when it enters the semilocal Weil form. Put*

$$\mathcal{S}_1(u) := \|u\|_2 + \|D_{\log} u\|_2 + |u(-a)| + |u(a)|.$$

Then the same zero-independent envelope as in (154) obeys

$$\boxed{\frac{|QW_\lambda(p_\lambda, u)|}{|B_\lambda|} \leq C \mathfrak{r}(\lambda) \mathcal{S}_1(u).} \quad (156)$$

The constant is independent of λ and u . In particular the fixed-physical-band hypothesis in (155) is a convenient way to bound the Sobolev and trace factors, not an essential spectral restriction on the continuum G1 pairing.

Proof. Repeat the proof of the cross-tail and pole–prime estimates without applying a Bernstein inequality. On the open logarithmic interval, differentiation of the correlation is placed on u , so Cauchy–Schwarz gives the factor $\|D_{\log} u\|_2$ in place of the earlier band constant T . The two jumps created by zero extension have coefficients exactly $u(-a)$ and $u(a)$, so Lemma 20.11 carries over with the endpoint-evaluation norm replaced by $|u(-a)| + |u(a)|$. Finally $|q(0)| \leq \|r_\lambda\|_2 \|u\|_2$, and

the archimedean numerator is controlled near the origin by the same H^1 correlation estimate. The Stieltjes integration-by-parts calculation leading to (152) is otherwise unchanged and gives

$$|QW_\lambda(k_{\lambda,\text{in}}, u)| \leq C|B_\lambda|\mathfrak{r}(\lambda)\mathcal{S}_1(u).$$

Simultaneous inversion preserves the L^2 norm, the logarithmic derivative norm, and the two endpoint traces. Averaging with the inverted vector as in Lemma 20.12 proves (156). $\square$

The ordinary Fourier finite-section issue requires more care. The relevant regularity is supplied directly by the arithmetic finite-sum formula, rather than by a global continuity assumption.

Lemma 20.14 (Arithmetic BV structure and absence of a periodic boundary jump). *Let $\tilde{h}_\lambda$ be the compactly supported exact-moment prolate source used above, and let*

$$\mathcal{E}(\tilde{h}_\lambda)(u) = u^{1/2} \sum_{n \geq 1} \tilde{h}_\lambda(nu)$$

be its co-Poisson/Riemann-sum vector. On the window $I_\lambda = [\lambda^{-1}, \lambda]$, the restriction of $\mathcal{E}(\tilde{h}_\lambda)$ is of bounded variation and has only finitely many possible interior jump points. Consequently

$$p_\lambda = \frac{1}{2}(I + J)k_{\lambda,\text{in}}, \quad Jf(u) = f(u^{-1}),$$

is also of bounded variation. Moreover the inversion identity $p_\lambda(u) = p_\lambda(u^{-1})$ implies

$$p_\lambda(\lambda^-) = p_\lambda((\lambda^{-1})^+) =: B_\lambda, \tag{157}$$

so the periodic representative in logarithmic coordinates has no jump at the identified boundary. No assertion of global continuity is needed.

Proof. The scale-invariant Riemann-sum formula $\mathcal{E}(f)(u) = u^{1/2} \sum_{n > 0} f(nu)$ is the standard co-Poisson map used in the prolate construction [13]. Because $\text{supp } \tilde{h}_\lambda \subset [-\lambda, \lambda]$, for $u \in [\lambda^{-1}, \lambda]$ only indices $n \leq \lambda/u \leq \lambda^2$ can occur. Thus the sum has only finitely many summands throughout the window. Between the finitely many threshold points at which an argument nu meets an endpoint or another piece boundary of the compact source, it is a finite sum of smooth functions; its one-sided limits are finite. Hence its total variation on the compact window is finite. The fixed smooth exact-moment repair used above does not alter this conclusion.

Inversion preserves bounded variation, so the symmetrized proxy p_λ is BV. Finally, by construction $p_\lambda = Jp_\lambda$. Taking one-sided limits at the two ends of I_λ gives (157). After the logarithmic endpoints are identified, this is exactly the absence of a periodic boundary jump. Interior arithmetic jumps remain allowed. $\square$

Thus for each *fixed* cutoff ordinary Fourier projection converges both in the closed semilocal Weil form norm and at the endpoint. This gives the following fixed-cutoff existence statement, but no quantitative diagonal compatible with the growing-band estimate is supplied.

Proposition 20.15 (Fixed-cutoff boundary-preserving Galerkin approximation). *Assume p_λ is of bounded variation on $[-a, a]$ and that its one-sided endpoint limits agree, with common nonzero value*

$$B_\lambda = p_\lambda(a^-) = p_\lambda((-a)^+).$$

Let P_N be ordinary Fourier projection onto the trigonometric space E_N . For each fixed λ ,

$$P_N p_\lambda \rightarrow p_\lambda$$

in the form norm of the closed semilocal Weil form and

$$(P_N p_\lambda)(a) \rightarrow B_\lambda.$$

Consequently, for every fixed λ and every $\varepsilon > 0$ there is an $N = N(\lambda, \varepsilon)$ such that

$$\|P_N p_\lambda - p_\lambda\|_{QW_\lambda} + |(P_N p_\lambda)(a) - B_\lambda| \leq \varepsilon |B_\lambda|.$$

No rate relating N to λ is asserted.

Proof. A bounded-variation function on the log interval has Fourier coefficients $\hat{p}_\lambda(n) = O_\lambda(|n|^{-1})$. The archimedean part of the semilocal Weil form has Fourier weight comparable to $1 + \log(1 + n^2)$, while at fixed λ the pole term and the finite prime-power sum are bounded perturbations. Hence

$$\sum_n (1 + \log(1 + n^2)) |\hat{p}_\lambda(n)|^2 < \infty,$$

and the Fourier tails tend to zero in the form norm. Since the two one-sided endpoint limits agree, the periodic representative has no boundary jump; the Dirichlet–Jordan theorem therefore gives convergence at the identified endpoint. Interior jumps do not affect this conclusion. $\square$

The lack of a rate is not cosmetic. The physical-band estimate proved above remains uniform only while the physical frequency grows sufficiently slowly compared with the quantitative PNT saving. A generic Fourier approximation of a function with an exponentially small endpoint value may require a much larger cutoff. Therefore Proposition 20.15 by itself does *not* transfer (154) to a diagonal $N(\lambda)$.

There is nevertheless a zero-independent finite-dimensional transfer of the *fixed-band residual*, and the quantitative envelope above makes its exact finite-section dependence transparent. For N large enough to contain the physical band, set

$$k_{\lambda,N} = P_N p_\lambda,$$

and define the relative boundary and band-form defects

$$\beta_{\lambda,N} := \frac{|\delta_N(k_{\lambda,N}) - B_\lambda|}{|B_\lambda|}, \tag{158}$$

$$\gamma_{\lambda,N,T} := \frac{1}{|B_\lambda|} \sup_{\substack{v \in E_{\lambda,T} \\ \|v\|_2=1}} |QW_\lambda(k_{\lambda,N} - p_\lambda, D_{\log} v)|. \tag{159}$$

Both quantities are defined entirely from the prolate/co-Poisson proxy and the finite Weil form; no zeta zero is used.

Proposition 20.16 (Quantitative and cofinal finite-section transfer of G1). *Fix $T > 0$. Whenever $\beta_{\lambda,N} < 1$,*

$$\boxed{\frac{\|P_T D_{\log} W_{\lambda,N} k_{\lambda,N}\|_2}{|\delta_N(k_{\lambda,N})|} \leq \frac{T \mathcal{E}_T(\lambda) + \gamma_{\lambda,N,T}}{1 - \beta_{\lambda,N}}}. \tag{160}$$

For each fixed λ ,

$$\beta_{\lambda,N} \longrightarrow 0, \quad \gamma_{\lambda,N,T} \longrightarrow 0 \quad (N \rightarrow \infty). \tag{161}$$

Consequently, for all sufficiently large λ there is a zero-independent threshold $N_{G1}(\lambda, T)$ such that for every $N \geq N_{G1}(\lambda, T)$,

$$\beta_{\lambda, N} \leq \mathfrak{r}(\lambda), \quad (162)$$

$$\gamma_{\lambda, N, T} \leq \mathfrak{r}(\lambda), \quad (163)$$

and hence

$$\sup_{N \geq N_{G1}(\lambda, T)} \frac{\|P_T D_{\log} W_{\lambda, N} k_{\lambda, N}\|_2}{|\delta_N(k_{\lambda, N})|} \leq C'_T \mathfrak{r}(\lambda) \rightarrow 0. \quad (164)$$

In particular one may choose any diagonal $N(\lambda) \geq N_{G1}(\lambda, T)$; G1 does not require an economical or finely tuned Fourier cutoff.

Proof. Hermiticity of the finite Weil matrix and self-adjointness of $D_{\log}$ on the periodic Fourier section give

$$\|P_T D_{\log} W_{\lambda, N} k_{\lambda, N}\|_2 = \sup_{\substack{v \in E_{\lambda, T} \\ \|v\|_2 = 1}} |QW_{\lambda}(k_{\lambda, N}, D_{\log} v)|.$$

Split $k_{\lambda, N} = p_{\lambda} + (k_{\lambda, N} - p_{\lambda})$. Since $D_{\log} v \in E_{\lambda, T}$ and $\|D_{\log} v\|_2 \leq T$, Lemma 20.12 bounds the first part by $T|B_{\lambda}|\mathcal{E}_T(\lambda)$, while the second is exactly bounded by $|B_{\lambda}|\gamma_{\lambda, N, T}$. Also

$$|\delta_N(k_{\lambda, N})| \geq |B_{\lambda}|(1 - \beta_{\lambda, N}).$$

This proves (160).

For fixed λ , Proposition 20.15 gives the boundary convergence and form-norm convergence of $P_N p_{\lambda} - p_{\lambda}$, proving the first limit in (161). To obtain the second without an unspecified pointwise continuity constant, choose m_{λ} so that

$$QW_{\lambda}(u, u) + m_{\lambda}\|u\|_2^2$$

is a positive closed form. Its Cauchy–Schwarz inequality, together with the finite-dimensional boundedness of $\{D_{\log} v : v \in E_{\lambda, T}, \|v\|_2 = 1\}$ in the corresponding form norm, shows that the supremum in (159) tends to zero with the form norm of $P_N p_{\lambda} - p_{\lambda}$.

The convergence in (161) is convergence of the full sequence, so for each fixed λ and the positive tolerance $\mathfrak{r}(\lambda)$ there is a threshold after which (162)–(163) hold for all larger N . Since $\mathfrak{r}(\lambda) \rightarrow 0$, it is $< 1/2$ for large λ . Insert these inequalities and $\mathcal{E}_T(\lambda) \leq C_T \mathfrak{r}(\lambda)$ into (160) to obtain (164). $\square$

Quantitative high–low decay and an exact-boundary G1 repair

Endpoint recovery of the ordinary Fourier partial sums in Proposition 20.16 was initially nonquantitative. Proposition 21.7 below now supplies an asymptotic polynomial cutoff for the fixed source. For the G1 residual itself one can also bypass endpoint selection. The key observation is that the explicit CCM matrix has an additional off-diagonal factor, so only bounded variation of the co-Poisson proxy is needed.

Write $U_n = L^{-1/2} e^{2\pi i n x/L}$ for the normalized logarithmic Fourier basis and $V_n = \kappa(U_n)$ for its multiplicative image, and set

$$\tau_{n, m}^{(\lambda)} := QW_{\lambda}(V_n, V_m).$$

Lemma 20.17 (Zero-independent high–low decay of the Weil matrix). *There is an absolute constant C such that, for $\lambda \geq 2$, $L = 2 \log \lambda$, and $n \neq m$,*

$$\boxed{|\tau_{n,m}^{(\lambda)}| \leq C \frac{\lambda}{|n-m|}.} \quad (165)$$

The constant uses no information about the nontrivial zeros of ζ .

Proof. For $n \neq m$, the explicit convolution kernel of Connes–Consani–Moscovici [7] is

$$q(U_n, U_m)(y) = \frac{\sin(2\pi my/L) - \sin(2\pi ny/L)}{\pi(n-m)}, \quad 0 \leq y \leq L,$$

so $q(U_n, U_m)(0) = 0$ and $\|q(U_n, U_m)\|_\infty \leq 2/(\pi|n-m|)$. The pole contribution is therefore bounded by

$$\frac{C}{|n-m|} \int_0^L (e^{y/2} + e^{-y/2}) dy \leq C \frac{\lambda}{|n-m|}.$$

For the nonarchimedean contribution, partial summation together with the classical Chebyshev bound $\psi(x) \ll x$ gives

$$\sum_{1 < k \leq e^L} \frac{\Lambda(k)}{\sqrt{k}} \ll e^{L/2} = \lambda,$$

and hence the same bound.

For the archimedean term set

$$\rho(y) = \frac{e^{y/2}}{e^y - e^{-y}}, \quad J_L(r) = \int_0^L \sin(2\pi ry/L) \rho(y) dy.$$

Because the off-diagonal kernel vanishes at the origin, the CCM formula reduces to

$$W_{\mathbb{R}}(V_n, V_m) = \frac{J_L(m) - J_L(n)}{\pi(n-m)}.$$

Uniformly for $L \geq 2 \log 2$ and integers r , $J_L(r) = O(1)$: on $0 < y \leq 1$, write $\rho(y) = (2y)^{-1} + O(1)$ and use the uniform boundedness of the sine integral, while on $y \geq 1$ use $\rho(y) \ll e^{-y/2}$. Thus the archimedean piece is $O(|n-m|^{-1})$. Combining the three contributions proves (165). $\square$

Corollary 20.18 (Complete Fourier-matrix bound). *For all integers n, m and $\lambda \geq 2$,*

$$\boxed{|\tau_{n,m}^{(\lambda)}| \leq \begin{cases} C\lambda/|n-m|, & n \neq m, \\ C(\lambda + 1 + \log(2 + |n|/L)), & n = m. \end{cases}} \quad (166)$$

The constant is absolute and uses no nontrivial-zero information.

Proof. The off-diagonal case is Lemma 20.17. The diagonal is a separate CCM formula, not the limit of the off-diagonal divided difference: for $0 \leq y \leq L$,

$$q(U_n, U_n)(y) = 2 \left(1 - \frac{y}{L}\right) \cos \frac{2\pi ny}{L}, \quad q(U_n, U_n)(0) = 2.$$

Thus the pole term is $O(\lambda)$ and, by the classical Chebyshev estimate, the prime-power term is also $O(\lambda)$.

It remains to control the archimedean term. Put $t_n = 2\pi n/L$. On $0 < x \leq 1$ the numerator in the logarithmic Weil kernel is, up to an absolute constant,

$$\left(1 - \frac{x}{L}\right) \cos(t_n x) - e^{-x/2}.$$

Since $\rho(x) = e^{x/2}/(e^x - e^{-x}) \ll x^{-1}$ there,

$$\left|\left(1 - \frac{x}{L}\right) \cos(t_n x) - e^{-x/2}\right| \leq C\{\min(1, t_n^2 x^2) + x\}.$$

Consequently

$$\int_0^1 \left|\left(1 - \frac{x}{L}\right) \cos(t_n x) - e^{-x/2}\right| \rho(x) dx \leq C(1 + \log(2 + |t_n|)).$$

For $x \geq 1$ the kernel is $O(e^{-x/2})$ and the diagonal convolution is bounded, so the remaining integral is $O(1)$. Therefore the archimedean contribution is $O(1 + \log(2 + |n|/L))$, proving (166). $\square$

A closed Weil operator and its omitted Fourier complement

The entrywise bound above can be strengthened to an operator statement. Here the closed operator is the canonical semilocal Weil operator; it is not the arithmetic generator whose realization is conditional in Section 18. Fix $\lambda > 1$, put $L = 2 \log \lambda$, $t_n = 2\pi n/L$, and write

$$P_\lambda = \sum_{1 < m \leq \lambda^2} \frac{\Lambda(m)}{\sqrt{m}}, \quad \rho(y) = \frac{e^{y/2}}{2 \sinh y}, \quad h_L = \sinh^2(L/4).$$

For later constants use the complete archimedean diagonal

$$A_n = \int_0^L \rho(y) \{2(1 - y/L) \cos(t_n y) - 2e^{-y/2}\} dy + \log(4\pi) + \gamma_E + \log \tanh(L/2). \quad (167)$$

The cancellation at zero makes this an ordinary convergent integral.

Proposition 20.19 (Bounded commutator and the operator core). *In the normalized logarithmic Fourier basis the canonical semilocal Weil operator W_λ is*

$$W_\lambda = D_d + [M_b, H], \quad H_{nm} = \begin{cases} (n - m)^{-1}, & n \neq m, \\ 0, & n = m, \end{cases} \quad (168)$$

where D_d is real diagonal, M_b is multiplication by the real odd sequence

$$b_n = \frac{32Lh_L n}{L^2 + 16\pi^2 n^2} + \frac{1}{\pi} \left\{ \int_0^L \rho(y) \sin(t_n y) dy + \sum_{1 < m \leq \lambda^2} \frac{\Lambda(m)}{\sqrt{m}} \sin(t_n \log m) \right\},$$

$$d_n = \frac{32Lh_L(L^2 - 16\pi^2 n^2)}{(L^2 + 16\pi^2 n^2)^2} - 2 \sum_{1 < m \leq \lambda^2} \frac{\Lambda(m)}{\sqrt{m}} (1 - \log m/L) \cos(t_n \log m) - A_n. \quad (169)$$

In particular the diagonal is not filled in by a divided-difference limit. One has

$$\|b\|_\infty \leq B_\lambda^{\text{mat}} := \frac{4h_L + P_\lambda + 1 + \pi/4}{\pi}, \quad \|[M_b, H]\| \leq 2\pi B_\lambda^{\text{mat}}. \quad (170)$$

At fixed λ , $d_n = \log |n| + O_\lambda(1)$ as $|n| \rightarrow \infty$. Consequently

$$\begin{aligned}\mathcal{D}(W_\lambda) &= \left\{ x : \sum_n \log^2(2 + |n|)|x_n|^2 < \infty \right\}, \\ \mathcal{D}[W_\lambda] &= \left\{ x : \sum_n \log(2 + |n|)|x_n|^2 < \infty \right\}.\end{aligned}\tag{171}$$

Finite Fourier sums form an operator core. The operator is lower bounded and has compact resolvent.

Proof. The exact CCM entries give $\tau_{nm} = (b_n - b_m)/(n - m)$ for $n \neq m$ and $\tau_{nn} = d_n$; see also (347). For $n > 0$, expansion $\rho(y) = \sum_{k \geq 0} e^{-(2k+1/2)y}$ and $t_n L = 2\pi n$ give

$$S_n := \int_0^L \rho(y) \sin(t_n y) dy = \sum_{k \geq 0} \frac{(1 - e^{-(2k+1/2)L})t_n}{(2k + 1/2)^2 + t_n^2}.$$

Interchange is justified by integrability of $\rho(y)|\sin(t_n y)|$. The summands are nonnegative. The $k = 0$ term is at most 1. For $k \geq 1$ discard the factor $1 - e^{-(2k+1/2)L} \leq 1$ and compare the decreasing majorant $t_n/\{(2k + 1/2)^2 + t_n^2\}$ with an integral; this gives at most $\pi/4$. Thus $0 \leq S_n \leq 1 + \pi/4$. Oddness treats negative n , and $S_0 = 0$. The pole part of b_n has absolute value at most $4h_L/\pi$; the prime sine sum is bounded by P_λ . This proves the first bound in (170).

On $\ell^2(\mathbb{Z})$, H is the discrete Hilbert transform. The Fourier series $\sum_{k \neq 0} e^{ik\theta}/k = i(\pi - \theta)$ for $0 < \theta < 2\pi$ proves $H^* = -H$ and $\|H\| = \pi$. Hence $[M_b, H]$ is bounded self-adjoint and has the asserted norm. Moreover

$$A_n = A_0 - \int_0^L w_L(y)(1 - \cos(t_n y)) dy, \quad w_L(y) = 2\rho(y)(1 - y/L).$$

Since $w_L(y) - 1/y$ is integrable on $(0, L)$, the last integral is $\log |t_n| + O_L(1)$. For example, split $\int_0^{|t_n|L} (1 - \cos v) dv/v$ at 1 and integrate its oscillatory tail by parts. The other terms in d_n are bounded at fixed λ , proving its logarithmic asymptotic.

The diagonal operator is self-adjoint on the first domain in (171), has finite sequences as an operator core, and has compact resolvent. Bounded self-adjoint perturbation preserves these statements and gives the displayed form domain. This identifies the actual physical Weil realization: the finite sums are a form core for its lower-bounded closed form by [7, Propositions 3.2–3.4], and both forms have exactly the same entries there. Uniqueness of the form closure proves equality, without imposing a new boundary condition. $\square$

Every periodic BV proxy, including the repaired p_λ , therefore belongs to $\mathcal{D}(W_\lambda)$ at fixed λ : its Fourier coefficients are $O_\lambda((1 + |n|)^{-1})$. Furthermore $P_N p_\lambda \rightarrow p_\lambda$ in this operator graph norm, by the diagonal description and boundedness of the commutator. This fixed-window statement supplies no endpoint-normalized uniform rate, and does not remove the separate sampler graph defect.

Proposition 20.20 (An explicit positive high-frequency tail). *Set $a = \min(1, L)$ and*

$$\begin{aligned}C_\lambda^{\text{tail}} &:= |A_0| + 4 + \pi/2 + a + 2a/L + 2P_\lambda + 2 \sinh(L/2) - L, \\ \Gamma_{\lambda, N} &:= \log \frac{2\pi(N+1)a}{L} - C_\lambda^{\text{tail}}.\end{aligned}\tag{172}$$

For every vector f in the form domain supported on Fourier indices $|n| > N$,

$$QW_\lambda(f, f) \geq \Gamma_{\lambda, N} \|f\|_2^2. \quad (173)$$

In particular this omitted complement is strictly positive whenever $\Gamma_{\lambda, N} > 0$. The constants use no nontrivial-zero data.

Proof. On $0 < y < a$, $2\rho(y) \geq e^{-y/2}/y \geq 1/y - 1/2$, whence

$$w_L(y) \geq 1/y - (1/2 + 1/L).$$

Since $w_L \geq 0$ on the full interval and $F(T) := \int_0^T (1 - \cos v) dv/v \geq \log T - 2$ for $T > 0$,

$$-A_n \geq \log(|t_n|a) - |A_0| - 2 - a - 2a/L \quad (n \neq 0).$$

The bound for F is immediate for $T \leq 1$; for $T \geq 1$ the integral of $\cos v/v$ on $[1, T]$ has absolute value at most 2 by integration by parts. The off-diagonal archimedean operator has norm at most $2 + \pi/2$, using $b_n^{\text{arch}} = S_n/\pi$ in the preceding commutator proof.

The complete prime operator is a sum of negative weighted truncated translations and their adjoints, so its lower bound is $-2P_\lambda I$. In centered logarithmic coordinates $x \in [-L/2, L/2]$, the pole form is

$$2|\langle f, \cosh(x/2) \rangle|^2 - 2|\langle f, \sinh(x/2) \rangle|^2 \geq -\{2 \sinh(L/2) - L\} \|f\|_2^2.$$

Combining these three estimates gives (173) first for finite Fourier sums and then on the tail form domain by closure. $\square$

A sharper positive Fourier complement at the certified window

The previous bound loses constants by bounding each place on the full space. On the omitted Fourier tail one can retain the limiting archimedean commutator, the physical incidence of prime shifts, and the decay of the negative pole vector. These are bounds for the same Weil operator and its original Fourier projection.

For $L = 2 \log \lambda$ put

$$\begin{aligned} R_L &= \frac{e^{-L/2}}{1 - e^{-2L}}, & C_L &= \max\{1, 1/4 + R_L\}, \\ E_L(t) &= \frac{1}{15t} + \frac{7}{2t^2} + \frac{\pi}{2Lt} + \frac{2 + 2R_L}{Lt^2}, & t &> 0. \end{aligned} \quad (174)$$

Define the weighted prime degree

$$M_\lambda = \text{ess sup}_{0 < x < L} \left\{ \sum_{\substack{1 < m < \lambda^2 \\ \log m \leq x}} \frac{\Lambda(m)}{\sqrt{m}} + \sum_{\substack{1 < m < \lambda^2 \\ \log m \leq L-x}} \frac{\Lambda(m)}{\sqrt{m}} \right\}. \quad (175)$$

The strict upper cut omits only a possible full-length translation, which is zero on $L^2([0, L])$.

Proposition 20.21 (A practical positive omitted complement). *For $N \geq 1$ let $t_* = 2\pi(N+1)/L$, and set*

$$\begin{aligned} \Gamma_{\lambda, N}^{\text{even}} &= \log \frac{N+1}{L} - M_\lambda - \frac{2C_L}{t_*} - E_L(t_*), \\ \Gamma_{\lambda, N}^{\text{all}} &= \Gamma_{\lambda, N}^{\text{even}} - \frac{\pi}{2} - \frac{4 \sinh^2(L/4)L}{\pi^2 N}. \end{aligned} \quad (176)$$

For every form-domain vector f whose Fourier coefficients vanish on $|n| \leq N$,

$$QW_\lambda(f, f) \geq \Gamma_{\lambda, N}^{\text{all}} \|f\|_2^2.$$

For an inversion-even f , the stronger lower bound with $\Gamma_{\lambda, N}^{\text{even}}$ holds. In particular, at the actual window $\lambda = 3$,

$$\Gamma_{3, 256}^{\text{all}} > 0.0148, \quad \Gamma_{3, 256}^{\text{even}} > 1.5867, \quad \Gamma_{3, 512}^{\text{all}} > 0.7085. \quad (177)$$

These inequalities control the entire omitted complement, not merely a finite further block.

Proof. Write $a_k = 2k + 1/2$ and use the exact Fourier-grid series for S_n in the proof of Proposition 20.19. For $t > 0$, the function $x \mapsto t/(x^2 + t^2)$ decreases on $x > 0$. Comparing its values on $1/2 + 2\mathbb{N}$ with its integral gives

$$\frac{\pi}{4} - \frac{1}{4t} \leq \sum_{k \geq 0} \frac{t}{a_k^2 + t^2} \leq \frac{\pi}{4} + \frac{1}{t}.$$

The exponentially weighted part of S_n is nonnegative and at most $R_L/|t_n|$. Oddness therefore proves

$$\left| S_n - \frac{\pi}{4} \operatorname{sgn} n \right| \leq \frac{C_L}{|t_n|} \quad (n \neq 0). \quad (178)$$

On the tail, the archimedean off-diagonal multiplier is consequently $b_n^{\text{arch}} = \operatorname{sgn}(n)/4 + e_n$ with $\|e\|_\infty \leq C_L/(\pi t_*)$. Since the compressed discrete Hilbert transform has norm at most π , its commutator has norm at most $\pi/2 + 2C_L/t_*$. For even coefficients the limiting commutator is instead positive: under the normalized identification with positive indices its matrix is

$$K_{nm} = \frac{1}{2(n+m)}, \quad n, m > N.$$

Positivity follows from $1/(n+m) = \int_0^1 x^{n+m-1} dx$. Thus only $2C_L/t_*$ is lost in the even sector. This uses the unshifted Fourier basis of (168); translation to centered physical coordinates conjugates by $(-1)^n$ and preserves both parity and the positive quadratic form.

Next retain the complete archimedean diagonal. Its exact series, already used in the independent matrix certificate, gives for $z = 1/4 + it/2$ and $t = |t_n| > 0$,

$$-A_n = \Re \psi_0(z) - \log \pi + \frac{\Re \psi_1(z)}{2L} - \frac{2}{L} \sum_{k \geq 0} e^{-a_k L} \frac{a_k^2 - t^2}{(a_k^2 + t^2)^2}. \quad (179)$$

Here ψ_0 and ψ_1 are digamma and trigamma. Binet's exact integral formula [20, eq. 5.9.15] at $w = z + 1 = 5/4 + it/2$ has remainder of modulus at most

$$2 \int_0^\infty \frac{u du}{|u^2 + w^2|(e^{2\pi u} - 1)} \leq \frac{8}{5t} \int_0^\infty \frac{u du}{e^{2\pi u} - 1} = \frac{1}{15t},$$

since $|u^2 + w^2| \geq 5t/4$. The recurrence $\psi_0(z) = \psi_0(z+1) - 1/z$, together with $\Re \log w \geq \log(t/2)$, $\Re(1/(2w)) \leq 5/(2t^2)$, and $\Re(1/z) \leq 1/t^2$, yields

$$\Re \psi_0(z) \geq \log(t/2) - \frac{1}{15t} - \frac{7}{2t^2}.$$

The convergent trigamma series and a decreasing-integrand comparison give

$$|\psi_1(z)| \leq \sum_{k \geq 0} \frac{1}{(k + 1/4)^2 + t^2/4} \leq \frac{4}{t^2} + \frac{\pi}{t}.$$

The last term in (179) has modulus at most $2R_L/(Lt^2)$. Hence

$$-A_n \geq \log \frac{|n|}{L} - E_L(|t_n|). \quad (180)$$

Its right side increases with $|n|$, so $|n| = N + 1$ supplies the uniform tail bound. No logarithmic constant or finite- L diagonal term was discarded from the exact identity.

For the prime part, set $w_m = \Lambda(m)/\sqrt{m}$ and $a_m = \log m$. The elementary inequality $2\Re(z\bar{w}) \leq |z|^2 + |w|^2$ gives

$$-2 \sum_m w_m \Re \int_0^{L-a_m} f(x+a_m) \overline{f(x)} dx \geq - \int_0^L d_\lambda(x) |f(x)|^2 dx \geq -M_\lambda \|f\|_2^2,$$

where d_λ is the function inside braces in (175). This applies equally to complex vectors.

In centered physical coordinates the negative pole term is $-2|\langle f, \sinh(x/2) \rangle|^2$. Direct Fourier integration gives

$$|\widehat{\sinh(x/2)}(n)|^2 = \frac{4 \sinh^2(L/4) t_n^2}{L(t_n^2 + 1/4)^2}.$$

Consequently

$$2\|Q_N \sinh(x/2)\|_2^2 \leq \frac{4 \sinh^2(L/4) L}{\pi^2} \sum_{n>N} n^{-2} \leq \frac{4 \sinh^2(L/4) L}{\pi^2 N}.$$

It vanishes identically on the even sector; the positive pole term can be retained or omitted in a lower bound. Combining these estimates proves (176) first on finite Fourier sums and then on the tail form domain by Proposition 20.19.

For the numerical constants at $\lambda = 3$, symmetry reduces the prime degree to $0 < x < \log 3$. Its breakpoints, expressed in e^x , are $9/8, 9/7, 9/5, 2, 9/4$. With $P = w_2 + w_3 + w_4 + w_5 + w_7 + w_8$, the six interval values are

$$P, \quad P - w_8, \quad P - w_8 - w_7, \quad w_2 + w_3 + w_4, \quad 2w_2 + w_3 + w_4, \quad 2w_2 + w_3.$$

All are at most P , using $w_5 > w_2$ for the fifth value, and the first occurs on an interval of positive length. Thus $M_3 = P$ exactly. Also $\sinh^2(L/4) = 1/3$, $R_L = 27/80$ and $C_L = 1$. Outward interval evaluation of the displayed elementary expressions gives (177); the reproducibility bundle contains the 256-bit scalar certificate. These are not new finite-matrix eigenvalue computations. $\square$

A positive weight for the prime shifts and a diagonal inverse bound

The flat degree bound can be sharpened without changing the Weil operator. Let T_{pr} denote the sum of the truncated translations and their adjoints with weights $w_m = \Lambda(m)/\sqrt{m}$. Its coefficients are nonnegative, and the prime contribution to the Weil form is $-\mathsf{T}_{\text{pr}}$. For every bounded function ϕ bounded away from zero and every complex f ,

$$|\langle \mathsf{T}_{\text{pr}} f, f \rangle| \leq \int_0^L \frac{(\mathsf{T}_{\text{pr}} \phi)(x)}{\phi(x)} |f(x)|^2 dx \leq M_\phi \|f\|_2^2, \quad M_\phi := \operatorname{ess\,sup}_{0 < x < L} \frac{(\mathsf{T}_{\text{pr}} \phi)(x)}{\phi(x)}. \quad (181)$$

Here ϕ is required to be strictly positive. This follows edge by edge from

$$2|f(x+s)f(x)| \leq \frac{\phi(x)}{\phi(x+s)} |f(x+s)|^2 + \frac{\phi(x+s)}{\phi(x)} |f(x)|^2.$$

Thus the argument is a weighted Schur bound on the translation sum; it assumes no sign for the complete Weil form.

Proposition 20.22 (A certified weighted prime bound at $\lambda = 3$). *At $L = 2 \log 3$, take $\phi(x) = \cosh(x - \log 3)$. With $w_m = \Lambda(m)/\sqrt{m}$, put*

$$\mathbf{m}_3 = \frac{85}{116}w_2 + \frac{65}{87}w_3 + \frac{193}{232}w_4 + \frac{137}{145}w_5 + \frac{35}{29}w_7. \quad (182)$$

Then $M_\phi = \mathbf{m}_3$ and

$$2.6890557719319011796 < \mathbf{m}_3 < 2.6890557719319011797.$$

In particular $\|\mathbf{T}_{\text{pr}}\| \leq \mathbf{m}_3$. Replacing M_3 by $\mathbf{m}_3$ in Proposition 20.21 gives

$$\tilde{\Gamma}_{3,256}^{\text{all}} > 0.4970, \quad \tilde{\Gamma}_{3,256}^{\text{even}} > 2.0690, \quad \tilde{\Gamma}_{3,512}^{\text{all}} > 1.1907. \quad (183)$$

These are omitted-complement bounds, with no assertion yet about the sign of the head/complement Schur matrix.

Proof. Write $u = e^x \in [1, 9]$. The ratio for a forward shift by $\log m$ is

$$\frac{mu^2 + 9/m}{u^2 + 9},$$

and for its backward shift it is $(u^2/m + 9m)/(u^2 + 9)$. On an interval with fixed active shifts, their weighted sum has the form $A + B \tanh(x - \log 3)$ and is monotone or constant. The endpoints are $u = 1, 9, m, 9/m$, for $m = 2, 3, 4, 5, 7, 8$. By reflection it suffices to examine the six intervals in $1 < u < 3$, whose successive upper endpoints are

$$9/8, \quad 9/7, \quad 9/5, \quad 2, \quad 9/4, \quad 3.$$

On each such interval the ratio is increasing. Its largest one-sided value is approached from $u < 9/7$ and is

$$\sum_{m=2,3,4,5,7} w_m \frac{9m^2 + 49}{58m} = \mathbf{m}_3.$$

The finite comparisons with the other endpoint values use rational coefficients and are certified at 256-bit Arb precision by `certify_weighted_prime_tail.py`; its exact coefficient lists and interval enclosures are included in the reproducibility bundle. The $m = 8$ term is retained on every interval where that shift is active. The $m = 9$ translation has full interval length and is zero in L^2 . This proves the asserted essential supremum and norm bound by (181). Substitution in the explicit formulas of Proposition 20.21, with $h_L = 1/3$, $R_L = 27/80$ and $C_L = 1$, gives (183); the same scalar interval certificate verifies the strict inequalities. $\square$

Proposition 20.23 (Diagonal domination of the tail inverse). *Continue at $\lambda = 3$ and retain E_L and C_L from (174). Fix $N \geq 1$ and set $t_* = 2\pi(N + 1)/L$. For the even sector define*

$$\kappa_{\text{even}} = \mathbf{m}_3 + 2C_L/t_*;$$

for the whole space, and hence also the odd sector, define

$$\kappa_{\text{all}} = \kappa_{\text{even}} + \pi/2 + 4h_L L/(\pi^2 N).$$

In the selected sector put

$$g_n = \log \frac{|n|}{L} - E_L(2\pi|n|/L) - \kappa, \quad |n| > N. \quad (184)$$

If $g_{N+1} > 0$, the canonical omitted-tail operator T satisfies the form inequality $T \succeq D_g$, where D_g is this diagonal operator, and therefore

$$T^{-1} \preceq D_g^{-1}. \quad (185)$$

The same weights apply in the normalized positive-index parity basis.

Proof. The proof of Proposition 20.21 bounds the archimedean diagonal at each index by $\log(|n|/L) - E_L(2\pi|n|/L)$ before any infimum is taken. The remaining archimedean error has norm at most $2C_L/t_*$. Its limiting commutator is nonnegative in the even sector and is bounded below by $-\pi/2$ on the whole tail. Apply Proposition 20.22 to the prime term and the existing negative-pole tail estimate to the whole-space case. These bounds prove $T \succeq D_g$ on finite tail sums and hence on the common logarithmic form domain. The weights g_n increase with $|n|$, since $E_L(t)$ is decreasing, so $D_g \succeq g_{N+1}I > 0$. The inverse inequality follows, for example, from the variational identity

$$\langle T^{-1}v, v \rangle = \sup_w \{2\Re \langle v, w \rangle - \mathfrak{t}[w]\} \leq \sup_w \left\{ 2\Re \langle v, w \rangle - \sum_n g_n |w_n|^2 \right\} = \sum_n \frac{|v_n|^2}{g_n}.$$

The quadratic quantities are real in the fixed first-slot-linear convention. Restriction to a parity sector is isometric and leaves the diagonal weights unchanged. $\square$

The preceding tail estimates alone do not establish full Weil positivity. The complete signed certificate below also controls the effective head. Its absolute arithmetic bound still gives a sufficient cutoff exponential in $O(\lambda)$; it does not reach the natural prolate scale. For the signed correction use $Q_N = I - P_N$ and let $\Gamma_{\lambda,N} > 0$ denote any of the proved tail bounds. Write

$$W_\lambda = \begin{pmatrix} F & B^* \\ B & T \end{pmatrix}, \quad F = P_N W_\lambda P_N, \quad B = Q_N W_\lambda P_N, \quad T \succeq \Gamma_{\lambda,N} I.$$

The cross operator B is bounded by (168). The remaining exact sign problem is the finite matrix $K = F - B^* T^{-1} B$. There is a useful verified-solve alternative to bounding B alone. For any $Z : E_N \rightarrow \mathcal{D}(T)$ put $R = B - TZ$ and $K_Z = F - B^* Z - Z^* B + Z^* T Z$. Then

$$K = K_Z - R^* T^{-1} R \succeq K_Z - \Gamma_{\lambda,N}^{-1} R^* R \succeq K_Z - \frac{\|R\|^2}{\Gamma_{\lambda,N}} I. \quad (186)$$

Expansion proves the identity and $T^{-1} \preceq \Gamma_{\lambda,N}^{-1} I$ proves both bounds. Retaining the residual Gram matrix $R^* R$ preserves directional cancellation lost by its scalar-norm majorant. If the final finite lower bound is $-\varepsilon I$, $\varepsilon \geq 0$, completing the form square gives $W_\lambda \succeq -\varepsilon I$. Proving such a signed bound along growing windows is still open; positivity of F alone is insufficient.

A directional certificate for the entire Fourier complement

The residual Gram in (186) can be enclosed without replacing its small directions by a scalar norm. We retain the actual matrix (169), including its separate diagonal. All adjoints below are the usual coordinate adjoints in the fixed orthonormal Fourier basis; quadratic-form order is unchanged by our first-slot-linear convention.

Proposition 20.24 (Moment enclosure of the remote residual). *Fix λ , integers $1 \leq M < J$, and $r \geq 1$. Let $G : \mathbb{C}^d \rightarrow E_M$ be any coefficient matrix, with row G_m at Fourier index m , and choose $B_* \geq \sup_n |b_n|$, for example $B_* = B_\lambda^{\text{mat}}$ from (170). Define the r -by- d moment matrices by*

$$S_j = \sum_{m=-M}^M (m/M)^j G_m, \quad T_j = \sum_{m=-M}^M (m/M)^j b_m G_m, \quad 0 \leq j < r, \quad (187)$$

using $0^0 = 1$. Set

$$\begin{aligned} H_{jk} &= M^{j+k} \{1 + (-1)^{j+k}\} \zeta(j+k+2, J+1), \\ c_{M,r} &= \sum_{m=-M}^M (|m|/M)^{2r}, \\ \delta_{M,J,r}^2 &= \frac{8B_*^2 c_{M,r} M^{2r} \zeta(2r+2, J+1)}{(1 - M/(J+1))^2}, \end{aligned} \quad (188)$$

where $\zeta(s, a) = \sum_{k \geq 0} (k+a)^{-s}$ is the Hurwitz zeta function. For every $\tau > 0$, the explicitly finite matrix

$$U_J = (1 + \tau) \{2B_*^2 S^* H S + 2T^* H T\} + (1 + \tau^{-1}) \delta_{M,J,r}^2 G^* G \quad (189)$$

satisfies

$$(Q_J W_\lambda G)^* (Q_J W_\lambda G) \preceq U_J.$$

The assertion also holds on either normalized parity sector, provided the moments are those of the corresponding full Fourier coefficients.

Proof. Since $|n| > J > M \geq |m|$, no diagonal entry occurs. The finite geometric identity for $(n-m)^{-1}$ gives exactly

$$\begin{aligned} (W_\lambda G)_n &= A_n + E_n, \\ A_n &= \sum_{j=0}^{r-1} \frac{M^j}{n^{j+1}} (b_n S_j - T_j), \\ E_n &= \sum_{m=-M}^M \left(\frac{m}{n}\right)^r \frac{b_n - b_m}{n - m} G_m. \end{aligned}$$

The matrix H is the Gram matrix of the sequences $M^j n^{-j-1}$ on $|n| > J$; pairing the two signs of n proves the formula in (188) and $H \succeq 0$. For a column x , apply $|b_n a_n - t_n|^2 \leq 2B_*^2 |a_n|^2 + 2|t_n|^2$ and sum over the remote indices. This bounds the Gram of A by $2B_*^2 S^* H S + 2T^* H T$.

For the remainder, Cauchy–Schwarz in m yields

$$|E_n x|^2 \leq \frac{4B_*^2 (M/|n|)^{2r} c_{M,r}}{(|n| - M)^2} \|Gx\|^2.$$

Use $|n| - M \geq |n|(1 - M/(J+1))$ and sum over both tails. The resulting bound is $\sum_{|n| > J} |E_n x|^2 \leq \delta_{M,J,r}^2 \|Gx\|^2$; this accounts for the factor 8 in (188). Finally apply $|a+e|^2 \leq (1+\tau)|a|^2 + (1+\tau^{-1})|e|^2$ before summing. Every estimate is a quadratic-form inequality for arbitrary x , so it proves the stated matrix order, not merely entrywise bounds.

For completeness, an even normalized parity vector has full coordinates $G_0 = v_0$, $G_m = G_{-m} = v_m/\sqrt{2}$ for $m > 0$; an odd vector has $G_0 = 0$, $G_m = -G_{-m} = v_m/\sqrt{2}$. As b is odd, $S_j = 0$ for odd j and $T_j = 0$ for even j in the even sector, with the opposite zeros in the odd sector. These zeros are exact. The two tails are already counted in H ; no further factor of two is inserted in the parity calculation. $\square$

Proposition 20.25 (Diagonal-weighted Schur certificate). *Write $P = P_N$, $Q = I - P$ and $F = PW_\lambda P$, $B = QW_\lambda P$, $T = QW_\lambda Q$. Suppose an independent form estimate gives*

$$T \succeq D_g := \text{diag}_{|n|>N}(g_{|n|}) \succeq \gamma I > 0,$$

where g_k is nondecreasing for $k > N$; the estimate of Proposition 20.23 is one source of such a sequence. Let $N < M < J$, let $Z : E_N \rightarrow E_M \cap Q\mathcal{H}$ be any finite matrix, and put

$$G = I_{E_N \hookrightarrow E_M} - Z, \quad K_Z = F - B^*Z - Z^*B + Z^*TZ.$$

For $N < |n| \leq J$ compute the rows $R_n = (W_\lambda G)_n$ using the complete actual matrix, and form

$$\mathcal{L}_Z = K_Z - \sum_{N < |n| \leq J} \frac{R_n^* R_n}{g_{|n|}} - \frac{U_J}{g_{J+1}}, \quad (190)$$

*with U_J from Proposition 20.24. Then the exact Schur complement satisfies $F - B^*T^{-1}B \succeq \mathcal{L}_Z$. In particular, a certified inequality $\mathcal{L}_Z \succeq -\varepsilon I$, $\varepsilon \geq 0$, implies $W_\lambda \succeq -\varepsilon I$ on the entire form domain. The same implication holds after restriction to either parity sector.*

Proof. The cross operator B is bounded by Proposition 20.19; finite Z has range in the operator domain. Thus $R = B - TZ = QW_\lambda G$ is well-defined. Inverse order gives $T^{-1} \preceq D_g^{-1}$. For unbounded positive operators this follows, for example, by comparing the variational formulas for their inverse quadratic forms; the positive lower bound γ makes both inverses bounded. The exact verified-solve identity therefore gives

$$F - B^*T^{-1}B = K_Z - R^*T^{-1}R \succeq K_Z - R^*D_g^{-1}R.$$

Split the last quadratic form into $N < |n| \leq J$ and $|n| > J$. Monotonicity gives $g_{|n|}^{-1} \leq g_{J+1}^{-1}$ on the remote part, where (189) applies. This proves (190). Writing $\mathfrak{w}_\lambda$ for the closed form of W_λ , completion of the square gives, for head x and form-domain tail y ,

$$\mathfrak{w}_\lambda[x + y] = \|T^{1/2}(y + T^{-1}Bx)\|^2 + \langle (F - B^*T^{-1}B)x, x \rangle.$$

Interpreted as the closed form on its form domain, this is at least $-\varepsilon\|x\|^2 \geq -\varepsilon\|x + y\|^2$. For a calculation confined to one parity sector, all ingredients are taken in that sector and the same proof applies. $\square$

The use of $g_{|n|}$ on the intervening rows, and of the full moment Grams on the remote rows, is substantive: replacing either by a single operator norm can destroy the small head directions that the certificate must preserve. The finite rows in (190) include the true diagonal d_n ; only the disjoint remote rows use the divided-difference expansion. These propositions supply a finite sufficient test, not a uniform-in- λ positivity theorem. A passed test at one window does not close G2 or justify any transfer without its existing graph defect.

A certificate for the complete fixed-window form

The preceding estimates now permit a signed certificate that includes the entire Fourier complement. This is a computer-assisted result for one physical window; its finite arithmetic and its analytic tail bounds are both part of the proof.

Proposition 20.26 (Complete positivity at $\lambda = 3$). *For the canonical closed Weil form of Proposition 20.19, at $\lambda = 3$ there is a constant $\epsilon_3 > 0$ such that*

$$QW_3(f, f) \geq \epsilon_3 \|f\|_2^2$$

for every vector in its form domain. The result includes both parity sectors and all complex vectors. No numerical value of ϵ_3 is asserted here. In particular, the LDL pivots below are not lower bounds for the smallest eigenvalue.

Analytic reduction and interval certificate. Use the actual unshifted orthonormal Fourier basis on $[0, L]$, $L = 2 \log 3$, and its orthonormal parity combinations e_0 , $(e_m + e_{-m})/\sqrt{2}$ and $(e_m - e_{-m})/\sqrt{2}$. The real matrix commutes with reflection, so positivity in both sectors implies positivity for the whole complex space. For each sector split the head at N , use an intermediate inverse-action support through M , and compute residual rows through J , with the following exact parameters:

Sector	Head dimension	N	M	J	Moment order r
Even	257	256	512	4096	80
Odd	512	512	1024	4096	100

The tail lower functions g_n are those of Proposition 20.23, with the common base cut N retained in every loss term. In particular their smallest values exceed 2.0690 in the even case and 1.1907 in the odd case.

Let T_M be the compression to $N < |n| \leq M$, and let B_M be the corresponding head-to-tail block. A 768-bit Arb interval solve encloses $T_M^{-1}B_M$; freeze its midpoint as the exact dyadic matrix Z . The certificate then uses the complete expression

$$K_Z = F - B_M^* Z - Z^* B_M + Z^* T_M Z,$$

and computes every residual row $N < |n| \leq J$ from $R_n = (W_3 G)_n$, where $G = I_{\text{head}} - Z$. This retains the nonzero intermediate residual from freezing Z . The correct separate logarithmic diagonal is used whenever $n = m$. For the remote rows, use Proposition 20.24 with

$$B_* = \frac{4/3 + P_3 + 1 + \pi/4}{\pi}, \quad P_3 = \sum_{1 < m \leq 9} \frac{\Lambda(m)}{\sqrt{m}}, \quad \tau = 10^{-6}.$$

The optional full-length term in this conservative B_* does not enter the actual matrix. It only enlarges the absolute sequence bound. The squared geometric remainder factors satisfy

$$\delta_{\text{even}}^2 < 1.656 \cdot 10^{-148}, \quad \delta_{\text{odd}}^2 < 3.288 \cdot 10^{-124}.$$

The full matrix majorant U_{remote} , including its leading moment terms, is retained; these small remainder factors alone are not estimates for the whole remote residual.

Proposition 20.25 gives the finite lower form

$$\mathcal{L} = K_Z - \sum_{N < |n| \leq J} \frac{R_n^* R_n}{g_n} - \frac{U_{\text{remote}}}{g_{J+1}} \preceq F - B^* T^{-1} B. \quad (191)$$

In a parity basis the sum uses one normalized row per positive index; the remote formula is evaluated in the isometric full-index embedding. There is no additional factor two or omitted opposite-sign

tail. Interval LDL decomposition certifies every pivot of $\mathcal{L}$ to be strictly positive. The smallest pivot enclosures are contained in

$$(9.17498, 9.17499) \cdot 10^{-9} \quad (257 \text{ even pivots}), \quad (9.68026, 9.68027) \cdot 10^{-8} \quad (512 \text{ odd pivots}).$$

The reproducibility bundle records every pivot, the exact dyadic witnesses, their hashes, the matrix assembly, and all analytic error constants. Replaying both identical witnesses at 896-bit precision again certifies all pivots positive.

For completeness, the coefficient evaluation does not truncate an uncontrolled archimedean series. With $a_k = 2k + 1/2$, the exact digamma and trigamma expressions from (179) leave exponentially weighted sums. After $k = 0, \dots, 127$, put $a_{128} = 256 + 1/2$ and $e_{128} = 3^{-513}$. The radii added to the sine sequence and complete archimedean diagonal respectively are

$$\frac{e_{128}}{2a_{128}(1 - 1/81)}, \quad \frac{2e_{128}}{La_{128}^2(1 - 1/81)}.$$

They follow from $t/(a^2 + t^2) \leq 1/(2a)$ and $|a^2 - t^2|/(a^2 + t^2)^2 \leq 1/a^2$, uniformly in frequency. The full-length $m = 9$ prime translation is exactly zero; every shorter prime-power shift is present. All remaining function evaluations and matrix operations use outward ball enclosures.

Thus the exact Schur complement is positive definite in each finite head and $T \succeq g_{N+1}I > 0$ on its entire complement. Completing the form square proves coercivity, since the triangular map $(x, y) \mapsto (x, y + T^{-1}Bx)$ is boundedly invertible and the finite positive Schur matrix has a positive smallest eigenvalue. The established Fourier form core extends the result to the complete closed form domain. No zero-location or Weil-positivity assumption is used. $\square$

Fixed-window spectral ordering and ordinary ground overlap

The fixed-window computer-assisted positivity certificate can be shifted to count its low eigenvalues. The thresholds in the two reflection sectors need not coincide. In particular, an even trial vector only sees the even spectral gap when its overlap with the ground state is estimated.

Proposition 20.27 (A fixed-window simple even ground state). *Let $\mu_0 \leq \mu_1 \leq \dots$ be the eigenvalues, counted with multiplicity, of the complete canonical operator $\mathbb{W}_3$. Then its ground state is simple and even, and*

$$0 < \mu_0 < 3.644 \cdot 10^{-38}, \quad \mu_1 > 10^{-36}. \quad (192)$$

Within the even sector alone, every eigenvalue other than μ_0 exceeds 10^{-34} . These statements concern the complete operator, including its infinite Fourier complement, at the single window $\lambda = 3$.

Shifted inertia and a matching trial direction. For a threshold $a > 0$ replace every diagonal d_n by $d_n - a$ and every tail lower weight g_n by $g_n - a$. Keep the same exact dyadic witness Z and the same head, intermediate and remote cutoffs as in Proposition 20.26. If $T_a = T - aI$, then

$$K_{Z,a} = K_{Z,0} - a(I + Z^*Z), \quad R_a = B - T_a Z = R_0 + aZ.$$

Thus all intermediate residual rows change and are recomputed. The remote moment bound is unchanged, because those rows have no diagonal entry and $G = \begin{pmatrix} I_{\text{head}} \\ -Z \end{pmatrix}$ is unchanged. The diagonal inverse comparison gives

$$S_a := F - aI - B^*T_a^{-1}B \succeq L_a := K_{Z,a} - \sum_{N < n \leq J} \frac{R_{a,n}^* R_{a,n}}{g_n - a} - \frac{U_J}{g_{J+1} - a}.$$

The sum uses normalized parity rows. All denominators are strictly positive by the earlier entire-tail bounds.

The signed interval LDL certificate for the even lower matrix at $a_e = 10^{-34}$ has 257 nonzero pivots, exactly one of which is negative (zero-based index 24). For the odd lower matrix at $a_o = 10^{-36}$ all 512 pivots are strictly positive. The complete signed elimination retains negative pivots with their signs; it does not stop at the first negative pivot. The exact dyadic witnesses and every pivot enclosure are retained in the bundle; `certify_shifted_inertia.py` recomputes all shifted residual rows. The exact parameters are those of Proposition 20.26: $(N, M, J, r) = (256, 512, 4096, 80)$ for the even sector and $(512, 1024, 4096, 100)$ for the odd sector. Both certificates pass at 768 bits and again at 896 bits using identical dyadic witnesses. By finite-dimensional min-max (see also [21, Section 4.3]), S_{a_e} has at most one nonpositive eigenvalue, because the second eigenvalue of its lower matrix is strictly positive. The odd Schur matrix is positive definite.

A matching negative even direction is supplied by the existing finite-supported candidate of Proposition A.1. More precisely, in that $N = 64$ certificate let u be its normalized even vector, let Q project onto its even orthogonal complement. Set $W_{64} = P_{64}W_3P_{64}$, $\alpha = \langle W_{64}u, u \rangle$, $C = QW_{64}Q$ on that complement, and $r = QW_{64}u$. Write $A = C - \alpha I$, $z = A^{-1}r$, $q = \|z\|$ and $E = \langle z, r \rangle$. The vector

$$v = \frac{u - z}{\sqrt{1 + q^2}}$$

has norm one and belongs to the finite Fourier core. Direct expansion gives its exact complete-form Rayleigh value

$$\beta = QW_3(v, v) = \alpha - \frac{E}{1 + q^2}.$$

The previously certified enclosures imply

$$\alpha < 5.312382 \cdot 10^{-38}, \quad E > 1.668982 \cdot 10^{-38}, \quad q < 0.000216, \quad \beta < 3.644 \cdot 10^{-38}.$$

The last strict scalar comparison is separately checked by `certify_ground_overlap_constants.py`. Because v has finite support, its finite-compression Rayleigh value equals its complete-form value exactly. No approximation of the omitted output coefficients is used in this identity. In particular $\beta < a_e$.

On the closed form domain, square completion reads

$$q_a[x + y] = \langle S_a x, x \rangle + \left\| T_a^{1/2}(y + T_a^{-1}Bx) \right\|^2.$$

The map $(x, y) \mapsto (x, y + T_a^{-1}Bx)$ is boundedly invertible and preserves that domain. It transfers the negative index to S_a ; also $\ker(W_3 - aI)$ is isomorphic to $\ker S_a$. The negative even trial direction therefore makes the first even Schur eigenvalue strictly negative, while its second is strictly positive. There is no even eigenvalue exactly at a_e , and exactly one even eigenvalue lies below it, counted with multiplicity. Every odd eigenvalue exceeds a_o . The established compact resolvent makes these full spectral statements. Finally Proposition 20.26 supplies $\mu_0 > 0$, and min-max applied to v supplies $\mu_0 \leq \beta$. Since $\beta < a_o < a_e$, the unique even low eigenvalue is the simple ground state of the whole operator. This proves (192) and the stronger even-sector threshold. $\square$

Corollary 20.28 (An ordinary ground-overlap bound at one window). *For the normalized rational candidate u of Proposition A.1 and the complete unit ground vector ξ_0 at $\lambda = 3$,*

$$\sin \angle(u, \xi_0) < 0.01931.$$

The same strict upper bound holds for the normalized exact $N = 64$ source projection after the already certified normalization error is included. This is an ordinary Hilbert-space bound; it asserts no endpoint comparison with ξ_0 .

Proof. The improved trial v in the proof above is even. In its spectral expansion, every non-ground component has energy greater than $a_e = 10^{-34}$, and the ground energy is positive. Hence

$$\sin \angle(v, \xi_0) < \sqrt{3.644 \cdot 10^{-38} / 10^{-34}}.$$

Also $z \perp u$, so the norm of the difference of the two rank-one orthogonal projections onto u and v equals $q/\sqrt{1+q^2} < 0.000216$. The triangle inequality for these projectors proves

$$\sin \angle(u, \xi_0) < 0.000216 + \sqrt{3.644 \cdot 10^{-4}} < 0.019305264 < 0.01931.$$

For the unnormalized rational candidate c , Proposition 20.78 gives $\|P_{64}p_3 - c\| < \varepsilon := 2.60 \cdot 10^{-36}$, while the exact rational norm calculation gives $\|c\| > 0.65172555$. Writing $\rho = \varepsilon/\|c\|$, elementary ball geometry yields

$$\left\| \frac{P_{64}p_3}{\|P_{64}p_3\|} - \frac{c}{\|c\|} \right\|^2 \leq \frac{2\rho^2}{1 + \sqrt{1 - \rho^2}} < (4 \cdot 10^{-36})^2.$$

Indeed the real inner product of these two unit vectors is positive and at least $\sqrt{1 - \rho^2}$, by minimizing the squared distance from c along the positive ray through $P_{64}p_3$. This normalization error fits within the final margin. The ordinary norm controls projector distance but not the physical endpoint functional; the earlier endpoint obstruction is therefore unchanged. $\square$

This proves parity, simplicity and quantitative ordinary overlap at one fixed window. It does not identify the unprojected source with the ground vector. There is still no uniform family of spectral thresholds or source overlap estimates along unbounded windows. The physical endpoint is resolved below by a separate regularity argument; it does not follow from this ordinary overlap bound. The sampler graph defect and full-strip requirements remain separate; G2 and RH are not closed by the fixed-window spectral result.

The physical endpoint of the complete Weil eigenfunctions

The endpoint of an actual eigenfunction is not determined by ordinary norm convergence of finite eigenvectors. For the complete operator it can instead be determined from the singular archimedean kernel. Throughout this subsection $\Omega = (0, L)$, physical functions are extended by zero outside Ω , and Fourier frequency is normalized as in $U_n(x) = L^{-1/2}e^{2\pi i n x/L}$.

Proposition 20.29 (Restricted logarithmic Laplacian decomposition). *Let $\mathcal{A} = \frac{1}{2}L_{\Delta}^{\text{Dir}}$ be the restricted, exterior-Dirichlet logarithmic Laplacian on Ω , whose full-line symbol is $\log |t|$. This is not the spectral logarithm of the Dirichlet Laplacian. The actual closed Weil operator satisfies*

$$W_{\lambda} = \mathcal{A} + \mathcal{K}_{\lambda}, \quad \mathcal{D}(W_{\lambda}) = \mathcal{D}(\mathcal{A}), \quad (193)$$

where, writing $k(y) = \rho(y) - (2y)^{-1}$, $a_m = \log m$, and $w_m = \Lambda(m)/\sqrt{m}$,

$$\begin{aligned} (\mathcal{K}_{\lambda}f)(x) = & -\log(2\pi)f(x) - \int_0^L k(|x-y|)f(y) dy \\ & - \sum_{1 < m < \lambda^2} w_m \{ \mathbf{1}_{x+a_m < L} f(x+a_m) + \mathbf{1}_{x>a_m} f(x-a_m) \} \\ & + 2 \int_0^L \cosh((x-y)/2) f(y) dy. \end{aligned} \quad (194)$$

The operator $\mathcal{K}_\lambda$ is real and self-adjoint on L^2 . For every $1 \leq p \leq \infty$ it is bounded on $L^p(\Omega)$, with

$$\|\mathcal{K}_\lambda\|_{p \rightarrow p} \leq C_\lambda := \log(2\pi) + 2 \int_0^L |k(y)| dy + 2P_\lambda + 4 \sinh(L/2). \quad (195)$$

Proof. In dimension one the integral normalization of the logarithmic Laplacian [22, Theorem 1.1] gives

$$\mathcal{A}f(x) = \int_0^\infty \frac{2f(x)\mathbf{1}_{y < 1} - f(x+y) - f(x-y)}{2y} dy - \gamma_E f(x)$$

on compactly supported smooth tests. The actual negative-archimedean part of the Weil operator is

$$\int_0^\infty \rho(y) \{2e^{-y/2} f(x) - f(x+y) - f(x-y)\} dy - \{\log(4\pi) + \gamma_E\} f(x).$$

Its Fourier diagonal is precisely the negative of (167); the diagonal integral over $y > L$ equals $-\log \tanh(L/2)$. Since

$$\int_0^\infty \left\{ \frac{1}{\sinh y} - \frac{\mathbf{1}_{y < 1}}{y} \right\} dy = \log 2,$$

subtracting $\mathcal{A}$ leaves $-\log(2\pi)I$ and the kernel $-k$. The integral follows by evaluating $\log \tanh(y/2)$ with a positive lower cutoff and then taking its limit. In particular the constant part of the logarithmic diagonal has been retained. The prime shifts and the pole kernel $2 \cosh((x-y)/2)$ give the remaining terms of (194). The latter kernel is the difference of the centered rank-one $2 \cosh \otimes \cosh$ and $2 \sinh \otimes \sinh$ terms, so no pole sign has been removed.

The function k extends continuously to zero, with $k(0) = 1/4$. The integral-kernel row and column bounds, and the contraction of each truncated translation, prove (195). For the pole kernel the largest row integral is $4 \sinh(L/2)$. Symmetry proves self-adjointness.

It remains to identify the closed forms, without imposing an unproved endpoint condition. Smooth compactly supported functions are a form core for the restricted logarithmic Laplacian [22, Theorem 3.1]. They are also a form core for the operator in Proposition 20.19. Indeed, approximate each finite periodic Fourier sum f by smooth cutoffs that remove endpoint strips of width ϵ . The discarded periodic function has L^2 norm $O(\epsilon^{1/2})$ and H^1 norm $O(\epsilon^{-1/2})$, hence H^s norm $O(\epsilon^{1/2-s})$ for any fixed $0 < s < 1/2$. This tends to zero in the logarithmically weighted form norm. The two closed lower-bounded forms therefore agree on a common form core. Bounded perturbation by $\mathcal{K}_\lambda$ proves (193), including its operator-domain assertion. $\square$

Theorem 20.30 (Zero boundary trace of every complete Weil eigenfunction). *Fix $\lambda > 1$. Every L^2 eigenfunction $W_\lambda u = \mu u$ has a bounded representative whose zero extension is continuous on $\mathbb{R}$. With*

$$\ell(r) = \frac{1}{|\log(\min\{r, 0.1\})|}, \quad d(x) = \min\{x, L-x\},$$

there is a finite constant C such that

$$|u(x)| \leq C \sqrt{\ell(d(x))} \quad (0 < x < L). \quad (196)$$

In particular its actual physical endpoints satisfy $u(0) = u(L) = 0$. No uniform bound for C as λ grows is asserted.

Proof. First we prove boundedness, without assuming a positive Sobolev gain. For $0 < \epsilon < \min\{1, L\}$ let $\mathcal{A}_\epsilon$ be the nonnegative killed small-jump generator

$$\mathcal{A}_\epsilon f(x) = \int_0^\epsilon \frac{2f(x) - f(x+y) - f(x-y)}{2y} dy$$

in its closed Dirichlet-form realization, and put

$$J_\epsilon f(x) = \int_{\substack{y \in \Omega \\ |x-y| \geq \epsilon}} \frac{f(y)}{2|x-y|} dy.$$

There is an exact decomposition

$$\mathcal{A} = \mathcal{A}_\epsilon + \{\log(1/\epsilon) - \gamma_E\}I - J_\epsilon.$$

The small-jump form is Markovian, including its nonnegative exterior killing term. Consequently its resolvent is consistent on L^2 and L^∞ , and $\|(\mathcal{A}_\epsilon + aI)^{-1}\|_{p \rightarrow p} \leq a^{-1}$ for $a > 0$, $p = 2, \infty$. One may obtain this directly by first cutting off jumps smaller than a positive number, using the contraction resolvent of that finite-activity killed jump generator, and then passing to the increasing closed forms. Also Cauchy–Schwarz gives

$$\|J_\epsilon\|_{2 \rightarrow \infty} \leq (2\epsilon)^{-1/2}.$$

Choose ϵ so that $a = \log(1/\epsilon) - \gamma_E - \mu > C_\lambda$. The eigen equation becomes

$$(\mathcal{A}_\epsilon + aI + \mathcal{K}_\lambda)u = J_\epsilon u.$$

The inverse on the left exists on both L^2 and L^∞ by the Neumann series for $[I + (\mathcal{A}_\epsilon + aI)^{-1}\mathcal{K}_\lambda]^{-1}(\mathcal{A}_\epsilon + aI)^{-1}$, with norm at most $(a - C_\lambda)^{-1}$. Since $J_\epsilon u \in L^2 \cap L^\infty$, the two series agree. Uniqueness in L^2 identifies the bounded solution with u , and proves

$$\|u\|_\infty \leq \frac{\|u\|_2}{(a - C_\lambda)\sqrt{2\epsilon}}.$$

Thus boundedness has not been inferred from the logarithmic operator domain alone.

By Proposition 20.29, the zero extension lies in the logarithmic Dirichlet form space and satisfies

$$L_\Delta u = 2(\mu u - \mathcal{K}_\lambda u) \in L^\infty(\Omega), \quad u = 0 \quad \text{outside } \Omega.$$

The interval satisfies an exterior uniform sphere condition. The boundary regularity theorem of Hernández-Santamaría, López Ríos and Saldaña [23, Theorem 1.1] therefore gives continuity of the zero extension and (196). For complex u , apply the real theorem to its real and imaginary parts. This proves the endpoint assertion for every eigenfunction, independently of its spectral order. $\square$

Corollary 20.31 (Filtered endpoint decay and the nonzero-source obstruction). *For an eigenfunction as above and $N \geq 4$, define its Fejér endpoint in the actual normalized Fourier coordinates by*

$$\sigma_{N-1}u(0) = L^{-1/2} \sum_{|n| < N} (1 - |n|/N)\hat{u}(n).$$

Then

$$|\sigma_{N-1}u(0)| \leq C\sqrt{\ell(L/\sqrt{N})} + \frac{\|u\|_\infty}{2\sqrt{N}} = O_{\lambda,\mu,u}((\log N)^{-1/2}). \quad (197)$$

For the exact repaired source with $B_\lambda = p_\lambda(0) \neq 0$, any complete-operator eigenfunction u and scalar c satisfy

$$\frac{|p_\lambda(0) - cu(0)|}{|B_\lambda|} = 1. \quad (198)$$

Proof. The positive Fejér kernel in physical coordinates is $L^{-1}F_N(2\pi x/L)$, where $F_N(\theta) = N^{-1}\{\sin(N\theta/2)/\sin(\theta/2)\}^2$; it has integral one. Put $h = L/\sqrt{N} \leq L/2$. On $d(x) < h$, (196) bounds the convolution by $C\sqrt{\ell(h)}$. On $d(x) \geq h$, $\sin(\pi x/L) \geq 2d(x)/L$ gives

$$L^{-1}F_N(2\pi x/L) \leq \frac{L}{4Nd(x)^2}, \quad \int_{d(x) \geq h} L^{-1}F_N(2\pi x/L) dx \leq \frac{L}{2Nh} = \frac{1}{2\sqrt{N}}.$$

This proves (197). The mismatch identity follows from $u(0) = 0$. $\square$

Theorem 20.30 applies in particular to the simple even ground at $\lambda = 3$. It disproves a nonzero physical endpoint comparison between the repaired source and the complete operator's ground, while leaving their ordinary-norm overlap intact. It does not identify the endpoint of a finite-compression eigenvector on a joint growing-window and growing-cutoff diagonal. Nor does continuity alone prove absolute Fourier convergence or convergence of sharp Fourier endpoint sums: the quantitative statement (197) concerns the stated positive filter. The finite endpoint and its graph-defect terms therefore remain separate from the complete eigenfunction's trace.

Proposition 20.32 (The Weil graph norm does not control the physical trace). *At every fixed $\lambda > 1$, physical endpoint evaluation on finite Fourier polynomials has no continuous extension to $\mathcal{D}(W_\lambda)$ with its operator graph norm. Indeed there are real even finite polynomials h_N with*

$$\delta(h_N) = 1, \quad \|h_N\|_2 + \|W_\lambda h_N\|_2 = O_\lambda\left(\frac{\log(2+N)}{\sqrt{N}}\right) \rightarrow 0.$$

The same failure holds with $W_\lambda - \mu I$ for any fixed $\mu \in \mathbb{R}$ and with the weaker closed-form norm.

Proof. In the physical logarithmic basis $U_n = L^{-1/2}e^{2\pi i n x/L}$ set

$$h_N = \frac{\sqrt{L}}{2N} \sum_{N < |n| \leq 2N} U_n.$$

There are exactly $2N$ terms. At either physical endpoint all their phases are one, so $\delta(h_N) = 1$ and $\|h_N\|_2 = \sqrt{L/(2N)}$. The exact decomposition in Proposition 20.19 gives

$$\|W_\lambda h_N\|_2 \leq \left(\max_{N < |n| \leq 2N} |d_n| + 2\pi B_\lambda^{\text{mat}} \right) \sqrt{\frac{L}{2N}}.$$

Since $d_n = \log|n| + O_\lambda(1)$, this proves the estimate. A fixed scalar shift changes only the bounded constant. Graph convergence implies form convergence for a lower-bounded self-adjoint operator. In fact the trace is not closable on the polynomial core in the graph norm: this sequence tends to zero while its trace tends to one. $\square$

Let

$$\mathcal{V}_\lambda := \text{Var}(p_\lambda)$$

be the total variation of the periodic logarithmic representative of the inversion-even proxy. Let

$$M_T(\lambda) := \max\{m \geq 0 : 2\pi m/L \leq T\}.$$

Proposition 20.33 (Explicit BV rate for the fixed-band Galerkin defect). *Fix $T > 0$. If $N \geq 2M_T(\lambda) + 1$, then*

$$\boxed{\gamma_{\lambda,N,T} \leq C_T \frac{\lambda L \mathcal{V}_\lambda}{|B_\lambda| N}}. \quad (199)$$

In particular the form defect in G1 has a fully quantitative, zero-independent rate; no C^2 or higher regularity of p_λ is used.

Proof. Use the translated logarithmic coordinate $[0, L]$. Since p_λ has bounded variation and matching periodic endpoint values, Stieltjes integration by parts gives, for $n \neq 0$,

$$|\widehat{p}_\lambda(n)| \leq \frac{\sqrt{L} \mathcal{V}_\lambda}{2\pi|n|}.$$

Let $g = D_{\log} v$ with $v \in E_{\lambda,T}$ and $\|v\|_2 = 1$. Then $\|g\|_2 \leq T$ and $g_m = 0$ for $|m| > M_T(\lambda)$. If $|n| > N \geq 2M_T + 1$, Lemma 20.17 and Cauchy–Schwarz imply

$$\left| \sum_{|m| \leq M_T} \tau_{n,m}^{(\lambda)} g_m \right| \leq C_T \frac{\lambda \sqrt{L}}{|n|}.$$

The resulting tail is absolutely summable, and hence

$$|QW_\lambda((P_N - I)p_\lambda, g)| \leq C_T \lambda L \mathcal{V}_\lambda \sum_{|n| > N} \frac{1}{n^2} \leq C_T \frac{\lambda L \mathcal{V}_\lambda}{N}.$$

Divide by $|B_\lambda|$ and take the supremum over v . □

A compatibility-friendlier filtered finite section

The exact-boundary repair below is useful for G1 alone, but it adds a new high-frequency vector. For the later same-object G1/G2 comparison it is preferable to stay inside a canonical smoothing of the literal prolate/co-Poisson proxy. A de la Vallée–Poussin filter does this while preserving all sufficiently low Fourier modes exactly.

Work in the translated logarithmic coordinate $[0, L]$, with the endpoints identified. Let $\widehat{p}_\lambda(n)$ be the normalized Fourier coefficients and define the multiplier

$$\nu_N(n) := \begin{cases} 1, & |n| \leq N, \\ 2 - |n|/N, & N < |n| < 2N, \\ 0, & |n| \geq 2N. \end{cases} \quad (200)$$

The de la Vallée–Poussin mean is

$$\mathbf{V}_N p_\lambda := \sum_{n \in \mathbb{Z}} \nu_N(n) \widehat{p}_\lambda(n) U_n \in E_{2N}. \quad (201)$$

Equivalently its convolution kernel is the difference of two Fejér kernels, $2F_{2N} - F_N$. In particular its L^1 norm is bounded by an absolute constant and, for $0 < \delta \leq L/2$, the kernel mass outside the boundary arc $|x| < \delta$ is $O(L/(N\delta))$.

Define the endpoint modulus

$$\omega_\lambda(\delta) := \sup_{0 \leq t \leq \delta} \max\{|p_\lambda(t) - B_\lambda|, |p_\lambda(L - t) - B_\lambda|\}. \quad (202)$$

The no-boundary-jump property gives $\omega_\lambda(\delta) \rightarrow 0$ as $\delta \downarrow 0$ for every fixed λ .

Lemma 20.34 (Quantitative endpoint recovery by de la Vallée–Poussin filtering). *For every $N \geq 1$ and $0 < \delta \leq L/2$,*

$$\boxed{|\delta_{2N}(\mathbf{V}_N p_\lambda) - B_\lambda| \leq C \left(\omega_\lambda(\delta) + \frac{\mathcal{V}_\lambda L}{N\delta} \right)}. \quad (203)$$

The constant is absolute. Hence the endpoint recovery is quantitative in terms of the explicit boundary modulus of the chosen prolate/co-Poisson proxy and uses no zero information.

Proof. Let K_N^{VP} be the period- L convolution kernel of $\mathbf{V}_N$. Since $|K_N^{\text{VP}}| \leq 2F_{2N} + F_N$, its L^1 norm is $O(1)$ and the standard Fejér estimate gives

$$\int_{|x| \geq \delta} |K_N^{\text{VP}}(x)| dx \leq C \frac{L}{N\delta}.$$

On the boundary arc use $|p_\lambda(x) - B_\lambda| \leq \omega_\lambda(\delta)$. On its complement, bounded variation and the absence of a periodic boundary jump give $|p_\lambda(x) - B_\lambda| \leq \mathcal{V}_\lambda$. Splitting the convolution integral proves (203). $\square$

For later bookkeeping it is useful to note that the endpoint modulus is itself zero-independently computable from the source. The compact support of the prolate source makes the co-Poisson sum finite on the window; away from the finitely many arithmetic thresholds it is a finite sum of smooth functions. Thus there are explicit numbers $\Delta_\lambda > 0$ and $H_\lambda < \infty$, determined only by the chosen source and λ , for which

$$\omega_\lambda(\delta) \leq H_\lambda \delta, \quad 0 < \delta \leq \Delta_\lambda. \quad (204)$$

For example one may take Δ_λ to be half the logarithmic distance from the identified endpoint to the nearest interior arithmetic threshold; H_λ is then the supremum of the ordinary logarithmic derivative on those two endpoint arcs. No uniform-in- λ estimate for H_λ is being asserted here.

Proposition 20.35 (Filtered BV transfer with explicit boundary control). *Fix $T > 0$ and suppose $N \geq 2M_T(\lambda) + 1$. Set*

$$k_{\lambda,N}^{\text{VP}} := \mathbf{V}_N p_\lambda \in E_{2N}, \quad \beta_{\lambda,N}^{\text{VP}} := \frac{|\delta_{2N}(k_{\lambda,N}^{\text{VP}}) - B_\lambda|}{|B_\lambda|}.$$

Then

$$\boxed{\frac{1}{|B_\lambda|} \sup_{\substack{v \in E_{\lambda,T} \\ \|v\|_2=1}} |QW_\lambda(k_{\lambda,N}^{\text{VP}} - p_\lambda, D_{\log} v)| \leq C_T \frac{\lambda L \mathcal{V}_\lambda}{|B_\lambda| N}}. \quad (205)$$

Whenever $\beta_{\lambda,N}^{\text{VP}} < 1$,

$$\boxed{\frac{\|P_T D_{\log} W_{\lambda,2N} k_{\lambda,N}^{\text{VP}}\|_2}{|\delta_{2N}(k_{\lambda,N}^{\text{VP}})|} \leq \frac{T\mathcal{E}_T(\lambda) + C_T \lambda L \mathcal{V}_\lambda / (|B_\lambda| N)}{1 - \beta_{\lambda,N}^{\text{VP}}}}. \quad (206)$$

Moreover, if $H_\lambda > 0$ and $N \geq C\mathcal{V}_\lambda L / (H_\lambda \Delta_\lambda^2)$, then

$$\boxed{\beta_{\lambda,N}^{\text{VP}} \leq \frac{C}{|B_\lambda|} \sqrt{\frac{H_\lambda \mathcal{V}_\lambda L}{N}}}. \quad (207)$$

Consequently one obtains a deterministic, zero-independent one-scale G1 diagonal by taking $N = N_\lambda$ large enough that

$$N_\lambda \geq C_T \max \left\{ 2M_T(\lambda) + 1, \frac{\lambda L \mathcal{V}_\lambda}{|B_\lambda| \mathfrak{r}(\lambda)}, \frac{H_\lambda \mathcal{V}_\lambda L}{|B_\lambda|^2 \mathfrak{r}(\lambda)^2}, \frac{\mathcal{V}_\lambda L}{H_\lambda \Delta_\lambda^2} \right\}, \quad (208)$$

with the last term omitted when it is vacuous. Along such a diagonal,

$$\boxed{\frac{\|P_T D_{\log} W_{\lambda, 2N_\lambda} k_{\lambda, N_\lambda}^{\text{VP}}\|_2}{|\delta_{2N_\lambda}(k_{\lambda, N_\lambda}^{\text{VP}})|} \leq C'_T \mathfrak{r}(\lambda) \rightarrow 0.} \quad (209)$$

Proof. The multiplier (200) is exactly one for $|n| \leq N$. Therefore the Fourier coefficients of $k_{\lambda, N}^{\text{VP}} - p_\lambda$ vanish in that range and, for $|n| > N$, have modulus at most $|\widehat{p}_\lambda(n)|$. The Stieltjes integration-by-parts bound from Proposition 20.33 and the high-low estimate of Lemma 20.17 therefore give (205) with the same $O(N^{-1})$ tail as the ordinary Galerkin projection.

Equation (206) is the same Hermitian fixed-band argument as (160). Lemma 20.34, followed by (204) and the choice $\delta \asymp (\mathcal{V}_\lambda L / (H_\lambda N))^{1/2}$, gives (207) once that δ lies inside the endpoint arc. The cutoff (208) makes both the form defect and boundary defect $O_T(\mathfrak{r}(\lambda))$; inserting the continuum envelope (154) proves (209). $\square$

The filtered vector has another advantage over an external boundary repair: it changes no Fourier coefficient below N . Its derivative can still grow, because the literal co-Poisson proxy is only BV across its arithmetic interior thresholds, but the growth is intrinsic rather than injected by a separate corrector. The coefficient estimate used above gives the explicit bound

$$\|D_{\log} k_{\lambda, N}^{\text{VP}}\|_2 \leq C \mathcal{V}_\lambda \sqrt{\frac{N}{L}}. \quad (210)$$

There is in fact a sharp obstruction to improving the derivative price by pushing an exact boundary repair farther into the Fourier tail.

Lemma 20.36 (High-frequency boundary repair has unavoidable square-root derivative cost). *Let*

$$c(x) = \sum_{j=N+1}^K w_j \cos(2\pi jx/L), \quad c(0) = \sum_{j=N+1}^K w_j = 1.$$

Then

$$\boxed{\|D_{\log} c\|_2^2 \geq \frac{2\pi^2}{L \sum_{j=N+1}^K j^{-2}} \geq \frac{2\pi^2 N}{L}.} \quad (211)$$

The first lower bound is sharp: equality is attained by weights $w_j \propto j^{-2}$.

Proof. Orthogonality of the sine modes gives

$$\|D_{\log} c\|_2^2 = \frac{2\pi^2}{L} \sum_{j=N+1}^K j^2 |w_j|^2.$$

Weighted Cauchy–Schwarz and $\sum w_j = 1$ give

$$1 \leq \left(\sum j^2 |w_j|^2 \right) \left(\sum j^{-2} \right).$$

This proves the first bound, with equality precisely for $w_j \propto j^{-2}$. Finally $\sum_{j>N} j^{-2} \leq \int_N^\infty x^{-2} dx = N^{-1}$. $\square$

Lemma 20.36 shows that the square-root derivative growth seen in the shell construction is not merely an artifact of equal weights: it is unavoidable for every even exact-boundary corrector supported entirely above frequency N . For a norm-level same-object estimate, the filtered literal proxy of Proposition 20.35 is therefore still the preferred G1 candidate. This derivative lower bound is not, however, a pairing-level no-go: the shell estimates in Section 31 below proves that the shell’s explicit G1 and scalar-shift pairings against the higher-order Hardy sampler can be made arbitrarily small by moving the shell outward.

The endpoint defect can also be removed exactly without asking for a quantitative Dirichlet–Jordan theorem. Put

$$q_{\lambda,N} = P_N p_\lambda, \quad d_{\lambda,N} = B_\lambda - q_{\lambda,N}(0),$$

and, for an even integer $K > 2N$, define the even high-frequency shell corrector

$$c_{\lambda,K}^{\text{sh}}(x) := \frac{2}{K} \sum_{j=K/2+1}^K \cos(2\pi jx/L) \quad \text{in logarithmic coordinates.} \quad (212)$$

Every summand has boundary value one, so $c_{\lambda,K}^{\text{sh}}(0) = 1$. Moreover all of its Fourier modes lie above N . Then set

$$k_{\lambda;N,K}^{\text{bc}} := q_{\lambda,N} + d_{\lambda,N} c_{\lambda,K}^{\text{sh}} \in E_K. \quad (213)$$

By construction it is real, inversion-even, and

$$\delta_K(k_{\lambda;N,K}^{\text{bc}}) = B_\lambda \quad (214)$$

exactly.

Theorem 20.37 (Constructive finite fixed-band G1 closure). *Use the explicit repaired source above and fix $T > 0$. For $N \geq 2M_T(\lambda) + 1$ and an even integer $K > 2N$,*

$$\boxed{\frac{\|P_T D_{\log} W_{\lambda,K} k_{\lambda;N,K}^{\text{bc}}\|_2}{|B_\lambda|} \leq T \mathcal{E}_T(\lambda) + C_T \frac{\lambda L}{N} \frac{\mathcal{V}_\lambda}{|B_\lambda|} + C_T \frac{\lambda L}{K} \left(1 + \sqrt{\frac{2N+1}{L}} \frac{\|p_\lambda\|_2}{|B_\lambda|} \right)}. \quad (215)}$$

Consequently there are completely zero-independent, deterministic cutoffs $N_\lambda < K_\lambda$ for which

$$\boxed{\frac{\|P_T D_{\log} W_{\lambda,K_\lambda} k_{\lambda;N_\lambda,K_\lambda}^{\text{bc}}\|_2}{|\delta_{K_\lambda}(k_{\lambda;N_\lambda,K_\lambda}^{\text{bc}})|} \leq C'_T \mathfrak{r}(\lambda) \longrightarrow 0.} \quad (216)$$

One admissible prescription is to choose, after enlarging C_T if necessary,

$$N_\lambda \geq \max \left\{ 2M_T(\lambda) + 1, C_T \frac{\lambda L}{\mathfrak{r}(\lambda)} \frac{\mathcal{V}_\lambda}{|B_\lambda|} \right\}, \quad (217)$$

$$K_\lambda \text{ even,} \quad K_\lambda > 2N_\lambda, \quad K_\lambda \geq C_T \frac{\lambda L}{\mathfrak{r}(\lambda)} \left(1 + \sqrt{\frac{2N_\lambda+1}{L}} \frac{\|p_\lambda\|_2}{|B_\lambda|} \right). \quad (218)$$

All quantities on the right are explicit functions of the chosen prolate/co-Poisson proxy and λ ; no nontrivial zeta zero enters the choice.

Proof. For $v \in E_{\lambda,T}$ with $\|v\|_2 = 1$, Hermiticity gives

$$|\langle P_T D_{\log} W_{\lambda,K} k_{\lambda;N,K}^{\text{bc}}, v \rangle| = |QW_{\lambda}(k_{\lambda;N,K}^{\text{bc}}, D_{\log} v)|.$$

Split the candidate as

$$k_{\lambda;N,K}^{\text{bc}} = p_{\lambda} + (P_N - I)p_{\lambda} + d_{\lambda,N} c_{\lambda,K}^{\text{sh}}.$$

The first term is at most $T|B_{\lambda}|\mathcal{E}_T(\lambda)$ by Lemma 20.12; Proposition 20.33 controls the second. For each mode in the shell, Lemma 20.17 and Cauchy–Schwarz on the fixed physical band give a bound $C_T \lambda L/j$; since $K/2 < j \leq K$ and the weights in (212) sum to one, averaging yields

$$|QW_{\lambda}(c_{\lambda,K}^{\text{sh}}, D_{\log} v)| \leq C_T \frac{\lambda L}{K}.$$

Finally the endpoint-evaluation norm on E_N is $\sqrt{(2N+1)/L}$, so

$$\frac{|d_{\lambda,N}|}{|B_{\lambda}|} \leq 1 + \sqrt{\frac{2N+1}{L}} \frac{\|p_{\lambda}\|_2}{|B_{\lambda}|}.$$

These estimates prove (215). The choices (217)–(218), together with $\mathcal{E}_T(\lambda) \leq C_T \mathfrak{r}(\lambda)$, make each of the last two terms $O_T(\mathfrak{r}(\lambda))$. Equation (214) removes the denominator defect exactly, proving (216). $\square$

Theorem 20.37 transfers the proved continuum G1 bound to a finite vector with exact endpoint normalization. It is a weak fixed-band estimate. Increasing the cutoff does not make the full derivative of the endpoint repair small: its squared norm is

$$|d_{\lambda,N}|^2 \frac{8\pi^2}{K^2 L} \sum_{j=K/2+1}^K j^2 \leq 4\pi^2 |d_{\lambda,N}|^2 K/L.$$

The growing derivative is relevant to any later graph argument. The scalar-shift identity remains exact,

$$D_{\log}(W - \alpha I)k = D_{\log} Wk - \alpha D_{\log} k,$$

but a weak bound on the first term supplies no norm bound on the second.

For filtered endpoint recovery, if the local Lipschitz constant is zero, the near-endpoint error is identically zero; the cutoff is chosen from the far-tail error alone. A formula that divides by that Lipschitz constant is used only in its positive case. All other displayed cutoff thresholds depend on the specified source, its variation, and the desired tolerance, without using a list of zeta zeros.

The next section restores an explicit weighted-sampler estimate from the earlier manuscript. The subsequent sampler construction then separates ordinary norm, physical endpoint, Weil form, and graph behavior. These four notions of compatibility cannot be interchanged.

Ordinary source mass and a full operator residual

The Sobolev G1 estimate and the polynomial Fourier tail also give an absolute ordinary-norm operator residual for this fixed source. This supplies the Rayleigh and residual prerequisites in the spectral-overlap criterion; it does not supply the overlap itself.

Lemma 20.38 (A nonzero ordinary-norm limit of the repaired proxy). *Put*

$$h_\infty(u) = \frac{4\pi}{\sqrt{3}} u^2 (2\pi u^2 - 3) e^{-\pi u^2}, \quad f_\infty = \mathcal{E}(h_\infty).$$

Extend p_λ by zero outside I_λ . Then

$$\|p_\lambda - f_\infty\|_{L^2(du/u)} = O(\lambda^{-1/2}), \quad Jf_\infty = f_\infty. \quad (219)$$

In particular $0 < \|f_\infty\| < \infty$. For $N = N_\lambda = \lceil \lambda^8(1+L) \rceil$ and $k_\lambda = P_N p_\lambda$, both $\|p_\lambda\|$ and $\|k_\lambda\|$ tend to $\|f_\infty\|$ and are bounded above and away from zero. A valid eventual common lower bound is

$$m_* := \frac{1}{4} \left(\int_1^2 h_\infty(u)^2 du \right)^{1/2} > 0. \quad (220)$$

No numerical threshold in λ is asserted.

Proof. The uniform fixed-mode Hermite approximation in [7, Lemma 7.2] gives suitably normalized $q_{j,\lambda}$ with $\sup_{|u| \leq \lambda} |q_{j,\lambda}(u) - H_j(u)| = O(\lambda^{-2})$, $j = 0, 4$, where H_j has unit $L^2(\mathbb{R})$ norm. This rate survives the manuscript's exact unit normalization: expanding the squared norm, the cross term is $O(\lambda^{-2}) \|H_j\|_1$, the squared-error term is $O(\lambda^{-3})$, and the omitted Hermite tail is Gaussian. Thus $\|q_{j,\lambda}\| = 1 + O(\lambda^{-2})$.

The limits of the coefficients in (132) are $-H_4(0) = -\sqrt{3}/(2 \cdot 2^{1/4})$ and $H_0(0) = 2^{1/4}$, with error $O(\lambda^{-2})$. Substitution of the normalized Hermite polynomials gives exactly the displayed h_∞ , including its scalar factor. The fixed repair has exponentially small amplitude by (135). Consequently

$$\sup_{|u| \leq \lambda} |\tilde{h}_\lambda(u) - h_\infty(u)| \leq C\lambda^{-2}.$$

The limiting source is a Fourier-invariant combination of orders 0, 4, with zero value at zero and zero integral. Poisson summation therefore gives $Jf_\infty = f_\infty$. Its Gaussian decay at infinity, followed by inversion at zero, gives square integrability.

On the physical window there are at most λ/u compact-source summands. Their accumulated approximation error is at most $C\lambda^{-1}u^{-1/2}$. Its squared Haar norm is bounded by

$$C\lambda^{-2} \int_{1/\lambda}^\lambda u^{-2} du \leq C\lambda^{-1}.$$

The omitted Gaussian summands $nu > \lambda$ contribute an exponentially smaller norm: bound their decreasing polynomial-Gaussian tail by its first term plus u^{-1} times its tail integral. The same estimate and inversion control f_∞ outside I_λ . Inversion averaging is an L^2 contraction, proving (219). For $u \geq 1$, all summands $h_\infty(nu)$ are positive. Thus $\|f_\infty\|^2 \geq \int_1^2 h_\infty(u)^2 du > 0$. Finally (302) gives

$$\|(I - P_N)p_\lambda\| \leq C|B_\lambda|\lambda\sqrt{L/N} \rightarrow 0,$$

so the projected vectors have the same nonzero norm limit. Choosing a common sufficiently large threshold proves (220). $\square$

Proposition 20.39 (A full residual after ordinary norm normalization). *Use the same N_λ, k_λ and the canonical closed operator W_λ of Proposition 20.19. For all sufficiently large λ , with $c = 2\pi\lambda^2$,*

$$\|W_\lambda p_\lambda\| + \|W_\lambda k_\lambda\| + \|W_{\lambda, N_\lambda} k_\lambda\| \leq C\lambda^6 e^{-c/3}. \quad (221)$$

For either source normalized to ordinary norm one, its Rayleigh value and its full centered residual tend to zero at this rate. The finite matrix assertion uses that same normalization and the original physical Fourier cutoff.

Proof. All source vectors belong to $\mathcal{D}(\mathbf{W}_\lambda)$ by Proposition 20.19. The weighted Chebyshev estimate $P_\lambda = O(\lambda)$, together with $h_L = O(\lambda)$, gives $\|[M_b, H]\| = O(\lambda)$ in (170). Corollary 20.18, including its separately proved diagonal, gives uniformly

$$|d_n| \leq C\{\lambda + 1 + \log(2 + |n|/L)\}.$$

Let $N = N_\lambda$ and $e = (I - P_N)p_\lambda$. Using (302) and comparing $\sum_{n>N} \log^2(2 + n/L)/n^2$ with its integral yields

$$\begin{aligned} \|\mathbf{W}_\lambda e\| &\leq C|B_\lambda|\lambda\sqrt{L/N}\{\lambda + 1 + \log(2 + N/L)\} \\ &\leq C|B_\lambda|\lambda^{-2}. \end{aligned} \tag{222}$$

Both the diagonal part and the bounded commutator are retained.

Next split the *output* at an auxiliary integer $M > 2N$. This does not change the candidate or its finite matrix. For any $v = P_M v$ with $\|v\| = 1$,

$$\|D_{\log} v\| \leq 2\pi M/L, \quad |v(-a)| + |v(a)| \leq 2\sqrt{(2M+1)/L}.$$

The full Sobolev estimate (156), whose constant is independent of its test function, therefore implies

$$\|P_M \mathbf{W}_\lambda p_\lambda\| \leq C|B_\lambda|\mathfrak{r}(\lambda)\{1 + M/L + \sqrt{M/L}\}. \tag{223}$$

This includes the zero Fourier mode. It uses the already established compact-source radicality and centered pole–prime cancellation, with both source moments and inversion averaging intact. No fixed-band constant is extrapolated to a growing band.

The diagonal has no cross block between E_N and $E_M^\perp$. For the complete off-diagonal matrix, Lemma 20.17 gives the Hilbert–Schmidt bound

$$\|(I - P_M)\mathbf{W}_\lambda P_N\|^2 \leq C\lambda^2 \sum_{|m|\leq N} \sum_{|n|>M} \frac{1}{(n-m)^2} \leq C\lambda^2 N/M.$$

Lemma 20.38 and (222) now give

$$\|(I - P_M)\mathbf{W}_\lambda p_\lambda\| \leq C\lambda^5\sqrt{1+L}M^{-1/2} + C|B_\lambda|\lambda^{-2}. \tag{224}$$

Thus all prime, pole and archimedean pieces remain in the actual matrix; no divergent exterior pole or prime norm sum is estimated separately.

Take $M = \lceil e^{2c/3} \rceil$, which exceeds $2N$ eventually. By (133) and (142), $|B_\lambda| \leq C\lambda^{11/2}e^{-c}$, while $\mathfrak{r}(\lambda)$ is bounded. Equations (223)–(224) are therefore at most $C\lambda^6 e^{-c/3}$ in total. This proves the first term of (221); (222) proves the second, and compression contractivity proves the third. Finally ordinary normalization is legitimate by (220). For a unit vector u , letting $\alpha = \langle \mathbf{W}_\lambda u, u \rangle$, one has $|\alpha| \leq \|\mathbf{W}_\lambda u\|$ and $\|(\mathbf{W}_\lambda - \alpha)u\| \leq 2\|\mathbf{W}_\lambda u\|$. The same argument applies to the compressed matrix. $\square$

The preceding proposition supplies no endpoint-normalized operator residual: dividing its bound by $|B_\lambda|$ leaves an exponentially growing upper bound. Nor does it identify the spectral point near zero as the lowest one. The elementary family $\text{diag}(-1, \epsilon)$, with candidate $(0, 1)$, has Rayleigh value ϵ and centered residual zero while its lowest level stays at -1 . An independent quantitative ground-space overlap or signed complementary lower bound is still necessary. No estimate on the corrected sampler’s graph defect, or on a growing family of prolate modes, is inferred here.

G2 calibration from the screw-function realization

Recent work of Suzuki [12] gives a useful, but more limited, structural calibration of G2. Let A_a denote the lower-bounded self-adjoint operator associated with the semilocal Weil form on $(-a, a)$, and write

$$\lambda_a = \inf \sigma(A_a).$$

Suzuki proves that A_a has discrete spectrum bounded from below and that $a \mapsto \lambda_a$ is continuous. He also proves positivity, simplicity, and evenness of the ground state for *sufficiently small* a by a separate small-support asymptotic argument. Those simplicity/parity conclusions must not be extrapolated to arbitrary growing support; the large- a spectral ordering needed here remains open.

There is nevertheless an elementary global monotonicity which is important for claim discipline. If $a_2 > a_1$, extension by zero identifies the smaller compactly supported form domain with a subspace of the larger one, while the global Weil form is unchanged on such vectors. The min-max principle therefore gives

$$\boxed{\lambda_{a_2} \leq \lambda_{a_1}.} \quad (\text{G2.1})$$

In particular, once λ_a becomes negative it remains bounded above by that same negative value at every larger support. This is compatible with Suzuki's global continuity theorem and his observation that failure of RH is equivalent to the existence of a support with $\lambda_a < 0$.

Weak fixed-band G1 alone does not test the form on the changing proxy itself. The stronger operator estimate in Proposition 20.39 now supplies that missing Rayleigh prerequisite for the actual normalized source and its literal polynomial projection: $q_a(u_a, u_a) \rightarrow 0$. Hence monotonicity gives

$$\lambda_a \rightarrow 0 \iff \lambda_a \geq 0 \text{ for every finite } a.$$

The forward implication uses only monotonicity; the reverse combines positivity with the proved Rayleigh upper bound. The right side is the compact-support Weil positivity criterion. The new source estimate proves the upper bound, not positivity or the asserted limit of the bottom level.

A quantitative spectral-overlap formulation isolates one possible extra ingredient without assuming simplicity. Let $P_a^{(0)}$ be the orthogonal projection onto the (finite-dimensional) lowest eigenspace of A_a , let $p_a \in \mathcal{D}(A_a)$ be a normalized explicit proxy, and put

$$\alpha_a = \langle A_a p_a, p_a \rangle, \quad r_a = (A_a - \alpha_a) p_a, \quad \kappa_a = \|P_a^{(0)} p_a\|.$$

Spectral expansion gives the exact bound

$$\boxed{|\lambda_a - \alpha_a| \kappa_a \leq \|r_a\|.} \quad (\text{G2.3})$$

Proposition 20.39 now proves both $\alpha_a \rightarrow 0$ and $\|r_a\| \rightarrow 0$ for the normalized actual source, with $\lambda = e^a$ and an upper envelope

$$\varepsilon_\lambda = C \lambda^6 e^{-2\pi\lambda^2/3}.$$

Therefore an independently proved overlap bound $\kappa_a \geq \kappa_0 > 0$ would imply $\lambda_a \rightarrow 0$ and close G2. More generally it suffices that $\varepsilon_\lambda/\kappa_a \rightarrow 0$ along an unbounded sequence; for example, $\kappa_a \geq e^{-\theta c}$ with fixed $\theta < 1/3$ and $c = 2\pi\lambda^2$ would suffice. No such lower bound is proved. If instead $\lambda_a \leq -\delta < 0$, spectral projection of $A_a u_a$ gives $\delta\kappa_a \leq \|A_a u_a\|$. Thus a negative ground state can coexist with the new residual estimate by having exponentially small overlap with this source.

At small support Suzuki's positivity-improving analysis gives a simple positive ground state, but it does not furnish a uniform large-support overlap bound. Even positivity of both vectors would

imply only a nonzero overlap, without its required quantitative size. The model $\text{diag}(-1, \epsilon)$ with source $(0, 1)$ shows directly why a vanishing source residual does not determine the bottom sign.

This calibration is deliberately restrictive: continuity and monotonicity of the bottom level are structural, while large-support simplicity/parity and quantitative spectral overlap remain part of the open RH-strength content. A successful continuation must therefore find new large-support structure, not extrapolate the small-support Dirichlet-form theorem.

A Schur-complement formulation of the large-support obstruction

The preceding overlap criterion singles out one vector. A more robust formulation does not require identification of the ground state and should be stated at the level of the closed semilocal Weil form q_a , rather than on the operator domain of A_a . Let P_a be any finite-rank orthogonal projection whose range is contained in the form domain, and decompose

$$L^2(-a, a) = \mathcal{R}_a \oplus \mathcal{R}_a^\perp, \quad \mathcal{R}_a = \text{Ran } P_a.$$

The following elementary estimate isolates exactly the diagonal, cross, and complement quantities that a prolate near-radical construction would have to control.

Proposition 20.40 (Form-Schur lower bound). *Suppose that for all $r \in \mathcal{R}_a$ and all $s \in \mathcal{R}_a^\perp$ in the form domain,*

$$q_a(r, r) \geq -\alpha_a \|r\|^2, \quad |q_a(r, s)| \leq \beta_a \|r\| \|s\|, \quad q_a(s, s) \geq c_a \|s\|^2$$

with $c_a > 0$, $\alpha_a \geq 0$, and $\beta_a \geq 0$. Then

$$\boxed{\lambda_a \geq -\alpha_a - \frac{\beta_a^2}{c_a}.} \tag{G2.4}$$

In particular, $\alpha_a \rightarrow 0$ and $\beta_a^2/c_a \rightarrow 0$ imply $\liminf_{a \rightarrow \infty} \lambda_a \geq 0$.

Proof. For $x = r + s$ the hypotheses give

$$q_a(x, x) \geq -\alpha_a \|r\|^2 - 2\beta_a \|r\| \|s\| + c_a \|s\|^2.$$

Completing the square in $\|s\|$ yields

$$q_a(x, x) \geq -\left(\alpha_a + \frac{\beta_a^2}{c_a}\right) \|r\|^2 \geq -\left(\alpha_a + \frac{\beta_a^2}{c_a}\right) \|x\|^2.$$

Taking the infimum of the Rayleigh quotient proves the claim. $\square$

A complement bound without a positive spectral gap

The ordinary operator residual permits a different reduction from the coercive Schur estimate. It does not require a positive gap on the complement and therefore introduces no division by a collapsing level.

Proposition 20.41 (A gap-free near-radical block bound). *Let A be a lower-bounded self-adjoint operator with closed form q , and let P be a finite-rank orthogonal projection with $\text{Ran } P \subset \mathcal{D}(A)$. Put $Q = I - P$ and assume $\|AP\| \leq \epsilon$. Then Q preserves the form domain and*

$$|q(f, f) - q(Qf, Qf)| \leq \frac{2}{\sqrt{3}} \epsilon \|f\|^2 \quad (f \in \mathcal{D}(q)). \tag{225}$$

Consequently, if

$$q(g, g) \geq -\eta \|g\|^2 \quad (g \in \text{Ran } Q \cap \mathcal{D}(q)), \quad \eta \geq 0,$$

then

$$\inf \sigma(A) \geq -\eta - \frac{2}{\sqrt{3}}\epsilon. \quad (226)$$

The constant $2/\sqrt{3}$ in both statements is sharp.

Proof. Write $h = Pf$ and $g = Qf$. The vector h lies in the operator domain, so g belongs to the form domain and, with the fixed first-slot-linear convention,

$$q(f, f) - q(g, g) = \langle Ah, h \rangle + 2\Re \langle Ah, g \rangle = \Re \langle Ah, h + 2g \rangle.$$

Since $h \perp g$, the absolute value is at most

$$\epsilon \|h\| \sqrt{\|h\|^2 + 4\|g\|^2} \leq \frac{2}{\sqrt{3}}\epsilon(\|h\|^2 + \|g\|^2).$$

For the last inequality, the maximum of $t\sqrt{4-3t^2}$ on $0 \leq t \leq 1$ is $2/\sqrt{3}$. This proves (225); the complementary lower bound and $\|Qf\| \leq \|f\|$ prove (226). Sharpness already holds on $\mathbb{C}^2$: take P onto the first coordinate and

$$A = -\frac{\epsilon}{\sqrt{3}} \begin{pmatrix} 1 & \sqrt{2} \\ \sqrt{2} & 0 \end{pmatrix}.$$

Then $\|AP\| = \epsilon$, the complementary form is zero, and the lowest eigenvalue is $-2\epsilon/\sqrt{3}$. $\square$

For the rank-one projection onto either normalized source of Proposition 20.39, $\|W_\lambda P\| = \|W_\lambda u_\lambda\|$ already tends to zero with the proved ordinary residual bound. It therefore suffices to prove asymptotic nonnegativity $q_\lambda|_{u_\lambda^\perp} \succeq -\eta_\lambda I$ with $\eta_\lambda \rightarrow 0$ along an unbounded window sequence. Strict complementary coercivity is unnecessary for this version of the reduction. For a growing source block the required uniform $\|W_\lambda P_\lambda\| \rightarrow 0$ remains a separate open input. Neither complementary sign estimate is proved here.

The missing sign cannot be concealed by removing the near-radical block. Indeed, suppose a fixed unit compact test f has $q(f, f) = -\delta < 0$, and extend it by zero to every larger window. Then (225) gives

$$q_\lambda((I - P_\lambda)f, (I - P_\lambda)f) \leq -\delta + \frac{2}{\sqrt{3}}\epsilon_\lambda.$$

Whenever $\epsilon_\lambda < \sqrt{3}\delta/2$, the projected vector is nonzero and negative; its normalized Rayleigh value is bounded above by the same negative quantity, since its norm is at most one. Thus a negative direction persists in the complement of any uniformly near-radical block, regardless of its rank. By the existing cofinal Weil-positivity criterion, the desired asymptotic complementary sign for the proved rank-one source remains RH-strength. The reduction removes a gap denominator; it does not supply the arithmetic sign.

Strong cancellation persists in the exact source complement

The proved ordinary source residual does not mean that significant prime/non-prime cancellation is confined to one direction. The following fixed-window certificate tests that possible shortcut on the literal source, rather than on sampled basis or random vectors.

Proposition 20.42 (A certified cancelling direction off the source). *At $\lambda = 3$ there exists an inversion-even unit Fourier polynomial $v \in E_{64}$ exactly orthogonal to the repaired source p_3 such that*

$$\begin{aligned} \langle v, p_3 \rangle &= 0, & 0 < QW_3(v, v) < 3 \cdot 10^{-31}, \\ q_{\text{np}}(v) &> 0.4, & q_{\text{pr}}(v) < -0.4. \end{aligned} \tag{227}$$

Here $q_{\text{np}} = q_{\text{ar}} + q_{\text{po}}$ includes both the pure archimedean term and the pole term, and q_{pr} is the signed prime contribution, so $QW_3 = q_{\text{np}} + q_{\text{pr}}$.

Proof. Let V have the two real dyadic columns supplied in Appendix A, in the normalized even basis of E_{64} . Let c be the exact rational source candidate from Proposition 20.78, converted to the same parity coordinates. Set $a = V^*c$ and $x = (-a_2, a_1)^t$. The exact algebra gives $\langle Vx, c \rangle = 0$. The certified Gram matrix V^*V is positive definite and $\|Vx\| > 0$, so put $w = Vx / \|Vx\|$. All vectors are real here; the stated Hermitian convention is unchanged.

Outward interval evaluation of the separate finite matrices gives

$$\begin{aligned} 2.5844134 \cdot 10^{-31} &< QW_3(w, w) < 2.5844135 \cdot 10^{-31}, \\ q_{\text{np}}(w) &> 0.4173237590, \\ q_{\text{pr}}(w) &< -0.4173237590. \end{aligned}$$

These are inequalities for exact dyadic/rational constructions; approximate eigenvectors were used only to propose the dyadic columns.

Write $u = P_{64}p_3 / \|P_{64}p_3\|$ and $\hat{c} = c / \|c\|$. The existing exact-source certificate, matched to c by its file hash, gives $\|P_u - P_{\hat{c}}\| < \delta = 4 \cdot 10^{-36}$. Since $w \perp c$, define

$$v = \frac{(I - P_u)w}{\|(I - P_u)w\|}.$$

Then $\|v - w\| < 2\delta$. Also $v \in E_{64}$, hence $v \perp u$ implies $v \perp p_3$, not just orthogonality to an approximate source. For any finite Hermitian matrix H ,

$$|\langle Hv, v \rangle - \langle Hw, w \rangle| \leq 2 \|H\| \|v - w\| < 4\delta \|H\|.$$

The certified all-row bounds for the compressed matrices are

$$\|W_{64}\| < 8.926, \quad \|H_{\text{np},64}\| < 4.176, \quad \|H_{\text{pr},64}\| < 5.490.$$

Applying these bounds to the full interval enclosures, before rounding, gives

$$2.5829853 \cdot 10^{-31} < QW_3(v, v) < 2.5858415 \cdot 10^{-31}$$

and preserves both strict 0.4 bounds. A finite-support polynomial's complete quadratic form equals its finite matrix quadratic form. No invariant-subspace or full operator-residual assertion is required. $\square$

Thus $QW_3(v, v)/q_{\text{np}}(v) < 7.5 \cdot 10^{-31}$ inside the exact source complement. This rules out an explanation in which substantial archimedean-plus-pole dominance eliminates the need for cancellation everywhere off that source. It does not rule out positivity on the complement, and exhibits neither a negative Weil direction nor an additional exact null vector. Positive basis-vector or typical random-vector energies cannot control these coherent linear combinations. The pole term must be included explicitly: calling q_{np} the pure archimedean contribution would change the asserted comparison.

Why a finite source block cannot reduce the unsigned prime norm

The arithmetic bound in the preceding tail estimates is not merely a poor choice of positive weight. There is an exact recurrence obstruction to replacing it by a small norm on the source complement.

Proposition 20.43 (Prime norm survives every finite-dimensional removal). *Fix $\lambda > 1$ and let T_{pr} be the positive-coefficient sum of truncated prime translations in (181). For every finite-rank orthogonal projection P on $L^2(0, L)$, $Q = I - P$, one has*

$$\|Q\mathsf{T}_{\text{pr}}Q\| = \|\mathsf{T}_{\text{pr}}\|. \quad (228)$$

For every compact operator C on this space,

$$\|Q(\mathsf{T}_{\text{pr}} - C)Q\| \geq \|\mathsf{T}_{\text{pr}}\|. \quad (229)$$

In particular the essential norm of T_{pr} equals its ordinary norm. The projection need not be a Fourier projection and may remove any prescribed finite source block.

Proof. For each positive integer k , let $M_k f(x) = e^{2\pi i k x/L} f(x)$. Simultaneous recurrence of the finitely many phases gives integers $k_j \rightarrow \infty$ such that

$$e^{2\pi i k_j \log m/L} \rightarrow 1 \quad \text{for every active } m.$$

For completeness, partition the unit cube in the finitely many phase coordinates into H^d cubes and compare $H^d + 1$ successive multiples. Two lie in one cube, giving a positive return with coordinate errors at most $1/H$. If such returns have an unbounded subsequence, use it; otherwise a fixed positive return has all coordinate errors zero, and its increasing multiples give the required sequence. No rational independence of prime logarithms is assumed.

Conjugating a truncated translation by M_k only multiplies it by its corresponding phase. Since each translation has norm at most one,

$$\|M_{k_j}^* \mathsf{T}_{\text{pr}} M_{k_j} - \mathsf{T}_{\text{pr}}\| \leq 2 \sum_m w_m |e^{2\pi i k_j \log m/L} - 1| \rightarrow 0. \quad (230)$$

For every fixed $f \in L^2(0, L)$, $M_{k_j} f \rightarrow 0$: each scalar product is a Fourier coefficient of an L^1 function. Thus $PM_{k_j} f \rightarrow 0$, $PM_{k_j} \mathsf{T}_{\text{pr}} f \rightarrow 0$, and $CM_{k_j} f \rightarrow 0$ strongly. Combining these statements with (230) gives

$$Q(\mathsf{T}_{\text{pr}} - C)QM_{k_j} f - M_{k_j} \mathsf{T}_{\text{pr}} f \rightarrow 0.$$

For unit f , the input has norm at most one, so taking norms and then the supremum over f proves (229). Putting $C = 0$ and using compression contractivity proves (228). Taking the infimum over compact C , with $P = 0$, proves the essential-norm assertion. $\square$

Proposition 20.44 (The sharp growing-window scale of the unsigned prime norm). *With the actual weights $w_m = \Lambda(m)/\sqrt{m}$ and $L = 2 \log \lambda$,*

$$\|\mathsf{T}_{\text{pr}}\| = (1 + o(1))\lambda \quad (\lambda \rightarrow \infty). \quad (231)$$

Consequently (231) also holds for $\|Q_\lambda \mathsf{T}_{\text{pr}} Q_\lambda\|$ for any family of finite-rank orthogonal projections $I - Q_\lambda$, with no restriction on their finite ranks or Fourier cutoffs.

Proof. Write

$$a(x) = e^{-x/2}, \quad b(x) = e^{-(L-x)/2}, \quad \phi(x) = a(x) + b(x),$$

and

$$\Psi(X) = \sum_{1 < m < X} \Lambda(m), \quad S(X) = \sum_{1 < m < X} \frac{\Lambda(m)}{m}.$$

The strict cutoff agrees with the truncated translations almost everywhere. Direct evaluation gives

$$\frac{(\mathsf{T}_{\text{pr}}\phi)(x)}{\phi(x)} = \frac{a(x)\{S(e^{L-x}) + \Psi(e^x)\} + b(x)\{\Psi(e^{L-x}) + S(e^x)\}}{a(x) + b(x)}. \quad (232)$$

The exact identity

$$a(x)e^x + b(x)e^{L-x} = \lambda\{a(x) + b(x)\}$$

identifies the leading constant. The classical prime number theorem [24], equivalently $\Psi(X) \sim X$, implies that for each $\delta > 0$ there is $C_\delta < \infty$ with

$$|\Psi(X) - X| \leq \delta X + C_\delta \quad (X \geq 1).$$

The Chebyshev bound and partial summation give $0 \leq S(X) \leq C(1 + \log X)$, with an absolute C . Therefore, uniformly for $0 < x < L$,

$$(1 - \delta)\lambda - C_\delta \leq \frac{\mathsf{T}_{\text{pr}}\phi}{\phi} \leq (1 + \delta)\lambda + C_\delta + C(1 + L).$$

At each fixed window ϕ is bounded and bounded away from zero. The upper estimate and the weighted Schur inequality (181) bound the operator norm from above. Integrating the lower estimate against ϕ^2 gives the same lower bound for its Rayleigh quotient and hence for the norm. Divide by λ , let $\lambda \rightarrow \infty$, and then let $\delta \downarrow 0$. This proves (231). The projection assertion follows from Proposition 20.43 at each window. $\square$

These statements concern the unsigned prime operator, not the sign of the full Weil form. They make precise a limitation of the particular place-by-place bound (176). Even replacing its prime constant by the exact tail norm cannot certify positivity unless

$$\log \frac{N+1}{L} > \|\mathsf{T}_{\text{pr}}\| = (1 + o(1))\lambda. \quad (233)$$

Here we refer to the whole-space bound, whose remaining subtracted terms are nonnegative. Thus this method still requires an exponential Fourier cutoff in λ , rather than the polynomial source cutoff. This is a limitation of that sufficient scalar bound, not a necessary cutoff for actual Weil positivity or for a frequency-weighted argument.

The physical pole operator has rank at most two. Equation (229) shows that combining it with the prime operator does not make their ordinary operator norm smaller than $\|\mathsf{T}_{\text{pr}}\|$, even after a finite source removal. The recurrence frequencies in the proof may be arbitrarily large; there the archimedean logarithmic energy also grows. No negative direction for the complete Weil operator is inferred. A scalable G2 estimate must control the arithmetic term together with that energy or retain more precise signed correlations, rather than assert a small unweighted norm on the complement. The required signed estimate remains open.

Retaining the archimedean energy in a one-sided estimate

Choose a literal Fourier tail on which $a_n := -A_n > 0$ and write $D = \text{diag}(a_n)$ there. By (168), the compressed closed form has $W = D + R$, with R bounded self-adjoint. It contains the actual signed prime and pole terms and the archimedean off-diagonal; no arithmetic diagonal is absorbed into D . Put

$$\mathcal{B} = D^{-1/2} R D^{-1/2}. \quad (234)$$

The operator $\mathcal{B}$ is compact self-adjoint, since $a_n \sim \log |n|$ at the fixed window. The form identity is

$$q_W[f] = \langle (I + \mathcal{B}) D^{1/2} f, D^{1/2} f \rangle.$$

Thus the exact positivity target is $\mathcal{B} \succeq -I$; the stronger two-sided condition $\|\mathcal{B}\| \leq 1$ can fail because of a harmless positive direction.

Proposition 20.45 (The scale of an approximate weighted sign). *Suppose $D \succeq d_0 I > 0$, $R \succeq -CI$, $C \geq 0$, and $D^{-1/2} R D^{-1/2} \succeq -(1 + \eta)I$, $\eta \geq 0$. Then, on the entire form domain,*

$$D + R \succeq -\frac{\eta C}{1 + \eta} I. \quad (235)$$

The constant is sharp. In particular, a dimensionless error $\eta_\lambda \rightarrow 0$ is not by itself an ordinary lower error $o(1)$ when C_λ grows.

Proof. The hypotheses give $D + R \succeq -\eta D$ and $D + R \succeq D - CI$. Add these inequalities with weights $1/(1 + \eta)$ and $\eta/(1 + \eta)$ respectively. For $C > 0$, the scalar example $D = C/(1 + \eta)$, $R = -C$ attains equality. Taking $C_\lambda = \lambda$ and $\eta_\lambda = 1/\lambda$ shows why the missing scale factor cannot be omitted. A sufficient conversion requires $\eta_\lambda C_\lambda \rightarrow 0$, or a more precise directional energy estimate. $\square$

There is also a limitation on how the compact weighted remainder can be estimated. A Hilbert–Schmidt budget for its whole infinite matrix is unavailable, despite the valid finite-column residual Grams above.

Proposition 20.46 (Weighted compactness without a square-summable matrix). *At any fixed $\lambda > \sqrt{2}$, let D be a positive diagonal on a fixed Fourier tail with $a_n \sim \log |n|$. The compact operator $D^{-1/2} \mathsf{T}_{\text{pr}} D^{-1/2}$ belongs to no Schatten class $\mathcal{S}_p$, $0 < p < \infty$. For the exact archimedean diagonal $a_n = -A_n$, the same assertion holds for $\mathcal{B}$ in (234).*

Proof. The prime diagonal is the nonzero finite trigonometric sequence

$$\tau_n = 2 \sum_{1 < m < \lambda^2} \frac{\Lambda(m)}{\sqrt{m}} \left(1 - \frac{\log m}{L} \right) \cos \frac{2\pi n \log m}{L}.$$

Collect equal unit-circle frequencies to write $\tau_n = \sum_{z \in F} c_z z^n$, with all $c_z > 0$. Finite geometric sums give

$$\frac{1}{X} \sum_{n=1}^X |\tau_n|^2 \longrightarrow \sum_{z \in F} c_z^2 > 0.$$

Boundedness then implies that $|\tau_n|$ exceeds a positive constant on a set of positive lower density. Consequently, for each $p \geq 1$, $\sum_n |\tau_n/a_n|^p = \infty$: the partial sum on that set is bounded below by a constant times $X/(\log X)^p$. For self-adjoint $K \in \mathcal{S}_p$, Jensen’s inequality in its spectral measure gives $\sum_n |\langle K U_n, U_n \rangle|^p \leq \text{Tr} |K|^p$. This is a contradiction. The inclusions $\mathcal{S}_p \subset \mathcal{S}_1$ for $0 < p < 1$

settle the remaining exponents. For the complete remainder its diagonal is $(-\tau_n + O_\lambda(n^{-2}))/a_n$, since the exact pole diagonal has this decay and the archimedean remainder has zero diagonal. The same argument applies. None of this invalidates the separated finite-column moment bounds, whose off-diagonal factor $(n - m)^{-1}$ is square summable in the remote row index. $\square$

Proposition 20.47 (Sharp fixed-window decay of the weighted prime tail). *Under the diagonal assumptions above, let Q_J remove $|n| \leq J$. At each fixed window,*

$$\lim_{J \rightarrow \infty} (\log J) \|Q_J D^{-1/2} \mathbf{T}_{\text{pr}} D^{-1/2} Q_J\| = \|\mathbf{T}_{\text{pr}}\|. \quad (236)$$

Subtracting any fixed compact operator from $\mathbf{T}_{\text{pr}}$ before weighting leaves this limit unchanged. No uniformity in λ is asserted.

Proof. The upper bound is $\|\mathbf{T}_{\text{pr}}\| / \inf_{|n| > J} a_n$. For the lower bound, fix $\epsilon > 0$ and choose a unit Fourier polynomial f , supported on $|m| \leq M$, with $|\langle \mathbf{T}_{\text{pr}} f, f \rangle| > \|\mathbf{T}_{\text{pr}}\| - \epsilon$. Returns of the finite vector of shift phases to any fixed neighborhood of the identity are relatively dense. Indeed its cyclic closure is a compact group; finitely many integer translates of that neighborhood cover it. The largest difference between the corresponding translation indices bounds the gap between return times. This also covers rational phase relations.

Choose the neighborhood so that the conjugation error in (230) is below ϵ . For every large J there is therefore a return $k \in [J + M + 1, J + M + 1 + K]$, with K independent of J . Set $g = M_k f$ and $x = D^{1/2} g$. Both are supported outside J , $\|x\|^2 = (1 + o(1)) \log J$, and the weighted Rayleigh numerator at x equals $\langle \mathbf{T}_{\text{pr}} g, g \rangle$, whose absolute value is greater than $\|\mathbf{T}_{\text{pr}}\| - 2\epsilon$. Let $J \rightarrow \infty$ and then $\epsilon \downarrow 0$. For compact C the change in the left side is at most

$$\frac{\log J}{\inf_{|n| > J} a_n} \|Q_J C Q_J\| \rightarrow 0.$$

The return-length bound K depends on the window and the tolerance. Thus one may not combine (236) with $\|\mathbf{T}_{\text{pr}}\| \sim \lambda$ to infer a bound on an arbitrary joint scale $J = J(\lambda)$. $\square$

Proposition 20.48 (A signed energy bound at the larger window $\lambda = 4$). *Let $L = 2 \log 4$, retain the exact diagonal $a_n = -A_n$, and let Q_{16} be the literal projection onto $|n| > 16$. For every complex vector in its complete closed form domain,*

$$QW_4(f, f) \geq 10^{-8} \sum_{|n| > 16} a_n |f_n|^2, \quad f = Q_{16} f. \quad (237)$$

Every weight in this sum is positive. This is an infinite-tail computer-assisted bound, not a claim about the remaining low modes or about an unbounded family of windows.

Proof. Apply the directional verified-solve proof to $W_4 - cD$, $c = 10^{-8}$, on Q_{16} . Only the archimedean diagonal is changed, to $(1 - c)a_n$; the exact off-diagonal coefficients b_n , both pole signs, and every prime power 2, 3, 4, 5, 7, 8, 9, 11, 13 remain. The full-length $m = 16$ translation is zero. The positive physical weight $\phi(x) = e^{-x/2} + e^{-(L-x)/2}$ gives the certified Schur bound

$$\|\mathbf{T}_{\text{pr}}\| \leq M_\phi < 4.624042316766478. \quad (238)$$

To verify it, set $u = e^x$. On each interval between 1, 16, m , $16/m$ the ratio $(\mathbf{T}_{\text{pr}} \phi) / \phi$ is a linear-fractional function of u , hence monotone or constant. Every one-sided endpoint value is evaluated with outward rounding; the chosen upper endpoint is checked against every candidate.

Use E_L, C_L, h_L from (174), with $C_L = 1$, $h_L = 9/16$ and $R_L = 64/255$. For $n > N$ the proved remote inverse bound applies with

$$g_n = (1 - c)\{\log(n/L) - E_L(2\pi n/L)\} - \kappa,$$

where $\kappa = M_\phi + 2C_L/t_*$ in the even sector and $\kappa = M_\phi + 2C_L/t_* + \pi/2 + 4h_LL/(\pi^2 N)$ in the odd sector, $t_* = 2\pi(N+1)/L$. These weights are positive and increasing for the choices below. Also the same diagonal estimate at $n = 17$ proves positivity of a_n on the whole claimed tail.

The finite head now consists of $17 \leq n \leq N$ in the normalized parity basis; it does not include the omitted physical low modes. The solve witness has support $N < n \leq M$. With $G = I - Z$, form the exact signed K_Z and every residual row up to J , and use the moment majorant U_J beyond J . The lower matrix is

$$\mathcal{L} = K_Z - \sum_{N < n \leq J} \frac{R_n^* R_n}{g_n} - \frac{U_J}{g_{J+1}}. \quad (239)$$

The same full-index parity conversion and both infinite tails as in Proposition 20.24 are used. Bounding its geometric-remainder constant by including the absent indices $1, \dots, 16$ is conservative. The certificate parameters are

sector	head size	N	M	J	moment order
even	496	512	1024	4096	48
odd	1520	1536	2048	4096	64

Both Z and an inverse-Cholesky congruence witness C are frozen to exact dyadic entries. Their numerical method of construction has no proof role: outward interval evaluation verifies every row of $C\mathcal{L}C^*$ has positive diagonal minus the sum of the absolute off-diagonal entries. Thus $C\mathcal{L}C^* \succ 0$ and $\mathcal{L} \succ 0$. Square completion with the positive remote operator proves (237) separately in each parity sector and hence for arbitrary complex vectors.

The saved witnesses, analytic series radii, all finite residual rows and the infinite moment bound are reproduced by `assembly_general.py` and `certify_weighted_tail.py`. Replaying the same witnesses at higher precision verifies the same strict inequalities. These are internal computer-assisted certificates, not independent external verification. No zero locations or RH assumptions enter them. $\square$

Proposition 20.49 (The two-sided weighted contraction fails at this cutoff). *For the same exact D and Q_{16} at $\lambda = 4$,*

$$\|D^{-1/2}Q_{16}(W_4 - D)Q_{16}D^{-1/2}\| > 1.$$

This does not contradict Proposition 20.48.

Proof. The archived exact dyadic even vector v , supported on $17 \leq |n| \leq 256$, has the rigorously enclosed quotient

$$2.05254409540345 < \frac{QW_4(v, v)}{\langle Dv, v \rangle} < 2.05254409540346.$$

Its positive numerator excess over $2\langle Dv, v \rangle$ exceeds 0.0525 in the saved normalization. The same analytic matrix assembly, independent of the floating eigenvector that suggested v , proves these interval inequalities. Testing the weighted remainder on $D^{1/2}v$ gives a Rayleigh value greater than one. This is a positive direction, not a negative Weil direction. $\square$

The larger-window tail sign reduces full positivity at $\lambda = 4$ to the signed effective matrix on E_{16} :

$$F - B^*T^{-1}B, \quad F = P_{16}W_4P_{16}, \quad B = Q_{16}W_4P_{16}, \quad T = Q_{16}W_4Q_{16}.$$

It has 17 even and 16 odd coordinates. Proposition 20.66 below settles the complete even sector; Proposition 20.70 completes the odd sector. It is not the raw finite Weil matrix: the inverse-tail correction must remain. The next growing-window estimate must also satisfy the ordinary-error scale in Proposition 20.45. A related recent preprint [25] studies certified compact-window positivity and a pointwise prime-comb envelope barrier. Our conclusion here concerns the actual Fourier-tail form and does not use that preprint's computational certificates. Neither fixed-window result supplies a uniform signed arithmetic estimate.

Reusing the signed tail certificate in the low-mode correction

The scalar conclusion $T \succeq 10^{-8}D$ discards most of the information in Proposition 20.48. The following inverse bound retains its finite signed factorization. All adjoints below use the fixed first-slot-linear convention.

Proposition 20.50 (A structured inverse bound for a certified tail). *Let a closed positive tail operator have a finite-head decomposition*

$$T = \begin{pmatrix} F & B^* \\ B & U \end{pmatrix}, \quad U \succeq D_g \succeq \gamma I > 0,$$

where B is bounded and D_g is a positive diagonal operator. Let Z map the finite head into $\mathcal{D}(U)$, and set

$$R = B - UZ, \quad K_Z = F - B^*Z - Z^*B + Z^*UZ.$$

Suppose a finite Hermitian matrix $\underline{L}$ and invertible C satisfy

$$0 \prec \underline{L} \preceq K_Z - R^*D_g^{-1}R, \quad C\underline{L}C^* \succeq \mu I, \quad \mu > 0. \quad (240)$$

Then T is bounded below by a strictly positive constant and, for any tail right-hand side $r = (h, k)$,

$$\langle T^{-1}r, r \rangle \leq \|D_g^{-1/2}k\|^2 + \mu^{-1} \|C(h - Z^*k - R^*D_g^{-1}k)\|^2. \quad (241)$$

The same statement holds as a Gram inequality for finitely many right-hand sides simultaneously.

Proof. Write the original variable as $(x, y - Zx)$. Its exact form is

$$\langle K_Z x, x \rangle + 2\Re \langle Rx, y \rangle + q_U[y],$$

which is bounded below by

$$\langle \underline{L}x, x \rangle + \|D_g^{1/2}(y + D_g^{-1}Rx)\|^2.$$

The first change preserves the original form domain because $Zx \in \mathcal{D}(U)$. The displayed lower form is defined on the possibly larger domain consisting of the finite head and $\mathcal{D}(D_g^{1/2})$. Its triangular change is bounded and invertible there because $D_g^{-1}Rx \in \mathcal{D}(D_g)$ and the head is finite dimensional. Thus that lower form is closed and strictly positive; the original form dominates it on its own domain. In its inverse variational formula the head part of the linear functional is $h - Z^*k - R^*D_g^{-1}k$. Consequently its inverse quadratic form equals the first term of (241) plus the quadratic form of $\underline{L}^{-1}$ on that head vector. Inverse order and (240) give $\underline{L}^{-1} \preceq \mu^{-1}C^*C$. This proves the bound. Applying it to every linear combination of the right-hand sides proves the Gram version. The argument concerns closed forms and does not replace the logarithmic operator by a bounded matrix. $\square$

If the saved factors instead certify $T - cD$, with $cD \succeq 0$, they still bound T^{-1} because $T^{-1} \preceq (T - cD)^{-1}$. The factors and residual in the right side must then be those of $T - cD$ throughout; a shifted residual is not an unshifted residual. In particular, the existing $\lambda = 4$ certificates are usable without losing their directional information to the coarse upper bound $10^8 D^{-1}$.

Here is a finite-row version that also retains every omitted mode. When Z has finite support, take J past that support and split the inner far-tail rows at J . For a matrix of right-hand sides define

$$H_J = h - Z^*k - \sum_{n \leq J} R_n^* k_n / g_n, \quad V_J = \sum_{n \leq J} k_n^* k_n / g_n.$$

The sums run over the inner far tail in normalized parity coordinates; both Fourier signs are included by that decomposition. Suppose certified positive matrices satisfy

$$\sum_{n > J} R_n^* R_n / g_n \preceq U_R, \quad \sum_{n > J} k_n^* k_n / g_n \preceq U_k, \quad \rho \geq \|CU_R C^*\|.$$

Then, for every $\tau > 0$,

$$r^* T^{-1} r \preceq V_J + U_k + \frac{(1 + \tau) H_J^* C^* C H_J + (1 + \tau^{-1}) \rho U_k}{\mu}. \quad (242)$$

Indeed, if $E = \sum_{n > J} R_n^* k_n / g_n$, then $E^* C^* C E \preceq \rho U_k$ by the operator Cauchy–Schwarz inequality. Apply the matrix Young inequality to $C(H_J - E)$. The trace of $CU_R C^*$ is a permissible choice of ρ ; its possible growth with the inner head dimension must then be retained. These are finite-column remote Grams, as in Proposition 20.24, not an invalid Hilbert–Schmidt estimate for the whole weighted Weil operator. A joint remote Gram can be sharper than separating this mixed term.

Proposition 20.51 (A convergent arithmetic refinement of the far inverse). *If $D_g \preceq U \preceq mD_g$, $D_g \succeq \gamma I > 0$, $m > 1$, let $A = D_g^{-1/2} U D_g^{-1/2}$ denote the bounded operator represented by the preconditioned form, and set*

$$S = I - A/m, \quad P_k(A) = \frac{1}{m} \sum_{j=0}^{k-1} S^j + S^k, \quad k \geq 0.$$

The empty sum gives $P_0 = I$. Then

$$A^{-1} \preceq P_{k+1}(A) \preceq P_k(A) \preceq I, \quad 0 \preceq P_k(A) - A^{-1} \preceq (1 - 1/m)^k I. \quad (243)$$

Thus $D_g^{-1/2} P_k(A) D_g^{-1/2}$ are decreasing upper bounds for U^{-1} , with error at most $(1 - 1/m)^k D_g^{-1}$ in form order. In particular,

$$U^{-1} \preceq D_g^{-1} - m^{-1} D_g^{-1} (U - D_g) D_g^{-1}. \quad (244)$$

Products involving U in this formula are understood through the bounded preconditioned form.

For the actual far operators used in Proposition 20.48, including their shift $W_4 - 10^{-8} D$, one can take

$$m = 67 \quad \text{on the even tail } n > 512, \quad m = 335 \quad \text{on the odd tail } n > 1536.$$

These bounds cover all omitted Fourier modes at this fixed window.

Proof. The spectral theorem gives $I \preceq A \preceq mI$ and $0 \preceq S \preceq (1 - 1/m)I$. The geometric identity is

$$P_k(A) - A^{-1} = S^k(I - A^{-1}).$$

Its factors commute and are positive. This proves the error bound; $P_k - P_{k+1} = m^{-1}S^k(A - I) \succeq 0$ proves monotonicity. Conjugate by $D_g^{-1/2}$, using the inverse identity for the positive closed form U . The case $k = 1$ is (244).

To verify the two concrete constants, first strengthen (180) to

$$|a_n - \log(|n|/L)| \leq E_L(|t_n|), \quad n \neq 0. \quad (245)$$

For the upper bound use the same Binet formula at $w = 5/4 + it/2$. It gives

$$\Re \log w - \log(t/2) = \frac{1}{2} \log(1 + 25/(4t^2)) \leq 25/(8t^2),$$

while $-\Re(1/(2w)) - \Re(1/z) \leq 0$, $z = 1/4 + it/2$. The Binet remainder is at most $1/(15t)$ in absolute value. Because $25/8 < 7/2$, the existing absolute trigamma and exponential series bounds give the same E_L for the upper error.

Write $W_4 = D + R_0$. From (169)–(170), a valid bound is

$$\|R_0\| \leq C_R := 2\pi B_* + 32h_L/L + 2P_4.$$

Here P_4 may omit the zero full-length translation $m = 16$. The pole diagonal bound is $32h_L/L$; its factor L is essential. The commutator bound is taken before the parity compression, so it also bounds either normalized parity sector. With $c = 10^{-8}$, $g_n = (1 - c)\{\log(n/L) - E_L(t_n)\} - \kappa$, and $t_* = 2\pi(N + 1)/L$, the already proved lower bound and (245) imply

$$D_g \preceq U \preceq D_g + \{\kappa + 2(1 - c)E_L(t_*) + C_R\}I.$$

Since g_n is positive and increasing, it suffices that

$$m \geq 1 + \frac{\kappa + 2(1 - c)E_L(t_*) + C_R}{g_{N+1}}. \quad (246)$$

Outward interval evaluation gives $C_R < 34.450462$. The right side of (246) is below 66.76640 for the even $N = 512$ choice, and below 334.71708 for the odd $N = 1536$ choice. The same prime bound, diagonal errors and inverse weights as the tail certificate are used. Strict interval checks at two precisions verify the stated integers. $\square$

For a matrix of residuals Y , the error of the k th inverse bound is at most

$$(1 - 1/m)^k Y^* D_g^{-1} Y.$$

Along growing windows this part of the error is therefore controlled if $\|D_{g,\lambda}^{-1/2} Y_\lambda\|^2 e^{-k\lambda/m_\lambda} \rightarrow 0$, with the actual values of m_λ retained. The explicit integers above apply only at $\lambda = 4$. Proposition 20.52 below improves these constants to 20 and 90 using the actual parity signs. Evaluating the signed powers and their complete remote contributions, and proving the remaining low-head sign, remain separate obligations. The inverse iteration proves neither of those signs.

Parity signs and certified evaluation of the inverse refinement

The global remainder norm in the preceding inverse estimate loses useful signs. Retaining them improves the complete far-operator sandwich; it does not determine the sign of the low Schur matrix.

Proposition 20.52 (Sharper inverse constants at the fixed window). *For the actual shifted far operators and the unchanged weights D_g in Proposition 20.48, one has*

$$D_g \preceq U \preceq 20D_g \quad \text{in the even sector } n > 512, \quad D_g \preceq U \preceq 90D_g \quad \text{in the odd sector } n > 1536.$$

Thus the constants in Proposition 20.51 can be replaced by 20 and 90, respectively, with error factors $(19/20)^k$ and $(89/90)^k$. These are complete-operator bounds at $\lambda = 4$, with no assertion of growing-window uniformity.

Proof. Put $h = 9/16$, $c = 10^{-8}$, $L = 2 \log 4$ and $t_N = 2\pi(N+1)/L$. Let M_ϕ be the certified prime bound in (238). The limiting archimedean off-diagonal in the normalized parity basis is the matrix $1/[2(n+m)]$ in the even sector and its negative in the odd sector. It is positive and has norm at most $\pi/2$; the remaining archimedean commutator has norm at most $2/t_N$, by the proof of Proposition 20.21.

In centered coordinates the pole form is $2|\langle f, \cosh(x/2) \rangle|^2 - 2|\langle f, \sinh(x/2) \rangle|^2$. Only the positive term acts on even vectors, and only the negative term on odd vectors. Direct integration gives

$$|\widehat{\cosh(x/2)}(n)|^2 = \frac{h}{L(t_n^2 + 1/4)^2}, \quad 2\|Q_N \cosh(x/2)\|^2 \leq \frac{hL^3}{4\pi^4} \sum_{n>N} n^{-4} \leq \frac{hL^3}{12\pi^4 N^3}.$$

The factor two for the form and both Fourier signs are included. Consequently the remainder above the shifted archimedean diagonal has the following upper bounds:

$$\begin{aligned} C_e(N) &= M_\phi + 2/t_N + \pi/2 + hL^3/(12\pi^4 N^3), \\ C_o(N) &= M_\phi + 2/t_N. \end{aligned} \tag{247}$$

The odd limiting Hilbert matrix and odd pole term are nonpositive and may be omitted in an upper bound. The shift cD changes only the exact archimedean diagonal; none of these remainder terms is multiplied by $1 - c$.

For the same κ and increasing g_n as before, the two-sided diagonal estimate (245) now yields

$$U \preceq D_g + \{\kappa + 2(1 - c)E_L(t_N) + C_{e/o}(N)\}I.$$

It suffices to bound one plus the braced quantity divided by g_{N+1} . Outward interval evaluation gives values below 19.215607 for even $N = 512$, and below 89.840960 for odd $N = 1536$. The integer gates 20 and 90 pass at both 320 and 384 bits. The lower bounds and closed form domains are unchanged. $\square$

Proposition 20.53 (Complete residual bounds from a finite prefix). *Assume $D_g \preceq U \preceq mD_g$, with $D_g \succeq \gamma I > 0$. Put $A = D_g^{-1/2}UD_g^{-1/2}$, $H = A - I$, and let P be an orthogonal prefix projection, $Q = I - P$. Suppose $QHQ \preceq \nu Q$ with $\nu \geq 0$. For a residual k write*

$$y = D_g^{-1/2}k = p + q, \quad p = Py, \quad q = Qy, \quad v = \|p\|^2, \quad a = \langle Hp, p \rangle, \quad \|q\|^2 \leq E.$$

With $[s]_+ = \max(s, 0)$, the following scalar bound includes every omitted residual row:

$$\langle U^{-1}k, k \rangle \leq v + E - \frac{1}{m}[\sqrt{a} - \sqrt{\nu E}]_+^2. \quad (248)$$

For a matrix of residual columns $Y = D_g^{-1/2}\mathcal{R}$, put $Y_P = PY$, $Y_Q = QY$, $V = Y_P^*Y_P$ and $A_P = Y_P^*HY_P$. If $Y_Q^*Y_Q \preceq \mathcal{E}$, then, for every fixed $0 < \theta \leq 1$,

$$\mathcal{R}^*U^{-1}\mathcal{R} \preceq V + \mathcal{E} - \frac{1}{m}\{(1 - \theta)A_P - (\theta^{-1} - 1)\nu\mathcal{E}\}. \quad (249)$$

If a certified $0 \leq \rho < 1$ instead satisfies $\nu\mathcal{E} \preceq \rho^2 A_P$, the braced lower bound can be replaced by $(1 - \rho)^2 A_P$.

Proof. The bounded operator H is positive, so the reverse triangle inequality for $H^{1/2}p + H^{1/2}q$ gives

$$\langle Hy, y \rangle \geq [\sqrt{a} - \|H^{1/2}q\|]_+^2 \geq [\sqrt{a} - \sqrt{\nu E}]_+^2.$$

Combine this with the first inverse polynomial $A^{-1} \preceq I - H/m$ and $\|y\|^2 = \|p\|^2 + \|q\|^2 \leq v + E$. For the matrix version, apply the lower Young inequality to $H^{1/2}Y_P x + H^{1/2}Y_Q x$ for every head vector x . Since $1 - \theta^{-1} \leq 0$, the remote upper bound supplies the stated lower form. The relative variant follows from the scalar reverse triangle inequality in every head direction. No factor equal to the number of columns is inserted. The scalar positive part in (248) is not a matrix positive-part operation; the latter does not follow from these arguments. $\square$

In the actual parity tail based at N , the prefix can end at any $J > N$. The same proof as (247) gives the complete remote compression bound

$$Q_J H Q_J \preceq \nu_J Q_J, \quad \nu_J = \frac{\kappa_N + 2(1 - c)E_L(t_J) + C_{e/o}(J)}{g_{J+1}}. \quad (250)$$

Here $t_J = 2\pi(J + 1)/L$, whereas the constant κ_N in the original D_g is kept fixed. The upper numerator decreases and the positive denominator increases with the index. The moment bound of Proposition 20.24, divided by g_{J+1} , supplies $\mathcal{E}$. Thus these formulas evaluate a signed inverse improvement without setting the infinite residual to zero. All occurrences of U are interpreted through its bounded preconditioned form.

Proposition 20.54 (One complete far-residual comparison). *At $\lambda = 4$, take the archived exact dyadic even solve X on modes $17, \dots, 256$, and the archived exact dyadic head vector v of length 17 specified in Appendix A. Put*

$$f = \begin{pmatrix} v \\ -Xv \end{pmatrix}, \quad K = QW_4(f, f), \quad k = Q_{512}W_4 f.$$

For the complete shifted even far operator $U = Q_{512}(W_4 - 10^{-8}D)Q_{512}$ one has

$$\langle U^{-1}k, k \rangle < 5.758 \cdot 10^{-20} < 6.469 \cdot 10^{-20} < K. \quad (251)$$

This is one far-energy comparison. It does not include the mixed term of (252), and it is not a lower bound for the full low Schur matrix.

Outward interval evaluation with a complete remote bound. Use $m = 20$ and $J = 65536$ in Proposition 20.53. The frozen f has support through 256. Thus its retained Weil energy and residual rows are evaluated from the actual finite matrix coefficients, without truncating f . The outer residual is unshifted; the far inverse and its D_g are those of the shifted tail certificate. The following outward enclosures suffice:

Quantity	Certified enclosure
K	$(6.4692843941, 6.4692843942) 10^{-20}$
$v_J = \ P_J D_g^{-1/2} k\ ^2$	$(6.9332416283, 6.9332416284) 10^{-20}$
$a_J = \langle H P_J D_g^{-1/2} k, P_J D_g^{-1/2} k \rangle$	$(4.0338113722, 4.0338113723) 10^{-19}$
$\ Q_J D_g^{-1/2} k\ ^2$	$< 3.4797080873 10^{-21}$
$\ Q_J H Q_J\ $	< 1.987323730

The remote norm uses the complete order-64 moment majorant, divided by g_{J+1} ; the last row uses (250). Substitution into (248) gives the stricter computed upper bound $5.757887789 \cdot 10^{-20}$, proving the displayed claim. All gates replay with identical exact input witnesses at 768 and 896 bits. The prefix matrix action uses exact Arb polynomial products for its divided-difference and Hankel parts, as detailed in the reproduction appendix. No floating-point sign decision, zero-location assumption or omitted-residual cutoff enters the gate. $\square$

Proposition 20.55 (Refining both terms of the structured inverse). *Keep the factors $Z, R, C, \underline{L}, \mu$ of Proposition 20.50, and assume also $U \preceq m D_g$. Let Q_* be a bounded positive operator such that*

$$U^{-1} \preceq Q_* \preceq D_g^{-1}.$$

Then, for every $r = (h, k)$,

$$\langle T^{-1} r, r \rangle \leq \langle Q_* k, k \rangle + \mu^{-1} \|C(h - Z^* k - R^* Q_* k)\|^2. \quad (252)$$

The same inequality holds for the simultaneous residual Gram. In particular all inverse polynomials of Proposition 20.51 can be used while retaining the original certified C and μ .

Proof. The assumed sandwich implies $m^{-1} D_g^{-1} \preceq Q_* \preceq D_g^{-1}$. Consequently Q_* is injective and the closed form of Q_*^{-1} has the same domain as $D_g^{1/2}$, with equivalent form norms. Inverse order gives $U \succeq Q_*^{-1}$ as forms. Moreover

$$K_Z - R^* Q_* R \succeq K_Z - R^* D_g^{-1} R \succeq \underline{L}.$$

Repeat the lower-form square completion in Proposition 20.50, replacing the far form by Q_*^{-1} and its inverse by Q_* . Its exact inverse contains Q_* in both displayed terms, proving the assertion. The change preserves the lower form domain since $Q_* R x$ belongs to the operator domain of Q_*^{-1} for every finite head vector. Applying the result to all linear combinations proves the Gram version. If the saved factors certify $T - cD$, use inverse order as before and retain those shifted factors throughout. $\square$

Improving only the first term while leaving the old mixed term unchanged is not justified. Positivity of the complete low Schur matrix requires the resulting full correction, on the entire head, to fit below K_X . The new estimates do not assert that comparison or convergence along growing windows.

Remark 20.56 (The simultaneous head acceptance test). For a positive definite trial matrix K_X , put $C_X = r^* T^{-1} r$. Then

$$\lambda_{\max}(K_X^{-1/2} C_X K_X^{-1/2}) \leq 1 \iff K_X - C_X \succeq 0.$$

Thus an eigenvalue or inverse-square-root estimate is unnecessary. It suffices to construct a Hermitian *upper enclosure in form order* $\bar{C}_X \succeq C_X$ for the complete residual Gram and certify $K_X - \bar{C}_X \succeq 0$, for example by interval LDL^* with strictly positive pivots. A fixed invertible dyadic congruence may improve conditioning without changing the sign. Strictly positive pivots prove the stronger strict comparison; an exact zero pivot requires the corresponding semidefinite factorization conditions, not an interval containing zero. An entrywise upper bound is not by itself an upper bound in form order. For $T_M = P_M T P_M$ and $r_M = P_M r$, the variational identity gives $r_M^* T_M^{-1} r_M \preceq C_X$, the opposite enclosure direction. All mixed and remote contributions must be enclosed before the LDL^* test proves the complete Schur sign.

Complete mixed correlations and a monotone head refinement

The successful far-energy comparison in Proposition 20.54 leaves a mixed term. The next estimate includes both the omitted input and the output outside the retained Fourier rows. These are different contributions.

Proposition 20.57 (A directional enclosure of the complete mixed term). *Use the factors of Proposition 20.55, and put*

$$A = D_g^{-1/2} U D_g^{-1/2}, \quad H = A - I, \quad Y = D_g^{-1/2} R C^*, \quad y = D_g^{-1/2} k, \quad d = C(h - Z^* k).$$

Here A is represented by its bounded closed form and $0 \preceq H \preceq (m-1)I$. Let P be an orthogonal Fourier prefix projection on the inner far tail, $Q = I - P$, $p = Py$ and $q = Qy$. Suppose

$$\|q\|^2 \leq E, \quad 0 \preceq QHQ \preceq \nu Q, \quad 0 \leq \nu < m.$$

Set $v = \|p\|^2$, $a = \langle Hp, p \rangle$ and

$$b = d - Y^*(I - H/m)y, \quad b_P = d - (PY)^*(I - PHP/m)p.$$

For any finite head vector z , let $\eta_z = \langle HPYz, PYz \rangle$ and suppose $\|QYz\|^2 \leq \rho_z$. Then

$$|z^*(b - b_P)| \leq \frac{\sqrt{\nu\eta_z E} + \sqrt{\nu a \rho_z}}{m} + \sqrt{\rho_z E} =: \epsilon_z. \quad (253)$$

In addition the complete first-polynomial far energy satisfies

$$\langle (I - H/m)y, y \rangle \geq v - \frac{a}{m - \nu}. \quad (254)$$

In particular, for $z \neq 0$ the complete scalar- μ majorant in (252) is at least

$$v - \frac{a}{m - \nu} + \frac{[|z^* b_P| - \epsilon_z]_+^2}{\mu \|z\|^2}. \quad (255)$$

Proof. Write $f = Yz = f_P + f_Q$. The difference of the mixed pairings is the sum of $m^{-1}f_P^*Hq$, $m^{-1}f_Q^*Hp$, and $-f_Q^*(I - QHQ/m)q$. Positive-form Cauchy–Schwarz bounds the first two by the corresponding square roots in (253). Since $0 \preceq QHQ \preceq \nu Q$ and $\nu < m$, the last factor is a contraction on the remote subspace. This proves the mixed estimate without setting either remote contribution to zero.

For $x = \|q\|$, positivity of H and its remote compression give

$$\langle (I - H/m)y, y \rangle \geq v - a/m - 2\sqrt{a\nu}x/m + (1 - \nu/m)x^2.$$

The minimum of this scalar quadratic over $x \geq 0$ is $v - a/(m - \nu)$. The directional lower bound on $\|b\|$ then proves (255). All pairings are bounded-operator pairings after preconditioning. No application of the unbounded U to an arbitrary Hilbert vector is used. $\square$

Proposition 20.58 (Failure of the first scalar-head majorant). *On the same exact even head direction and frozen support-256 trial as Proposition 20.54, let $\mathcal{B}_1$ denote the complete right side of (252) with $Q_* = D_g^{-1/2}(I - H/20)D_g^{-1/2}$ and the saved $\mu = 0.9999999999$. Then*

$$\mathcal{B}_1 > 6.7609 \cdot 10^{-20} > 6.4693 \cdot 10^{-20} > K_X(v, v). \quad (256)$$

Thus this particular complete majorant cannot certify that trial direction. This is not a negative Weil vector and does not exclude a better inverse polynomial or a stronger initial head certificate.

Complete interval certificate. Retain $J = 65536$, $m = 20$ and the complete remote bounds of Proposition 20.54. Choose the exact 160-bit dyadic head vector z specified in the reproduction appendix, obtained by rounding the direction of b_P . Its normalization is enclosed, not assumed exact. The following outward enclosures suffice in addition to the previous values of v, a, E, ν :

Quantity	Certified enclosure
η_z	(0.4276838592, 0.4276838593)
ρ_z	< 0.0672057785
$ z^*b_P $	$(1.7339321076, 1.7339321077) 10^{-10}$
$\ z\ $	$(1 - 10^{-45}, 1 + 10^{-45})$
ϵ_z	$< 2.961708367 10^{-11}$
complete far energy	$> 4.693812399 10^{-20}$
$\mu^{-1} \ b\ ^2$	$> 2.067157472 10^{-20}$

The bound on ρ_z uses the complete order-64 moment majorant on the finite vector $[I; -Z]C^*z$, whose support ends at 1024, divided by g_{J+1} . It includes every row beyond J . The scalar η_z and the retained pairings use the exact Arb polynomial action from the preceding appendix, with the unchanged archimedean-series remainder and parity normalization. Equation (255) gives $\mathcal{B}_1 > 6.760969872 \cdot 10^{-20}$, whereas the already certified $K_X(v, v) < 6.469284395 \cdot 10^{-20}$. The same frozen direction and solve replay at 768 and 896 bits. The outer residual is that of the unshifted Weil form; the inner inverse factors consistently belong to the certified shifted tail. Their inverse upper bound remains valid by order. $\square$

Proposition 20.59 (Retaining the strengthened head metric). *In the same setup, let a positive definite finite matrix M_0 satisfy*

$$CK_ZC^* - Y^*Y \succeq M_0 \succ 0.$$

The existing certificate permits $M_0 = \mu I$. Let $P_j(A)$ be the inverse upper polynomials of Proposition 20.51, and set

$$\begin{aligned} M_j &= M_0 + Y^*(I - P_j(A))Y, \\ b_j &= d - Y^*P_j(A)y, \\ \mathcal{E}_j(r) &= \langle P_j(A)y, y \rangle + b_j^*M_j^{-1}b_j. \end{aligned}$$

Then these are complete simultaneous upper bounds with

$$\langle T^{-1}r, r \rangle \leq \mathcal{E}_{j+1}(r) \leq \mathcal{E}_j(r). \quad (257)$$

They converge as $j \rightarrow \infty$. Their limit need not equal $\langle T^{-1}r, r \rangle$: any loss in replacing $CK_ZC^* - Y^*Y$ by M_0 remains. For the first polynomial, the head improvement is exactly Y^*HY/m . It must be retained together with the changed mixed vector if monotonicity of the complete bound is required.

Proof. After the finite triangular substitution and head congruence, the original form has blocks CK_ZC^* , RC^* and U , and the dual vector is (d, k) . Put $Q_j = D_g^{-1/2}P_j(A)D_g^{-1/2}$ and consider the lower form

$$\mathcal{L}_j = \begin{pmatrix} M_0 + Y^*Y & CR^* \\ RC^* & Q_j^{-1} \end{pmatrix}.$$

The assumed head comparison and $Q_j^{-1} \preceq U$ imply that the transformed original form dominates $\mathcal{L}_j$. Its Schur complement is $M_j \succeq M_0 \succ 0$. Because $m^{-1}D_g^{-1} \preceq Q_j \preceq D_g^{-1}$, all far forms have the same domain $\mathcal{D}(D_g^{1/2})$. The finite head and bounded coupling give closed, coercive lower forms by the same square completion as before.

Their inverse quadratic forms at (d, k) are precisely $\mathcal{E}_j(r)$. The polynomials decrease in operator order, so Q_j^{-1} increases. The head of $\mathcal{L}_j$ is fixed; therefore $\mathcal{L}_{j+1} \succeq \mathcal{L}_j$, proving the claimed inverse order. Norm convergence of $P_j(A)$ to A^{-1} , boundedness of Y , and the common positive floor of M_j give convergence of the displayed scalar and Gram expressions. The possible head loss explains the stated limitation of the limit. A certificate for the shifted tail is again transferred only by the already stated inverse order. $\square$

Corollary 20.60 (The first head update does not rescue this trial). *For the same saved data, choose $M_0 = \mu I$ in Proposition 20.59. Even its exact first head update satisfies*

$$\mathcal{E}_1(r) > 6.6588 \cdot 10^{-20} > K_X(v, v).$$

This excludes the first inverse degree with that initial certificate, including its full positive head update. It does not exclude a stronger initial M_0 or a higher inverse degree.

Proof. Use the same frozen z and complete tail bounds. Positive-form Cauchy gives

$$z^*M_1z \leq \mu \|z\|^2 + \frac{(\sqrt{\eta z} + \sqrt{\nu \rho z})^2}{20} < 1.051962238.$$

Metric Cauchy–Schwarz therefore gives

$$b_1^*M_1^{-1}b_1 \geq \frac{[|z^*b_P| - \epsilon_z]_+^2}{z^*M_1z} > 1.965049122 \cdot 10^{-20}.$$

Adding the complete far lower bound gives $\mathcal{E}_1(r) > 6.658861521 \cdot 10^{-20}$. Both precision replays include this gate with the same exact dyadic witnesses and all remote correlations. $\square$

Proposition 20.61 (Finite probes for the positive head improvement). *Keep H, Y, P, Q, ν as above. Let V have finitely many finite-prefix columns, suppose $G = V^*HV \succ 0$, and set $B_P = V^*HPY$. If $Y^*QY \preceq E_Y$, then for every $0 < \theta < 1$,*

$$Y^*HY \succeq (1 - \theta)B_P^*G^{-1}B_P - (\theta^{-1} - 1)\nu E_Y. \quad (258)$$

Consequently the right side, divided by m and added to M_0 , is a valid finite lower enclosure of the first improved head. It may be inverted only after its positivity is certified.

Proof. The Gram matrix of $H^{1/2}V$ and $H^{1/2}Y$ gives $Y^*HY \succeq B^*G^{-1}B$, where $B = V^*HY$. Write $B = B_P + E_B$ with $E_B = V^*HQY$. The projection onto the range of $H^{1/2}V$ shows

$$E_B^*G^{-1}E_B \preceq Y^*QH QY \preceq \nu E_Y.$$

Young's inequality in the G^{-1} metric proves (258). This keeps the omitted probe correlation explicitly. No unjustified matrix positive-part operation is applied. $\square$

These first-degree bounds do not settle the complete low-mode Schur sign. The later simultaneous and directional-complement certificates prove both parity signs at $\lambda = 4$. The first-degree obstruction now has a proof even with the induced head update and the saved initial certificate. Increasing the inverse degree, or strengthening the initial certificate, requires a different complete enclosure. Neither finite-prefix improvements nor convergence at this fixed window establishes uniformity as $\lambda \rightarrow \infty$.

A complete Schur-direction certificate from a finite trial

The first-degree obstruction above concerns a particular inverse majorant and saved trial. A finite iterative calculation can instead propose a new trial, after which the complete residual is verified afresh. This uses the same verified-solve identity; it does not identify a power of a finite compression with the full power.

Proposition 20.62 (A posteriori verification without a full-power claim). *Let the complete outer operator have finite head and coercive tail T , with Schur form $S_\infty = F - B^*T^{-1}B$. For a finite trial $g = (v, w)$ in its operator domain, set $e = Bv + Tw$. Then*

$$S_\infty(v, v) = QW(g, g) - \langle T^{-1}e, e \rangle. \quad (259)$$

Thus any certified complete upper bound $\mathcal{U}(e)$ for the last term gives

$$QW(g, g) - \mathcal{U}(e) \leq S_\infty(v, v) \leq QW(g, g).$$

The algorithm that proposed the frozen finite coefficients of w need not be an approximation to any particular infinite polynomial.

Proof. Expand $\langle T^{-1}(Bv + Tw), Bv + Tw \rangle$ and cancel the tail and cross terms in $QW(g, g)$. Coercivity justifies the inverse. Equivalently, for any residual r and finite correction z ,

$$\langle T^{-1}r, r \rangle = 2\Re \langle r, z \rangle - \langle Tz, z \rangle + \langle T^{-1}(r - Tz), r - Tz \rangle.$$

If the inner certificate belongs to $T - cD$, use $T^{-1} \preceq (T - cD)^{-1}$ only in bounding the new residual term. The energy and $e = P_{\text{tail}}Wg$ remain unshifted. This avoids an erroneous omission of a cDz residual or a shift energy when the two operators are mixed. $\square$

For completeness, let the new residual split as $e = (h, k)$ in the already certified inner decomposition. Past the support of the saved Z , retain rows through J and put

$$V_J = \sum_{n \leq J} |k_n|^2 / g_n, \quad b_J = C \left(h - Z^* k - \sum_{n \leq J} R_n^* k_n / g_n \right).$$

The sums run over the inner far rows. If complete remote bounds give $\sum_{n > J} |k_n|^2 / g_n \leq E$ and $\|D_g^{-1/2} Q_J R C^*\|^2 \leq \rho$, then the old structured inverse certificate supplies

$$\mathcal{U}(e) = V_J + E + \mu^{-1}(\|b_J\| + \sqrt{\rho E})^2. \quad (260)$$

Every row beyond J is included. For a matrix of trial columns, the simultaneous version retains their Gram matrices and uses (242); separate positive scalar tests are insufficient for positivity on their span.

Proposition 20.63 (One complete positive Schur direction at $\lambda = 4$). *Let v be the exact archived even head vector used in Propositions 20.54 and 20.58. For the complete even Schur form with head $0 \leq n \leq 16$,*

$$3.0930 \cdot 10^{-20} < S_\infty(v, v) < 3.2356 \cdot 10^{-20}. \quad (261)$$

This certifies one fixed direction. It does not assert that the whole even head, the odd head, or the full $\lambda = 4$ form is positive, and it provides no growing-window uniformity.

Frozen finite trial and complete residual certificate. Start with the previous support-256 vector $f = [v; -Xv]$. Subtract the exact dyadic tail correction z , supported on $17 \leq n \leq 4096$, specified in the reproduction appendix. The new vector $g = f - (0, z)$ has exactly the same head v . A finite preconditioned iteration proposed z ; neither its convergence nor its recursive residual is used in this proof.

Evaluate $QW_4(g, g)$ and the unshifted residual $e = Q_{16}W_4g$ through $J = 65536$. Retain the original shifted inner factors, $\mu = 0.999999999$, and the previously certified $\rho < 0.165102333$. Direct interval evaluation gives

Quantity	Certified enclosure
$QW_4(g, g)$	$(3.2355843481, 3.2355843482) 10^{-20}$
V_J	$< 3.742924324 10^{-22}$
$\ b_J\ ^2$	$< 1.617782100 10^{-23}$
E	$< 8.080848756 10^{-22}$
$\mathcal{U}(e)$	$< 1.424889 10^{-21}$

The bound E is the complete order-64 moment majorant applied to the finite vector g , with its actual support through 4096, divided by g_{J+1} . The saved remote ρ belongs to the same inner witness and cutoff. Substitution in (260) and (259) proves the lower bound. The upper bound follows from $T^{-1} \succeq 0$.

The frozen correction uses exact dyadic coefficients produced at 256 bits; the proof evaluates them anew at 768 and 896 bits. Both checks retain the zero-mode $\sqrt{2}$ factor, normalized parity, logarithmic diagonal and archimedean-series remainder. Selected rows, including zero and the support endpoint, are also checked by dense coefficient assembly. The independently expanded energy identity agrees with the direct energy. No finite power is used as a substitute for an infinite operator power. $\square$

For the *old* trial in this same direction the complete inverse correction is consequently bounded by

$$3.2337 \cdot 10^{-20} < \langle T^{-1}r, r \rangle < 3.3762 \cdot 10^{-20}.$$

In particular $S_\infty(v, v) > 0.4781K_X(v, v)$ on this one direction. This explains why the earlier majorant could fail without identifying an intrinsically difficult Schur direction. The next target is the simultaneous matrix comparison, including the remaining directions of small energy. This directional certificate cannot be extrapolated to that matrix or to RH.

A simultaneous complete Schur certificate

A finite trial may be chosen in coordinates adapted to its head energy; the final certificate must still use its actual head and every residual cross term. This distinction is essential when some ordinary head energies are extremely small.

Proposition 20.64 (Simultaneous trial and ordinary-error certificate). *Let the complete closed Weil form have finite head and coercive tail T , and let G be a matrix of finite Fourier trial columns. Write $V = P_{16}G$, $E = Q_{16}W_4G$ and $K = G^*W_4G$. Then*

$$V^*S_\infty V = K - E^*T^{-1}E. \quad (262)$$

*Suppose V is square and invertible, and a complete matrix upper bound $\mathcal{U} \succeq E^*T^{-1}E$ is certified. If $K - \mathcal{U} \succ 0$, the entire parity form is strictly positive. More generally, if $\varepsilon \geq 0$ and*

$$K - \mathcal{U} + \varepsilon V^*V \succeq 0,$$

then $S_\infty \succeq -\varepsilon I$ and the entire parity form is bounded below by $-\varepsilon I$ in ordinary norm.

Proof. Apply Proposition 20.62 to every linear combination of the columns. Invertibility of the actual head V transfers the resulting congruent inequality to S_∞ . For a head vector x and tail vector y , the closed form identity is

$$q_{W_4}[x, y] = \langle S_\infty x, x \rangle + \left\| T^{1/2}(y + T^{-1}Bx) \right\|^2.$$

The head-to-tail map B is bounded because the head is finite and its Fourier vectors belong to the operator domain. Thus the completion is valid on the form domain. The tail square is nonnegative, and $\|x\| \leq \|(x, y)\|$. Strict head and tail positivity also imply fixed-window coercivity by the bounded invertible triangular change in this identity. No quantitative ordinary spectral gap is identified with an LDL^* pivot. $\square$

Proposition 20.65 (Retaining a joint remote Gram). *In the finite-row structured inverse bound, put $Y = D_g^{-1/2}Q_JRC^*$ and $y = D_g^{-1/2}Q_Jk$, and let $H = C(h - Z^*k - R_J^*D_g^{-1}k_J)$ be the retained mixed matrix. Suppose one joint enclosure and a lower bound satisfy*

$$\Gamma = \begin{pmatrix} \Gamma_{YY} & \Gamma_{Yk} \\ \Gamma_{Yk}^* & \Gamma_{kk} \end{pmatrix} \succeq [Y, y]^*[Y, y], \quad 0 \preceq L_Y \preceq Y^*Y, \quad A_0 = \mu I + L_Y - \Gamma_{YY} \succ 0.$$

Then the complete inverse correction is bounded by

$$E^*T^{-1}E \preceq V_J + \Gamma_{kk} + (H - \Gamma_{Yk})^*A_0^{-1}(H - \Gamma_{Yk}). \quad (263)$$

*Here $V_J = k_J^*D_g^{-1}k_J$. One may take $L_Y = 0$. If $\Gamma_{YY} \preceq \rho I$ with $\rho < \mu$, replacing A_0^{-1} by $(\mu - \rho)^{-1}I$ gives a simpler valid bound.*

Proof. For an outer coefficient vector x , the remote portion of the old structured majorant is

$$\|yx\|^2 + \mu^{-1} \|Hx - Y^*yx\|^2 = \sup_z \{\|yx - Yz\|^2 - \|Yz\|^2 + 2\Re \langle Hx, z \rangle - \mu \|z\|^2\}.$$

Apply the joint upper Gram to $(-z, x)$ and the lower bound L_Y to $\|Yz\|^2$, then complete the finite A_0 square. This proves the claim for every x . The bounded finite-column maps justify all remote pairings. The cross block must come from the *same joint* Gram; unrelated diagonal upper bounds do not justify an invented mixed term. $\square$

Proposition 20.66 (The complete even form at $\lambda = 4$). *The canonical complete Weil form W_4 restricted to its even parity space is strictly positive on every nonzero vector in its form domain, and is bounded below by a positive constant at this fixed window.*

Frozen matrix certificate. Keep the outer head $0 \leq n \leq 16$ and the complete tail certified in Proposition 20.48. The archived matrix trial has 17 columns and support through $M = 4096$. Its exact dyadic base and correction coefficients define G ; the actual head $V = P_{16}G$ is certified invertible. The proposal first scales the old finite trial by its head-energy LDL^* factors, rounds that base to exact dyadics, and uses 16 preconditioned finite CG steps at 512 bits for each column. Neither CG convergence nor its recursive residual enters the proof.

Recompute the unshifted $K = G^*W_4G$ and $E = Q_{16}W_4G$ at 768 and 896 bits from the same frozen inputs. Retain all cross-Grams through $J = 65536$ and apply the order-64 moment bound to the entire trial for every omitted row. The saved shifted inner factors have $N = 512$, support 1024, $\mu = 0.9999999999$ and complete remote bound $\rho < 0.165102333$. Inverse order is used only for the residual inverse, so the trial energy and residual stay unshifted. For $\tau = 1/10$, equation (242) gives the complete matrix upper bound $\mathcal{U}_\tau$. All 17 interval LDL^* pivots of $K - \mathcal{U}_\tau$ are positive; the smallest exceeds 0.7645944101 in these saved head coordinates. Both precision replays give this strict gate. This number certifies sign and is not an ordinary-norm eigenvalue bound.

The verifier checks the actual head determinant, exact dyadic reconstruction, inner-certificate provenance, positivity of the far diagonal, and nine independently assembled physical rows for every column. The complete remote moment includes the zero-mode normalization and both Fourier signs through the parity basis. Proposition 20.64 therefore proves positivity of the entire even form, not only a Fourier compression. $\square$

A common remote Gram for the padded inner source $[I; -Z]C^*$ and this trial also supplies the cross centre in Proposition 20.65. Its scalar-denominator certificate has all 17 pivots positive, the smallest greater than 0.7892519172; using the full finite denominator improves that smallest pivot to more than 0.7892519657. This is an additional fixed-witness check, not needed for the preceding even sign proof. The whole-operator conclusion comes from a simultaneous matrix inequality and complete tail control; separately positive directions would not suffice.

The comparison also has a certified dimensionless margin:

$$\mathcal{U}_{1/10} \preceq \left(1 - \frac{62629}{100000}\right) K. \quad (264)$$

Interval LDL^* verifies the equivalent positive matrix at both precisions. Thus the generalized relative margin exceeds 0.62629; unlike a smallest ordinary eigenvalue or a coordinate pivot, this ratio is invariant under an invertible head congruence. On the original physical head direction from Proposition 20.63, this new matrix trial has certified headroom greater than 12.1284. Its trial is different from the earlier single-direction trial, whose headroom was about 22.71; changing the trial changes this diagnostic while leaving $S_\infty(v, v)$ unchanged.

The analogous initial odd-head bounds are inconclusive. The next subsection completes the odd sector by combining a complete complement certificate with a larger-support directional trial. No finite list of such fixed-window statements implies the cofinal estimate needed for RH.

A complete complement bound and the full sign at $\lambda = 4$

The $\lambda = 4$ odd computation can be reduced to one direction without inferring a matrix sign from separately positive vectors. The reduction uses positivity of the *exact inverse correction*, which is already known from the coercive complete tail.

Proposition 20.67 (One repaired subspace and its complete complement). *Let K be a positive definite Hermitian form on a finite-dimensional complex space, let $\mathcal{C} \succeq 0$, and put $S = K - \mathcal{C}$. Let E be a subspace and write $F = E^\perp_K$. Suppose that $s \leq 1$, $\sigma \geq 0$, and, simultaneously on the indicated subspaces,*

$$S|_E \succeq sK|_E, \quad \mathcal{C}|_F \preceq \sigma K|_F.$$

Then

$$S \succeq (s - \sigma)K.$$

In particular, when $E = \text{span}\{v\}$ and $K(v, v) = 1$, the first premise is the single inequality $S(v, v) \geq s$.

Proof. For $x = e + f$ with $e \in E$ and $f \in F$, Cauchy–Schwarz for the positive semidefinite form $\mathcal{C}$ gives

$$\begin{aligned} \mathcal{C}(x, x) &\leq (\sqrt{\mathcal{C}(e, e)} + \sqrt{\mathcal{C}(f, f)})^2 \\ &\leq (\sqrt{1 - s} \|e\|_K + \sqrt{\sigma} \|f\|_K)^2 \\ &\leq (1 - s + \sigma)(\|e\|_K^2 + \|f\|_K^2) = (1 - s + \sigma)K(x, x). \end{aligned}$$

Subtracting from $K(x, x)$ proves the claim. Degeneracy of $\mathcal{C}$ causes no problem, since Cauchy–Schwarz holds for its induced seminorm. The constant cannot be improved under these premises alone: take $K = I$ on $\mathbb{C}^2$, $E = \text{span}\{(1, 0)\}$, and $\mathcal{C} = uu^*$ with $u = (\sqrt{1 - s}, \sqrt{\sigma})^T$. $\square$

Corollary 20.68 (Ordinary error localized to the unresolved subspace). *Under the hypotheses of Proposition 20.67, assume $0 \leq \sigma < 1$, let P_E be the K -orthogonal projection onto E , and set*

$$\eta = \frac{(\sigma - s)_+}{1 - \sigma}.$$

Then

$$S \succeq -\eta P_E^* K P_E. \tag{265}$$

If S is the complete Schur form of a trial with invertible physical head V and coercive complete tail, the full form has ordinary lower error at most

$$\varepsilon^{\text{loc}} = \eta \left\| K^{1/2} P_E V^{-1} \right\|^2. \tag{266}$$

The congruence cost is thus restricted to the unresolved subspace; it is not discarded.

Proof. Write $x = e + f$, $e \in E$, $f \in E^{\perp_K}$, and $a = \|e\|_K$, $b = \|f\|_K$. Positivity of $\mathcal{C}$ gives

$$S(x, x) \geq sa^2 + (1 - \sigma)b^2 - 2\sqrt{(1 - s)\sigma}ab.$$

Completing the square in b bounds this below by $(s - \sigma)a^2/(1 - \sigma)$, which proves (265). For physical head $y = Vx$, use $a^2 = \|K^{1/2}P_EV^{-1}y\|^2$. The positive complete-tail square and $\|y\| \leq \|\text{full vector}\|$ give (266) on the closed form domain. $\square$

Along a cofinal window family, establishing all these hypotheses and $\varepsilon_{\lambda}^{\text{loc}} \rightarrow 0$ would satisfy Proposition 32.1, using its fixed-support identification with the same Weil form. This is a sufficient target which can tolerate a negative restricted estimate; no such convergence is proved here. Its factors must be controlled for the complete operator at every window of that family.

Use with two finite trials. For a coercive complete tail T and any finite trial G with invertible physical head V , let $K = G^*WG$, $r = QWG$, and $\mathcal{C} = r^*T^{-1}r$. The exact Schur matrix in these head coordinates is $S = K - \mathcal{C}$. A second finite trial with the same physical head gives the same S . Consequently a directional estimate from the second trial may be combined with a complete complement estimate from the first, provided that both are normalized by the *same first-trial metric* K . An available complete upper bound $\mathcal{C} \preceq U$ permits the complement premise to be verified as $Z^*(\sigma K - U)Z \succeq 0$, where Z spans the exact K -orthogonal complement. No numerical eigenvector assumption is needed.

Proposition 20.69 (Complete odd complement). *At $\lambda = 4$, let G_0 be the archived odd 16-column trial with support through 4096, physical head $V = P_{16}G_0$, $K = G_0^*W_4G_0$, and complete correction $\mathcal{C} = (Q_{16}W_4G_0)^*T^{-1}(Q_{16}W_4G_0)$. There is a specified nonzero exact dyadic vector $v \in \mathbb{R}^{16}$ for which*

$$K \succ 0, \quad \mathcal{C}|_{v^{\perp_K}} \preceq 10^{-3}K|_{v^{\perp_K}}. \quad (267)$$

This is a complete inverse-correction estimate on the indicated subspace, with every omitted Fourier row retained.

Outward certificates and the complete-form passage. Use the same odd inner factors as Proposition 20.48, with $N = 1536$, inner solve support 2048, verification cutoff $J = 65536$ and remote moment order 64. The frozen trial has 16 proposal iterations per column; its construction is not a proof input. Reconstructing it at 768 bits gives the complete matrix upper $\mathcal{U}_\tau$ of Proposition 20.64. The exact frozen dyadics defining v are included in the bundle.

Put $a = v^*Kv > 0$ and $P = I - v(v^*K)/a$. Delete column 12 of P to obtain Z . The corresponding coordinate of v is nonzero, so Z has rank 15 and range $v^{\perp_K}$: the kernel of P is $\text{span}\{v\}$, which has no nonzero vector with coordinate 12 zero. The projection uses the exact K ; its interval enclosure is not frozen at a midpoint. For $\tau = 1/20$, interval LDL^* proves

$$Z^*(10^{-3}K - \mathcal{U}_{1/20})Z \succ 0,$$

with all 15 pivots positive and the smallest exceeding 0.0004086079. Since $\mathcal{C} \preceq \mathcal{U}_{1/20}$, this proves (267). The scalar direction was suggested numerically, but its proposed eigenvector property is neither assumed nor needed. $\square$

Proposition 20.70 (Complete positivity at $\lambda = 4$). *The canonical complete Weil form W_4 is strictly positive on every nonzero vector in its form domain and is bounded below by a positive constant at this fixed window. In particular its nonnegative-error formulation has $\varepsilon_4 = 0$ exactly.*

A *larger-support directional certificate*. Keep G_0, K, V, v and the complete odd tail T from Proposition 20.69; put $a = v^* K v$. The archived exact dyadic odd trial f has support through 8192 and physical head exactly Vv . Thus, with $r_f = Q_{16} W_4 f$,

$$S(v, v) = QW_4(f, f) - \langle T^{-1} r_f, r_f \rangle, \quad S = K - \mathcal{C}.$$

This identity uses the same complete Schur form and the old normalization a , although the finite trial has changed.

The same physical diagonal, inner certificate, $J = 65536$ and order-64 remote moment give the following conservative outward intervals, reproduced at 1024 and 1280 bits on the same frozen witness:

quantity	enclosing interval
a	(1, 1.0000000000000002)
$QW_4(f, f)$	(0.9016760830, 0.9016760831)
finite far energy upper V_f	(0.3125027254, 0.3125027255)
remote source majorant E_f	(0.0608540372, 0.0608540373)
finite mixed squared norm H_f	(0.0701478147, 0.0701478148)
complete correction upper U_f	(0.4725352935, 0.4725352936)
$(QW_4(f, f) - U_f)/a$	(0.4291407894, 0.4291407895)

Here $\mu = 0.9999999999$ and $0.0412001510 < \rho < 0.0412001511$ are the certified odd inner constants. Proposition 20.64, or its scalar Cauchy–Schwarz form, gives

$$\langle T^{-1} r_f, r_f \rangle \leq U_f := V_f + E_f + \mu^{-1}(\sqrt{H_f} + \sqrt{\rho E_f})^2.$$

The bound includes every row past J , not only the finite residual. The verifier checks exact equality of all physical head coefficients, provenance of the frozen inner factors, and eight independently assembled rows of the unshifted physical operator. The physical head equality is also checked with independent exact rational arithmetic; its largest dyadic mantissa has 912 bits. The finite proposal method supplies no hypothesis about its convergence.

The outward directional inequality proves $S(v, v) \geq (429/1000)a$. Combining it with (267) in Proposition 20.67 yields the simultaneous odd-head estimate

$$S \succeq \frac{428}{1000} K \succ 0. \quad (268)$$

Since V is invertible and the complete tail is coercive, square completion proves positivity of the entire odd form. The triangular change is bounded with bounded inverse, so the finite positive head and coercive tail also give a positive ordinary lower constant at this fixed window. The orthogonal parity decomposition and Proposition 20.66 prove the full claim. The relative constant $428/1000$ is not an ordinary spectral-gap estimate, and no congruence factor is needed to pass the exact sign $\varepsilon_4 = 0$. $\square$

The reduced directional argument closes this fixed-window obligation. It gives no bound for the heads, inverse factors or residuals at unbounded λ . In particular, the constants above are tied to $L = 2 \log 4$ and are not transferred to other windows.

What the present diagonal majorant costs as the window grows

Proposition 20.71 (Exponential cutoff cost of the scalar far bound). *Fix $0 \leq c < 1$, put $L = 2 \log \lambda$, and use the present far weights*

$$g_{N+1} = (1 - c) \{ \log((N + 1)/L) - E_L(2\pi(N + 1)/L) \} - \kappa_{\lambda, N}, \quad \kappa_{\lambda, N} \geq M_{\phi, \lambda} \geq \|\mathbf{T}_{\text{pr}, \lambda}\|.$$

If $g_{N+1} > 0$, then

$$N + 1 > L \exp\left(\frac{M_{\phi, \lambda}}{1 - c}\right). \quad (269)$$

Consequently, along any unbounded family using these positive weights,

$$\liminf_{\lambda \rightarrow \infty} \frac{\log((N(\lambda) + 1)/L)}{\lambda} \geq \frac{1}{1 - c}.$$

This is a cost of this particular scalar diagonal majorant, not a lower bound on the cutoff required by every possible argument.

Proof. The error E_L is positive, so positivity of g_{N+1} gives $(1 - c) \log((N + 1)/L) > \kappa_{\lambda, N} \geq M_{\phi, \lambda}$. Use Proposition 20.44, which proves $\|\mathbf{T}_{\text{pr}, \lambda}\|/\lambda \rightarrow 1$, for the limit. $\square$

For $c = 10^{-8}$, freshly evaluating the physical weighted prime bound and the parity constants at each window gives the following smallest integer N for which the stated g_{N+1} is positive. These are certificates for the remote block only; they do not certify the intervening inner head or its inverse factors.

λ	$M_{\phi, \lambda}$ (rounded)	even N	odd N
3	2.980079	44	209
4	4.624042	283	1360
5	6.145139	1502	7224
6	7.644648	7488	36021
8	10.306338	124442	598623

Each threshold has positive interval margin, while its preceding integer has negative margin; monotonicity in N proves minimality for this formula. The displayed prime values are rounded summaries, not the outward endpoints used by the verifier.

The appropriate whole-head diagnostic is $1 - \lambda_{\max}(\mathcal{U}, K)$, with $K \succ 0$ and a *complete* inverse upper bound $\mathcal{U}$. The separately useful ratios $\lambda_{\min}(K - \mathcal{U})/\lambda_{\min}(K)$ and $\|\mathcal{U}\|/\lambda_{\min}(K)$ depend on the chosen coordinates; they should be recorded with the head dimension and condition of its congruence. A strong single direction cannot distinguish all possible causes of a weak whole-head margin. For example,

$$K = \text{diag}(1, \epsilon), \quad U = \text{diag}(\delta, \epsilon(1 - \eta)), \quad 0 < \epsilon, \delta, \eta < 1,$$

has generalized margin $\min(1 - \delta, \eta)$, while the headroom in its first direction is $1/\delta$ and the condition of a normalizing congruence is $\epsilon^{-1/2}$. These parameters may vary independently. The actual error terms α_λ , e_λ^2 and t_λ^2/μ_λ still require complete window-specific witnesses. The same numerical cutoff or moment order may be used at different windows only after its inequalities are verified afresh.

Proposition 20.72 (An ordinary-error target for growing windows). *For each window, let the complete form have an outer finite head and positive complement T_λ , with bounded coupling B_λ . For a finite-support solve X_λ put*

$$\begin{aligned} K_{X,\lambda} &= F_\lambda - B_\lambda^* X_\lambda - X_\lambda^* B_\lambda + X_\lambda^* T_\lambda X_\lambda, \\ r_\lambda &= B_\lambda - T_\lambda X_\lambda. \end{aligned}$$

Assume the inner tail admits the factors of Proposition 20.50, and write $r_\lambda = (h_\lambda, k_\lambda)$ in that inner decomposition. If, in operator norm on the entire outer head,

$$\begin{aligned} K_{X,\lambda} &\succeq -\alpha_\lambda I, \quad \|D_{g,\lambda}^{-1/2} k_\lambda\| \leq e_\lambda, \\ \|C_\lambda(h_\lambda - Z_\lambda^* k_\lambda - R_\lambda^* D_{g,\lambda}^{-1} k_\lambda)\| &\leq t_\lambda, \quad \alpha_\lambda \geq 0, \end{aligned}$$

then the whole closed form satisfies

$$W_\lambda \succeq -(\alpha_\lambda + e_\lambda^2 + t_\lambda^2/\mu_\lambda) I. \quad (270)$$

Thus convergence of the displayed error to zero along a cofinal window family is sufficient for the large-support sign criterion. No such convergence is asserted here.

Proof. The exact outer Schur matrix is

$$S_\lambda = K_{X,\lambda} - r_\lambda^* T_\lambda^{-1} r_\lambda.$$

Apply (241) to each outer head vector. Completing the outer form square leaves S_λ on the head and a nonnegative complementary square. Since the head projection is orthogonal, its norm is at most the full ordinary norm, proving (270). $\square$

Two scales cannot be suppressed in applying this criterion. If only individual bounds $\|D^{-1/2} r e_j\| \leq \delta$ are known for a head of dimension d , the valid conclusion is $\|D^{-1/2} r\|^2 \leq d\delta^2$. Equal residual columns attain this factor. In particular, a single source estimate or columnwise decay does not prove the required growing-head bound. Likewise, $C\mathcal{L}C^* \succeq -\eta I$ implies only

$$\mathcal{L} \succeq -\eta C^{-1} C^{-*} \succeq -\eta \|C^{-1}\|^2 I.$$

Congruence preserves positivity, but it does not preserve the size of an ordinary negative error. Using merely $T \succeq cD$ yields the coarser correction $c^{-1} \|D^{-1/2} r\|^2$, and its division by c is sharp already for a scalar tail $T = cD$.

For clarity, the remote truncation itself has an explicit uniform budget. If G has Fourier support $|m| \leq M$, $J > M$, and $|b_n| \leq B_*$, the actual off-diagonal divided difference gives

$$(Q_J W_\lambda G)^* (Q_J W_\lambda G) \preceq \frac{8B_*^2(2M+1)}{J-M} G^* G. \quad (271)$$

Its proof bounds the squared Hilbert–Schmidt norm of the separated row/column block by $4B_*^2(2M+1) \sum_{|n|>J} (|n| - M)^{-2}$ and uses the integral bound $\sum_{n>J} (n - M)^{-2} \leq 1/(J - M)$. The diagonal is absent. If the diagonal lower weights g_n are increasing, dividing by g_{J+1} gives the weighted Gram bound. For $G = [I; -X]$ its exact final factor is $I + X^* X$, with no extra number-of-columns factor. The moment bound is usually much sharper, but a growing moment order controls only its geometric remainder; it does not force the leading moment Gram to vanish.

These estimates control certification of the low-mode correction. The simultaneous certificate above settles the complete even form at $\lambda = 4$. The complete odd sign is established below, while the corresponding growing-window head/inverse estimates remain unproved; positivity of a remote tail alone does not supply them.

The time–frequency geometry suggests a canonical family of candidate projections P_a . Put $\lambda = e^a$ and consider the Slepian concentration operator at time–bandwidth parameter $c \asymp \lambda^2$. Connes–Consani construct corrected co-Poisson “prolate vectors” from the highly concentrated PSWFs and observe that the number of extremely small semilocal Weil eigenvalues grows on the same $\asymp \lambda^2$ scale [13]. Quantitative Slepian eigenvalue bounds sharpen the geometric picture. If $\theta_j(c)$ are the concentration eigenvalues and one chooses

$$\delta_\lambda = \lambda^{-K},$$

then the number of indices in the plunge region

$$\delta_\lambda < \theta_j(c) < 1 - \delta_\lambda$$

is at most

$$O\left(\log c \log \frac{1}{\delta_\lambda}\right) = O_K(a^2), \quad (\text{G2.5})$$

using the standard quantitative transition estimates for one-dimensional time–frequency limiting operators [14]. Thus the concentration geometry has three distinct pieces:

$$\boxed{\mathcal{H}_a = \mathcal{R}_a^{\text{deep}} \oplus \mathcal{T}_a^{\text{plunge}} \oplus \mathcal{C}_a^{\text{low}},} \quad (\text{G2.6})$$

where $\dim \mathcal{R}_a^{\text{deep}} \asymp \lambda^2$, the plunge block has dimension $O_K(a^2)$, and $\mathcal{C}_a^{\text{low}}$ is an infinite-dimensional low-concentration complement.

Equation (154), now proved using Theorem 20.1, controls fixed-band pairings for the fixed-order proxy treated above. It does *not* prove the operator-norm estimate $\|A_a P_a\| \rightarrow 0$ uniformly on the full deep block. A possible route would combine the exact global-radical identity with the concentration defect, but it needs a uniform bound on the truncation cost in the relevant form or graph norm, as well as normalized endpoint control for the growing family. Neither follows from the fixed-source approximation theorem, whose constants may depend on the source. Thus the growing-block estimate remains an open subgate, not a consequence of the weak G1 closure.

More importantly, the small plunge dimension does *not* dispose of the complement. Time–frequency concentration alone supplies no sign for the Weil form on $\mathcal{C}_a^{\text{low}}$. One sufficient G2 route seeks a coercivity estimate

$$\boxed{P_{\mathcal{C}_a^{\text{low}}} A_a P_{\mathcal{C}_a^{\text{low}}} \succeq c_a I} \quad (\text{G2.7})$$

together with control of the couplings to the deep and plunge blocks. Proposition 20.41 gives an alternative: a uniform operator residual on the removed block and asymptotic nonnegativity of its whole complement suffice, without a strictly positive gap. Propositions 20.43–20.44 make one limitation exact: finite source removal leaves the unsigned prime norm unchanged, and that norm is $(1 + o(1))e^a$. Subtracting this scalar from the minimum archimedean tail diagonal forces the expensive cutoff (233). This does not exclude a frequency-weighted or directional estimate; the recurrence vectors restoring the prime norm may have much larger archimedean energy than that minimum.

This three-block formulation is useful mainly because it localizes the unresolved content. Classical prolate theory controls the geometry and shows that the genuine transition region is only polynomial

in a ; the RH-strength issue is the semilocal Weil coercivity of the low-concentration complement and the uniform residual control of the entire deep near-radical block. Neither statement follows from Slepian concentration by itself.

Finite trial positivity and the exact omitted-tail correction

The trial-head term in Proposition 20.72 must be distinguished from the exact Schur complement. A finite-support certificate can settle the former without settling the latter.

Proposition 20.73 (Finite-support trial forms and Galerkin Schur bounds). *Let the outer head be finite dimensional, let $T \succeq cI$ with $c > 0$ be a positive self-adjoint tail operator, and let B be the bounded head-to-tail coupling. For X with range in $\mathcal{D}(T)$, define K_X, r as in Proposition 20.72 and put $S_\infty = F - B^*T^{-1}B$. Then*

$$K_X - S_\infty = (X - T^{-1}B)^*T(X - T^{-1}B) \succeq 0, \quad (272)$$

$$S_\infty = K_X - r^*T^{-1}r. \quad (273)$$

Let P_M be the orthogonal projection onto a finite tail space in $\mathcal{D}(T)$, and write $T_M = P_M T|_{\text{Ran } P_M}$, $B_M = P_M B$ and $S_M = F - B_M^* T_M^{-1} B_M$. If $X = P_M X$, then

$$K_X = \begin{bmatrix} I \\ -X \end{bmatrix}^* \begin{bmatrix} F & B_M^* \\ B_M & T_M \end{bmatrix} \begin{bmatrix} I \\ -X \end{bmatrix}. \quad (274)$$

In particular, $S_M \succeq 0$ implies $K_X \succeq 0$ for every such X , and strict positivity transfers as well. This is the complete operator's K_X , not a truncation of that trial form.

For the exact finite solve $X_M = T_M^{-1} B_M$, embedded in the complete tail, one has $K_{X_M} = S_M$ and

$$\boxed{S_\infty = S_M - r_M^* T^{-1} r_M, \quad r_M = B - T X_M.} \quad (275)$$

If the finite tail spaces are nested and their union is a form core for T , then $S_M \downarrow S_\infty$ in matrix norm.

Proof. Expanding the first square gives $K_X - F + B^*T^{-1}B$. Since $r = -T(X - T^{-1}B)$, its value is $r^*T^{-1}r$, which proves both identities. In (274) every trial lies in the head plus the retained finite tail space, so compression changes none of the quadratic terms. Finite square completion gives

$$K_X = S_M + (X - T_M^{-1}B_M)^*T_M(X - T_M^{-1}B_M).$$

Thus $K_X \succeq S_M$, and the graph map $h \mapsto (h, -Xh)$ is injective. Substitution of the exact finite solve proves (275); its retained residual rows vanish, but its omitted rows need not vanish.

For each head vector h , the variational identity is

$$\langle B^*T^{-1}Bh, h \rangle = \sup_{y \in \mathcal{D}(T^{1/2})} \{2\Re \langle Bh, y \rangle - \|T^{1/2}y\|^2\}.$$

Restricting the supremum to $\text{Ran } P_M$ gives $\langle B_M^*T_M^{-1}B_Mh, h \rangle$. Nestedness and form-core density make these quantities increase to the full supremum. Polarization and finite dimensionality of the head give matrix-norm convergence of the Schur complements in the stated decreasing order. $\square$

A finite list of positive S_M therefore does not prove positivity of S_∞ . Positivity for every member of a cofinal sequence at a fixed window would imply $S_\infty \succeq 0$, but strict positivity would still require a positive limiting margin. The structured inverse bounds concern this same T ; a comparison metric or a shifted lower operator does not redefine the target Schur complement.

Remark 20.74 (The semidefinite range condition). For $T \succeq 0$, the notation $T^{-1}B$ requires an additional condition. A form version is available if $\text{Ran } B \subset \text{Ran } T^{1/2}$. Define the reduced inverse $C = T^{-1/2}B$, with values orthogonal to $\ker T$. For X with range in the form domain, use $(T^{1/2}X)^*T^{1/2}X$ in the definition of K_X . Then

$$K_X - (F - C^*C) = (T^{1/2}X - C)^*(T^{1/2}X - C) \succeq 0.$$

Orthogonality to $\ker T$, or membership in the closure of its range, does not suffice when the range is not closed. At the certified $\lambda = 4$ split there is no such issue: Proposition 20.48 gives $T \succeq 10^{-8}D \succeq 1.7940 \cdot 10^{-8}I$ on Q_{16} .

Proposition 20.75 (A certified local zero head-error term). *For the actual $\lambda = 4$ Weil form with outer head E_{16} , $K_X \succ 0$ in both parity sectors for $X = 0$ and for the frozen dyadic solves supported on modes $17, \dots, 256$ specified in Appendix A. Consequently $\alpha_4 = 0$ is admissible in Proposition 20.72 for each of these choices, on the complete operator.*

Proof. Assemble the exact finite entries with the separate logarithmic diagonal, normalized parity basis and $L = 2 \log 4$. Outward interval LDL^* factorization of K_X verifies all 17 even pivots and all 16 odd pivots strictly positive, both for $X = 0$ and for the specified dyadics. The identical witnesses pass at 768 and 896 bits; Appendix A gives the files and gates. Real symmetric positivity also proves positivity on the complex parity spaces. Equation (274) identifies these matrices with the complete trial forms. No infinite Schur sign or small residual is used in this argument. $\square$

This result does not certify $\alpha_\lambda = 0$ at untested windows or for larger unsupported solves. Positivity of an exact Schur complement would guarantee $K_X \succeq 0$ for every trial; without that premise the least admissible α can depend on X . For example, $F = 0$, $B = T = 1$ gives $S = -1$, $K_0 = 0$ and $K_1 = -1$. Even $K_0 \succ 0$ need not help the signed target: its residual is the entire coupling B .

With a proved $\alpha_\lambda = 0$ along a growing family, the precise remaining sufficient estimates would be

$$e_\lambda \longrightarrow 0, \quad t_\lambda / \sqrt{\mu_\lambda} \longrightarrow 0. \quad (276)$$

The second condition is stronger than $t_\lambda \rightarrow 0$ when $\mu_\lambda \rightarrow 0$. Geometric convergence of an inverse approximation improves an enclosure of $r_\lambda^* T_\lambda^{-1} r_\lambda$; it does not force that residual energy to vanish for a fixed trial. These growing-window estimates remain unproved. Failure of this sufficient bound would not itself refute Weil positivity.

An inverse-weighted finite spectral certificate

The complement estimate can also identify a single candidate without dividing its entire residual by the smallest spectral gap. The following finite statement is independent of any PSWF approximation.

Proposition 20.76 (Inverse-weighted finite certificate). *Let W be a finite Hermitian matrix, u a unit vector, and P the orthogonal projection onto $u^\perp$. Put*

$$\alpha = \langle Wu, u \rangle, \quad r = PWu, \quad C = PWP|_{u^\perp}, \quad A = C - \alpha I.$$

If $A \succeq \gamma I$ for some $\gamma > 0$, then W has a simple lowest eigenvalue $\lambda_0 \leq \alpha$. Set $q = \|A^{-1}r\|$ and $E = \langle A^{-1}r, r \rangle$. For the angle θ between u and the lowest eigenspace,

$$\tan \theta \leq q, \quad \sin \theta \leq \frac{q}{\sqrt{1+q^2}}, \quad \alpha - E \leq \lambda_0 \leq \alpha. \quad (277)$$

If W commutes with reflection and u is even, its ground state is even. For any approximate solve $z \in u^\perp$ one also has

$$q \leq \|z\| + \frac{\|r - Az\|}{\gamma}. \quad (278)$$

Proof. Min-max gives $\lambda_0 \leq \alpha$ and $\lambda_1 \geq \inf \sigma(C) > \alpha$. Write a ground vector as $bu + h$, where $h \perp u$ and $b \neq 0$. The complementary equation gives $h = -b(C - \lambda_0 I)^{-1}r$. With $t = \alpha - \lambda_0 \geq 0$, the scalar equation is $t = \langle (A + tI)^{-1}r, r \rangle$. Spectral calculus for the same positive operator A gives $t \leq E$ and $\|(A + tI)^{-1}r\| \leq \|A^{-1}r\|$. These prove (277). An odd ground state would be orthogonal to the even u , contradicting the complement bound. Finally $A^{-1}r = z + A^{-1}(r - Az)$ proves (278). $\square$

For certified approximations $\tilde{A}, \tilde{r}$, suppose $\|A - \tilde{A}\| \leq \delta_A$, $\|r - \tilde{r}\| \leq \delta_r$ and $\tilde{A} \succeq \tilde{\gamma}I$ with $\tilde{\gamma} > \delta_A$. Then the usable error budget is

$$q \leq \|z\| + \frac{\delta_r + \delta_A \|z\| + \|\tilde{r} - \tilde{A}z\|}{\tilde{\gamma} - \delta_A}.$$

For a fixed exact candidate and matrix error $\|W - \tilde{W}\| \leq \eta$, one may take $\delta_r = \eta$ and $\delta_A = 2\eta$. True-source approximation error is a separate input. The condition $\gamma > 0$ is sufficient, not necessary, for source-to-ground convergence: in $W = \text{diag}(0, d, 1)$, the candidate $u = \sqrt{1 - \tau^2}e_0 + \tau e_2$ has angle tending to zero but $\gamma \leq d - \tau^2 < 0$ when $0 < d < \tau^2$.

Appendix A certifies the hypotheses for one explicit rational candidate at $\lambda = 3$, $N = 64$, using the actual arithmetic Weil matrix. That certificate alone proves no uniform complement theorem or exact-source error bound. The following argument supplies the latter at its fixed parameters; endpoint and graph transfer remain separate.

From a finite prolate approximation to the exact source

The preceding certificate treats its input vector as exact. To identify that vector with a projection of the manuscript's source, the omitted Legendre tail must also be controlled. Here this can be done at fixed parameters, including pointwise values rather than only an L^2 error.

Lemma 20.77 (Certified angular eigenfunction error). *On $L^2(-1, 1)$ let $e_\ell(x) = \sqrt{(2\ell + 1)/2} P_\ell(x)$, $De_\ell = \ell(\ell + 1)e_\ell$, and $J_c = D + c^2 X^2$, where $Xf(x) = xf(x)$ and $\mathcal{D}(J_c) = \mathcal{D}(D)$. Restrict to the even subspace. Let J_M be its compression to $e_0, e_2, \dots, e_{2M-2}$, let b_M be the link from its last coordinate to e_{2M} , and put $d_M = 2M(2M + 1)$. For $x < d_M$, the number of eigenvalues of J_c below x is between the negative inertia counts of*

$$J_M - xI, \quad J_M - xI - \frac{b_M^2}{d_M - x} E_{M-1, M-1}. \quad (279)$$

If both matrices are nonsingular with the same count, that count is exact.

Suppose these counts certify exactly one eigenvalue ν in $(\xi - g, \xi + g)$ and no other spectrum there, with $g > 0$. Let $\xi \in \mathbb{R}$, let v be an exact normalized real vector in this even finite span, and

suppose $\|(J_c - \xi)v\| \leq \epsilon < g$. For the normalized eigenfunction ψ at ν , with sign chosen so that $\langle v, \psi \rangle > 0$, put $\delta = \sqrt{2}\epsilon/g$. Then

$$\begin{aligned} |\nu - \xi| &\leq \epsilon, & \|v - \psi\|_2 &\leq \delta, \\ \|v - \psi\|_\infty &\leq 2\epsilon + (1 + |\xi| + c^2)\delta. \end{aligned} \quad (280)$$

The residual includes the first omitted coordinate $b_M v_{M-1}$. The gap g is measured from ξ to the other exact eigenvalues.

Proof. The diagonal operator D is self-adjoint with compact resolvent; $c^2 X^2$ is bounded, positive and self-adjoint. Its angular realization is the regular PSWF operator (the differential spectral parameter is shifted by c^2 relative to the convention in [19]). The omitted compression T satisfies $T \succeq d_M I$. For $x < d_M$ its positive inverse gives the finite Schur complement

$$J_M - xI - b_M^2 \left\langle (T - xI)^{-1} e_{2M}, e_{2M} \right\rangle E_{M-1, M-1}.$$

The scalar lies between 0 and $(d_M - x)^{-1}$. Inertia and min-max therefore prove (279). Completing the square in the quadratic form justifies the Schur argument for the unbounded T .

The spectral theorem places an eigenvalue within ϵ of ξ ; the certified interval identifies it as ν . All other coefficients of v have squared mass at most $(\epsilon/g)^2$. Choosing positive overlap gives $\|v - \psi\|_2 \leq \sqrt{2}\epsilon/g$. For $e = v - \psi$,

$$J_c e = \xi e + (\xi - \nu)\psi + (J_c - \xi)v,$$

so $\|(I + D)e\|_2 \leq 2\epsilon + (1 + |\xi| + c^2)\delta$. Finally $|P_\ell(x)| \leq 1$ on $[-1, 1]$ and

$$\sum_{\ell \geq 0} \frac{(2\ell + 1)/2}{[1 + \ell(\ell + 1)]^2} \leq \frac{1}{2} + \frac{1}{2} \sum_{\ell \geq 1} \left(\frac{1}{\ell^2} - \frac{1}{(\ell + 1)^2} \right) = 1.$$

Cauchy-Schwarz gives uniform convergence of the Legendre expansion and $\|e\|_\infty \leq \|(I + D)e\|_2$, including both endpoint traces. $\square$

Here is the source error budget used below. Set $\lambda = 3$ and let $\hat{h}_j = \lambda^{-1/2} v_j(\cdot/\lambda)$ approximate $h_{j,\lambda}$ for $j = 0, 4$. Let d_j and u_j bound their physical L^2 and uniform errors. Define

$$\hat{a}_0 = - \int \hat{h}_4, \quad \hat{a}_4 = \int \hat{h}_0, \quad \hat{h}^\circ = \hat{a}_0 \hat{h}_0 + \hat{a}_4 \hat{h}_4.$$

Both this polynomial and the exact h° have zero integral. With $d_{a_0} = \sqrt{2\lambda} d_4$ and $d_{a_4} = \sqrt{2\lambda} d_0$, a valid uniform error is

$$\eta = \sum_{j=0,4} \{(|\hat{a}_j| + d_{a_j})u_j + d_{a_j} \|\hat{h}_j\|_\infty\}. \quad (281)$$

Use the fixed repair with $b = 3/4$, $\phi(t) = \exp(1 - 1/(1 - (t/b)^2))$ for $|t| < b$, zero outside, and $\psi(t) = \phi(t)(1 - Ct^2)$ where $C = \int_0^b \phi / \int_0^b t^2 \phi$. Since $\phi \leq 1$ everywhere and $\phi \geq 1/2$ on $[b/4, b/2]$, $Cb^2 \leq 128$ and $\|\psi\|_\infty \leq 129$. Consequently, with $Z = |\hat{h}^\circ(0)|$,

$$\|\tilde{h} - \hat{h}^\circ\|_\infty \leq U := \eta + 129(Z + \eta). \quad (282)$$

This bounds the entire exact repair without assuming a numerical quadrature value for C .

Let $\hat{p}$ be the inversion-even co-Poisson proxy of $\hat{h}^\circ$ on the same window. At almost every $u \in [\lambda^{-1}, \lambda]$, at most λ/u summands occur, so the physical factor $\sqrt{u}$ gives $|\mathcal{E}(\tilde{h} - \hat{h}^\circ)(u)| \leq \lambda^{3/2}U$. Inversion averaging is an $L^2(du/u)$ contraction. Bessel's inequality and the strict interior endpoint formula therefore give

$$\|P_{Np_3} - P_N \hat{p}\|_2 \leq \sqrt{\lambda^3 L} U, \quad L = 2 \log 3, \quad (283)$$

$$|B_3 - \hat{B}| \leq \frac{1}{2} \left(\sqrt{\lambda} \eta + 8\lambda^{-1/2} U \right), \quad (284)$$

where $\hat{B} = \frac{1}{2} \{ \sqrt{\lambda} \hat{h}^\circ(\lambda) + \lambda^{-1/2} \sum_{m=1}^8 \hat{h}^\circ(m/\lambda) \}$. The $m = 9$ term is excluded from this lower interior trace; the repair vanishes at the upper endpoint. These are fixed-parameter estimates, not a growing-window spectral theorem.

Proposition 20.78 (Exact-source angle and endpoint at one window). *Use the exact repaired source of Proposition 20.5 with the bump just specified. At $\lambda = 3, N = 64$, let c^* be the unnormalized rational coefficient vector in Proposition A.1, and let v^* be the coefficient vector of the exact P_{64p_3} in the same translated Fourier basis. The interval certificate in Appendix A proves*

$$\|v^* - c^*\|_2 < 2.60 \cdot 10^{-36}, \quad \sin \angle(v^*, \text{ground state of } W_{3,64}) < 0.000216.$$

It also proves

$$5.58669 \cdot 10^{-19} < B_3 < 5.58670 \cdot 10^{-19}, \quad 1.374 < \frac{|\delta_{64}(P_{64p_3}) - B_3|}{B_3} < 1.375. \quad (285)$$

Thus the small ordinary ground-state angle coexists with a relative physical-endpoint error greater than one for this exact projected source.

Proof. The appendix certifies all error budgets above, including the full infinite angular residuals, and the exact polynomial Fourier moments. It gives the first bound and the displayed endpoint intervals. For nonzero vectors v^*, c^* with error at most $e < \|c^*\|$,

$$\|P_{Cv^*} - P_{Cc^*}\| = \sin \angle(v^*, c^*) \leq \frac{e}{\|c^*\| - e}.$$

This is below $4 \cdot 10^{-36}$ here. The triangle inequality for rank-one orthogonal projections and the sharper enclosed sine bound from Proposition A.1 prove the angle claim. No new positive-complement estimate for the perturbed candidate is assumed. Equation (285) concerns physical endpoint evaluation, not the equal-coefficient functional without its $L^{-1/2}$ factor. $\square$

21 Weighted G1 transfer and the arithmetic jump train

Proposition 21.1 (Cofinal finite G1 bound on a weighted sampler class). *Use the explicit repaired source of Section 20. Let $R > 3$, let $K \geq 2N$, and write*

$$x = \sum_{|m| \leq K} x_m U_m \in E_K, \quad \|x\|_{\mathcal{H}_{R,L}} := \sup_{|m| \leq K} \sqrt{L} \left(1 + \frac{|m|}{L} \right)^R |x_m|.$$

If $N \geq 2L$, then the filtered proxy of Proposition 20.35 satisfies

$$\left| \frac{\langle D_{\log} W_{\lambda,K} k_{\lambda,N}^{\text{VP}}, x \rangle}{|B_\lambda|} \right| \leq C_R \|x\|_{\mathcal{H}_{R,L}} \left[\mathfrak{r}(\lambda) + \frac{\lambda \mathcal{V}_\lambda}{|B_\lambda|} \left\{ \frac{L}{N} + \left(\frac{L}{N} \right)^{R-1} (1 + \log(2N)) \right\} \right]. \quad (286)$$

The bound is uniform in the ambient cutoff K . Thus, once the arithmetic core cutoff N has been fixed, enlarging the Hardy-sampler cutoff cannot spoil this G1 estimate.

Proof. Put $e_N = k_{\lambda,N}^{\text{VP}} - p_\lambda$. Because the Vallee–Poussin multiplier is one for $|n| \leq N$, one has $\widehat{e}_N(n) = 0$ there, while for $|n| > N$ the BV estimate gives

$$|\widehat{e}_N(n)| \leq |\widehat{p}_\lambda(n)| \leq \frac{\sqrt{L}\mathcal{V}_\lambda}{2\pi|n|}.$$

The coefficient hypothesis and $R > 3$ imply, by comparison with Riemann sums,

$$\|D_{\log x}\|_2 + \|D_{\log x}^2\|_2 + |(D_{\log x})(-a)| + |(D_{\log x})(a)| \leq C_R \|x\|_{\mathcal{H}_{R,L}}.$$

Indeed the two L^2 sums reduce to $L^{-1} \sum_m (|m|/L)^{2j} (1 + |m|/L)^{-2R}$, $j = 1, 2$, and each endpoint trace to $L^{-1} \sum_m (|m|/L)(1 + |m|/L)^{-R}$. Proposition 20.13, applied to $D_{\log x}$, therefore gives

$$|QW_\lambda(p_\lambda, D_{\log x})| \leq C_R B_\lambda |\mathfrak{r}(\lambda)| \|x\|_{\mathcal{H}_{R,L}}.$$

For the projection error separate the off-diagonal and diagonal matrix entries. Lemma 20.17 gives, uniformly in K ,

$$\begin{aligned} |QW_\lambda(e_N, D_{\log x})|_{\text{off}} &\leq C_R \lambda \mathcal{V}_\lambda \|x\|_{\mathcal{H}_{R,L}} \sum_{|n| > N} \frac{1}{|n|} \sum_{m \neq n} \frac{a_m}{|n - m|}, \\ a_m &:= \frac{|m|}{L} \left(1 + \frac{|m|}{L}\right)^{-R}. \end{aligned}$$

For nonzero integer $n - m$ one may replace $|n - m|^{-1}$ by $2(1 + |n - m|)^{-1}$. For $|n| > N \geq 2L$, split the inner sum into $|m| \leq |n|/2$, $|n|/2 < |m| \leq 2|n|$, and $|m| > 2|n|$. Since $\sum_m a_m \leq C_R L$, the first range is $O_R(L/|n|)$. On the middle range, $a_m \leq C_R(|n|/L)^{1-R}$ and the harmonic sum contributes $O(1 + \log(2|n|))$. On the last range the denominator is $\gg |m|$, and summation of m^{-R} gives $O_R((|n|/L)^{1-R})$. Hence

$$\sum_{m \neq n} \frac{a_m}{|n - m|} \leq C_R \left[\frac{L}{|n|} + \left(\frac{|n|}{L}\right)^{1-R} (1 + \log(2|n|)) \right],$$

and therefore

$$|QW_\lambda(e_N, D_{\log x})|_{\text{off}} \leq C_R \lambda \mathcal{V}_\lambda \|x\|_{\mathcal{H}_{R,L}} \left[\frac{L}{N} + \left(\frac{L}{N}\right)^{R-1} (1 + \log(2N)) \right]. \quad (287)$$

The diagonal must be handled separately. Corollary 20.18, the BV coefficient estimate, and the definition of $\mathcal{H}_{R,L}$ give

$$\begin{aligned} |QW_\lambda(e_N, D_{\log x})|_{\text{diag}} &\leq C \frac{\mathcal{V}_\lambda}{L} \|x\|_{\mathcal{H}_{R,L}} \\ &\quad \times \sum_{N < |n| \leq K} (\lambda + 1 + \log(2 + |n|/L)) \left(1 + \frac{|n|}{L}\right)^{-R}. \end{aligned}$$

Comparison with the corresponding decreasing integral, using $N \geq 2L$ and $R > 3$, yields

$$|QW_\lambda(e_N, D_{\log x})|_{\text{diag}} \leq C_R \mathcal{V}_\lambda \|x\|_{\mathcal{H}_{R,L}} (\lambda + 1 + \log(2N)) \left(\frac{L}{N}\right)^{R-1}. \quad (288)$$

Because $\lambda \geq 2$, (288) is absorbed by the second term on the right of (287). Thus the projection-error contribution has exactly the rate stated in (286), despite the logarithmic diagonal growth. Finally $\langle D_{\log} W_{\lambda,K} k^{\text{VP}}, x \rangle = QW_\lambda(k^{\text{VP}}, D_{\log x})$ because both finite vectors lie in E_K . Combining the two pieces proves (286). $\square$

Exact extraction of the arithmetic jump train

The BV estimate above treats all variation of the prolate/co-Poisson proxy in one lump. That is unnecessarily expensive after normalization by the tiny endpoint amplitude. The actual discontinuities have a much more rigid arithmetic form and can be separated exactly.

Let $y = \log(\lambda u) \in [0, L]$ and let $B_\lambda^+ = \sqrt{\lambda} \tilde{h}_\lambda(\lambda^-)$ denote the upper source boundary amplitude. Put

$$\mathcal{M}_\lambda := \{m \in \mathbb{N} : 1 < m < \lambda^2\}.$$

The fixed smooth exact-moment repair used above is supported strictly inside the source interval, so it creates no endpoint jumps.

Lemma 21.2 (Exact arithmetic jump train). *There is a zero-mean periodic BV function j_λ on the logarithmic circle such that its distributional derivative is*

$$dj_\lambda = \frac{B_\lambda^+}{2} \sum_{m \in \mathcal{M}_\lambda} m^{-1/2} (\delta_{\log m} - \delta_{L - \log m}). \quad (289)$$

Moreover

$$c_\lambda := p_\lambda - j_\lambda$$

is continuous and periodic, is piecewise smooth, and its first logarithmic derivative has bounded variation. If

$$t_n := \frac{2\pi n}{L}, \quad \mathcal{D}_\lambda(t) := \sum_{m \in \mathcal{M}_\lambda} m^{-1/2} \sin(t \log m),$$

then for every $n \neq 0$,

$$\widehat{j}_\lambda(n) = -\frac{B_\lambda^+}{t_n \sqrt{L}} \mathcal{D}_\lambda(t_n). \quad (290)$$

Consequently, for all sufficiently large λ ,

$$|\widehat{j}_\lambda(n)| \leq C |B_\lambda| \frac{\lambda \sqrt{L}}{|n|}. \quad (291)$$

Finally, if

$$\mathcal{W}_\lambda := \|D_y^2 c_\lambda\|_{\mathcal{M}(\mathbb{T}_L)}$$

denotes the total variation of the distributional second derivative of the continuous remainder, then

$$|\widehat{c}_\lambda(n)| \leq \frac{L^{3/2} \mathcal{W}_\lambda}{4\pi^2 n^2}, \quad n \neq 0. \quad (292)$$

All quantities in the statement are determined by the explicit source and λ ; no zeta zero is used.

Proof. For $f_\lambda(u) = \mathcal{E}(\tilde{h}_\lambda)(u)$, the summand with index m disappears as u increases through $u = \lambda/m$. Its one-sided jump is therefore

$$-\left(\frac{\lambda}{m}\right)^{1/2} \tilde{h}_\lambda(\lambda^-) = -\frac{B_\lambda^+}{\sqrt{m}}.$$

In the y coordinate this occurs at $L - \log m$. The inverted function $f_\lambda(u^{-1})$ has the opposite jump at $\log m$. Since $p_\lambda = \frac{1}{2}(f_\lambda + Jf_\lambda)$ on the window, its complete interior jump measure is exactly

(289). The total mass of that measure is zero, so it is the derivative of a periodic BV function; fixing zero mean makes j_λ unique. Subtracting it removes every function jump. The arithmetic sum has only finitely many thresholds and is smooth between them, so c_λ is continuous, piecewise smooth, and $D_y c_\lambda$ is BV.

For $n \neq 0$, periodic Stieltjes integration by parts gives

$$\widehat{j_\lambda}(n) = \frac{1}{it_n \sqrt{L}} \int_0^L e^{-it_n y} dj_\lambda(y).$$

Because $e^{-it_n L} = 1$, (289) yields

$$\int_0^L e^{-it_n y} dj_\lambda(y) = \frac{B_\lambda^+}{2} \sum_{m \in \mathcal{M}_\lambda} m^{-1/2} \{m^{-it_n} - m^{it_n}\} = -iB_\lambda^+ \mathcal{D}_\lambda(t_n),$$

which is (290). The elementary estimate

$$|\mathcal{D}_\lambda(t)| \leq \sum_{m < \lambda^2} m^{-1/2} \leq 2\lambda$$

and $B_\lambda^+ \asymp B_\lambda$ from Corollary 20.9 prove (291).

For the continuous remainder there is no first-order boundary term. A second periodic Stieltjes integration by parts, now against the finite measure $D_y^2 c_\lambda$, gives

$$|\widehat{c_\lambda}(n)| \leq \frac{\mathcal{W}_\lambda}{\sqrt{L} t_n^2} = \frac{L^{3/2} \mathcal{W}_\lambda}{4\pi^2 n^2},$$

proving (292). □

The exact sine sum in (290) is potentially useful in its own right: any unconditional exponential-sum improvement immediately sharpens the cutoff below. Even the trivial bound already removes the factor $\mathcal{V}_\lambda/|B_\lambda|$ from the discontinuous arithmetic part.

Proposition 21.3 (Jump/continuous split for the weighted G1 transfer). *Let $R > 3$, $K \geq 2N$, $N \geq 2L$, and let $x \in E_K$ be measured by $\|x\|_{\mathcal{H}_{R,L}}$ as in Proposition 21.1. Then*

$$\begin{aligned} & \frac{|QW_\lambda((V_N - I)p_\lambda, D_{\log} x)|}{|B_\lambda|} \\ & \leq C_R \|x\|_{\mathcal{H}_{R,L}} \left[\lambda^2 \left\{ \frac{L}{N} + \left(\frac{L}{N} \right)^{R-1} (1 + \log(2N)) \right\} \right. \\ & \quad \left. + \frac{\lambda \mathcal{W}_\lambda}{|B_\lambda|} \left\{ \left(\frac{L}{N} \right)^2 + \left(\frac{L}{N} \right)^R (1 + \log(2N)) \right\} \right]. \end{aligned} \tag{293}$$

The first bracket is the entire contribution of the arithmetic function jumps and has no small-boundary denominator. The only surviving endpoint-normalized regularity cost belongs to the continuous remainder and gains one additional power of N^{-1} .

Proof. Write $p_\lambda = j_\lambda + c_\lambda$ and use that the Vallee–Poussin multiplier is exactly one on $|n| \leq N$. For the jump part, (291) replaces the BV coefficient estimate used in Proposition 21.1 by

$$|(\widehat{V_N - I})j_\lambda(n)| \leq C|B_\lambda| \frac{\lambda \sqrt{L}}{|n|}, \quad |n| > N.$$

Repeating the off-diagonal convolution estimate there therefore replaces $\lambda \mathcal{V}_\lambda / |B_\lambda|$ by $C\lambda^2$. The diagonal bound in Corollary 20.18 is absorbed by the same second term because $\lambda \geq 2$.

For the continuous remainder use (292). With

$$a_m = \frac{|m|}{L} \left(1 + \frac{|m|}{L}\right)^{-R},$$

the off-diagonal part is bounded by

$$C_R \lambda L \mathcal{W}_\lambda \|x\|_{\mathcal{H}_{R,L}} \sum_{|n| > N} \frac{1}{n^2} \left[\frac{L}{|n|} + \left(\frac{|n|}{L}\right)^{1-R} (1 + \log(2|n|)) \right].$$

Comparison with decreasing integrals gives

$$C_R \lambda \mathcal{W}_\lambda \|x\|_{\mathcal{H}_{R,L}} \left[\left(\frac{L}{N}\right)^2 + \left(\frac{L}{N}\right)^R (1 + \log(2N)) \right].$$

Using the diagonal estimate from Corollary 20.18 with the n^{-2} coefficient bound produces no larger term. Division by $|B_\lambda|$ proves (293). $\square$

Corollary 21.4 (The arithmetic jumps are resolvable at near-Slepian scale). *Fix $R > 3$ and put*

$$N_\lambda^{\text{jump}} := \left\lceil C_R \frac{\lambda^2 L}{\mathfrak{r}(\lambda)} \right\rceil. \quad (294)$$

For C_R sufficiently large, the jump contribution in (293) is $O_R(\mathfrak{r}(\lambda))$ uniformly for every ambient cutoff $K \geq 2N_\lambda^{\text{jump}}$. Moreover

$$\boxed{N_\lambda^{\text{jump}} = \lambda^{2+o(1)}}. \quad (295)$$

At the same cutoff, the continuous-remainder contribution is

$$O_R \left(\frac{\mathcal{W}_\lambda}{|B_\lambda|} \frac{\mathfrak{r}(\lambda)^2}{\lambda^3} \right) \quad (296)$$

(up to a smaller R -dependent logarithmic term). Thus the concrete zero-independent condition

$$\boxed{\frac{\mathcal{W}_\lambda}{|B_\lambda|} \ll_R \frac{\lambda^3}{\mathfrak{r}(\lambda)}} \quad (297)$$

would upgrade the filtered $G1$ transfer to the natural $\lambda^{2+o(1)}$ Fourier scale. Condition (297) is not proved here; it is a newly isolated regularity subgate rather than an RH assumption.

Proof. The first term in (293) is made $O_R(\mathfrak{r}(\lambda))$ by (294); the second high-order term is smaller because $R > 3$ and $\mathfrak{r}(\lambda) < 1$ for large λ . Since

$$\log \frac{1}{\mathfrak{r}(\lambda)} = O(\sqrt{L} + \log L) = o(L)$$

while $\lambda^2 = e^L$, one has $\log N_\lambda^{\text{jump}} = L + o(L)$, proving (295). Finally $L/N_\lambda^{\text{jump}} \asymp \mathfrak{r}(\lambda)/\lambda^2$; substitution in the continuous term of (293) gives (296). Requiring it to be $O_R(\mathfrak{r}(\lambda))$ gives (297). $\square$

A polynomial cutoff for the literal source

The small physical endpoint need not force an exponentially large Fourier cutoff for this fixed two-mode source. The additional ingredient is an exterior norm estimate for a finite entire extension of the arithmetic sum. It controls the endpoint derivative jets without bounding the whole interior variation by the endpoint amplitude.

Lemma 21.5 (Polynomial approximation with a growing derivative order). *Fix a compact real interval I of positive length. Suppose*

$$F(z) = \int_{-\Omega}^{\Omega} e^{izt} d\mu(t), \quad \Omega \leq A_0 c, \quad \|\mu\| \leq c^{A_1}.$$

For each fixed $H > 0$, there is $A > 0$ such that a polynomial T of degree $D = \lceil Ac \rceil$ satisfies, simultaneously for $0 \leq j \leq \lceil c \rceil$,

$$\left\| (F - T)^{(j)} \right\|_{L^\infty(I)} \leq e^{-Hc} (Cc)^j. \quad (298)$$

For all sufficiently large c , the constants are independent of c and j . Moreover,

$$\|T\|_\infty \leq C(D+1) \|T\|_2, \quad \left\| T^{(j)} \right\|_\infty \leq (CD^2)^j \|T\|_\infty$$

on I .

Proof. Take the degree- D Taylor polynomial about the midpoint of I under the finite measure. If d is the half-length of I , its differentiated remainder is at most

$$\|\mu\| \Omega^j e^{\Omega d} \frac{(\Omega d)^{D+1-j}}{(D+1-j)!}.$$

The inequality $k! \geq (k/e)^k$, with A sufficiently large, proves (298) for the entire stated range of j . The first polynomial inequality follows from normalized Legendre expansion, $|P_k| \leq 1$, and $\sum_{k=0}^D (2k+1) = (D+1)^2$. The second is Markov's inequality iterated on the derivatives, with the fixed interval scaling. Real and imaginary parts give the same bounds for complex polynomials. $\square$

Lemma 21.6 (Endpoint jets of the finite arithmetic sum). *Put $c = 2\pi\lambda^2$, $r = \lceil c \rceil$, and let p_λ° be the inversion-even proxy before the fixed value-at-zero repair. If J_j is the sum of absolute jumps of its j th ordinary logarithmic derivative, including the jump at the identified periodic endpoint, then*

$$J_j \leq C|B_\lambda^+| \lambda^3 (C\lambda^4)^j, \quad 1 \leq j < r. \quad (299)$$

Its zeroth-order periodic boundary jump is zero. On its smooth pieces, for fixed constants $A, C, \rho > 0$,

$$\left\| (p_\lambda^\circ)_{\text{ac}}^{(r)} \right\|_1 \leq CL\lambda^3 r! \rho^{-r} e^{Ac}. \quad (300)$$

Proof. Write

$$g(x) = \sqrt{\lambda} h_\lambda^\circ(\lambda x) = \sum_{j=0,4} a_{j,\lambda} \psi_{j,c}(x), \quad g(1) = B_\lambda^+.$$

The exact finite Fourier eigenrelation used in Lemma 20.4 represents g by a measure supported on $[-c, c]$ with total variation $O(\lambda)$; this is also the bandlimited extension in [19, §30.15]. Proposition 20.2 and endpoint noncancellation give $|g(x)| \leq C|B_\lambda^+|$ for $x \geq 1$.

Set $M = \lceil 2\lambda^2 \rceil$ and $G(v) = \sum_{m=1}^M g(mv/\lambda^2)$. Its measure has bandwidth $2\pi M = O(c)$ and total variation $O(\lambda^3)$. For $1/2 \leq v \leq 1$,

$$G(v) = \lambda v^{-1/2} f_\lambda^\circ(v/\lambda) + \sum_{\substack{m \leq M \\ mv \geq \lambda^2}} g(mv/\lambda^2)$$

almost everywhere. The last sum is $O(|B_\lambda^+|\lambda^2)$. Corollary 20.9, which includes the unrepaired source, and $dv/v = du/u$ therefore give

$$\|G\|_{L^2(1/2,1)} \leq C|B_\lambda^+|\lambda^2.$$

Since $|B_\lambda^+| \asymp c^{11/4}e^{-c}$, apply Lemma 21.5 with fixed sufficiently large H . Its polynomial inequalities imply

$$\begin{aligned} |G^{(j)}(1)| &\leq C|B_\lambda^+|\lambda^4(C\lambda^4)^j, \\ \sup_{1 \leq x \leq 3} |g^{(j)}(x)| &\leq C|B_\lambda^+|(C\lambda^4)^j, \quad 0 \leq j \leq r. \end{aligned}$$

For the second bound approximate g on $[1, 3]$ and use its already bounded sup norm, so the extra Legendre degree factor is unnecessary.

At the strict interior lower trace $v = 1+$, subtract from G precisely the terms with $m \geq \lambda^2$, including equality when λ^2 is an integer, and multiply by $\sqrt{v}/\lambda$. Their arguments lie in $[1, 3]$, their number is $O(\lambda^2)$, and differentiation costs at most 3^j . Thus the logarithmic trace jets are bounded by $C|B_\lambda^+|\lambda^3(C\lambda^4)^j$. To justify the conversion uniformly in j , use $(v\partial_v)^j = \sum_k S(j, k)v^k\partial_v^k$ and $S(j, k) \leq \binom{j}{k}j^{j-k}$; hence $\sum_k S(j, k)A^k \leq (A+j)^j$. Here $j = O(\lambda^2)$ and $A = C\lambda^4$. For the product factor, $|(\sqrt{v})^{(k)}|_{v=1} \leq k!$ and $\sum_{k=0}^j \binom{j}{k}k!A^{j-k} \leq 2A^j$ when $A \geq 2j$; thus its cost is uniform in j as well. At the upper trace only the first summand survives, giving the stronger bound without λ^3 .

Each interior derivative jump is a source endpoint jet times $m^{-1/2}$, with the reflected sign. Summing $m < \lambda^2$ costs $O(\lambda)$; the preceding trace estimates control the periodic jump. This proves (299). Inversion symmetry gives the zero function jump at the periodic boundary. Finally extend each smooth piece using its fixed finite collection of entire summands. On a complex logarithmic disk of fixed small radius ρ , every active source argument has imaginary part bounded by a fixed constant. The Fourier representation bounds the extension by $C\lambda^3e^{Ac}$. Cauchy's estimate, integrated over total length L , proves (300). No asymptotic remainder was differentiated, and the smooth compact repair is not used in this step. $\square$

Proposition 21.7 (Polynomial physical-endpoint recovery). *For the source of Proposition 20.5, repaired as in Corollary 20.6, set*

$$N_\lambda = \lceil \lambda^8(1+L) \rceil, \quad k_\lambda = P_{N_\lambda}p_\lambda.$$

For all sufficiently large λ ,

$$\frac{|\delta_{N_\lambda}(k_\lambda) - B_\lambda|}{|B_\lambda|} \leq \frac{C}{\lambda}, \quad (301)$$

$$|\widehat{p}_\lambda(n)| \leq C|B_\lambda| \frac{\lambda\sqrt{L}}{|n|}, \quad |n| > N_\lambda. \quad (302)$$

This is an asymptotic estimate with no certified numerical threshold or optimized exponent. The vector is the literal ordinary Fourier projection; no endpoint shell is added.

Proof. First omit the compact repair. The periodic proxy is BV and continuous at the identified endpoint, so the Dirichlet–Jordan theorem identifies its symmetric Fourier sum there with its common one-sided trace. We may therefore estimate the actual endpoint error by its symmetric Fourier tail. In the jump train of Lemma 21.2, the endpoint tail contributed by a threshold m is, up to sign,

$$\frac{B_\lambda^+}{\pi\sqrt{m}} \sum_{n>N} \frac{\sin(2\pi n \log m/L)}{n}.$$

Dirichlet summation and the bounded sine-series partial sums give

$$\left| \sum_{n>N} \frac{\sin(n\theta)}{n} \right| \leq C \min \left(1, \frac{1}{N|\sin(\theta/2)|} \right).$$

Separate the largest integer $m_0 < \lambda^2$: its contribution is $O(|B_\lambda^+|/\lambda)$ even when λ^2 approaches an integer. For every remaining m , $\lambda^2 - m \geq 1$. Splitting at $m = \lambda$ and $m = \lambda^2/2$ gives

$$\sum_{\substack{1 < m < \lambda^2 \\ m \neq m_0}} \frac{m^{-1/2}}{|\sin(\pi \log m/L)|} \leq C\lambda L(1 + \log \lambda).$$

Near the upper end use $\log(\lambda^2/m) \geq (\lambda^2 - m)/\lambda^2$ and sum the harmonic denominator. Thus the entire function-jump endpoint error divided by $|B_\lambda^+|$ is at most

$$C \left\{ \lambda^{-1} + \frac{\lambda L(1 + \log \lambda)}{N} \right\}. \quad (303)$$

This retains, rather than assumes away, near-threshold arithmetic resonance.

Integrate by parts $r = \lceil c \rceil$ times on the smooth pieces, retaining the jumps of every derivative and the periodic derivative jumps. With $t_n = 2\pi n/L$, the j th jump term in a normalized Fourier coefficient has magnitude at most $J_j/(\sqrt{L}|t_n|^{j+1})$. Its physical-endpoint tail is at most

$$\frac{2J_j}{L} \sum_{n>N} |t_n|^{-j-1} \leq \frac{CJ_j}{j} \left(\frac{L}{2\pi N} \right)^j.$$

By Lemma 21.6, summing $1 \leq j < r$ gives $C|B_\lambda^+|\lambda^7 L/N$ whenever $C\lambda^4 L/N < 1/2$. The final smooth-piece remainder, after endpoint summation and division by $|B_\lambda^+|$, is bounded by

$$C\lambda^A(1+L)^A e^{Ac} r! \left(\frac{CL}{N} \right)^{r-1}. \quad (304)$$

Here A, C are fixed and independent of r, λ . At $N = N_\lambda$, $r! \leq (C\lambda^2)^r$, so the logarithm of this bound is at most $-6c \log \lambda + O(c + \log \lambda + \log L)$. It is smaller than every fixed inverse power of λ . Combining this with (303) proves the first claim for p_λ° .

The same expansion gives, at every $|n| > N_\lambda$,

$$|\widehat{p}_\lambda^\circ(n)| \leq \frac{C|B_\lambda^+|}{\sqrt{L}} \left(\frac{\lambda}{|t_n|} + \frac{\lambda^7}{|t_n|^2} \right) + |R_n|.$$

The second term and remainder are absorbed by the first. For the remainder, its ratio to $|B_\lambda^+|\lambda/(\sqrt{L}|t_n|)$ decreases with $|n|$ and is already smaller than every fixed inverse power at N_λ . This proves (302) before repair.

Finally let $e = p_\lambda - p_\lambda^\circ$ and $a = h_\lambda^\circ(0)$. Direct counting gives $\|e\|_\infty \leq C\lambda^{3/2}|a|$. The Fourier Lebesgue constant is $O(\log(N+2))$, so

$$\frac{|(P_N - I)e(0)|}{|B_\lambda|} \leq C\lambda^{3/2} \log(N+2) \frac{|a|}{|B_\lambda|} = O_A(\lambda^{-A})$$

along this polynomial diagonal, by (135). For the coefficient assertion a direct variation estimate is stronger: if the fixed repair bump is supported in $[-b, b]$, then

$$\int_{1/\lambda}^\lambda |D_{\log} \mathcal{E}(\psi)(u)| \frac{du}{u} \leq \sum_{m \leq b\lambda} m^{-1/2} \int_0^b x^{-1/2} (\tfrac{1}{2}|\psi(x)| + x|\psi'(x)|) dx \leq C\sqrt{\lambda}.$$

Inversion averaging preserves this bound and has no periodic function jump. Hence $|\widehat{e}(n)| \leq C|a|\sqrt{\lambda L}/|n|$, which is negligible. Use $B_\lambda^+ \asymp B_\lambda$ to finish. $\square$

Corollary 21.8 (Weak G1 on the same polynomial-cutoff vector). *Let $R > 3$, $N = N_\lambda$, $K \geq N$, and $x \in E_K$, measured by the weighted norm of Proposition 21.1. For all sufficiently large λ ,*

$$\frac{|\langle D_{\log} W_{\lambda, K} k_\lambda, x \rangle|}{|\delta_K(k_\lambda)|} \leq C_R \|x\|_{\mathcal{H}_{R,L}} \left[\mathfrak{r}(\lambda) + \lambda^2 \left\{ \frac{L}{N} + \left(\frac{L}{N} \right)^{R-1} (1 + \log(2N)) \right\} \right]. \quad (305)$$

The bound is uniform in K and tends to zero on bounded weighted test families. This is a fixed-source weak pairing statement, not control in operator norm on a growing source block.

Proof. The coefficients of $(P_N - I)p_\lambda$ vanish for $|n| \leq N$. Insert (302) into the absolute convolution proof of Proposition 21.1: on the only frequencies used there, $\mathcal{V}_\lambda$ is replaced by $C|B_\lambda|\lambda$. Both its off-diagonal estimate and its separately treated logarithmic diagonal therefore give the displayed λ^2 error. The sums converge absolutely for $R > 3$; the fixed-window domain identification in Proposition 20.19 also justifies the passage from finite Fourier sums to p_λ . The continuum term is Proposition 20.13. Finally $|\delta_K(k_\lambda)| \geq (1 - C/\lambda)|B_\lambda|$ and, with the first-slot-linear convention, $\langle D_{\log} W_{\lambda, K} k_\lambda, x \rangle = QW_\lambda(k_\lambda, D_{\log} x)$. Unlike the filtered vector, $P_N p_\lambda$ lies in E_N , so $K \geq N$ suffices. $\square$

These results assert no positive Weil complement along this diagonal, a small inverse-weighted spectral residual, or control of the corrected sampler's graph defect. Those obligations remain separate. The fixed orders 0, 4 are essential; no uniform claim over a growing prolate block is made.

Use on the centered and corrected sampler

The weighted estimate is unchanged by the factors $(-1)^n$, because its proof bounds coefficient magnitudes. For any fixed finite root family, sufficient Hardy smoothing therefore controls the centered sampler and any fixed finite number of its derivatives. The source input is now proved in Section 20; constants may depend on the test family. It supplies no uniformity for a derivative order growing with L .

For the later even endpoint shell, the zeroth residual pairing is preserved exactly by parity: an even candidate k has odd $D_{\log} Wk$ and hence zero pairing with that shell. Higher residual jets do not all have the same parity, so no automatic preservation is claimed for every jet. A fixed finite Gramian made from pairings that tend to zero also tends to zero, by summing their squared magnitudes. Such a Gramian cannot by itself supply a uniformly positive metric on that root space.

Lemma 21.9 (A legitimate common-cutoff selection). *Suppose a normalized G1 error is at most ϵ for every $N \geq N_0(L, \epsilon)$ and every ambient cutoff $K \geq 2N$. Suppose finitely many additional errors are at most ϵ for every $K \geq K_0(L, N, \epsilon)$. Then one can choose a common cutoff satisfying all these bounds. The same statement holds with nested eventual thresholds $N \geq N_0$, $K \geq K_0(N)$ when that is what has actually been proved.*

Proof. Choose an admissible core cutoff first, then choose the ambient cutoff above all the finitely many thresholds. Apply this with a positive tolerance sequence tending to zero. Each error must be normalized by the actual endpoint or scale used in the target expression before taking thresholds. Merely knowing that two sets of acceptable cutoffs are unbounded would not suffice: two unbounded sets can be disjoint. $\square$

The explicit corrected-sampler rule below can be enlarged to meet additional proved eventual thresholds: its error bounds still vanish when K grows beyond that rule. A formula can be independent of the zero list while a proof of its convergence on a fixed hypothetical window has window-dependent constants. Neither observation proves a missing graph-pairing estimate.

22 A consistent finite sampling problem

The immediate question is whether a finite sampler can simultaneously retain the Cauchy evaluation, use the actual CCM endpoint functional, and converge in the same finite Weil form. The graph identity required by the wider G1/G2 argument is a fourth property. It is kept separate here because it does not follow from the other three.

Fix $L \geq 2$, $\lambda = e^{L/2}$, $A = L/2$, and

$$t_n = 2\pi n/L, \quad s_n = \frac{1}{2} + it_n.$$

In the coordinate $x = \log(\lambda u) \in [0, L]$, use

$$V_n(x) = L^{-1/2} e^{it_n x}, \quad D_L V_n = t_n V_n.$$

Interval vectors are extended by zero outside the interval when evaluating the global Weil form. Write $W_{L,N}$ for its compression to $|n| \leq N$. This is the same compression previously denoted $W_{\lambda,N}$, with $\lambda = e^{L/2}$; $\mathcal{Q} = QW$ denotes the same global Weil form in the fixed first-slot-linear convention. The coordinate change is unitary. The Hermitian product is linear in its first slot. The physical endpoint is

$$\delta_{L,N} \left(\sum a_n V_n \right) = L^{-1/2} \sum a_n, \quad \eta_{L,N} = \sqrt{L} \delta_{L,N}. \quad (306)$$

This is the equal-coefficient endpoint of the original matrix, not an interior evaluation. The basis and matrix realization are those of [7]; its opposite sesquilinear convention is translated here.

To remove the Fourier-sign ambiguity, define

$$H_F(s) = \int_{\mathbb{R}} g_F(y) e^{-(s-1/2)y} dy, \quad g_F(y) = \frac{1}{2\pi} \int_{\mathbb{R}} H_F\left(\frac{1}{2} + it\right) e^{ity} dt. \quad (307)$$

Thus the critical-line transform has the usual negative Fourier sign. Reversing this sign reverses physical tails and Fourier indices, but does not remove the half-period character identified below.

For the manuscript's Hardy output put

$$H_F(s) = \kappa(s) F(s), \quad \kappa(s) = \frac{\sigma(\sigma + a)^M}{\sigma - 1} \frac{s - 1}{(s + a)^M (\sigma - s)}, \quad \mathcal{E}(F) = \sigma F(\sigma). \quad (308)$$

Here F ranges over a fixed finite-dimensional space of root functions $\zeta(s)/(s - \rho)^j$, with the denominator canceled by a genuine zeta zero of order at least j . For the one-sided application all these zeros have real part greater than $1/2$. This is a conditional model: their existence is not asserted. The additional output $H = \kappa\zeta$ will provide an unconditional numerical test of the sampling mechanism.

Choosing M sufficiently large gives uniform polynomial vertical decay of any fixed order needed below on the relevant bounded strip. Contour shifts, after cancellation at $s = 1$ and at the selected zeros, imply the following useful facts. They also follow directly by Fourier inversion:

$$g_F(y) = \mathcal{E}(F)e^{qy} \quad (y < 0), \quad q = \sigma - \frac{1}{2}, \quad |g_F^{(j)}(y)| \leq Ce^{-ry} \quad (y > 0), \quad j \leq 2, \quad (309)$$

where any fixed $r < a + 1/2$ can be used after sufficient smoothing. Constants are uniform on a finite-dimensional unit sphere. In particular, $g_F(0) = \mathcal{E}(F)$. To verify the negative tail, close the inverse contour to the right for $y < 0$. The only pole crossed is σ , whose residue is $-\sigma F(\sigma)$; the contour orientation gives (309). For the positive tail shift left to $\Re s = 1/2 - r > -a$. Using a rate strictly below $a + 1/2$ avoids hiding the polynomial associated with the order- M pole at $-a$.

Both transfer and the uncentered nonconvergence theorem use $q, r > 1/2$. These margins are available for every fixed $\sigma > 1$ and $a > 0$, after choosing $r < a + 1/2$ and taking M sufficiently large. No conclusion below requires moving these parameters with L .

In the convention (307), the first-slot-linear zero formula is

$$\mathcal{Q}(f, g) = \sum_{\rho \in \mathcal{Z}} H_f(\rho) \overline{H_g(\rho^\sharp)}, \quad \rho^\sharp = 1 - \bar{\rho}, \quad (310)$$

with nontrivial zeros counted with multiplicity. The symmetries of the zero set make this consistent with the Fourier form of the explicit formula. All sums used below are absolutely convergent. The standard bound $N(T) = O(T \log T)$ implies $\sum_\rho (1 + |\Im \rho|)^{-2} < \infty$.

23 The two samplers are different finite vectors

Let $P_L g(y) = \sum_{k \in \mathbb{Z}} g(y + kL)$. Its Fourier series is $L^{-1} \sum_n H(s_n) e^{it_n y}$. Define

$$J_{L,N}^u F = L^{-1/2} \sum_{|n| \leq N} H_F(s_n) V_n, \quad (311)$$

$$J_{L,N}^c F = L^{-1/2} \sum_{|n| \leq N} (-1)^n H_F(s_n) V_n. \quad (312)$$

The superscripts mean uncentered and centered, respectively. For fixed L , their full Fourier limits on $[0, L]$ are

$$u_{L,F}(x) = P_L g_F(x), \quad c_{L,F}(x) = P_L g_F(x - A).$$

Translation of $c_{L,F}$ back to $[-A, A]$ gives the centered fundamental representative. Translation invariance of the global form therefore applies to $c_{L,F}$; it does not identify $u_{L,F}$ with that representative.

Proposition 23.1 (Endpoint and interior evaluations). *With Fourier truncations taken sufficiently far,*

$$\delta(J^u F) \longrightarrow \mathcal{E}(F), \quad \delta(J^c F) \longrightarrow 0 \quad (L \rightarrow \infty).$$

On the centered sampler the evaluation at $x = A$, rather than at the cut, converges to $\mathcal{E}(F)$.

Proof. For fixed L , Fourier convergence gives $\delta(J^u F) \rightarrow P_L g_F(0)$ and $\delta(J^c F) \rightarrow P_L g_F(A)$. Equation (309) gives $P_L g_F(0) = \mathcal{E}(F) + O(e^{-\min(q,r)L})$ and $P_L g_F(A) = O(e^{-\min(q,r)L/2})$. Evaluation of (312) at A cancels $(-1)^n$. $\square$

The diagonal unitary $P = \text{diag}((-1)^n)$ commutes with D_L . A pure change of basis must replace W by PWP and the boundary covector by its transformed covector. Applying P to the samples while retaining the original W and δ is instead a different physical vector. That is precisely the change used in (312).

24 Centered Weil transfer with the boundary terms retained

Lemma 24.1 (A sufficient strip estimate). *Suppose $|g^{(j)}(y)| \leq Ce^{-\mu|y|}$ for $j \leq 2$, with $\mu > 1/2$. Let $h_L = 1_{[-A,A]}P_L g$ and let H_L be its transform. Then, uniformly for $0 \leq \Re s \leq 1$,*

$$|H_L(s) - H(s)| \leq \frac{Ce^{-(\mu-1/2)L/2}}{1 + |\Im s|}. \quad (313)$$

The estimate includes the jumps of the zero-extended representative.

Proof. Set $e_L = h_L - g$ and $d = \Re s - 1/2$. Outside $[-A, A]$ use the tails of g ; inside use the nonzero periodic images. Integrating the resulting geometric series, including their first derivatives, gives

$$\|e_L e^{-d \cdot}\|_1 + \text{Var}(e_L e^{-d \cdot}) \leq Ce^{-(\mu-1/2)A}, \quad |d| \leq 1/2.$$

The variation includes both endpoint jumps, whose weighted sizes obey the same bound. The L^1 bound handles bounded ordinates. One integration by parts in the sense of a finite derivative measure handles large ordinates, giving (313). $\square$

Arbitrary inverse powers of the ordinate are not claimed. A nonzero endpoint jump normally produces a $1/t$ term. That decay is sufficient: each Weil summand is a product of two transforms.

Theorem 24.2 (Centered form transfer). *On any fixed finite-dimensional family satisfying the preceding decay bounds and $|H_F(s)| \leq C(1 + |\Im s|)^{-1}$ on the strip,*

$$\mathcal{Q}(c_{L,F}, c_{L,G}) \longrightarrow \mathcal{Q}(g_F, g_G)$$

uniformly on bounded sets of F, G . The same conclusion holds for $\langle W_{L,N} J_{L,N}^c F, J_{L,N}^c G \rangle$ with a sufficiently large common Fourier cutoff. On a one-sided root space the self-form limit is zero. For reflected top-root outputs, the nonzero global cross-form also transfers.

Proof. Let $\epsilon_L = e^{-(\mu-1/2)L/2}$. In (310), replace each transform by $H_F + E_{L,F}$. Lemma 24.1 bounds each difference by $C\epsilon_L/(1 + |\Im \rho|)$, and the unchanged transform has the same ordinate decay without ϵ_L . Hence the difference of the forms is at most $C(\epsilon_L + \epsilon_L^2) \sum_\rho (1 + |\Im \rho|)^{-2}$. Translation of h_L to $[0, L]$ does not change the paired form.

For completeness, Fourier truncation can be justified directly, without inferring convergence of these particular projections from abstract core density. The correct CCM matrix bounds are

$$|W_{nm}| \leq C\lambda/|n - m| \quad (n \neq m), \quad |W_{nn}| \leq C\{\lambda + 1 + \log(2 + |n|/L)\}. \quad (314)$$

The diagonal uses $2(1 - y/L) \cos(t_n y)$, not a limit of the off-diagonal kernel. Near zero its archimedean numerator is bounded by $C\{\min(1, t_n^2 y^2) + y\}$; integration against $O(1/y)$ gives the logarithm in (314). The prime and pole pieces are $O(\lambda)$.

If $|H_F(s_n)| \leq C(1 + |n|/L)^{-R}$ with $R \geq 2$, then the normalized coefficients a_n have

$$\sum_n |a_n| \leq C\sqrt{L}, \quad \sum_{|n| > N} |a_n| \leq C\sqrt{L}(1 + N/L)^{1-R}.$$

The absolute double matrix series therefore converges. Bounding off-diagonal entries by $C\lambda$ and treating the logarithmic diagonal separately gives, for $N \geq L$,

$$|\mathcal{Q}(c_{L,F}, c_{L,G}) - \langle W_{L,N} J_{L,N}^c F, J_{L,N}^c G \rangle| \leq C\lambda L(1 + N/L)^{1-R}. \quad (315)$$

The constants are uniform on the fixed family. This also identifies the matrix series with the semilocal form, using its closed realization.

For a one-sided root space, H_F is nonzero at at most finitely many zeros, all on one side, so each paired summand in its self-form is zero. If F and G are the top root functions at ρ and $\rho^\sharp$, their cross-form is

$$m_\rho \kappa(\rho) \frac{\zeta^{(m_\rho)}(\rho)}{m_\rho!} \overline{\kappa(\rho^\sharp) \frac{\zeta^{(m_\rho)}(\rho^\sharp)}{m_\rho!}},$$

which is nonzero. This is the first-slot-linear convention. $\square$

25 The uncentered sampler has an explicit cut defect

The failure of the original proof is more than an omitted phase in its notation. It is possible to compute the form created by the different cut. In this section assume $q, r > 1/2$, and abbreviate $E = \mathcal{E}(F)$, $H = H_F$, $z_\rho = \rho - 1/2$. Let $g_+ = 1_{(0,\infty)}g$ and $g_- = 1_{(-\infty,0)}g$. Set

$$f_L = g_+ + g_-(\cdot - L).$$

The negative half has been moved by L . Since $H_-(s) = E/(\sigma - s)$, its full transform is

$$\widehat{f}_L(s) = H(s) + E \frac{e^{-(s-1/2)L} - 1}{\sigma - s}. \quad (316)$$

Lemma 25.1 (Comparison with the actual periodic cut). *For $u_L = 1_{[0,L]}P_L g$ and some $\nu > 0$,*

$$\mathcal{Q}(u_L, u_L) = \mathcal{Q}(f_L, f_L) + O(e^{-\nu L}).$$

One may take $\nu = \min(q, r) - 1/2 > 0$.

Proof. Write $s = 1/2 + d + it$, $|d| \leq 1/2$, and $p_0 = \min(q, r)$. On $[0, L]$ the negative images sum exactly to $Ee^{q(x-L)}/(1 - e^{-qL})$. Their transform differs from the transform of $g_-(\cdot - L)$ by exactly

$$\frac{Ee^{-qL}}{1 - e^{-qL}} \frac{e^{-(s-1/2)L} - 1}{\sigma - s}.$$

The extra positive images are $O(e^{-rL}e^{-rx})$ on $[0, L]$; the removed positive tail lies in $x > L$. Their weighted L^1 norms and variations, including endpoint jumps, give the same type of bound. Consequently

$$|\widehat{u}_L(s) - \widehat{f}_L(s)| \leq \frac{Ce^{-p_0L}(1 + e^{-dL})}{1 + |t|}, \quad |\widehat{f}_L(s)| \leq \frac{C(1 + e^{-dL})}{1 + |t|}.$$

It matters that the partner has real displacement $-d$. Products in the form error contain $(1 + e^{-dL})(1 + e^{dL}) \leq 4e^{L/2}$, rather than two independent worst-case factors. Expanding the zero sum therefore bounds the error by $Ce^{-(p_0-1/2)L} \sum_\rho (1 + |\Im \rho|)^{-2}$, as required. $\square$

For the one-sided root family, let S be the finite set of zeros at which H does not vanish. Sums over S below include their multiplicities. For $H = \kappa\zeta$, take S empty.

Proposition 25.2 (Cut-defect formula). *Under the stated hypotheses,*

$$\begin{aligned} \mathcal{Q}(u_L, u_L) = \mathcal{Q}(g, g) + 2\Re \sum_{\rho \in S} \frac{H(\rho)\overline{E}}{q + z_\rho} (e^{z_\rho L} - 1) \\ + 2|E|^2 \sum_{\rho \in \mathcal{Z}} \frac{1 - e^{z_\rho L}}{q^2 - z_\rho^2} + O(e^{-\nu L}). \end{aligned} \quad (317)$$

Both displayed sums are well defined without RH; the second is absolutely convergent for each fixed L .

Proof. Insert (316) into (310). The two mixed sums are complex conjugates after replacing ρ by $\rho^\sharp$. The product of the two added terms has numerator $2 - e^{z_\rho L} - e^{-z_\rho L}$ and denominator $q^2 - z_\rho^2$. The symmetry $\rho \mapsto 1 - \rho$ makes the two exponential sums equal, giving (317). Lemma 25.1 handles the actual cut. Absolute convergence follows from $|\Re z_\rho| < 1/2$ and the quadratic denominator at large ordinates. $\square$

Theorem 25.3 (Failure of the claimed full-scale raw transfer). *Suppose $E \neq 0$, and H vanishes at every critical-line zero and at all but finitely many other nontrivial zeros. Under the preceding decay hypotheses, $\mathcal{Q}(u_L, u_L)$ does not converge to any finite limit as real $L \rightarrow \infty$. In particular it cannot converge to $\mathcal{Q}(g, g) = 0$ for a one-sided root output with nonzero E .*

Proof. Choose a critical-line zero $\rho_0 = 1/2 + i\gamma_0$ of multiplicity m_0 , with $\gamma_0 \neq 0$. Such zeros exist unconditionally by Hardy's theorem [16]. The coefficient of $e^{i\gamma_0 L}$ in the infinite sum in (317) is

$$-\frac{2|E|^2 m_0}{q^2 + \gamma_0^2} \neq 0. \quad (318)$$

The finite mixed terms have exponents strictly off the imaginary axis, so do not contribute at this exponent.

Here is an argument that does not assume away possible off-line zeros. Take the Laplace transform in $L \geq L_0$ of (317), initially for $\Re p > 1/2$. Its series of rational-exponential terms continues meromorphically across the imaginary axis: the denominators at large ordinates are $O(|\Im \rho|^3)$, giving locally normal convergence away from the discrete poles. The exponentially decaying error is analytic in $\Re p > -\nu$. At $p = i\gamma_0$ the continued transform has residue (318); the lower integration limit does not change the residue.

If $\mathcal{Q}(u_L, u_L)$ had a finite limit, it would be bounded and its Laplace transform would be holomorphic for $\Re p > 0$. All apparent poles in that half-plane would therefore have to cancel. Moreover bounded convergence, after subtracting the finite limit, would give

$$\lim_{\epsilon \downarrow 0} \epsilon \int_{L_0}^{\infty} e^{-(\epsilon + i\gamma_0)L} \mathcal{Q}(u_L, u_L) dL = 0.$$

For the constant part the same limit is zero since $\gamma_0 \neq 0$. But the meromorphic expression gives the nonzero residue (318). This is a contradiction. $\square$

Remark 25.4 (Exact scope). The theorem concerns the full real-scale limit. It does not exclude specially chosen subsequences L_j on which the form happens to approach zero. Nor does it exclude

different sampling weights or a changed matrix. It applies to Galerkin approximations whenever their error relative to this continuum cut tends to zero; choosing a larger faithful Fourier cutoff cannot remove the continuum defect.

For any nonempty hypothetical one-sided root window, a basis vector has $E = \sigma\zeta(\sigma)/(\sigma-\rho)^j \neq 0$. Thus the original full-scale transfer and the original unphased sampling convention are incompatible in that window. This is a conditional diagnosis, not an assertion that an off-line zero exists or an unconditional disproof of an implication with a potentially empty premise. The output $H = \kappa\zeta$ is an unconditional example of the same failure in the larger analytic output class.

On $\ker \mathcal{E}$, the negative half in (309) vanishes, and this cut obstruction disappears. That codimension-one restriction does not recover the whole nonempty root window.

26 A single finite object with form transfer and exact endpoint

For an integer $K \geq 1$ use the ambient space $|n| \leq 2K$, and define the even endpoint shell

$$c_{L,K} = \frac{\sqrt{L}}{2K} \sum_{K < |n| \leq 2K} V_n. \quad (319)$$

There are exactly $2K$ terms. Consequently

$$\delta(c_{L,K}) = 1, \quad \|c_{L,K}\|^2 = \frac{L}{2K}, \quad \|D_L c_{L,K}\|^2 = \frac{\pi^2}{3L} (14K + 9 + K^{-1}). \quad (320)$$

Define a linear functional and a corrected sampler by

$$d_{L,K}(F) = \mathcal{E}(F) - \delta(J_{L,2K}^c F), \quad \tilde{J}_{L,K} F = J_{L,2K}^c F + c_{L,K} d_{L,K}(F). \quad (321)$$

Theorem 26.1 (Three-property compatibility). *On a fixed finite-dimensional Hardy-output family as above, choose*

$$K(L) = \lceil \lambda^4 L^4 \rceil, \quad \lambda = e^{L/2},$$

and choose M sufficiently large to give $R \geq 2$ in (315). The same finite matrix, sampler, and endpoint obey

$$\delta_{L,2K}(\tilde{J}_{L,K} F) = \sigma F(\sigma) \quad \text{exactly,} \quad (322)$$

$$\|\tilde{J}_{L,K} F - J_{L,2K}^c F\| \rightarrow 0, \quad (323)$$

$$\langle W_{L,2K} \tilde{J}_{L,K} F, \tilde{J}_{L,K} G \rangle \rightarrow \mathcal{Q}(g_F, g_G), \quad (324)$$

$$\langle \tilde{J}_{L,K} F, \tilde{J}_{L,K} G \rangle \rightarrow \langle g_F, g_G \rangle. \quad (325)$$

The limits are uniform on bounded subsets of the fixed family. The construction requires no zero list in the matrix, shell, cutoff rule, or formula for the endpoint correction. It is not an economical cutoff claim.

Proof. The endpoint identity follows immediately from (320). Uniform decay of the critical-line samples gives $\|J^c F\| \leq C\|F\|$, and the corresponding Riemann sum converges to $\|g_F\|^2$. The same coefficient bound and the endpoint result give $\|d_{L,K}\| \leq C$ for large L and the stated cutoffs. Therefore $\|\tilde{J} F - J^c F\| \leq C\sqrt{L/K}\|F\|$.

The Schur row-sum estimate applied to the correct matrix bounds yields

$$\|W_{L,2K}\| \leq C\lambda(1 + \log(4K + 1)). \quad (326)$$

Expansion of the two corrected arguments bounds the change in their form pairing by

$$C\lambda(1 + \log(4K + 1)) \left(\sqrt{L/K} + L/K \right) \|F\| \|G\|. \quad (327)$$

For the displayed cutoff this is $O(\lambda^{-1}L^{-1/2})$. The centered Fourier-truncation error in (315) is at most $O(\lambda^{-3}L^{-2})$ when $R = 2$, and smaller for larger R . Theorem 24.2 proves (324). The ordinary norm correction tends to zero, so the Riemann-sum limit also proves (325). $\square$

If $F \mapsto g_F$ is injective on the fixed finite-dimensional space, the last limit gives a uniform lower ordinary-norm bound there. In particular, on a one-sided root space the corrected self-form tends to zero, and adding any fixed positive scalar to W gives restricted coercivity for sufficiently large cutoffs. The nonzero reflected cross-form also transfers to this corrected sampler. These are legitimate consequences of this particular construction; they do not identify its graph transport with the old one.

27 The graph identity is the remaining compatibility cost

Suppose the root-space operator satisfies

$$BF = (s - \tfrac{1}{2})F - \eta_B(F)\zeta.$$

The centered samples satisfy the exact coefficientwise identity

$$iD_L J^c - J^c B = z_{L,2K}^c \eta_B, \quad z_{L,2K}^c = L^{-1/2} \sum_{|n| \leq 2K} (-1)^n \kappa(s_n) \zeta(s_n) V_n. \quad (328)$$

For the corrected sampler, however,

$$iD_L \tilde{J} - \tilde{J} B = z_{L,2K}^c \eta_B + \mathcal{R}_{L,K}, \quad \mathcal{R}_{L,K} F = iD_L c_{L,K} d_{L,K}(F) - c_{L,K} d_{L,K}(BF). \quad (329)$$

This is an exact rank-at-most-two defect. Parity gives $\langle D_L c_{L,K}, c_{L,K} \rangle = 0$, so

$$\|\mathcal{R}_{L,K} F\|^2 = |d_{L,K}(F)|^2 \frac{\pi^2(14K + 9 + K^{-1})}{3L} + |d_{L,K}(BF)|^2 \frac{L}{2K}. \quad (330)$$

For $\mathcal{E}(F) \neq 0$, this diverges along the displayed construction. Smallness of a correction in ordinary norm or Weil form therefore does not make it small in the graph identity.

Proposition 27.1 (Rigidity and a general trace obstruction). *For a fixed one-sided off-line root space, the equation $iD_L J - JB = z\eta_B$ has at most one solution J for each prescribed z . Moreover, any family of corrections T_L with $\|T_L\| \rightarrow 0$ and $\delta(T_L F) \rightarrow \mathcal{E}(F) \neq 0$ for some fixed F has $\|iD_L T_L F - T_L B F\| \rightarrow \infty$.*

Proof. All eigenvalues of B have positive real part, whereas those of iD_L are purely imaginary. Row by row, the homogeneous equation is $\ell_n(it_n I - B) = 0$, and $it_n I - B$ is invertible. This proves uniqueness.

For a periodic H^1 function b on $[0, L]$, choose a point where $|b|^2$ does not exceed its average and integrate the derivative of $|b|^2$. Cauchy–Schwarz gives

$$|b(0)|^2 \leq L^{-1} \|b\|^2 + 2 \|b\| \|D_L b\|. \quad (331)$$

Apply this to $b = T_L F$. Its norm tends to zero while its endpoint tends to a nonzero value, so $\|D_L T_L F\| \rightarrow \infty$. Since B is bounded on the fixed finite-dimensional space, $\|T_L B F\| \rightarrow 0$. The reverse triangle inequality proves the claim. $\square$

There is also a useful exact description at the weaker pairing level. Here η denotes the boundary covector $\eta_{L,2K}$, distinct from the root-space functional η_B . In physical scaling the finite commutator is

$$[D_L, W] = |\beta\rangle\eta - |\eta\rangle\beta^*, \quad \eta(c_{L,K}) = \sqrt{L}, \quad \beta^*(c_{L,K}) = 0,$$

where $\beta = \beta^{\log}$ includes the physical factor $2\pi/L$ and is odd and $c_{L,K}$ is even. Here the rank-one notation means $|a\rangle b^* : x \mapsto a \langle x, b \rangle$, and the vector representing η is understood. Therefore for every finite test v ,

$$\begin{aligned} \langle W \mathcal{R}_{L,K} F, v \rangle &= i d_{L,K}(F) \langle D_L W c_{L,K}, v \rangle - i \sqrt{L} d_{L,K}(F) \langle \beta, v \rangle \\ &\quad - d_{L,K}(BF) \langle W c_{L,K}, v \rangle. \end{aligned} \quad (332)$$

Thus even where the first and last terms can be made small, the boundary commutator term has not disappeared. No unproved asymptotic value for this term is used in this manuscript.

One part of G1 is preserved particularly simply. The original finite matrix preserves parity; if the G1 candidate k is even, $D_L W k$ is odd. Hence

$$\langle D_L W k, \tilde{J} F \rangle = \langle D_L W k, J^c F \rangle. \quad (333)$$

This is exact and requires no estimate on the shell derivative. Establishing the centered right-hand G1 bound still requires the independently audited continuum G1 inputs. Equation (333) does not cancel (332) in the later graph calculation.

28 The rebuilt finite metric and reflected-partner argument

Throughout this section $J = \tilde{J}_{L,K}$, $W = W_{L,2K}$, $D = D_L$, $z = z_{L,2K}^c$, and $\mathcal{R} = \mathcal{R}_{L,K}$ are the objects of (321) and (329) on the *same* finite space. Write $\eta_\partial = \sqrt{L}\delta$ and $\beta = \beta^{\log}$. Then

$$[D, W] = |\beta\rangle\eta_\partial - |\eta_\partial\rangle\beta^*.$$

The rank-one convention is $|u\rangle v^* : x \mapsto u \langle x, v \rangle$. In particular the root functional η_B is not identified with either finite endpoint functional. Theorem 19.2 gives the full finite-core metric budget without a global closed-operator assumption. The source-annihilating projection of Proposition 19.3 is a separate positive benchmark on this same finite object, not a replacement for W in the identities below or a claim of small metric defect. Proposition 19.7 also shows why inserting that sharp band projection into a Weil pairing does not automatically preserve the earlier holomorphic-strip transfer. Proposition 19.8 further shows that reflection alone does not cancel the leading edge term; control of the entire zero sum is still required. Proposition 19.9 supplies a different, full-space projection that kills the source and both shell directions while preserving ordinary coercivity and Weil transfer through a proved full-strip product bound. Its exact commutator remains in (113). Corollary 19.10 shows why adding a fixed positive scalar to its projected Weil form still fails the small-defect

criterion. Neither result licenses meromorphic resolvent transfer or changes the endpoint of J . With the stronger fixed tail margin $p > 1$, Proposition 19.12 now controls $WSJF$ in ordinary norm by an explicit finite exponential sum. Corollary 19.13 audits the globally positive candidate SW^2S : top-root mass grows, while lower jets at multiple zeros lose unscaled coercivity. The full squared-metric commutator is still required.

Proposition 28.1 (Restricted positive shifts after corrected transfer). *On any fixed one-sided root space, for every fixed $a_0 > 0$ there is $c > 0$ such that, for sufficiently large L ,*

$$\langle (W + a_0 I) JF, JF \rangle \geq c \|F\|^2.$$

This is restricted coercivity, not positivity on the full finite space.

Proof. The output map $F \mapsto g_F$ is injective, so its Gram matrix has a positive minimum eigenvalue on the fixed space. Theorem 26.1 transfers that ordinary Gram matrix, while the Weil Gram matrix tends to zero by one-sided isotropy. The positive scalar term therefore dominates. $\square$

If a negative compact Weil test is used to bound the lowest finite form eigenvalue away from zero under failure of RH, its particular Fourier approximants must also converge in the closed form. For a smooth compact test this follows from the correct matrix bound and rapid coefficient decay. This optional calibration is not needed for the proposition. In either formulation coercivity alone does not imply a small Lyapunov defect.

Proposition 28.2 (Lyapunov identity with the graph defect retained). *Let $BF = \mu F$, $d = \Re \mu > 0$, $\|F\| = 1$, and put*

$$x = JF, \quad e = \eta_B(F), \quad y = ez + \mathcal{R}F, \quad b = \eta_\partial(x).$$

For every self-adjoint finite matrix S ,

$$d \langle Sx, x \rangle = -\Re \langle Sy, x \rangle - \frac{i}{2} \langle [D, S]x, x \rangle. \quad (334)$$

For $T = W + a_0 I$ this becomes

$$2d \langle Tx, x \rangle = -2\Re \langle T(ez + \mathcal{R}F) + ib\beta, x \rangle. \quad (335)$$

If $\langle Tx, x \rangle \geq c$ and $\|x\| \leq C$, then

$$\|T(ez + \mathcal{R}F) + ib\beta\| \geq dc/C.$$

Proof. The exact graph equation is $iDx - \mu x = y$. Pair it in both slots with Sx . Self-adjointness gives $2d \langle Sx, x \rangle = -i \langle [D, S]x, x \rangle - 2\Re \langle Sy, x \rangle$. For $S = T$ the scalar part commutes with D . The rank-two commutator has expectation $b \langle \beta, x \rangle - \overline{b \langle \beta, x \rangle}$, which gives (335). Apply Cauchy-Schwarz for the bound. $\square$

For this sampler $b = \sqrt{L} \mathcal{E}(F)$ exactly. The vector $\mathcal{R}F$ cannot be omitted on the ground that $J - J^c$ tends to zero in ordinary norm: (330) proves the opposite behavior for its derivative. On a left root window the signs reverse, with the same absolute-value conclusion.

Proposition 28.3 (Candidate comparison and scalar cancellation). *For an even candidate k with $a = \eta_\partial(k) \neq 0$, put $r = DWk$, $q = ib/a$, and $m = y - qDk$. Then*

$$Ty + ib\beta = Tm + q(r + a_0 Dk), \quad (336)$$

$$d \langle Wx, x \rangle = -\Re \langle Wm, x \rangle - \Re \langle q \langle r, x \rangle \rangle. \quad (337)$$

More generally, a self-adjoint correction commuting with D satisfies $d \langle Sx, x \rangle = -\Re \langle Sy, x \rangle$ exactly.

Proof. Parity gives $[D, W]k = a\beta$, hence $WDk = r - a\beta$. Since $qa = ib$, expansion proves the first identity. Use Proposition 28.2 with $S = W$ for the second. The final statement is (334) with zero commutator. $\square$

Thus scalar coercivity balances its own source and graph terms. It does not independently exclude an off-line root. For a noncommuting positive S , the exact combined quantity

$$\mathfrak{X}_S = \Re \langle S(ez + \mathcal{R}F), x \rangle + \frac{i}{2} \langle [D, S]x, x \rangle = -d \langle Sx, x \rangle$$

still needs an independent upper estimate before its lower bound can be used in a contradiction. With $S = W^2$, $[D, W^2] = [D, W]W + W[D, W]$ has rank at most four. Form smallness alone does not give norm smallness.

Proposition 28.4 (Reflected top-root lower bound for the corrected sampler). *Suppose ρ is an off-line zero of exact multiplicity m and $\Re \rho > 1/2$. Set $\rho^\sharp = 1 - \bar{\rho}$, $F_+ = \zeta/(s - \rho)^m$, and $F_- = \zeta/(s - \rho^\sharp)^m$. On the common finite construction, with $x = JF_+$ and $x_- = JF_-$,*

$$\langle Wx, x \rangle \rightarrow 0, \quad \langle Wx, x_- \rangle \rightarrow C_{\rho, m} \neq 0, \quad \liminf \|Wx\| > 0,$$

where the first-slot-linear constant is

$$C_{\rho, m} = m \kappa(\rho) \frac{\zeta^{(m)}(\rho)}{m!} \overline{\kappa(\rho^\sharp) \frac{\zeta^{(m)}(\rho^\sharp)}{m!}}. \quad (338)$$

Proof. The functional equation and conjugation preserve zero multiplicity, so $\rho^\sharp$ also has exact order m . The top root output is nonzero only at its selected zero in the global value-level zero sum. Thus the global cross form is exactly (338); its factors are nonzero. Both outputs meet the holomorphic strip hypotheses of Theorem 26.1, which transfers the self and cross forms. Ordinary norms of x_- are bounded. The inequality $\|Wx\| \|x_- \| \geq |\langle Wx, x_- \rangle|$ proves the lower bound. $\square$

For $m > 1$ this is a bound on the top generalized vector. The genuine eigenvector $\zeta/(s - \rho)$ vanishes at every zero when $m > 1$, and the value-level Weil formula supplies no analogous bound on it. No claim of coercivity on an entire multiple root block follows from this proposition.

Boundary-annihilating tests: the exact surviving identity

Write $\mu = \rho - 1/2$, $A_\rho = iD - \mu I$, and $x_j = JF_{\rho, j}$. With $x_0 = 0$, the corrected graph relation gives

$$A_\rho x_m = u_m, \quad u_m = x_{m-1} + \eta_B(F_{\rho, m})z + \mathcal{R}F_{\rho, m}.$$

In the canonical chain $\eta_B(F_{\rho, 1}) = 1$ and the functional vanishes on higher chain vectors. Thus the old radical predecessor now has the additional term $\mathcal{R}F_{\rho, m}$. Define $v^\pm = (A_\rho^*)^{-1}x_\pm$, with $x_+ = x_m$. The inverse has norm at most $1/d$, where $d = \Re \rho - 1/2$. If $\delta(v^+) \neq 0$, set

$$\alpha = \delta(v^-)/\delta(v^+), \quad \tilde{v} = v^- - \alpha v^+, \quad \tilde{y} = x_- - \alpha x_m = A_\rho^* \tilde{v}.$$

These are finite definitions, not assertions of a bounded limiting ratio. For $b_m = \eta_\partial(x_m)$, commutation gives exactly

$$\langle Wu_m + ib_m \beta, \tilde{v} \rangle = \langle Wx_m, \tilde{y} \rangle. \quad (339)$$

Indeed $A_\rho Wx_m = Wu_m + i[D, W]x_m$, and $\eta_\partial(\tilde{v}) = 0$ removes the other boundary channel. If the ratio α is bounded along a chosen sequence, the right side tends to $C_{\rho, m}$ by Proposition 28.4 and

conjugate-linearity in the second slot. Corollary 28.7 below shows that this bounded-ratio premise actually fails for the corrected sampler. Even a hypothetical bounded-ratio construction would also require, before isolating the beta term,

$$\langle W(x_{m-1} + \eta_B(F_{\rho,m})z), \tilde{v} \rangle + \langle W\mathcal{R}F_{\rho,m}, \tilde{v} \rangle \longrightarrow 0. \quad (340)$$

Neither term is disposed of by the holomorphic form-transfer theorem. The first involves a potentially meromorphic adjoint-resolvent test; the second is the new graph correction. Equation (332) shows its explicit boundary contribution.

What is withdrawn. The previous pole-aware transfer proof controlled the singular zero but did not bound the product error over all zeros. Its conclusion is not used here. A valid replacement must give a summable majorant for the paired transforms on the entire strip, including cut jumps and slow reflected tails, on the actual finite test family. Consequently the old nonzero limit for $\sqrt{L} \langle \beta, \tilde{v} \rangle$ is not an established result of this version. The exact arithmetic formula for that quantity remains valid for every finite test, as shown in Section 29.

The endpoint ratio and a limitation on fixed test spaces

The bounded-ratio premise above can now be decided for the corrected sampler. Fix a hypothetical zero $\rho = 1/2 + \mu$, where $\mu = d + i\gamma$ and $0 < d < 1/2$, and set $p = \rho^\sharp = 1/2 - \bar{\mu}$. Let $\mathcal{F}$ be a fixed finite-dimensional space of Hardy inputs satisfying the holomorphy, tail and vertical-decay hypotheses of Sections 22–28. One may include any fixed finite set of genuine root functions and ζ itself, choosing the smoothing order sufficiently large. Fix any norm on $\mathcal{F}$. Use the common cutoff

$$K \geq \lceil e^{2L} L^4 \rceil, \quad J = \tilde{J}_{L,K}, \quad V_L G = (A_\rho^*)^{-1} JG.$$

The letter V_L here denotes a resolvent map; V_n still denotes a Fourier basis vector. Choose r as in (309) and put $\nu = \min(q, r, 1) > 1/2$. All constants below may depend on the fixed space and on ρ , but not on L , K , or a bounded input.

Lemma 28.5 (Endpoint of the corrected adjoint resolvent). *Uniformly for $G \in \mathcal{F}$,*

$$\delta V_L G = -H_G(p) \frac{e^{-\bar{\mu}L/2}}{1 - e^{-\bar{\mu}L}} + O(e^{-\nu L/2}) \|G\|. \quad (341)$$

Consequently the rescaled endpoint functional satisfies, in the dual norm of this fixed space,

$$\ell_L(G) := e^{\bar{\mu}L/2} (1 - e^{-\bar{\mu}L}) \delta V_L G = -H_G(p) + O(e^{-(\nu-d)L/2}) \|G\|. \quad (342)$$

Proof. The multiplier of $(A_\rho^*)^{-1}$ is $(-it - \bar{\mu})^{-1}$. Thus the full centered Fourier series is the periodization of the inverse transform of

$$H_{v,G}(s) = -\frac{H_G(s)}{s - p}.$$

This transform is used only for Fourier inversion, not in a Weil zero-sum formula. Its possible pole at p has residue $-H_G(p)$. Contour shifts with the same decay bounds as (309) give

$$h_G(y) = \begin{cases} -\mathcal{E}(G)(\sigma - p)^{-1} e^{qy}, & y < 0, \\ -H_G(p) e^{-\bar{\mu}y} + O(e^{-ry}) \|G\|, & y > 0. \end{cases}$$

Summing these tails at the half-period points $(j + 1/2)L$ gives

$$\begin{aligned} \delta V_L^{c,\infty} G &= -H_G(p) \frac{e^{-\bar{\mu}L/2}}{1 - e^{-\bar{\mu}L}} \\ &\quad - \frac{\mathcal{E}(G)}{\sigma - p} \frac{e^{-qL/2}}{1 - e^{-qL}} + O(e^{-rL/2}) \|G\|. \end{aligned}$$

If $|H_G(1/2 + it)| \leq C(1 + |t|)^{-R} \|G\|$, $R \geq 2$, the omitted Fourier endpoint tail is at most $C(1 + K/L)^{-R} \|G\|$, because the resolvent supplies one further inverse power. The endpoint correction has coefficient $d_{L,K}(G) = \mathcal{E}(G) - \delta J_{L,2K}^c G = O(\|G\|)$. For its shell $c = c_{L,K}$, pairing the positive and negative indices gives the exact identity

$$\begin{aligned} \delta(A_\rho^*)^{-1} c &= \frac{1}{2K} \sum_{K < |n| \leq 2K} \frac{1}{-it_n - \bar{\mu}} \\ &= -\frac{\bar{\mu}}{K} \sum_{n=K+1}^{2K} \frac{1}{t_n^2 + \bar{\mu}^2} = O_\mu(L^2/K^2). \end{aligned} \tag{343}$$

For large L , K/L exceeds every fixed multiple of $|\mu|$, which justifies the last bound despite the complex denominator. Both cutoff errors are $O(e^{-4L} L^{-6})$ at the stated minimum cutoff and decrease upon enlarging it. They are absorbed by $O(e^{-\nu L/2})$. This proves (341); multiplication by the displayed scale proves (342). $\square$

Lemma 28.6 (Uniform lower norm bound for the resolvent map). *There are $c, C > 0$ such that for all sufficiently large L ,*

$$c\|G\| \leq \|V_L G\| \leq C\|G\| \quad (G \in \mathcal{F}).$$

Proof. The Gram form of the centered resolvent samples is a Riemann sum converging, uniformly on the fixed finite-dimensional space, to

$$\frac{1}{2\pi} \int_{\mathbb{R}} \frac{|H_G(1/2 + it)|^2}{|-it - \bar{\mu}|^2} dt.$$

This form is positive definite: a zero integral forces the analytic output, and hence G , to vanish. Its minimum eigenvalue is positive. The shell changes the resolvent vector by at most $d^{-1}|d_{L,K}(G)|\sqrt{L/(2K)} = o(\|G\|)$. Finite-dimensional compactness proves both bounds. $\square$

Corollary 28.7 (Divergence of the proposed endpoint ratio). *For the top-root pair in Proposition 28.4, put $v^\pm = V_L F_\pm$. Wherever $\delta v^+ \neq 0$ and L is sufficiently large, the proposed ratio obeys*

$$\left| \frac{\delta v^-}{\delta v^+} \right| \geq c e^{(\nu-d)L/2}.$$

Thus it cannot be bounded along any sequence tending to infinity on which it is defined. At zeros of the denominator the proposed formula for the test is undefined. The vectors $v^- - (\delta v^-/\delta v^+)v^+$ have unbounded ordinary norm.

Proof. Genuine zero cancellation gives

$$H_{F_+}(p) = 0, \quad H_{F_-}(p) = \kappa(p) \frac{\zeta^{(m)}(p)}{m!} \neq 0.$$

The preceding endpoint lemma bounds the denominator by $Ce^{-\nu L/2}$ and the numerator below by $ce^{-dL/2}$. The inputs F_+ and F_- are linearly independent, and Lemma 28.6 then gives norm divergence of the displayed combination. $\square$

Theorem 28.8 (Loss of the reflected signal on a fixed test space). *Include F_+ in $\mathcal{F}$ and let $G_L \in \mathcal{F}$. Suppose $v_L = V_L G_L$ is bounded in ordinary norm and $\delta v_L = 0$. Then*

$$\langle W J F_+, A_\rho^* v_L \rangle = \langle W J F_+, J G_L \rangle \longrightarrow 0. \quad (344)$$

This conclusion uses holomorphic form transfer only. No pole-aware Weil transfer is assumed.

Proof. Lemma 28.6 makes G_L bounded. Equation (342) and $\delta v_L = 0$ imply $H_{G_L}(p) \rightarrow 0$. On the other hand, the global value-level zero formula and the top-root cancellation give

$$\mathcal{Q}(g_{F_+}, g_G) = m H_{F_+}(\rho) \overline{H_G(p)}.$$

Theorem 26.1 transfers this identity uniformly for bounded $G \in \mathcal{F}$. Substitution of G_L proves the result. Equivalently, for every bounded family of inputs,

$$\langle W J F_+, J G_L \rangle = -m H_{F_+}(\rho) \overline{\ell_L(G_L)} + o(1). \quad (345)$$

$\square$

In particular, normalizing the divergent two-vector test does not retain the desired nonzero cross limit. The endpoint constraint suppresses the very reflected evaluation detected by the Weil form. This is a limitation of a fixed finite-dimensional input space with the specified resolvent and sampler. It does not exclude test spaces that grow or change with L , nor other ways to handle the boundary channel.

A remaining alternative and its cost. A finite test can instead be corrected directly:

$$w_L = v^- - \delta(v^-) c_{L,K}, \quad \delta w_L = 0.$$

It is bounded and differs from v^- by a norm-vanishing vector, but $A_\rho^* w_L = x_- - \delta(v^-) A_\rho^* c_{L,K}$ is no longer the image of a bounded input in the same fixed test space. The exact price is

$$\langle W J F_+, A_\rho^* w_L \rangle = \langle W J F_+, x_- \rangle - \overline{\delta(v^-)} \langle W J F_+, A_\rho^* c_{L,K} \rangle.$$

For this particular large shell, $\|A_\rho^* c_{L,K}\|^2 = \|D c_{L,K}\|^2 + |\mu|^2 \|c_{L,K}\|^2$ by symmetry. Consequently its graph correction has size comparable to $e^{-dL/2} \sqrt{K/L}$, which diverges at the specified cutoff. A different endpoint correction or a smaller auxiliary shell inside the same ambient space may change that cost. Either choice requires an independent bound for the displayed Weil pairing and the complete graph source. The present calculation establishes no such bound.

A parity benchmark that uses ordinary norm transfer only

Let $PV_n = V_{-n}$ and $P_+ = (I + P)/2$. Then $P_+ \succeq 0$ and $[D, P_+] = DP$. For the fixed one-sided root family the even projection of the Hardy output is injective. To see this, a vanishing even part implies $\kappa(s)F(s) + \kappa(1-s)F(1-s) = 0$ by analytic continuation. Writing $F = \zeta R_F$ and using the functional equation gives a meromorphic identity whose rational principal parts lie in disjoint right and left windows. The coefficients at each pole on the right must vanish, so $F = 0$. Finite-dimensional

compactness gives a positive lower norm bound. Ordinary Riemann-sum convergence transfers it to J^c ; the centered phase commutes with reflection, and the norm-small endpoint correction preserves the bound for J . Thus P_+ is a bounded noncommuting positive benchmark. Its exact Lyapunov source is $P_+(ez + \mathcal{R}F)$, however, and the new graph term is still present. Positivity by itself supplies no small upper bound.

29 The arithmetic boundary observable

Proposition 29.1 (Exact arithmetic realization of the CCM boundary channel). *Let $L = 2 \log \lambda$, $t_n = 2\pi n/L$, and let $\mathfrak{b}_\lambda(n)$ for $n \in \mathbb{Z}$ denote the real odd coefficient in the CCM rank-two commutator before physical rescaling, so that*

$$\beta_{\lambda,N}^{\log} = \frac{2\pi}{L} \sum_{|n| \leq N} \mathfrak{b}_\lambda(n) V_n. \quad (346)$$

With

$$\rho_\infty(y) = \frac{e^{y/2}}{e^y - e^{-y}}, \quad \mathcal{V}(y) = 2(e^{y/2} - 1) - \sum_{2 \leq k \leq e^y} \frac{\Lambda(k)}{\sqrt{k}}, \quad 0 \leq y \leq L,$$

where $\mathcal{V}$ is exactly the centered prime cumulative already used in (151), one has the finite, zero-independent identity

$$\begin{aligned} \mathfrak{b}_\lambda(n) &= \frac{32L \sinh^2(L/4) n}{L^2 + 16\pi^2 n^2} + \frac{1}{\pi} \int_0^L \sin(t_n y) \rho_\infty(y) dy \\ &\quad + \frac{1}{\pi} \sum_{1 < k \leq e^L} \frac{\Lambda(k)}{\sqrt{k}} \sin(t_n \log k) \end{aligned} \quad (347)$$

$$= \boxed{\frac{t_n}{\pi} \int_0^L \mathcal{V}(y) \cos(t_n y) dy + \frac{1}{\pi} \int_0^L (\rho_\infty(y) - e^{-y/2}) \sin(t_n y) dy.} \quad (348)$$

Equivalently, if

$$\mathcal{E}_L(t) := \sum_{1 < k \leq e^L} \frac{\Lambda(k)}{\sqrt{k}} \sin(t \log k) - \int_0^L e^{y/2} \sin(ty) dy,$$

then at the lattice frequencies

$$\boxed{\mathfrak{b}_\lambda(n) = \frac{\mathcal{E}_L(t_n)}{\pi} + \frac{1}{\pi} \int_0^L \sin(t_n y) \rho_\infty(y) dy - \frac{4t_n(1 - \lambda^{-1})}{\pi(1 + 4t_n^2)}.} \quad (349)$$

Thus the same centered pole–prime discrepancy that drives $G1$ also generates the boundary vector appearing in the graph calculation; no nontrivial-zero data occur in (347)–(349).

More precisely, write an arbitrary finite vector as

$$v = L^{-1/2} \sum_{|n| \leq N} h_n V_n, \quad H_v(y) = \frac{1}{L} \sum_{|n| \leq N} h_n e^{it_n y}, \quad A_v(y) = \overline{H_v(-y)} - \overline{H_v(y)}.$$

Then, for the first-slot-linear convention used throughout this manuscript,

$$\boxed{i\sqrt{L} \langle \beta_{\lambda,N}^{\log}, v \rangle = \int_0^L \mathcal{V}(y) A'_v(y) dy + \int_0^L (\rho_\infty(y) - e^{-y/2}) A_v(y) dy.} \quad (350)$$

Since A_v is L -periodic and odd, one also has the exact folded form

$$\begin{aligned} i\sqrt{L} \langle \beta_{\lambda,N}^{\log}, v \rangle &= \int_0^{L/2} (\mathcal{V}(y) + \mathcal{V}(L-y)) A'_v(y) dy \\ &+ \int_0^{L/2} [\rho_\infty(y) - e^{-y/2} - \rho_\infty(L-y) + e^{-(L-y)/2}] A_v(y) dy. \end{aligned} \quad (351)$$

Proof. Connes–Consani–Moscovici show that the truncated Weil matrix has $\tau_{ij} = (b_i - b_j)/(i - j)$ off the diagonal and that its commutator with the integer Fourier generator is the corresponding rank-two matrix [7]. Their point contribution

$$W_{0,2}(V_n, V_m) = \frac{32L \sinh^2(L/4)(L^2 - 16\pi^2 mn)}{(L^2 + 16\pi^2 m^2)(L^2 + 16\pi^2 n^2)}$$

therefore contributes $32L \sinh^2(L/4)n/(L^2 + 16\pi^2 n^2)$ to b_n . For $n \neq m$, the kernel already used in Lemma 20.17 is

$$q(U_n, U_m)(y) = \frac{\sin(t_m y) - \sin(t_n y)}{\pi(n - m)}.$$

The signs in the semilocal Weil decomposition then give the positive prime sine sum and the positive archimedean sine integral in (347). This proves the first identity directly from the finite Weil matrix.

The point term also has the integral representation

$$\frac{32L \sinh^2(L/4)n}{L^2 + 16\pi^2 n^2} = -\frac{1}{\pi} \int_0^L (e^{y/2} + e^{-y/2}) \sin(t_n y) dy.$$

In Stieltjes form,

$$d\mathcal{V}(y) = e^{y/2} dy - \sum_{1 < k \leq e^L} \frac{\Lambda(k)}{\sqrt{k}} \delta_{\log k}(dy).$$

Substitution in (347) gives

$$\mathfrak{b}_\lambda(n) = -\frac{1}{\pi} \int_0^L \sin(t_n y) d\mathcal{V}(y) + \frac{1}{\pi} \int_0^L (\rho_\infty(y) - e^{-y/2}) \sin(t_n y) dy.$$

The boundary term in Stieltjes integration by parts vanishes because $t_n L = 2\pi n$, giving (348). Formula (349) follows from

$$\int_0^L e^{-y/2} \sin(t_n y) dy = \frac{4t_n(1 - \lambda^{-1})}{1 + 4t_n^2}.$$

Finally, (346) and the first-slot-linear inner product give

$$\sqrt{L} \langle \beta_{\lambda,N}^{\log}, v \rangle = \frac{2\pi}{L} \sum_{|n| \leq N} \mathfrak{b}_\lambda(n) \overline{h_n}.$$

But

$$A_v(y) = \frac{2i}{L} \sum_{|n| \leq N} \overline{h_n} \sin(t_n y), \quad A'_v(y) = \frac{2i}{L} \sum_{|n| \leq N} t_n \overline{h_n} \cos(t_n y),$$

which proves (350). Periodicity gives $A_v(L-y) = -A_v(y)$ and $A'_v(L-y) = A'_v(y)$; splitting at $L/2$ proves (351). $\square$

Proposition 29.2 (The present absolute PNT remainder is quantitatively insufficient for the G2 boundary channel). *Fix $T > 0$. The unconditional estimate (151), inserted absolutely into the exact formula (348), gives constants $c_T, C_T > 0$ such that*

$$\sup_{|t_n| \leq T} |\mathbf{b}_\lambda(n)| \leq C_T(1 + \lambda e^{-c_T \sqrt{L}}). \quad (352)$$

In particular this already-audited PNT input, used only by absolute values, does not imply that the boundary observable $\sqrt{L} \langle \beta^{\log}, v \rangle$ on a varying test family tends to zero. The extra factor $e^{-L/2} = \lambda^{-1}$ that closes the G1 pole–prime estimate comes from the special co-Poisson cross-tail kernel and is absent from the periodic Hardy boundary kernel in (350). This records the limitation of this absolute-value estimate. It is not a lower bound on the actual error, nor a proof that signed cancellation is impossible.

Proof. The second integral in (348) is $O_T(1)$: near zero $\rho_\infty(y) = (2y)^{-1} + O(1)$ while $\sin(t_n y) = O_T(y)$, and for $y \geq 1$ the kernel decays exponentially. For the first integral, (151) gives

$$|t_n| \int_1^L |\mathcal{V}(y)| dy \ll_T \int_1^L e^{y/2 - c\sqrt{y}} dy \ll_T \lambda e^{-c_T \sqrt{L}},$$

after splitting at $L/4$ and decreasing c_T if necessary. The bounded initial interval is absorbed into the constant. This proves (352). $\square$

30 Analytic filters, moving phases, and the reflected pole

The following continuum identities remain useful diagnostics. They do not identify the endpoint or graph behavior of the corrected finite sampler without a further transfer argument.

The exact meromorphic normal form

For an off-line zero ρ of order m , the critical-line adjoint resolvent multiplier is $-(s - \rho^\sharp)^{-1}$. Define

$$H_{v+}(s) = -\frac{\kappa(s)\zeta(s)}{(s - \rho)^m(s - \rho^\sharp)}, \quad H_{v-}(s) = -\frac{\kappa(s)\zeta(s)}{(s - \rho^\sharp)^{m+1}}.$$

Their continuum values at the uncentered cut follow by the right-hand Cauchy pole at σ . They give

$$\alpha_\infty = \left(\frac{\sigma - \rho}{\sigma - \rho^\sharp} \right)^m, \quad H_{\tilde{v}}(s) = \frac{\kappa(s)\zeta(s)}{s - \rho^\sharp} \left[\frac{\alpha_\infty}{(s - \rho)^m} - \frac{1}{(s - \rho^\sharp)^m} \right].$$

The bracket vanishes at σ , canceling the Hardy Cauchy pole. At ρ the zeta zero cancels the apparent pole. At $\rho^\sharp$ there remains exactly one simple pole, with residue

$$-\kappa(\rho^\sharp) \frac{\zeta^{(m)}(\rho^\sharp)}{m!} \neq 0. \quad (353)$$

This is an algebraic statement about the uncentered continuum transform. For the corrected finite resolvent tests, Corollary 28.7 proves that the endpoint ratio instead diverges in magnitude; it cannot tend to this finite α_∞ .

Adding a transform holomorphic at $\rho^\sharp$ cannot change this residue. If such a correction w is cross-invisible to the top root, then, wherever the paired form is well-defined,

$$QW(g_{\rho,m}, A_\rho^*(cv^- + w)) = \bar{c} C_{\rho,m}.$$

The conjugate on c is required by the first-slot-linear convention. This corrects the opposite convention used in the previous proof.

Proposition 30.1 (The bounded-filter tradeoff). *Let Φ_L be bounded and holomorphic on $\Re s > 1/2$ and admit continuation near $\rho^\sharp$. Put $M_L = \|\Phi_L\|_{H^\infty(\Re s > 1/2)}$. If $|\Phi_L(\rho)\overline{\Phi_L(\rho^\sharp)}| \geq c_0 > 0$, then*

$$|\Phi_L(\rho^\sharp)| \geq c_0/M_L.$$

In particular residue damping by $e^{-\epsilon L}$ while retaining that product forces $M_L \geq c_0 e^{\epsilon L}$.

Proof. Since ρ is in the right half-plane, $|\Phi_L(\rho)| \leq M_L$. The conclusion follows by dividing the product lower bound by M_L . $\square$

To interpret the product as a global Weil cross form, impose in addition global transform admissibility and the strip/tail estimates required for that form and its transfer. Local continuation at two points is not sufficient. A boundary-annihilation ratio also requires $\Phi_L(\sigma) \neq 0$. Under these extra conditions the cross constant is multiplied by $\Phi_L(\rho)\overline{\Phi_L(\rho^\sharp)}$ and the residue by $\Phi_L(\rho^\sharp)$.

The absolute PNT envelope applied to a reflected image of size $|\Phi_L(\rho^\sharp)|e^{-dL}$ yields an upper estimate with scale $|\Phi_L(\rho^\sharp)|e^{(1/2-d)L-c\sqrt{L}}$. Making this particular estimate tend to zero by amplitude damping alone calls for exponentially small residue and hence an exponentially large available Hardy norm. The existing G1 estimate pays uniform Sobolev constants and does not cover that regime. This is a limitation of this proof strategy. A large upper bound on an error does not prove that the actual error is large, and the calculation does not exclude signed cancellation or a new estimate that avoids that norm cost.

Moving phases and the omitted periodic image

For real τ the multiplier $H_F^{(\tau)}(s) = e^{-\tau(s-1/2)}H_F(s)$ translates the inverse transform to $g_F(y - \tau)$. Global cross forms are invariant, because the two reflected factors cancel. The uncentered Cauchy value at zero is $e^{-q\tau}\mathcal{E}(F)$ for $\tau > 0$. These are exact continuum statements, and the coefficientwise graph identity changes its source by the same phase.

The earlier claimed periodic kernel bound proportional to $e^{-\tau/2}$ omitted the nearest reflected-pole image. A pole at real displacement $-d$, $0 < d < 1/2$, contributes a term on the scale $e^{-d(L-\tau)}$ at the opposite end of the period. For example, $d = 0.1$ and $\tau = 0.26L$ give $e^{-0.074L}$, larger than $e^{-0.13L}$. Restricting $\tau < L/2$ does not repair that comparison.

The historical threshold comparison

$$\frac{d}{d+q} < \frac{1}{2q+1} \iff d < 1/2 \quad (q > 0)$$

is algebraically correct. It proves a conflict only if both proposed asymptotic thresholds have first been justified, with remainders small relative to the relevant periodic image. An absolute $O(e^{-cL})$ remainder is not enough if the image decays faster. The previous unconditional exclusion of the whole moving-phase route is therefore withdrawn. Moving σ likewise requires uniform smoothing, cutoff, and transfer estimates; the scalar relation $\sigma F(\sigma) \rightarrow \eta_B(F)$ on a fixed root space is not such a uniform theorem.

31 Candidate changes, exact matching, and their scope

Projective candidate invariance is an exact finite identity

For any candidate k with $a = \eta_\partial(k) \neq 0$, parity is not needed for the full commutator identity

$$W \frac{Dk}{a} = \frac{DWk}{a} - \beta + \frac{\beta^*(k)}{a} \eta_\partial.$$

When paired with a fixed test v having $\eta_\partial(v) = 0$, the last term vanishes. For two such candidates,

$$\langle W(Dk_1/a_1 - Dk_2/a_2), v \rangle = \langle DWk_1/a_1 - DWk_2/a_2, v \rangle.$$

Consequently a candidate change cannot alter that fixed paired mismatch when both normalized G1 pairings tend to zero. This does not assert a nonzero limit, and it does not cover changes to the tests, matrix, or sampler. Replacing the prolate candidate by an endpoint shell preserves this exact identity but does not import a disputed G2 lower limit.

An exactly matching sampler

On an algebraic one-sided root space put

$$R_F(s) = F(s)/\zeta(s), \quad R_{BF}(s) = (s - 1/2)R_F(s) - \eta_B(F).$$

For an even finite candidate k with physical endpoint $B_k \neq 0$, let $z^{(k)} = iDk/B_k$ and define

$$[J^{(k)}F]_n = [z^{(k)}]_n R_F(s_n). \quad (354)$$

Multiplication of the rational graph equation by $[z^{(k)}]_n$ proves

$$iDJ^{(k)} - J^{(k)}B = z^{(k)}\eta_B.$$

With $x = J^{(k)}F$ and $q = i\eta_\partial(x)/\eta_\partial(k)$,

$$\eta_B(F)z^{(k)} - qDk = \frac{iDk}{B_k}(\eta_B(F) - \delta(x)).$$

This removes a vector mismatch algebraically. Endpoint recovery is only one remaining requirement: injectivity, uniform ordinary norm bounds, Weil transfer, and weak G1 on these tests also need proof. No coercivity follows from the displayed coefficient identity alone.

Endpoint recovery is a periodic Laplace problem

Let k be a trigonometric polynomial on $[0, L]$, with $k(0) = k(L) = B_k$, and $\Re \alpha > 0$. Put

$$E_{L,k}(\alpha) = \frac{\alpha}{B_k(1 - e^{-\alpha L})} \int_0^L e^{-\alpha y} k(y) dy.$$

Solving $(\partial_y - \alpha)x = k'/B_k$ with periodic boundary condition gives $x(0) = 1 - E_{L,k}(\alpha)$: multiply the equation by $e^{-\alpha y}$, integrate, and integrate its right side by parts. Since $\partial_\alpha(\partial_y - \alpha)^{-1} = (\partial_y - \alpha)^{-2}$, higher Jordan coordinates satisfy

$$\eta_B(F_{\rho,j}) - \delta(J^{(k)}F_{\rho,j}) = \frac{\partial_\alpha^{j-1} E_{L,k}(\alpha)}{(j-1)!}, \quad \rho = 1/2 + \alpha.$$

This formula is exact for each finite candidate.

For the endpoint shell $k = c_{L,K}$ of (319), the coefficient of the first adapted root is $\sqrt{L}(2K)^{-1}it_n/(it_n - \alpha)$ on its $2K$ modes. If $K/L \rightarrow \infty$, its physical endpoint tends to one while its squared ordinary norm is $O(L/K) \rightarrow 0$. Higher root endpoints have the corresponding vanishing limits and their ordinary masses also vanish. Thus near-perfect endpoint matching can coexist with collapse of the whole sampled norm. The shell exact matcher is not a coercive transport.

Global divisibility and endpoint-jet expansions

For a smooth exact-moment source h supported up to λ , the Mellin identity is $M_0(Eh) = \zeta M_0h$. Set $a = \log \lambda$, $\varphi(y) = e^{y/2}h(e^y)$ for $y \leq a$, and $B^+ = \varphi(a) \neq 0$. With the positive Mellin convention used in the Sonine sections,

$$it M_0h(1/2 + it)/B^+ = e^{ita} - \frac{1}{B^+} \int_{-\infty}^a \varphi'(y)e^{ity} dy.$$

This is integration by parts. The negative Fourier convention of Section 22 represents the same formula after $y \mapsto -y$; the signs must be changed together. If $\varphi' \in L^1$, the bulk integral tends to zero for fixed λ and $|t| \rightarrow \infty$. Thus the global zeta quotient cancels but a nondecaying endpoint character remains. If $\varphi' \in L^2$, Plancherel gives bulk squared norm $\|\varphi'\|_2^2/|B^+|^2$.

For $r \geq 1$, assuming one-sided derivatives, vanishing lower traces, and $\varphi^{(r)} \in L^1$, repeated integration gives

$$it M_0h(1/2 + it)/B^+ = e^{ita} \sum_{j=0}^{r-1} \frac{(-1)^j \varphi^{(j)}(a)}{B^+(it)^j} + O\left(\frac{\|\varphi^{(r)}\|_1}{|B^+||t|^{r-1}}\right).$$

This is a fixed-cutoff high-frequency expansion. The terms $(it)^{-j}$ cannot be subtracted through $t = 0$ without a low-frequency prescription. On the lattice $t_n = 2\pi n/L$, $L = 2a$, the character is $(-1)^n$. Its appearance is consistent with the centered/unphased distinction and does not license treating the two samplers as one vector.

Shell estimates and the limited scalar-shift obstruction

The corrected logarithmic diagonal is sufficient for shell weak-pairing estimates. For the explicit shell and a bounded ordinary-norm test,

$$|\langle W_{c_{L,K}}, v \rangle| \leq C\lambda(1 + \log(4K + 1))\sqrt{L/K} \|v\|.$$

For a Hardy test with adequate weighted coefficient decay, the weighted convolution estimate controls $\langle DWc, v \rangle = \langle Wc, Dv \rangle$ similarly. One may enlarge a common ambient cutoff to make these shell pairings small, but (332) still contains its separate beta term. The old shell calculation is not a proof of graph compatibility.

Finally a scalar-shift shortcut has a limited, elementary obstruction. If on the same normalized candidate a full residual is $o(1)$, while a proposed shift has $|\alpha| \geq \alpha_0 > 0$ and normalized derivative norm bounded below, the reverse triangle inequality prevents the shifted full residual from also tending to zero. Weak fixed-band G1 supplies neither of those full-norm premises. The exact cancellation in Proposition 28.3 is the appropriate statement for the weaker paired argument.

32 The remaining fixed-space criterion and research goals

The original relative boundary criterion is unchanged. If, for every fixed nontrivial zero and prime p ,

$$\frac{\|P_R D_p(v_\rho^{pR} - v_\rho^R)\|}{\|v_\rho^R\|} \longrightarrow 0,$$

then contractivity of $T_{R,p}$ and (2) give $|p^{1/2-\rho}| \leq 1$. Reflection gives RH. The critical-line rate derived from Assumption 2.1 is consistent with this criterion; it does not establish the off-line estimate. Equivalently, the exact diagonal defect asks for an independent arithmetic sign on each zero evaluator. Full positivity on their span is an unnecessarily strong sufficient condition. Proposition 8.2 now removes a technical prerequisite from this alternative: the Fredholm remainder and its signed moment correction vanish after evaluator normalization without invoking Assumption 2.1. The precise outstanding target is (29) at the single prime 2. The elementary off-line upper bound suffices for this reduction; a new arithmetic argument must still orient the leading shell imbalance. The reduction does not infer that sign from the already known prime eigenvalue. Proposition 9.2 now writes the arithmetic constraint as the local distributional equation $\mathcal{A}y = A_y + B_y/t$ on the expanded source interval. The integral-corrected cutoff converges with the explicit bound in (35); the raw cutoff can instead diverge in norm. Corollary 9.3 justifies fixed finite source-Gram calculations, not inversion of a limiting source Gram operator. The next step is a signed quadratic estimate exploiting this equation together with both Sonine support conditions. Neither linear source orthogonality alone nor the power response (37) supplies that estimate.

The weak G1 source chain is now closed: Theorem 20.1 supplies compact-source radicality and the semilocal domain passage; Proposition 20.5 and Corollary 20.6 give the normalized source and exact repair; the subsequent tail, endpoint and correlation estimates give (154) and (156). The deterministic finite transfer in Theorem 20.37 is consequently unconditional for this source and a fixed physical test band.

Three distinct research tasks remain for the finite-Weil route. First, seek genuinely uniform estimates for a growing deep prolate block, or the separately stated continuous-remainder improvement needed for an economical near-Slepian cutoff. Neither is required for the proved weak G1 limit, and neither follows from its fixed-source domain passage. Second, estimate the corrected graph pairings on a new, fully defined reflected test family. The finite object is fixed by (321), but the former bounded endpoint ratio is ruled out by Corollary 28.7. Theorem 28.8 further shows that bounded resolvent tests from any fixed finite-dimensional input space lose the reflected cross signal under exact endpoint annihilation. A viable replacement must address that obstruction, for example through a changing test space or a direct endpoint correction with its additional Weil pairing controlled. For such a replacement, the full graph source, the relevant meromorphic transform errors, and any entire zero-sum estimate remain obligations. Third, prove a positive metric comparison whose Lyapunov defect tends to zero independently of the assumed off-line roots. Restricted scalar coercivity is already a consequence of form transfer and cannot serve as that independent input. Section 19 now sharpens this task: Theorem 19.2 supplies a finite-core criterion, and Proposition 19.3 constructs a positive, uniformly coercive metric that kills the zeta source and endpoint-shell graph terms exactly. Its remaining commutator has the strictly positive variance limit in Proposition 19.4; no independent vanishing signed pairing follows. Removing more source derivatives by bounded projections loses coercivity by Proposition 19.5. A sharp fixed-band Weil correction faces a separate transfer cost: Proposition 19.7 proves exponential off-line transform growth after source projection. Proposition 19.8 computes the reflected-pair term: unless its explicit endpoint product vanishes, it remains exponentially large and oscillatory. Any

proposed cancellation must therefore be checked over the entire zero sum or on the actual finite prime/pole/archimedean form; bounded ordinary norms and pairwise reflection do not supply it. There is now a transfer-preserving alternative: Proposition 19.9 removes the source and the two shell directions on the full expanding Fourier space. It proves a uniform summable strip-product estimate, fixed-core coercivity and the unchanged global Weil pullback. The next metric target is its remaining signed commutator, or an independently justified arithmetic correction to it. Reusing the same projected Weil form plus a fixed scalar is excluded by Corollary 19.10: the Weil pullback tends to zero on the one-sided root space and cannot cancel the nonzero scalar metric budget. A proposed correction must be tested against the full relative criterion before any positivity claim is made. Proposition 19.12 now supplies a stronger diagnostic: at tail rate $p > 1$, the actual finite Weil action converges in norm to the projected finite exponential sum of zero values. Consequently the positive square SW^2S has the top-root growth and lower-jet collapse in Corollary 19.13; it does not furnish the missing signed estimate. Further metrics should be checked for both the relative defect and multiplicity-sensitive coercivity, with any scale change accounted for explicitly.

The fixed-space route has separate foundational obligations: prove the quantitative evaluator estimates and give a complete closed-operator realization if the Riesz-window formulation is retained. These are not prerequisites for the finite-core exclusion criterion in Theorem 19.2, nor for the direct complete-Weil route below. Bypassing them does not establish them for the Riesz formulation. The shell-sign, uniform metric, large-support spectral-overlap, and Schur-complement formulations are alternative targets; none has been shown to follow from weak G1.

Proposition 32.1 (The direct cofinal Weil route). *Suppose $\lambda_j \rightarrow \infty$ and the complete, unshifted semilocal Weil forms satisfy*

$$QW_{\lambda_j}(f, f) \geq -\epsilon_j \|f\|^2 \quad (f \in \mathcal{D}(QW_{\lambda_j})), \quad \epsilon_j \geq 0, \quad \epsilon_j \rightarrow 0.$$

Then the Weil form is nonnegative on every smooth compactly supported test function, and RH follows by Weil's positivity criterion.

Proof. Fix such a test f . For all sufficiently large j , its multiplicative support lies in $[\lambda_j^{-1}, \lambda_j]$. Extension by zero preserves both its ordinary norm and the global Weil value, so

$$QW(f, f) = QW_{\lambda_j}(f, f) \geq -\epsilon_j \|f\|^2.$$

Sending j to infinity proves nonnegativity. Apply the compact-support Weil criterion recalled in [7]. $\square$

Consequently the full cofinal conclusion of Proposition 20.72, if its hypotheses and vanishing error are proved, would already yield RH by this direct route. The conditional Riesz realization and evaluator assumptions belong to other routes and are not additional premises of Proposition 32.1. A fixed-window certificate, a bound on a finite compression alone, or an incomplete transport estimate does not meet this proposition. The implication isolates the strength of the missing estimate; it does not establish that estimate.

Proposition 20.76 isolates an inverse-weighted residual that can be far smaller than the full residual divided by the smallest gap. Proposition A.1 certifies one finite case. Proposition 20.78 now identifies the exact projected source there, including its angular tail, and proves a relative physical-endpoint mismatch greater than one. Proposition 21.7 now recovers the exact source endpoint, with relative error $O(\lambda^{-1})$ at the unchanged polynomial Fourier cutoff. Corollary 21.8 puts weak G1 on this same finite vector, uniformly over the ambient Fourier cutoff in the weighted test norm. These results supply no practical certified threshold at $\lambda = 3$, and no endpoint

comparison to an actual ground vector. Proposition 20.19 closes the semilocal Weil operator-core issue and establishes fixed-window graph convergence for BV sources. This is distinct from the open arithmetic realization in Section 18. Proposition 20.21 sharpens the omitted-tail estimate to $N = 256$ at $\lambda = 3$. Proposition 20.39 additionally proves a small full ordinary residual and Rayleigh value for the actual source and its polynomial projection. It gives no endpoint-normalized operator estimate and no lowest-state overlap. Proposition 20.26 now closes the signed head/complement test at $\lambda = 3$ for both parity sectors and the entire closed form. Positive weighted prime bounds and the increasing tail diagonal are essential to its verified inverse-action estimate; the residual's directional moments control every omitted row. Proposition 20.27 now proves that the complete operator at this window has a simple even ground, with $0 < \mu_0 < 3.644 \cdot 10^{-38}$ and $\mu_1 > 10^{-36}$. Corollary 20.28 bounds the ordinary sine angle of the exact normalized $P_{64}p_3$ by 0.01931. The stronger 10^{-34} threshold for all other even states is essential here. The earlier $N = 64$ finite-matrix angle bound and this complete-operator bound are different statements.

The remaining G2 spectral task is a complete lower bound tending to zero along unbounded windows, or sufficient quantitative source-to-ground overlap on such a family. Proposition 32.1 shows why this is RH-strength: even a cofinal sequence of complete lower bounds $-o(1)$ implies nonnegativity of every compact test. A finite list of certified windows cannot replace that estimate. Proposition 20.41 removes an unnecessary positive-gap requirement from one route. The already proved rank-one source residual reduces the target to asymptotic nonnegativity of its entire orthogonal complement. A negative compact test would persist in that complement; the required arithmetic sign is still unproved. Growing source blocks also need an independent uniform operator-residual bound. Proposition 20.42 rules out the claim that substantial cancellation is confined to the single source: an exactly source-orthogonal finite test has non-prime energy greater than 0.4 but positive total energy below $3 \cdot 10^{-31}$. This does not disprove complementary positivity; it prevents using basis or random-vector samples as a uniform dominance argument.

The unsigned-prime shortcut is now excluded by Propositions 20.43–20.44: every finite-dimensional source or Fourier-head removal retains the full prime norm, asymptotic to λ . The rank-two pole term cannot make their combined ordinary operator norm smaller either. The next estimate must therefore use the archimedean energy, directional inverse weights or signed correlations; enlarging a finite source block alone cannot make that unweighted arithmetic norm small. This is an obstruction to a sufficient norm bound, not to full Weil positivity.

Proposition 20.48 now proves a signed lower bound on the entire $\lambda = 4$ tail $|n| > 16$, retaining its exact archimedean diagonal. Both complex parity sectors are included. The effective head $F - B^*T^{-1}B$ has 33 dimensions. Proposition 20.66 certifies its entire 17-dimensional even sector and hence the complete even form. Proposition 20.70 now completes the 16-dimensional odd sector and proves $\varepsilon_4 = 0$. The remaining target is complete ordinary-error control along an unbounded window family, not another finite compression at this already certified window. Corollary 20.68 gives a refined sufficient target: bound the complementary correction below its trial energy, then control the unresolved block with the physical factor $\left\| K_\lambda^{1/2} P_{E_\lambda} V_\lambda^{-1} \right\|^2$ retained. Its error convergence is not established. Proposition 20.75 now certifies the raw head and specified finite-support trial forms K_X positive in both parity sectors. This gives local $\alpha_4 = 0$, without an infinite-tail transfer for K_X . It still does not suffice for the exact Schur sign because the nonnegative residual inverse correction must be subtracted, as in (275). This leaves a separate growing-window target: a corresponding ordinary lower error tending to zero. Proposition 20.45 shows that a dimensionless weighted error must be multiplied by its arithmetic scale before it can serve that purpose.

Proposition 20.50 now reuses the signed factorization to bound the actual inverse-tail correction, including all remote rows through (242). Proposition 20.72 identifies the sufficient ordinary error

$\alpha_\lambda + e_\lambda^2 + t_\lambda^2/\mu_\lambda$. Its operator norms concern the entire growing head; individual source or column bounds do not suffice. Proposition 20.51 additionally gives convergent upper bounds for the complete far inverse, with certified constants at $\lambda = 4$, sharpened to 20 and 90 by Proposition 20.52. The complete far energy now fits below the trial budget on one certified head direction, by Proposition 20.54. This is not the full inverse correction. Proposition 20.58 now proves that its first-polynomial scalar-head bound fails after the complete mixed term is included. Corollary 20.60 also excludes its exact positive head update with the saved initial certificate. Proposition 20.63 proves a complete positive value on the saved head direction. Proposition 20.66 extends this to the entire even head through frozen energy-scaled trials and a simultaneous complete Gram bound. Proposition 20.70 completes the odd sector by a different finite trial in one certified direction and the whole complementary bound. Thus $W_4 \succeq 0$ exactly at this window. The primary next target is the cofinal ordinary-error estimate, rather than a longer finite list of certified windows.

Proposition 20.71 identifies a structural cost: with the present scalar far majorant, its inner cutoff must grow exponentially in λ . The fresh threshold table gives the window dependence explicitly. A viable family must either control the associated head dimensions, conditioning and residual Grams at that cost, or improve the signed far estimate before constructing the growing inner heads. Finite-only scaling diagnostics in the research log do not provide the missing complete witnesses.

The trial-head hypothesis is not yet verified along an unbounded window family. With $\alpha_\lambda = 0$, the remaining target is $e_\lambda \rightarrow 0$ and $t_\lambda/\sqrt{\mu_\lambda} \rightarrow 0$, retaining all ordinary-energy and dimension factors. Inverse approximation convergence alone does not establish those limits. The complete sign at one window closes a local obligation, not G2 or RH.

The successful one-sided bound does not require a norm contraction: Proposition 20.49 certifies that the latter fails at the very same cutoff. Nor can the whole weighted remainder be estimated by an infinite Hilbert–Schmidt sum, by Proposition 20.46. Finite-column residual Grams remain valid. The logarithmic weighted-tail asymptotic in Proposition 20.47 is strictly a fixed-window statement and supplies no uniform joint-limit obstruction.

The continuum endpoint is now settled by Theorem 20.30: every complete eigenfunction has continuous zero physical trace. Hence a nonzero endpoint comparison to the repaired source is false, with exact relative mismatch one. Corollary 20.31 supplies a fixed-window filtered endpoint-decay rate. It gives no sharp-cut or joint growing-window rate for finite-compression ground vectors. Those finite endpoint estimates, the sampler graph defect and full-strip normalization remain separate. The next spectral work should retain these distinctions while seeking an independent signed complementary lower bound, rather than imposing a nonzero continuum ground trace. G2 and RH remain open.

What is established by the sampler correction. The uncentered full-scale transfer in the previous version cannot be used under the stated nonzero-endpoint hypotheses. Centering repairs the form transfer; an explicit shell then restores the original physical endpoint on the same finite matrix. Ordinary norm coercivity and the reflected top-root cross form survive. The graph identity gains an explicit defect with unavoidable derivative growth. Strong graph convergence fails for norm-vanishing endpoint repairs, but that fact alone does not settle the weaker arithmetic pairing needed for the paper’s objective. The new resolvent calculation identifies a second obstruction: on a fixed test space the rescaled endpoint converges to minus the reflected evaluation, which is exactly the evaluation needed for the nonzero cross form. Killing that endpoint with bounded coefficients therefore kills that particular G2 signal.

Claim discipline. This manuscript claims neither RH nor a completed positivity transfer. The pole-aware transfer and the earlier nonzero critical beta limit have been removed from the established chain. The moving-phase exclusion is restricted to its proved algebra, pending relative alias estimates. Unspecified earlier source-quasimode, partner-blind, and certified nonorthogonality calculations are not used as theorems. Numerical checks below include exploratory illustrations, a rigorous finite-matrix certificate and its exact-source identification, complete fixed-window sign certificates, finite-trial positivity and an exact-source cancellation certificate. These do not settle the remaining uniform analytic obligations. A publication draft would need the listed inputs proved or its scope narrowed to the results that are already supported.

A Reproducible finite checks

The accompanying file `check_sampler.py` constructs the real symmetric CCM matrix with the separate logarithmic diagonal, using prime-power sums and numerical quadrature only for the smooth archimedean integrals. The test is $H(s) = \kappa(s)\zeta(s)$ with $\sigma = 2$, $a = 4$, $M = 6$. Its global zero-side form is exactly zero and $E = 2\zeta(2) = 3.289868133696453$. No hypothetical zero location is inserted. At a fixed cutoff $N = 512$:

L	Uncentered form	Centered form	Centered endpoint
4	0.763119	$1.174 \cdot 10^{-3}$	0.129293
6	1.036652	$1.085 \cdot 10^{-4}$	0.033565
8	0.669071	$4.000 \cdot 10^{-6}$	0.008042
10	1.337515	$4.078 \cdot 10^{-7}$	0.001816

For $L = 6$, endpoint correction is exact to the displayed floating-point precision, while its form price decreases and derivative price increases:

$N = 2K$	Corrected form	Correction norm	Derivative norm
128	1.400023	0.705010	72.5381
256	0.906192	0.498518	102.3284
512	0.492218	0.352505	144.5331

These moderate cutoffs do not yet make the corrected form small. The existence and rates in Theorem 26.1 come from the proof, not from extrapolating this table. Increasing the quadrature order on a representative matrix changed entries by less than $2 \cdot 10^{-12}$. The script uses 30-digit zeta evaluations followed by double-precision matrix arithmetic; the checks are not interval-certified bounds.

A rational check of the resolvent endpoint asymptotic

The file `check_resolvent_endpoints.py` tests the sign, half-period phase, shell contribution and residue scale in Lemma 28.5 using explicit rational functions:

$$H_{-}(s) = \frac{(\sigma + a)^M}{(\sigma - s)(s + a)^M}, \quad H_{+}(s) = (s - p)H_{-}(s).$$

Here $\sigma = 2$, $a = 4$, $M = 6$, $\mu = 0.2 + 1.3i$ and $p = 1/2 - \bar{\mu}$. These are model transforms, not zeta root functions. Their Cauchy tail coefficients are $E_{-} = 1$ and $E_{+} = \sigma - p$. The centered samples,

their adjoint resolvents and the endpoint shell correction are summed explicitly in Fourier pairs. For numerical tractability this check uses $K = \lceil e^{L/2} L^2 \rceil$, much smaller than the theorem cutoff. It does not compute a Weil form.

Write $e_{\pm} = \delta V_L H_{\pm}$, interpreting V_L here as the same sampling and resolvent operations applied directly to these transforms. The predicted reflected limit is

$$S_L := \frac{e^{\bar{\mu}L/2}(1 - e^{-\bar{\mu}L})e_-}{-H_-(p)} \longrightarrow 1.$$

The output, rounded from double-precision arithmetic, is:

L	K	$ S_L - 1 $	$ e_-/e_+ $
4	119	$7.347 \cdot 10^{-2}$	4.426
6	724	$5.792 \cdot 10^{-3}$	35.261
8	3495	$9.638 \cdot 10^{-4}$	282.874
10	14842	$2.276 \cdot 10^{-4}$	1869.438
12	58094	$7.847 \cdot 10^{-5}$	5886.940
14	214941	$1.872 \cdot 10^{-5}$	24924.379

This is a reproducible check of the analytic mechanism and conventions, not an interval certificate, a verification on hypothetical zeta zeros, or evidence for RH. The theorem and its application to genuine root functions depend on the contour and uniformity arguments in the text.

A high-precision check of the normalized prolate source

The script `check_prolate_source.py` constructs the even Legendre–Galerkin matrix for $-\partial_x((1 - x^2)\partial_x) + c^2x^2$ on $[-1, 1]$. It takes the first and third even eigenvectors, corresponding to indices 0 and 4, and normalizes them in $L^2(-1, 1)$. The compressed Fourier eigenvalue is recovered from the ratio of the integral to the central value, with the factor $\sqrt{c/(2\pi)}$ retained. Thus the source coefficients are those in (132), not an imposed endpoint fit.

Let S_{λ} denote the noncancellation ratio on the left of (134), and let $Z_{\lambda} = |h_{\lambda}^{\circ}(0)|/|B_{\lambda}^+|$. With 50 even Legendre modes and 85-decimal-digit arithmetic the rounded output is:

c	$b_4/(c^2b_0)$	S_{λ}	Z_{λ}
10	6.40490	1.00164349	$1.92330 \cdot 10^{-2}$
20	9.94110	1.00029407	$4.00922 \cdot 10^{-6}$
30	10.99058	1.00012034	$3.99136 \cdot 10^{-10}$
40	11.51074	1.00006514	$3.10538 \cdot 10^{-14}$

Repeating $c = 40$ with 65 modes and 105-digit arithmetic preserves the displayed figures. The script also reports the concentration loss divided by Fuchs’s leading term and the ratio $b_n^2/[c(1 - \vartheta_n)]$. Increased precision is important because concentration losses become exponentially small. This is a finite spectral approximation with no certified truncation or roundoff bound; the analytic proof, not this table, establishes the asymptotic estimates. No zeta zero is inserted.

A compact-source primitive convention check

The script `check_bv_radical_passage.py` tests the powers and sign in the primitive mechanism of Theorem 20.1. Take $h(u) = u^2 - \frac{5}{3}u^4$ on $[-1, 1]$, with zero extension. It has the two exact moments

and $h(1^-) = -2/3 \neq 0$. If $N = \lfloor 1/u \rfloor$ and $S_j(N) = \sum_{n=1}^N n^j$, an explicit logarithmic primitive is

$$P(u) = \frac{2}{5}u^{5/2}S_2(N) - \frac{10}{27}u^{9/2}S_4(N) + \frac{4}{135}\zeta(\tfrac{1}{2}, N+1),$$

where the last function is the Hurwitz zeta function. Away from a cut, $uP'(u) = \mathcal{E}(h)(u)$; the jump cancellation in this expression agrees with continuity of the primitive. At 70-digit working precision, testing $1/u = N + \theta$ for $N = 10, 100, 1000, 10000$ and $\theta = 0.2, 0.5, 0.8$ gives maximum derivative discrepancy less than $3.3 \cdot 10^{-69}$. The sampled values of $P(u)/u^{3/2}$ remain bounded. This is a toy-source convention check, not an interval certificate or a numerical proof of the uniform primitive bound, radicality, G1, or RH. Those conclusions use the analytic arguments in the text.

Finite-core metric and projection checks

The script `check_section19_metric.py` checks the exact budget (94), source/shell annihilation, (100), and the new projected graph source (102). It uses the rational source samples

$$z_n = \frac{(-1)^n}{\sqrt{L}(2 + it_n)^3}, \quad t_n = 2\pi n/L,$$

with a length-two Jordan chain at $\mu = 0.23 + 0.7i$ and a further eigenvector at $\nu = 0.37 + 1.4i$. These parameters are model eigenvalues, not zeta zeros. The tests include a complex endpoint-shell correction and the first-slot-linear convention. At $T = 3$ the rounded results are

L	$\lambda_{\min}(J^*SJ)$	$\ [D, S]\ $	$ \mathcal{L}_{11} /G_{11}$
8	$2.55980 \cdot 10^{-4}$	0.976536	0.460000
16	$8.40732 \cdot 10^{-4}$	0.997413	0.460000
32	$8.95576 \cdot 10^{-4}$	1.007014	0.460000

The nine algebra-check discrepancies are below $2.1 \cdot 10^{-15}$; the source-polynomial remainder identity is also checked on a grid. The final column equals $2\Re\mu$, as the exact identity predicts: source annihilation does not make the relative Lyapunov defect small. No Weil matrix or zeta zeros enter this model. The calculations are floating-point convention checks, not positivity certificates or proofs of any limiting estimate; Section 19 supplies the analytic arguments.

Sharp-band transform convention checks

The script `check_band_cut.py` uses the rational source $Z(t) = (2 + it)^{-3}$ and $R(t) = (it - \mu)^{-1}$ with $\mu = 0.23 + 0.7i$. It subtracts the exact sampled source projection, sets $T = 3$, and takes $L = 2\pi N/3$, $N = 8, 16, 32, 64, 128, 256$. The second-difference identity in Lemma 19.6 is checked to error below $3 \cdot 10^{-15}$, and direct physical integration at the smallest size agrees to below 10^{-14} . At $N = 128$ the ordinary squared norm is approximately 0.0156032, while the transform magnitudes at $s = 1$ and $s = 0$ are approximately $1.088 \cdot 10^{24}$ and $6.525 \cdot 10^{23}$, respectively. These large values are consistent with the proved cut mechanism, not evidence about zeta zeros or the sign of a Weil form. The experiment evaluates no Weil matrix and supplies no certified numerical error bound. It also checks the transform product at $v = \mu$ and $v^\sharp = -\bar{\mu}$ against the edge expression in Proposition 19.8. At $N = 256$ the relative discrepancy is approximately 0.0148 and the real pair sum is approximately $1.414 \cdot 10^{43}$; this is a rational-model convention check, not a zeta-zero or whole-Weil-form computation.

Full-space projection and holomorphic strip checks

The file `check_full_projection.py` tests Proposition 19.9 with the rational model

$$Z(v) = \frac{(v - \mu)^2(v - \nu)}{(v + 2)^8(2 - v)}, \quad \mu = 0.23 + 0.7i, \quad \nu = 0.37 + 1.4i.$$

The model root outputs $Z(v)/(v - \mu)$, $Z(v)/(v - \mu)^2$ and $Z(v)/(v - \nu)$ are holomorphic on the closed critical strip; no meromorphic-output assumption is hidden in this check. At $L = 8, 12, 16, 24, 32$ it uses the tractable cutoff $K = 4L^2$, not the theorem's exponential cutoff. Eight finite algebra checks, including the complete graph source, both shell directions and the commutator budget, have discrepancies below $4.2 \cdot 10^{-14}$. On a 25-point grid in the strip, the largest ordinate-weighted transform error decreases from approximately 0.02043 at $L = 8$ to $4.121 \cdot 10^{-6}$ at $L = 32$. The model first-eigenvector relative Lyapunov defect remains $0.46 = 2\Re\mu$. No zeta zero or Weil matrix enters this experiment. These are floating-point convention checks, not uniform strip certificates or evidence for RH.

An arithmetic-matrix check of the Weil action

The script `check_weil_action.py` uses the actual finite arithmetic matrix from `check_sampler.py`, including the separate logarithmic diagonal, and the first known critical-line zero $\rho_1 \simeq 1/2 + 14.1347251417i$. The transform is $\kappa(s)\zeta(s)/(s - \rho_1)$ with $\sigma = 2$, $a = 4$, $M = 6$. The root location and derivative are evaluated at 40-digit precision, then samples and matrices use double precision. This is a numerical simple-root check, not a new multiplicity certificate. After the full-space projection, it compares $WSJF$ with the exact finite Fourier projection of $\kappa(\rho_1)\zeta'(\rho_1)e^{(\rho_1 - 1/2)y}$. At the moderate cutoff $K = 256$, $N = 2K = 512$:

L	Absolute action error	Relative action error
4	$7.9105 \cdot 10^{-4}$	$5.7219 \cdot 10^{-2}$
6	$2.3776 \cdot 10^{-4}$	$1.4045 \cdot 10^{-2}$
8	$4.1955 \cdot 10^{-5}$	$2.1459 \cdot 10^{-3}$
10	$1.5078 \cdot 10^{-5}$	$6.8992 \cdot 10^{-4}$

The script also checks $K = 128$, the target Fourier coefficient by independent physical quadrature, and the last matrix action with 400 additional archimedean quadrature nodes. These finite cutoffs do not meet the theorem's exponential rule. No hypothetical off-line zero is inserted, no interval certification is claimed, and the table is not evidence for RH. It checks the action's sign, centered phase and physical normalization using an actual arithmetic matrix.

Signed Fredholm moment check

The accompanying `check_fredholm_shell.py` uses Gauss–Legendre Nyström matrices for the actual kernel $2\cos(2\pi xt)$ on $(0, a)$. It compares the two-channel inverse, the signed A_a formula and the constant-kernel resummation, using complex test profiles at $a = 0.2, 0.1, 0.05, 0.025$ and two quadrature orders. It separately checks that reversing one profile reverses the mixed term. The profiles are arbitrary interval data, not certified Sonine vectors or arithmetic evaluators. These floating-point checks test signs, scale factors and the remainder formula only; they do not test the arithmetic shell sign, certify a uniform estimate, or supply evidence for RH.

Regularized co-Poisson adjoint check

The script `check_copoisson_adjoint.py` uses $\phi(t) = (t-1)^3(2-t)^3$ on $[1, 2]$, extended by zero, and $h = \mathcal{D}\phi$. Both source moments vanish exactly. This toy source has sufficient piecewise smoothness for the cutoff estimates; it is not asserted to be a smooth arithmetic-core vector or an evaluator. At 45-digit precision, independent physical and adjoint integrals are compared for $y(x) = e^{-x}$ and $M = 4, 8, 16, 32$. The weak limit is checked against its convergent zeta-coefficient series. A separate power-response check uses $w = 0.7 + 1.1i$, not a zeta zero, and M up to 1024. These checks include $y(x) = e^{-(1+0.4i)x}$ to test the first-slot conjugation. The non-certified computations check counterterm signs and scales, not the shell inequality or RH. Results are recorded in the supplied JSON.

A certified finite Weil matrix and candidate angle

The following finite result is certified by interval arithmetic; the earlier numerical illustrations remain non-certified. Set $\lambda = 3$, $L = 2 \log 3$, $N = 64$ and retain all 129 indices $-64 \leq n \leq 64$ in the translated $[0, L]$ basis U_n . The matrix W is the same arithmetic Weil matrix used above, with all prime powers 2, 3, 4, 5, 7, 8, 9, the pole term and the separately computed logarithmic diagonal. No list of zeta zeros is used.

The candidate is defined exactly, rather than by an accuracy claim: the 65 decimal strings in `g2_finite_candidate.json` are interpreted as rational numbers $c_0, \dots, c_{64}$, with $c_{-n} = c_n$, and $u = c / \|c\|$. The file's SHA-256 is

6d326cafbe715752f89b90cd92eb1d5a68
ab0b78befb6afe074e797372b1a18e

The two lines concatenate to the identifier. The strings were generated from a 105-digit, 70-even-mode approximation of the repaired source, but the conclusions below depend only on their exact rational values. This candidate definition alone does not certify proximity to the true PSWF coefficients; Appendix A supplies that bound.

Proposition A.1 (Computer-assisted finite result). *For this exact W and rationally specified normalized candidate, the complement in Proposition 20.76 satisfies*

$$C - \alpha I \succ 10^{-34}I, \quad \|(C - \alpha I)^{-1}r\| < 0.000216.$$

The matrix is positive definite and has a simple even ground state, with

$$3.64 \cdot 10^{-38} < \lambda_0 < 5.32 \cdot 10^{-38}, \quad \sin \theta < 0.000216. \quad (355)$$

Certificate and verification. The supplied `certify_g2_finite.py` uses Python-FLINT 0.9.0 and FLINT 3.6.0 ball arithmetic [18] at 512 bits. It encloses all archimedean integrals, uses both exact reflection sectors, and constructs the complement by the Householder transformation determined by the exact normalized candidate. Orthogonality belongs to that underlying exact matrix, not to arbitrary independent choices from its entrywise balls. Interval LDL^T elimination gives 64 strictly positive pivot enclosures for each of

$$C_{\text{even}} - \alpha I - 10^{-34}I, \quad W_{\text{odd}} - \alpha I - 10^{-34}I.$$

An enclosed LU solve gives $q < 0.000216$ and verifies, by strict interval comparisons,

$$\alpha < 5.32 \cdot 10^{-38}, \quad \alpha - \langle (C - \alpha I)^{-1}r, r \rangle > 3.64 \cdot 10^{-38}.$$

Proposition 20.76 proves the claims.

The independent program `certify_g2_series.py` repeats the certificate at 768 bits using the analytic series below, with 128 terms and explicit tail bounds. It uses a rational, nonorthonormal complement in the full 129-dimensional space. For the unnormalized rational vector c , let $s = c^T c$ and choose $c_k \neq 0$. Columns $Q_j = e_j - (c_j/c_k)e_k$, $j \neq k$, span $c^\perp$ exactly. Put

$$G = Q^T Q, \quad H = Q^T (W - \alpha I) Q, \quad b = Q^T W c.$$

All 128 interval pivots of $H - 10^{-34}G$ are positive. For the enclosed solution $H z = b$, the physical quantities are $q^2 = z^T G z / s$ and $E = b^T z / s$. The Gram matrix G is retained. This independently verifies the same three strict inequalities. All 8321 parity-block entries also have overlapping enclosures in the two independently assembled matrices. The supplied JSON files retain the pivot and result intervals, input hash, versions and precision. No approximate floating-point solve is used for either certificate. $\square$

For reproducibility, write $\rho_\infty(y) = e^{y/2}/(2 \sinh y)$, $t_n = 2\pi n/L$ and let ψ_0, ψ_1 denote digamma and trigamma. The first calculation removes the apparent singularity at zero using $\text{sinc } z = \sin z/z$ and $\sinh y/y = \text{sinc}(iy)$. The second uses the following exact formulas for the sine integral and the *complete* archimedean diagonal:

$$\begin{aligned} S_n &:= \int_0^L \rho_\infty(y) \sin(t_n y) dy = -\frac{1}{2} \Im \psi_0(z_n) - \sum_{k \geq 0} e^{-a_k L} \frac{t_n}{a_k^2 + t_n^2}, \\ A_n^\infty &:= \int_0^L \rho_\infty(y) \{2(1 - y/L) \cos(t_n y) - 2e^{-y/2}\} dy \\ &\quad + \log(4\pi) + \gamma_E + \log \tanh(L/2) \\ &= \log \pi - \Re \psi_0(z_n) - \frac{\Re \psi_1(z_n)}{2L} + \frac{2}{L} \sum_{k \geq 0} e^{-a_k L} \frac{a_k^2 - t_n^2}{(a_k^2 + t_n^2)^2}, \end{aligned}$$

where $a_k = 2k + \frac{1}{2}$ and $z_n = \frac{1}{4} - it_n/2$. To verify them, expand $\rho_\infty(y) = \sum_{k \geq 0} e^{-a_k y}$. Integrate the convergent combinations $\rho_\infty \sin(t_n y)$, $\rho_\infty(\cos(t_n y) - e^{-y/2})$, and $y \rho_\infty \cos(t_n y)$ on $(0, \infty)$ and subtract their tails. The identity $t_n L = 2\pi n$ simplifies the endpoint terms. The odd exponential tail cancels $\log \tanh(L/2)$, while $\psi_0(1/2) = -\gamma_E - 2 \log 2$ leaves $\log \pi$. The digamma difference must not be split into two divergent sums. After retaining $k < K$, the absolute tail bounds are

$$R_S \leq \frac{e^{-a_K L}}{2a_K(1 - e^{-2L})}, \quad R_A \leq \frac{2e^{-a_K L}}{La_K^2(1 - e^{-2L})}.$$

These follow from $|t|/(a^2 + t^2) \leq 1/(2a)$ and $|a^2 - t^2|/(a^2 + t^2)^2 \leq 1/a^2$. At $\lambda = 3$ the weights are the exact rationals 3^{-4k-1} ; both tails are enclosed symmetrically.

Scope and endpoint limitation. This certificate concerns one finite matrix and one explicitly frozen candidate. It gives no lower bound on the omitted Fourier complement, no growing-window ordering theorem. Its fixed-source approximation prerequisite is addressed in Appendix A. The preparatory non-certified source pilot gives relative physical-endpoint errors about 1.375 at $N = 64$ and 0.1046 at $N = 4096$ for $\lambda = 3$; the latter calculation concerns source coefficients only, not a matrix of that dimension. Small ordinary ground-state angle therefore cannot substitute for endpoint recovery. An endpoint-shell correction would still require its derivative and graph terms. The full-strip transform cost and an independent uniform arithmetic estimate remain open. Neither G2 nor RH follows from this finite certificate.

Certified exact-source identification and endpoint error

The additional verifier `certify_pswf_source.py` upgrades the source comparison at this one window from numerical agreement to a computer-assisted proof. Its input `pswf_rational_input.json` contains two 70-coordinate exact decimal rational vectors and two exact decimal rational approximate eigenvalues. Its SHA-256 is

ac9dfa8c03f54a326e3ba7af59917ac435e
63d41e978310301e6df6e53e83076

The two lines concatenate. Approximate eigensolver output is used only to select this input; its accuracy is not assumed. Python-FLINT 0.9.0 / FLINT 3.6.0 at 768 bits verifies the following steps.

In the normalized Legendre basis, $Xe_\ell = A_\ell e_{\ell+1} + A_{\ell-1} e_{\ell-1}$ with $A_\ell = (\ell+1)/\sqrt{(2\ell+1)(2\ell+3)}$ and $A_{-1} = 0$. The even Jacobi diagonal and next diagonal are

$$2k(2k+1) + c^2(A_{2k}^2 + A_{2k-1}^2), \quad c^2 A_{2k} A_{2k+1}, \quad c = 18\pi.$$

The omitted diagonal lower bound is $140 \cdot 141 = 19740$. For mode zero the two bounding matrices in (279) both have inertia 0 at $\xi_0 - 1$ and 1 at $\xi_0 + 1$. For mode four their counts are 2 and 3. Every interval LDL^T pivot has a certified nonzero sign. Thus $g = 1$ is a valid separation from each ξ_j to all other exact even eigenvalues. The full residuals include the first omitted link, which dominates here. The following rounded numbers are strict upper bounds, not error estimates inferred from precision settings:

Quantity	Mode 0	Mode 4
Full residual ϵ	$5.913 \cdot 10^{-56}$	$4.796 \cdot 10^{-52}$
Angular L^2 error	$8.362 \cdot 10^{-56}$	$6.783 \cdot 10^{-52}$
Angular uniform error	$2.723 \cdot 10^{-52}$	$2.509 \cdot 10^{-48}$

The approximate physical central values exceed 1.187 and 0.716, respectively, by interval comparison with their errors, fixing the true sign convention $h_{j,3}(0) > 0$. The error budget gives

$$\eta < 1.725 \cdot 10^{-48}, \quad |\hat{h}^\circ(0)| < 2.611 \cdot 10^{-39}, \quad U < 3.368 \cdot 10^{-37}.$$

The exact moment repair is bounded by $\|\psi\|_\infty \leq 129$; no uncertified evaluation of its defining integral ratio is inserted.

The polynomial Fourier coefficients are also evaluated with enclosures. If $\hat{h}^\circ(v) = \sum_r d_r(v/\lambda)^{2r}$ and $t_n = 2\pi n/L$, the translated coefficients of its inversion average are exactly

$$\hat{c}_n = \frac{1}{\sqrt{L}} \Re \sum_r d_r \frac{\sqrt{\lambda} \sum_{m=1}^9 m^{-1/2-it_n} - \lambda^{-4r-1/2} \sum_{m=1}^9 m^{2r}}{2r+1/2+it_n}. \quad (356)$$

To check all scale factors, integrate each monomial on $-\log \lambda < x < \log(\lambda/m)$ with the sole co-Poisson factor $e^{x/2}$. Translation from the centered interval contributes $(-1)^n$, which cancels $e^{\pm it_n \log \lambda}$ at the integration limits. Taking the real part performs inversion averaging. At $m = 9$ the interval has zero length, so its two terms cancel exactly; this integral endpoint convention differs from the strict trace in (284).

The interval polynomial coefficient vector differs from the frozen rational vector by less than $6.422 \cdot 10^{-40}$ in ordinary norm. Adding (283) gives a bound below $2.595 \cdot 10^{-36}$, in particular the slightly looser strict threshold $2.60 \cdot 10^{-36}$ in Proposition 20.78. The normalized projector distance is below $4 \cdot 10^{-36}$; no division by the tiny Weil spectral gap is used in this final source replacement. The saved interval sine bound is less than 0.000215893.

Direct Legendre evaluation at 1 and $m/9$, $1 \leq m \leq 8$, together with (284), encloses B_3 . Physical evaluation of the projection is $L^{-1/2}(c_0 + 2 \sum_{n=1}^{64} c_n)$, and its propagated error is at most $\sqrt{129/L}$ times the coefficient error. Strict interval comparisons verify (285). Thus the endpoint mismatch is now proved for the exact repaired source, not just observed for a numerical approximation.

The JSON output records the input hashes, inertia counts, full residual and pointwise bounds, coefficient error, physical endpoint and relative error intervals. The earlier frozen-candidate certificate remains a separate dependency. These two source modes use the canonical regular angular realization; no closed arithmetic operator assumption is needed for this fixed-source computation. This source certificate alone supplies no omitted *Weil* Fourier-complement bound, growing-window sign, endpoint-shell graph estimate or full-strip normalization. G2 remains open.

Scale check for the explicit omitted-tail constant

The new operator-core and polynomial-endpoint arguments are analytic proofs, not extrapolations from numerical data. A small additional interval calculation evaluates the explicit constant in (172) at $\lambda = 3$. Using 256-bit Arb quadrature of the regularized zero-mode archimedean integral gives

$$17.69849 < C_3^{\text{tail}} < 17.69851, \quad \Gamma_{3,50\,000\,000} > 1.0797.$$

The supplied script `g2_deep_next/check_tail_constant.py` and its output record the complete intervals. No matrix of that size was assembled: only the scalar constant was enclosed. This bound illustrates the cost of the unsigned arithmetic estimate. It does not certify the intervening low modes, their Schur correction, or positivity of the whole window. In particular the earlier $N = 64$ certificate is not extended to the continuum by this calculation.

Sharper tail certificate and ordinary residual scope

The script `g2_signed_tail/check_sharp_tail_constants.py` evaluates the exact elementary constants of Proposition 20.21 with 256-bit Arb intervals. Its output gives

$$\Gamma_{3,256}^{\text{all}} > 0.0148329, \quad \Gamma_{3,512}^{\text{all}} > 0.7085077, \quad \Gamma_{3,256}^{\text{even}} > 1.5867887.$$

The interval arithmetic certifies these scalar evaluations of analytic bounds. That v1.15 calculation did not certify a head matrix or its inverse-action correction; the complete certificate below now does so at the same physical window. The ordinary residual theorem is an analytic asymptotic estimate with no claimed effective threshold at $\lambda = 3$. It concerns the same exact source and literal Fourier projection as the endpoint theorem. An independent internal adverse review checked the norm normalization, low-output duality, high-output matrix bound, graph tail, all scale factors and the remaining lack of ground overlap.

Complete fixed-window certificate and its reproduction

Proposition 20.26 is certified by `g2_schur_directional/certify_infinite_schur.py`, using the coefficient assembly in `assembly.py` and the scalar check `certify_weighted_prime_tail.py`. The bundle contains both exact dyadic inverse-action witnesses, all positive LDL pivot enclosures,

and 768-bit and 896-bit replay reports. Its reproduction guide gives the complete commands and analytic dependencies. The frozen witnesses are unchanged between the two precision runs.

All finite output rows through $J = 4096$ are enclosed, and the moment inequality bounds every row beyond J . Thus the result is not an extrapolation from finite eigenvalues. The finite Fourier entries through $N = 64$ were also checked against the earlier independent interval integration: all 8,321 even and odd entries overlap their reference enclosures. This consistency check supplements, but does not replace, the analytic coefficient identities and series-remainder proof.

The unshifted result certifies complete fixed-window positivity. The shifted extension below also establishes ground-state ordering. Neither certificate alone supplies an endpoint comparison or a bound for unbounded windows. The separate boundary theorem now determines the continuum eigenfunction trace to be zero. Internal adversarial review checked the inverse order, dyadic freezing residual, parity factors, remote Gram and form-domain passage; this is not external peer review or proof-assistant verification. Exploratory failures of weaker sufficient bounds are preserved in the research log and are not presented as negative Weil-form results.

Shifted inertia and complete ground-overlap reproduction

Proposition 20.27 uses `g2_ground_order/certify_shifted_inertia.py` with `shift = 10-34` in the even sector and `10-36` in the odd sector. The bundle retains the signed pivot reports at 768 and 896 bits. It reuses the unchanged exact witnesses from `g2_schur_directional`; their identity is checked by hash. The script shifts both the actual logarithmic diagonal and all inverse comparison weights, recomputes the middle residual and retains the full remote moment bound. A negative lower-matrix pivot is not itself a negative direction of the complete operator: the proved trial quotient supplies that direction for the shifted even operator.

The separate `certify_trial_reuse.py` checks the exact rational trial's full-form Rayleigh quotient, constructs the finite-supported correction and imports the proved exact-source error from Proposition 20.78. Its corrected Rayleigh enclosure is contained in

$$(3.64339965699233062, 3.64339965699233064) \cdot 10^{-38}.$$

The separate scalar script also verifies the conservative constants displayed in the theorem. The complete ground-overlap conclusion uses the independently proved even-sector spectral threshold. It does not reuse the finite matrix's gap as if it were a gap for the complete operator. All reproducibility commands and unsuccessful exploratory tests are recorded separately in the guide and research log.

Analytic endpoint and complement checks

The proof of Theorem 20.30 is analytic. It does not use a numerical endpoint extrapolation. Its three gates are the exact physical kernel identity with scalar $-\log(2\pi)$, a common closed-form core, and the simultaneous L^2/L^∞ resolvent argument that makes the boundary theorem applicable. The archived independent reviews check each gate, the retained pole signs and the distinction between a restricted logarithmic Laplacian and a spectral logarithm. The script `check_log_kernel_normalization.py` independently compares the original and decomposed constant-mode archimedean diagonal at $\lambda = 1.2, 2, 3, 10$, with 90-decimal working precision. Agreement is a diagnostic only; the scalar integral in the proof establishes the normalization exactly.

The Fejér bound retains the physical $L^{-1/2}$ Fourier factor and uses a positive kernel of total mass one. It does not replace the paper's literal Fourier cutoff in any earlier formula. Proposition 20.32 explicitly shows why graph convergence does not transfer the physical trace. The complement estimate in Proposition 20.41 is an exact Hilbert-space inequality with a sharp two-dimensional

example. Its use of the rank-one source invokes the proved ordinary operator residual; its arithmetic complementary-sign hypothesis remains open. These checks close the continuum endpoint question and improve the formulation of the growing-window target, without closing G2.

Prime norm recurrence and growing-window scale

Propositions 20.43–20.44 are analytic. The archived proof and adverse review distinguish a finite-codimension norm obstruction from the still-open signed estimate. The script `check_prime_norm.py` checks the positive trial $\phi = e^{-x/2} + e^{-(L-x)/2}$ using

$$\|\phi\|_2^2 = 2(1 - \lambda^{-2}) + 2L/\lambda, \quad C_\phi(s) = 2e^{-s/2}(1 - e^{-(L-s)}) + 2(L-s)\lambda^{-1} \cosh(s/2).$$

Thus its prime Rayleigh quotient is $2 \sum_m w_m C_\phi(\log m) / \|\phi\|_2^2$. Independent piecewise physical quadrature at $\lambda = 2, 3$ agrees with the autocorrelation formula to below 10^{-50} at 60-decimal precision. At $\lambda = 300$ the trial quotient divided by λ is approximately 1.026889; this is illustrative, not a norm certificate or an asymptotic proof.

The same script tests phase recurrence at $\lambda = 2$. Modulation by Fourier index 381074 has normalized P_{64} head norm about $9.52 \cdot 10^{-13}$, while its prime Rayleigh quotient differs from the unmodulated value by about $4.7 \cdot 10^{-14}$. These floating-point checks illustrate the analytic mechanism only. They give no negative complete-Weil direction, because the archimedean logarithmic energy grows with the modulation frequency.

A signed energy certificate at $\lambda = 4$

The directory `g2_weighted_signed` in the cumulative bundle contains the proof, internal adverse review, exact dyadic witnesses and interval reports for Proposition 20.48. `assembly_general.py` encloses the actual digamma/trigamma series with explicit exponential remainders. The separate pole diagonal is retained. `certify_weighted_tail.py` evaluates the verified-solve lower matrix, including every row up to 4096 and the infinite moment remainder, and verifies every Gershgorin row after an exact dyadic congruence. The saved solve and congruence witnesses are reused at higher precision; their origin is not a proof assumption.

`certify_weighted_counterexample.py` verifies the finite dyadic positive direction disproving the two-sided norm contraction. `weighted_pilot.py` is only a floating-point exploration: its small negative values at larger windows are unresolved numerical roundoff, not negative Weil directions. The analytic arguments for weighted compactness, non-Schatten behavior, fixed-window norm decay and ordinary-error scaling do not rely on these computations. The primary-source comparison with [25] is contextual; no external certificate or conjectured decay law is imported.

Structured low-mode inverse bounds and their scope

The low-mode investigation retains the exact $\lambda = 4$ matrix, E_{16} and the previously certified complete tail. The scripts `certify_low_schur.py` and `structured_low_schur.py` evaluate the coarse and structured inverse bounds, respectively. All original arithmetic diagonals, normalized parity factors and infinite residual moment bounds remain. The saved tail witnesses are used as exact dyadics; their hashes and previously certified congruence margins are checked before reuse. The outer residual belongs to W_4 , whereas the reused inner factors belong to $W_4 - 10^{-8}D$. Inverse order justifies that distinction.

These experiments have not certified the low Schur sign. Failed lower enclosures and their exact diagnostic witnesses are kept in the research log, not asserted as negative Weil directions.

The script `certify_far_inverse.py` verifies the strict far-inverse constants at 320 and 384 bits; the proof includes the upper archimedean diagonal bound and the exact pole scale. The analytic results of Propositions 20.50 and 20.72 are proved directly above and do not rely on the success of a numerical positivity gate.

Odd complement and complete fixed-window sign

The bundle adds `g2_odd_complement`. The script `direction_complement_certificate.py` binds the complete trial ingredients to their exact dyadic witness, forms the exact K -dependent complement projection with outward arithmetic, and checks all 15 complement pivots. The script `quantify_head_error.py` tests the ordinary-energy target using the actual physical head V , rather than discarding its congruence factor. The complete residual ingredients come from `simultaneous_trial.py`; they include every row beyond $J = 65536$ through the moment majorant. Their coordinate pivots are not ordinary spectral gaps.

The script `directional_tail_repair.py` reconstructs the same frozen 8192-support directional trial at 1024 and 1280 bits and bounds its complete inverse correction. The success certificate combines the exact rational thresholds $429/1000$ and $1/1000$ to give the relative matrix margin $428/1000$. The physical head is checked both by outward dyadic arithmetic and by a separate exact rational calculation. The additional complement calculation at 896 bits reuses the previously certified 768-bit ingredient enclosures; it is not represented as a fresh evaluation of those ingredients.

The proof and implementations have been checked adversarially. Earlier unsuccessful repairs, the intermediate ordinary error 10^{-43} , and the limits of these tests are retained in the research log. No accuracy assumption for a suggested eigenvector or convergence theorem for conjugate gradients is used. This internal computer-assisted certificate does not constitute independent external verification or a growing-window theorem.

Simultaneous even certificate and window scaling

The cumulative bundle adds `g2_simultaneous`. Its main `simultaneous_trial.py` verifier reconstructs the frozen 17-column even trial and evaluates all complete residual Grams; `complete_margin.py` certifies the exact rational generalized margin. Both use the same witness at 768 and 896 bits. The optional `joint_remote_gate.py` retains one common remote Gram and its cross block. The complete inner tail certificate and its original exact witnesses are retained unchanged.

The initial odd trials at support 4096 and the unscaled even trial are retained as inconclusive experiments in the log. Those earlier trials are not sign certificates. All trial heads are checked for full rank; midpoint proposals never replace outward evaluation in a proof gate. The recorded LDL^* pivots are coordinate pivots, not ordinary spectral gaps.

`window_scaling_probe.py` freshly evaluates the analytic far-cut thresholds at 320 and 384 bits. Its separate finite Galerkin study uses windows 3, 4, 5, 6, 8, heads through λ^2 and fixed retained/test cuts 256/384. It records the finite Schur correction, dimensionless margins and conditioning. Modes beyond 384 are absent from that diagnostic; it is not an execution of the complete-tail pipeline at the larger windows. Every finite tail used in its elimination must pass its own positive interval gate. Complete window-specific inner certificates remain necessary before interpreting such measurements as bounds for G2.

Complete verification of a new finite trial

The cumulative bundle adds `g2_posterior_trial`, containing `posterior_trial.py`, its imported helper scripts, the saved remote-Gram reports, the frozen dyadic corrections and the independent internal proof/code review. The main certificate uses `posterior_witness_M4096_s16.json.gz`; its 4080 dyadic pairs define the correction on indices 17 through 4096. The original head vector and support-256 solve are unchanged.

The proposal uses 16 steps of a finite preconditioned conjugate gradient calculation at 256 bits. This is only a way of choosing coefficients. The separate verification routine reconstructs the exact frozen vector, checks its dimensions and witness hashes, and evaluates the unshifted complete residual at 768 or 896 bits. Its final gate includes the full moment remainder beyond $J = 65536$ and the old complete inner metric. It also checks nine rows by independent dense assembly and checks the expanded energy identity. No convergence theorem for the proposing iteration is needed. A separate four-step trial also passes with lower bound $2.9028 \cdot 10^{-20}$; it is not asserted to be the old degree-4 inverse polynomial. All trial histories are retained in the log.

Complete mixed-correlation certificate and head refinement

The cumulative bundle retains the exact outer and inner witnesses from the preceding revision. The new files in `g2_mixed.complete` are:

- `mixed_refinement.py`, for retained mixed vectors, complete first-polynomial upper enclosures, and explicitly separate higher-degree finite-prefix diagnostics;
- `mixed_directional_check.py`, for the complete directional lower certificate of Proposition 20.58;
- `mixed_directional_witness_J65536.json`, specifying the exact 496-component probe by integer numerators over 2^{160} ;
- the 768- and 896-bit interval reports and the independent internal adverse review of the mixed, monotone-head and probe bounds.

The inner witness SHA-256 is checked against the saved positive-tail report. The prefix inverse and its head pairings use the actual unshifted residual and consistently shifted inner factors. The certified probe norm is retained rather than rounded to one. Both remote input and remote first-polynomial output enter the error estimate. Replaying the analytic and interval inequality is the sign certificate; a small floating residual is not.

The higher-polynomial experiments are retained in the research log. Their positive finite-prefix margins do not certify the corresponding complete corrections and are not used as propositions in this paper. The strengthened head bound is proved analytically. The complete $\lambda = 4$ sign is established by the later finite-trial and complement certificates; this does not certify those separate higher-polynomial experiments.

Parity inverse refinement and its directional gate

The cumulative bundle's `g2_inverse_refinement` directory contains `refine_inverse.py`, both parity-constant interval reports, both final directional-gate reports and the internal adverse review. The constants replay at 320 and 384 bits; the directional gate replays at 768 and 896 bits. The mathematical arguments of Propositions 20.53 and 20.55 are independent of a numerical positivity gate.

The exact direction uses the 17 numerators and denominator 2^{160} from the archived `g2_low_schur` directory:

- `structured_ceiling_counterwitness.json` defines v ;

- `low_witness_even_M256_b768.json.gz` defines the dyadic X .

These archived files define the inputs, not a new rounded eigenvector. The actual matrix assembly retains its analytic archimedean-series remainder and correct logarithmic diagonal.

For consecutive positive indices, four Arb polynomial products evaluate the Toeplitz divided difference and the Hankel matrix action. The first uses the convolution kernel $1/j$ for $j \neq 0$, with zero central coefficient, applied to z_n and $b_n z_n$. The second uses reversed input coefficients and the kernel $1/(n+m)$. Thus the output is exactly

$$((U_J z)_n) = (d_n - ca_n)z_n + \sum_{m \neq n} \frac{b_n - b_m}{n - m} z_m + \sum_m \frac{b_n + b_m}{n + m} z_m$$

in the even sector, including the opposite-sign contribution on its parity diagonal. The polynomial products retain outward ball errors; a dense 25-row evaluation independently checks their indexing. Only the computational method changes. The tail beyond $J = 65536$ is still included by the complete moment bound and (250). Earlier inconclusive cutoffs are retained in the research log. No full mixed-term, matrix-head or G2 sign certificate is asserted by this directional test.

Finite trial positivity and exact-source cancellation audit

The cumulative bundle’s `g2_schur_cancellation` directory contains the exact witnesses, interval outputs and internal adverse review for Propositions 20.75 and 20.42. These use the same actual matrix assembly as the earlier certificates, with all physical scale and parity factors retained. No externally reported F-68 or F-69 certificate is imported as a premise.

`certify_finite_K.py` reconstructs K_X from its definition for $\lambda = 4$, split 16 and solve support through 256. The files `finite_K_X.even.json.gz` and `finite_K_X.odd.json.gz` contain the frozen exact dyadics. Every interval LDL^* pivot is positive in both the 768- and 896-bit replays. Conservative lower bounds for the smallest pivot are listed below; they are *not* eigenvalue lower bounds.

Sector	Trial	Smallest pivot exceeds
Even	$X = 0$	$1.4513 \cdot 10^{-13}$
Odd	$X = 0$	$8.4162 \cdot 10^{-13}$
Even	Frozen solve	$1.2207 \cdot 10^{-58}$
Odd	Frozen solve	$1.3872 \cdot 10^{-56}$

This certifies K_X , not S_∞ or the omitted inverse-residual subtraction. The local $\alpha = 0$ conclusion therefore does not contradict previously failed lower Schur enclosures.

`check_cancellation_subspace.py` uses the exact columns in `cancellation_two_mode_witness.json`, each with denominator 2^{300} . The 2×2 Gram positivity, nonzero constructed vector, actual quadratic values and all matrix row-sum bounds are verified by outward ball arithmetic. The source candidate’s hash is checked against the archived `pswf_source_certificate.json`; the existing exact-source certificate is an explicit dependency. Projection to the exact source complement is justified analytically, as in the proof, rather than by a rounded orthogonality residual.

For clarity, the approximate intermediate energies are

Contribution	Source candidate	Orthogonal trial w
Pure archimedean	-1.4324413624	-0.4195793594
Pole	$+1.4877438175$	$+0.8369031184$
Signed prime	-0.0553024551	-0.4173237590
Archimedean plus pole	$+0.0553024551$	$+0.4173237590$
Complete Weil	$5.3123817 \cdot 10^{-38}$	$2.5844134 \cdot 10^{-31}$

The displayed rounded components cannot resolve the tiny totals by subtraction; the separate full-precision interval evaluations establish the strict inequalities. Both precision replays use identical witness hashes. The prior source identification and the finite-support equality then supply the exact-source statement in the main text.

These are internal computer-assisted proofs with reproducible inputs, not external peer verification. The witness proposals are numerical; the stated sign gates and perturbation margins are interval-certified. No growing-window convergence or new G2 sign conclusion follows.

B Integration ledger for the complete previous manuscript

The table accounts for all 72 numbered theorems, lemmas, propositions and corollaries in the supplied 89-page r11 source. Numbers in the first column are historical identifiers; theorem numbering in this version is new. A retained local implication does not certify an unproved upstream input. The supplied audits were treated as arguments to check, not as authority.

All original subject areas are represented: Sections 1–19 are revised in place, the original Section 20 is rebuilt with a proved compact-source passage and restored older matrix/weighted estimates, and the original Section 21 is replaced by the consistent sampler and the rebuilt downstream analysis. The original Sections 22–23 are replaced by the criterion and goals in Section 32. Invalid proofs are withdrawn instead of being preserved as established results in an appendix.

Old	Disposition in v1.26
4.1	Pass. Three-piece arithmetic source decomposition. Support covariance and a partition of unity identify the closure of the three-source sum; no closed-range theorem is asserted.
4.2	Pass. Arithmetic projection disappears in two adjoint channels. The three orthogonality channels remove the arithmetic projections exactly.
6.1	Pass. Finite-cutoff span positivity would imply RH. Full-span positivity is sufficient but stronger than the diagonal criterion; the research target is corrected to each evaluator.
12.1	Repaired. Exact multiplicity- m leakage factorization. Off-line hypothesis restored; compact division and finite-jet density prove the missing kernel inclusion. Quantitative persistence still uses the evaluator assumption.
13.1	Pass. No generic domination between cross-scale and fixed-space defects. The finite-dimensional examples establish that generic projection geometry gives no defect domination.
14.1	Repaired. Exact generator range defect. Compact division and finite-jet density supply a proof. The range is the algebraic source-core power before closure.
15.1	Pass. Exact zero-incidence criterion. Pole removability proves zero incidence under the stated ambient Sonine division framework. The skew-extension sign is corrected nearby.
17.1	Pass. Exact diagonal contractions force orthogonality. Dense unimodular eigenvectors of a contraction force a unitary and phase orthogonality.
17.2	Conditional. Finite-mode asymptotic calibration. Retained under the explicit quantitative evaluator assumption.

Old	Disposition in v1.26 (continued)
18.1	Conditional. Exact off-line spectrum of the fixed arithmetic operator. Retained under an explicit closed Hilbert realization, domain, graph-core and spectral-completeness assumption; the construction is not supplied.
19.1	Retained; finite-core alternative added. The Riesz-window implication retains its realization hypothesis. Theorem 19.2 bypasses it on algebraic root spaces. The full-space projection now preserves coercivity and Weil transfer while annihilating source and shell; a small signed defect is still missing, and a fixed scalar Weil shift fails.
20.1	Pass. Endpoint-normalized radial bound. The fixed-order radial estimate is retained with its cited asymptotic input.
20.2	Pass. Away-boundary and exterior-energy bounds. Away-boundary and exterior-energy estimates are retained for each fixed radial order.
20.3	Pass. Uniform far field with controlled c -dependence. The radial ODE and integrable far-field perturbation argument are retained.
20.4	Pass. Endpoint-normalized radial lattice bound. The lattice summation argument and resonance control are retained.
20.5	Pass for the explicit repaired source. Fixed-order co-Poisson cross-tail. The normalized source, repaired tail, inversion-endpoint comparison and correlation estimates are proved. Theorem 20.1 now supplies the global radical/semilocal-domain passage; no source assumption remains.
20.6	Pass. Quantitative endpoint-jump bound. The normalized source-tail bounds and the PNT remainder prove the rate; both translated-tail terms are retained.
20.7	Pass. Inversion symmetrization preserves the G1 envelope. Inversion preserves the chosen test class and transfers the proved envelope.
20.8	Pass for Sobolev pairings. Theorem 20.1 closes the remaining domain input. This does not supply a growing-block operator-norm estimate.
20.9	Pass. Arithmetic BV structure and absence of a periodic boundary jump. The finite arithmetic sum has BV regularity and matching periodic endpoint representatives.
20.10	Pass. Fixed-cutoff boundary-preserving Galerkin approximation. Fixed-cutoff BV Fourier/form approximation is retained, with the logarithmic form weight and endpoint convergence distinguished.
20.11	Pass. Quantitative and cofinal finite-section transfer of G1. The finite-band approximation transfers the proved continuum estimate; its eventual threshold need not be economical.
20.12	Pass. Zero-independent high-low decay of the Weil matrix. The off-diagonal matrix estimate is retained and supplemented by the separately restored diagonal bound.
20.13	Pass. Explicit BV rate for the fixed-band Galerkin defect. BV coefficients and the off-diagonal estimate give the summable high-low tail.
20.14	Pass. Quantitative endpoint recovery by de la Vallée-Poussin filtering. The near/far endpoint convolution estimate is retained.
20.15	Pass. Filtered BV transfer with explicit boundary control. The filtered transfer uses the now-proved continuum G1; zero local Lipschitz constant is handled without division.
20.16	Pass. High-frequency boundary repair has unavoidable square-root derivative cost. Weighted Cauchy-Schwarz gives the derivative cost; a general trace obstruction is added for the new sampler.
20.17	Pass in the fixed-band sense. Constructive zero-independent finite G1 closure. The exact-boundary construction transfers the now-proved continuum estimate with the stated deterministic cutoffs. No economical cutoff or growing-block uniformity is inferred.
20.18	Repaired. Form-Schur lower bound. The Schur error parameters are nonnegative. Rayleigh calibration and operator-domain overlap are separated from weak G1.
21.1	Retained. One-sided off-line Weil isotropy. Global one-sided isotropy is proved in the corrected convention within the centered transfer theorem.
21.2	Replaced. Off-line semilocal form transfer. The full-scale raw transfer is withdrawn. Centered strip transfer retains jumps; the explicit endpoint repair preserves its form limit.

Old	Disposition in v1.26 (continued)
21.3	Rebuilt. G2b under the off-line contradiction hypothesis. Restricted coercivity follows for the corrected sampler. A negative compact-test calibration is optional and needs its own form approximation.
21.4	Rebuilt. Restricted scalar-shift coercivity is automatic after form transfer. Positive scalar-shift coercivity follows from corrected form and ordinary Gram transfer; it is not independent positivity.
21.5	Repaired. Smooth-test continuity of the semilocal Weil pairing. The logarithmic diagonal is restored. Absolute matrix-series estimates and the weighted convolution proof replace the false uniform-diagonal bound.
21.6	Rebuilt. The proved G1 estimate applies to the full Hardy sampler. The restored weighted theorem gives weak G1 for centered Hardy samples using the now-proved source passage; the even endpoint correction is invisible to the zeroth pairing by parity.
21.7	Scoped. Finite G1 jet hierarchy on the full Hardy sampler. Sufficient smoothing controls each fixed finite derivative family. No growing-order uniformity or automatic parity preservation for all corrected jets is asserted.
21.8	Retained. Fixed residual-jet Gramians cannot supply G2 coercivity. A fixed finite Gramian of vanishing normalized residual pairings also vanishes; the fixed-family G1 input is now proved.
21.9	Rebuilt. Off-line coercivity forces a boundary-weighted sampler residual. The exact Lyapunov residual now includes the corrected graph term. Its coercive lower bound is retained.
21.10	Rebuilt. Exact sampler–prolate compatibility decomposition. The finite candidate decomposition uses the total source $e z$ plus the graph defect.
21.11	Rebuilt. Commuting-correction cancellation and the scalar-shift no-go. Commuting and scalar cancellation remain exact with the total source. The old raw-transfer asymptotic is removed.
21.12	Rebuilt. Exact noncommuting G2 excess. The noncommuting excess includes the graph defect and still requires an independent upper estimate.
21.13	Retained. W^2 benchmark: positivity moves the gap to norm transfer. The positive square and rank-four commutator are exact; form smallness does not imply norm smallness.
21.14	Repaired. Reflected-partner W^2 coercivity at arbitrary multiplicity. The reflected top-root lower bound is proved for the corrected sampler; conjugation is on the second factor. This is not full-block coercivity.
21.15	Rebuilt. A boundary-annihilating G2 test isolates the exact sampler–prolate mismatch. Boundary annihilation is an exact finite construction when its denominator is nonzero. The proposed bounded endpoint ratio is now ruled out by Corollary 28.7; bounded tests from a fixed input space lose the reflected signal by Theorem 28.8.
21.16	Withdrawn. Pole-aware radical transfer for the fixed Hardy sampler. The pole-aware transfer proof is insufficient. A replacement must bound paired errors over all zeros, including slow tails and actual cut boundaries.
21.17	Conditional only. The surviving simple-zero obstruction is a boundary commutator channel. The finite commutator decomposition survives; no nonzero beta limit is claimed.
21.18	Rebuilt. Multiplicity-robust reflected-partner block obstruction. The corrected top-Jordan relation contains an extra graph term. The cross-form lower bound survives. The original endpoint-annihilating test cannot retain it with bounded norm; a different test construction and graph estimates are required.
21.20	Scoped. Rank–two boundary traces exhaust the reflected G2 signal. Rank-two trace algebra survives, but it does not exhaust the signal without controlling the new graph pairing.
21.21	Withdrawn. Critical $L^{-1/2}$ scale of the surviving boundary trace. The critical nonzero beta limit is removed from the established chain. One former premise, a bounded ratio for the corrected resolvent, is disproved; the other graph and form obligations remain explicit.

Old	Disposition in v1.26 (continued)
21.22	Retained. Exact arithmetic realization of the surviving CCM boundary channel. The exact finite arithmetic identity is retained with physical scaling, integer index domain, and the first-slot convention.
21.23	Conditional only. G2 is an explicit finite prime/pole/archimedean forcing statement. The finite observable is explicit. Its asserted nonzero asymptotic is not established for the corrected test family.
21.24	Scoped. The present absolute PNT remainder is quantitatively insufficient for the G2 boundary channel. The absolute PNT estimate is retained as an upper-bound limitation, not a lower bound or impossibility theorem.
21.26	Retained. Hardy phase shifts preserve the global G2 signal while damping the Hardy endpoint. Global translation, Cauchy damping, and reflected cross-form invariance are retained in the fixed Fourier convention.
21.27	Withdrawn. A translated Hardy kernel recovers an absolute PNT gain at the continuum-kernel level. The periodic upper bound omitted a reflected image. The missing image and an explicit exponent comparison are now displayed.
21.28	Scoped. Reflected-pole alias obstruction to the moving Hardy phase. Threshold algebra is retained. A no-go conclusion requires relative alias remainder estimates that are not supplied.
21.30	Retained. Exact continuum normal form of the boundary-annihilated G2 test. The continuum meromorphic normal form and its residue are retained; it is not asserted to be the corrected finite endpoint-ratio limit.
21.31	Repaired. The reflected pole is forced by any cross-retaining holomorphic correction. Residue persistence is retained and second-slot scaling uses the conjugated scalar.
21.32	Repaired. Bounded Hardy filtering cannot suppress the reflected pole while retaining G2. The modulus inequality is proved; global admissibility and a nonzero endpoint factor are required for form interpretations.
21.33	Scoped. Absolute-PNT rescue by pole damping lies outside the audited G1 regime. Exponential norm cost describes this envelope-based estimate, not a necessary lower bound on actual transfer error.
21.35	Retained. Projective boundary-quotient invariance of the G1/G2 mismatch. Projective invariance is exact for a fixed endpoint-zero test. Its asymptotic consequence assumes both normalized G1 pairings vanish.
21.36	Scoped. The prolate core may be replaced by a pure boundary shell in the lower obstruction. A shell can replace the candidate in the exact paired identity; no shared nonzero G2 limit is imported.
21.37	Scoped. Moving Hardy evaluation aligns the graph functional. The moving Cauchy scalar converges to the graph functional on a fixed root space; moving-parameter uniform transfer remains open.
21.38	Retained. Candidate-adapted graph sampler: exact vector matching. Coefficientwise exact graph matching and scalar endpoint error are proved. Injectivity, norm, form and G1 compatibility are additional obligations.
21.39	Retained. Exact-radical factorization removes the zeta quotient but exposes an endpoint phase. Global Mellin divisibility and the endpoint/bulk integration-by-parts identity remain, with the Fourier-sign conversion explicit.
21.40	Scoped. Finite boundary-jet expansion of the exact matcher. The fixed-cutoff high-frequency jet expansion remains. Subtraction through zero frequency requires a prescription.
21.42	Retained. Exact periodic-resolvent formula for the adapted endpoint. The periodic ODE and parameter derivatives give the exact normalized Laplace-jet formula.
21.43	Retained. Boundary-perfect high-frequency matching has vanishing bulk mass. The shell exact matcher has asymptotically correct endpoints but vanishing ordinary mass, so it is not coercive.

Old	Disposition in v1.26 (continued)
21.45	Rebuilt. Parity projection gives a bounded noncommuting G2 benchmark. Parity coercivity uses only ordinary norm transfer and survives the corrected sampler. Its Lyapunov identity includes the graph defect.
21.48	Repaired. The exact-boundary shell is compatibility-efficient in the coercive pairing. Correct diagonal and row-sum bounds give shell weak estimates. They do not remove the commutator term in the graph pairing.
21.49	Scoped. One shell cutoff neutralizes both the G1 repair and its G2 pairing cost. Common-cutoff shell smallness is retained as an eventual-threshold argument; it controls auxiliary shell terms only.
21.50	Replaced. A joint exact-boundary G1/pairing diagonal. An explicit normalized common-cutoff selection lemma replaces ambiguous cofinality claims. It proves no missing core graph estimate.
21.51	Scoped. Scalar-shift positivity cannot remove the final defect. The reverse-triangle obstruction is retained only with its explicit full-residual and derivative-norm premises; weak G1 does not supply them.

Earlier-version material. The separate correct diagonal bound, weighted-sampler convolution estimate, arithmetic jump train, jump/continuous split, and the limited near-Slepian jump cutoff were recovered from the older 65-page source. The 32-page v10 was checked for useful earlier structure. Its unsupported exact jet Gram asymptotic, incomplete far-field comparison, and unseparated endpoint jumps were not restored. The more careful one-sided Gram bound, radial ODE argument, and endpoint-jump separation of the later source are retained.

Audit reconciliation. The supplied G1/G2 audit overstated the finite lower mechanism. The supplied Claude audit checked useful local algebra but did not settle the analytic transfers affected by the cut and diagonal errors. This revision adopts locally sound calculations, corrects their conventions and scope, and replaces the central transfer rather than treating either audit as a certification of the whole paper.

Authorship and assistance. AI tools assisted with drafting, exploratory calculations, and adversarial review. They are not mathematical authorities or sources of certification. Responsibility for the manuscript's claims and any eventual publication remains with its author. The reproducible finite checks are supplied separately. Exploratory computations are distinguished from the rigorous finite interval matrix and exact-source certificates in Appendix A. The new endpoint and operator arguments received internal adversarial review, including growing derivative orders, arithmetic threshold resonance, and identification of the physical form closure. This is not external referee verification.

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